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(Kim et al.)

(54) **MOTION VECTOR REFINEMENT FOR TEMPORALLY AMORTIZED SUPERSAMPLING**

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(71) Applicant: **Intel Corporation**, Santa Clara, CA (US)

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(73) Assignee: **Intel Corporation**, Santa Clara, CA (US)

(52) **U.S. Cl.**
CPC **G06T 3/4076** (2013.01); **G06T 1/20** (2013.01); **G06T 3/18** (2024.01); **G06T 3/4046** (2013.01); **G06T 15/503** (2013.01)

(21) Appl. No.: **19/188,633**

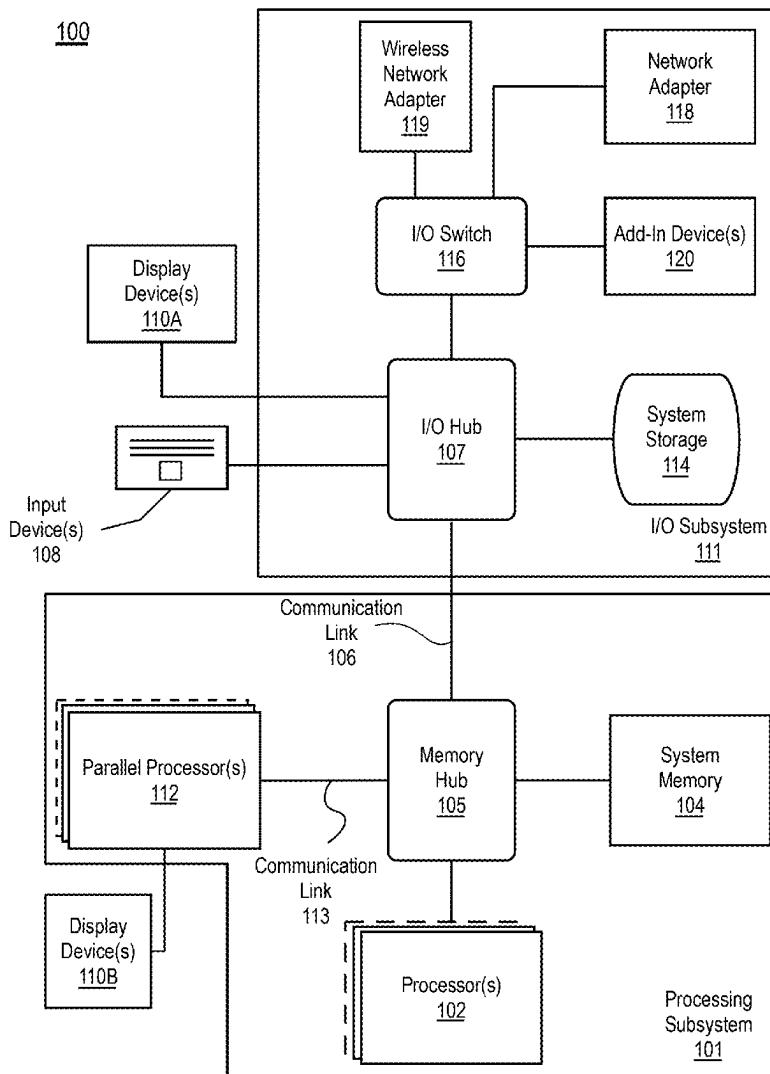
(22) Filed: **Apr. 24, 2025**

Related U.S. Application Data

(63) Continuation of application No. 17/888,253, filed on Aug. 15, 2022.

(57) **ABSTRACT**

A residual network is used to predict a set of residual motion vectors that provide additional motion data for portions of the frame for which motion vectors are not provided, such as animated textures, mirrored/reflected objects, and/or moving objects without motion information.



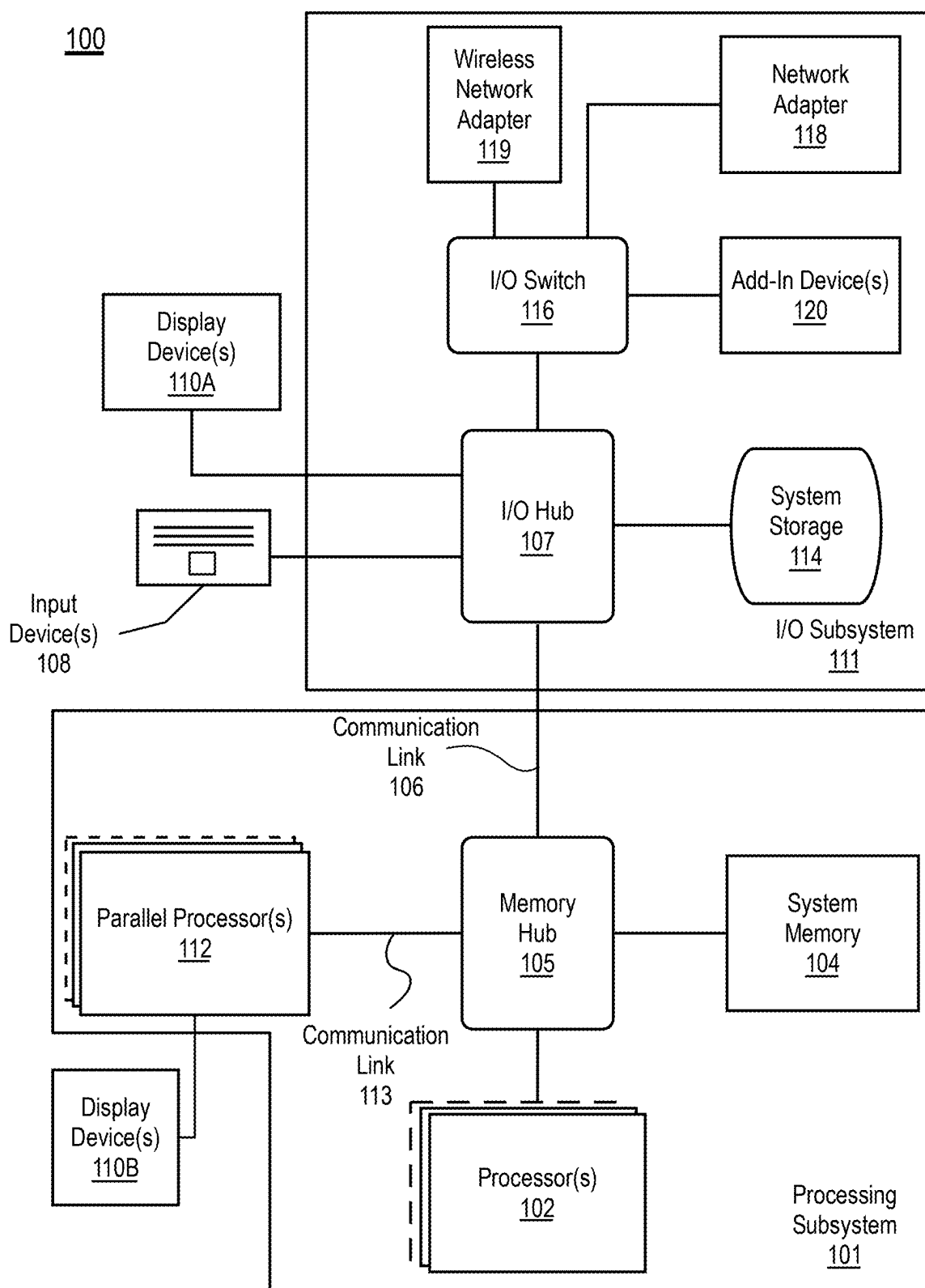


FIG. 1

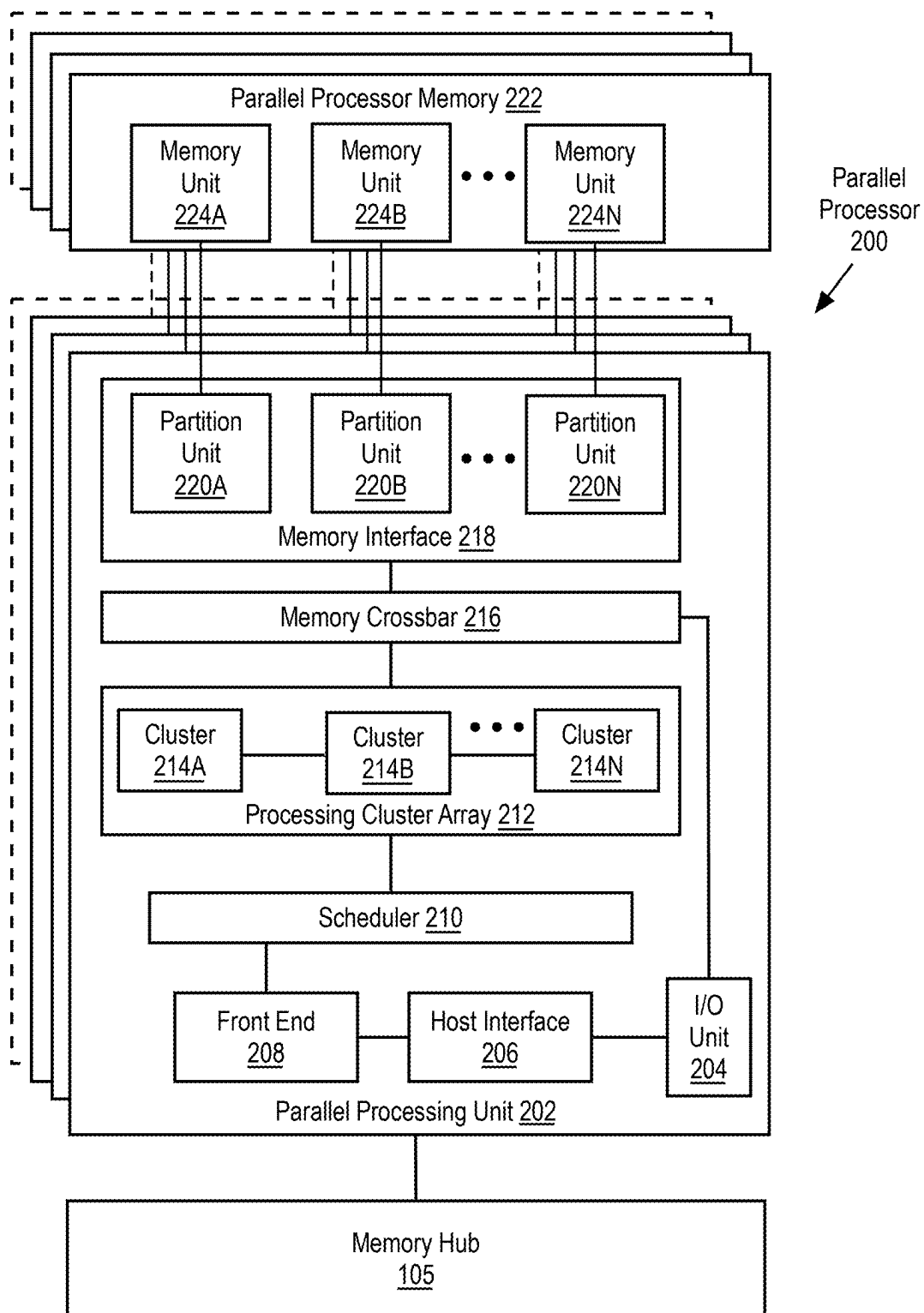
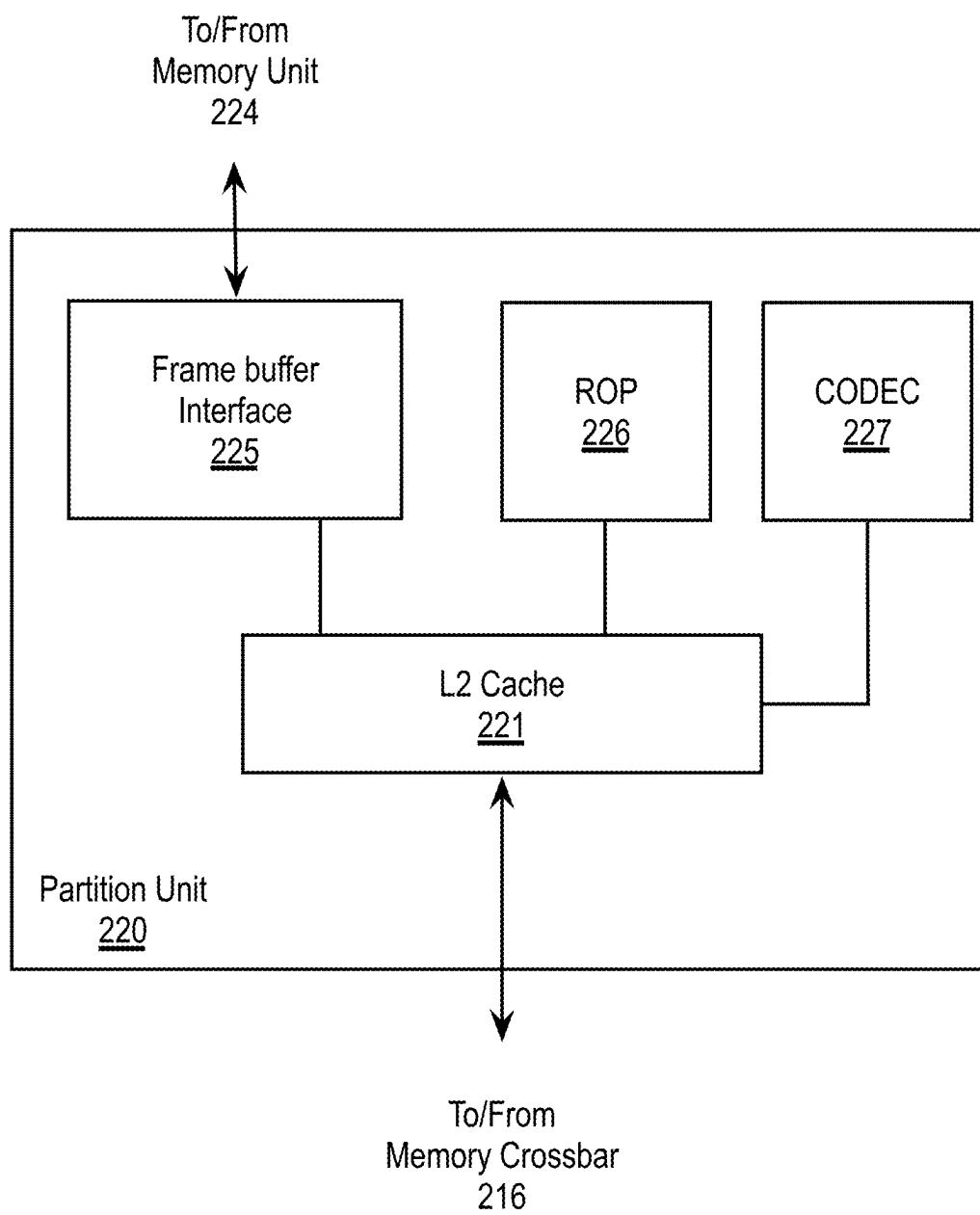


FIG. 2A

**FIG. 2B**

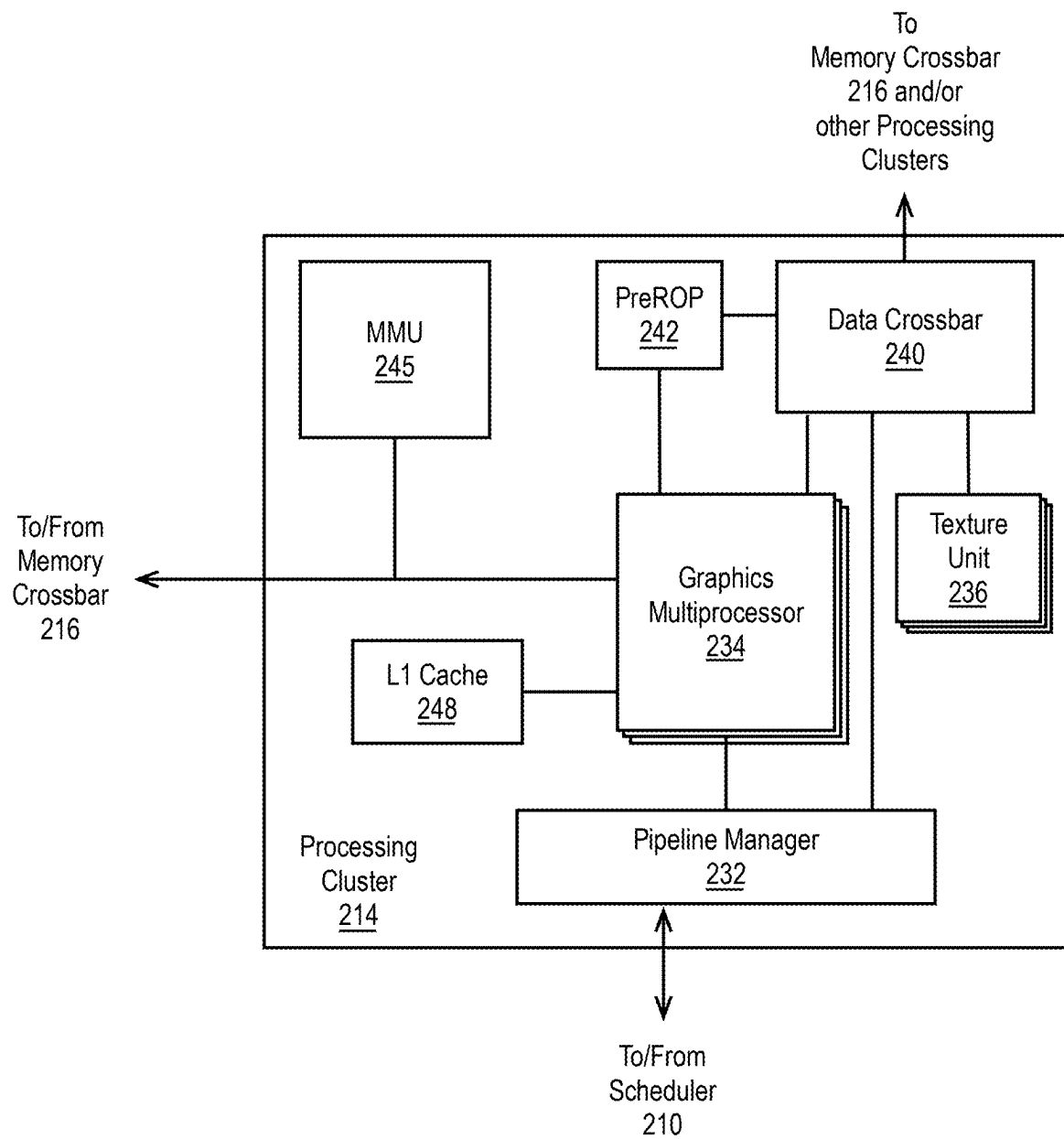


FIG. 2C

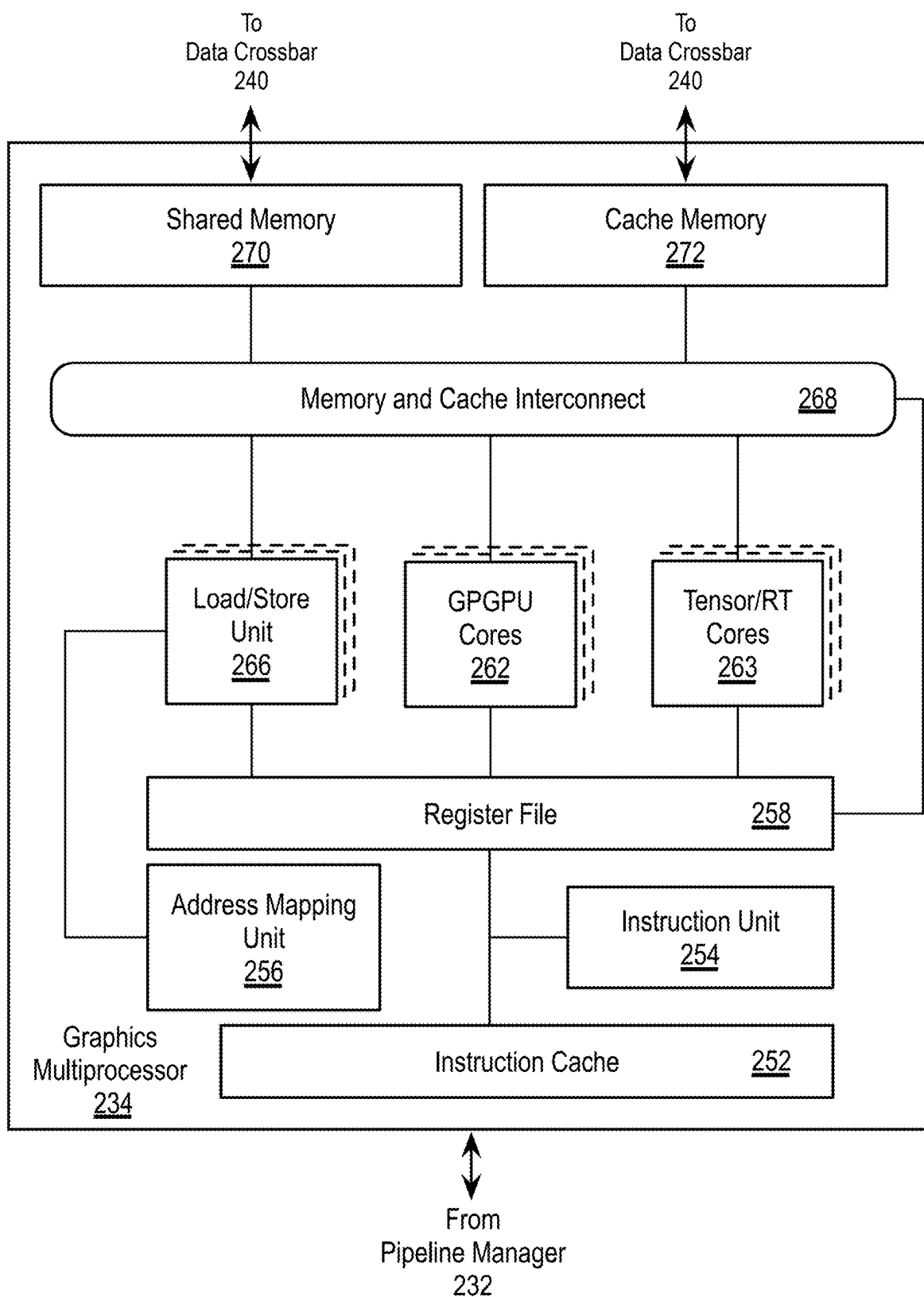
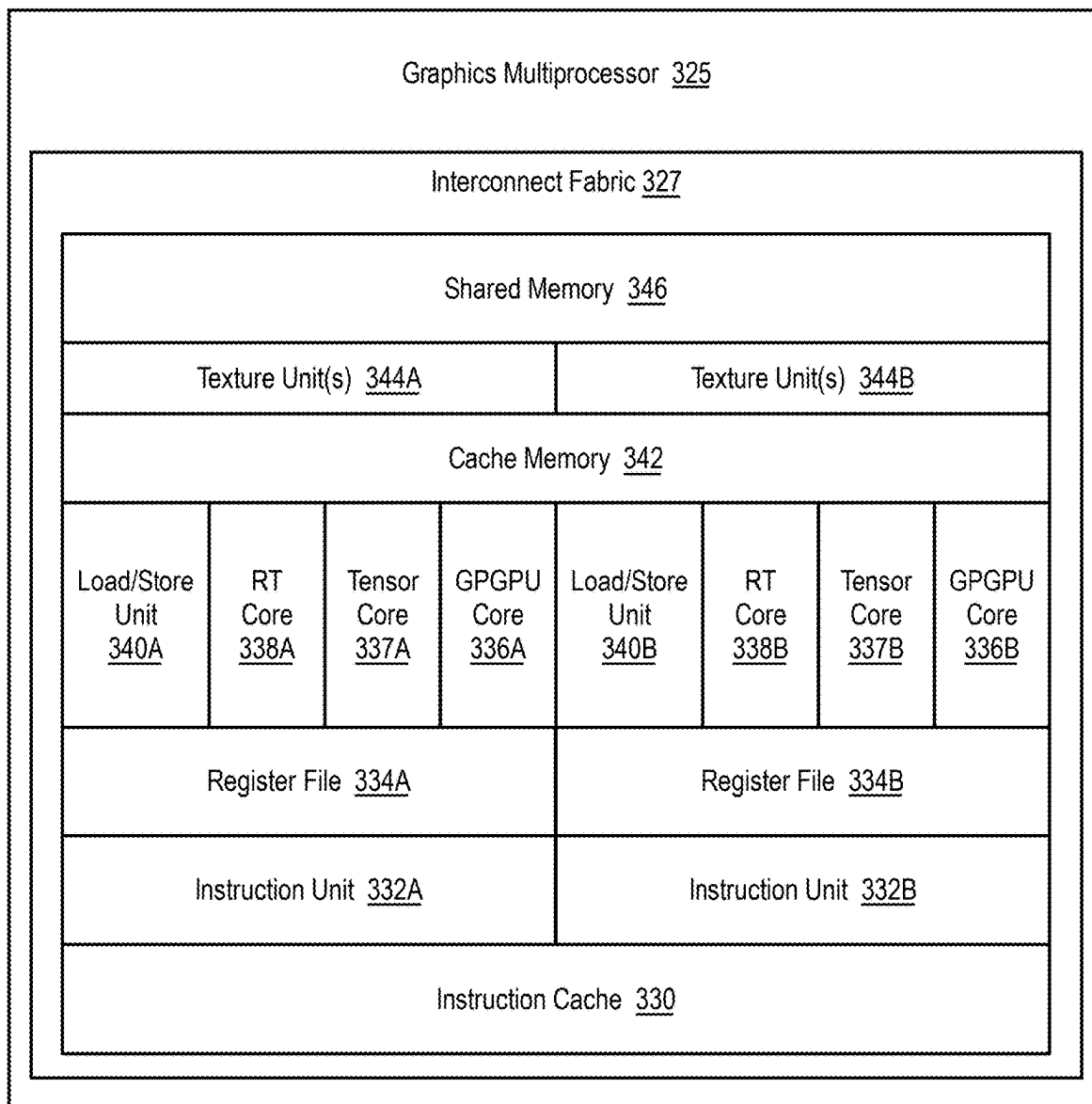
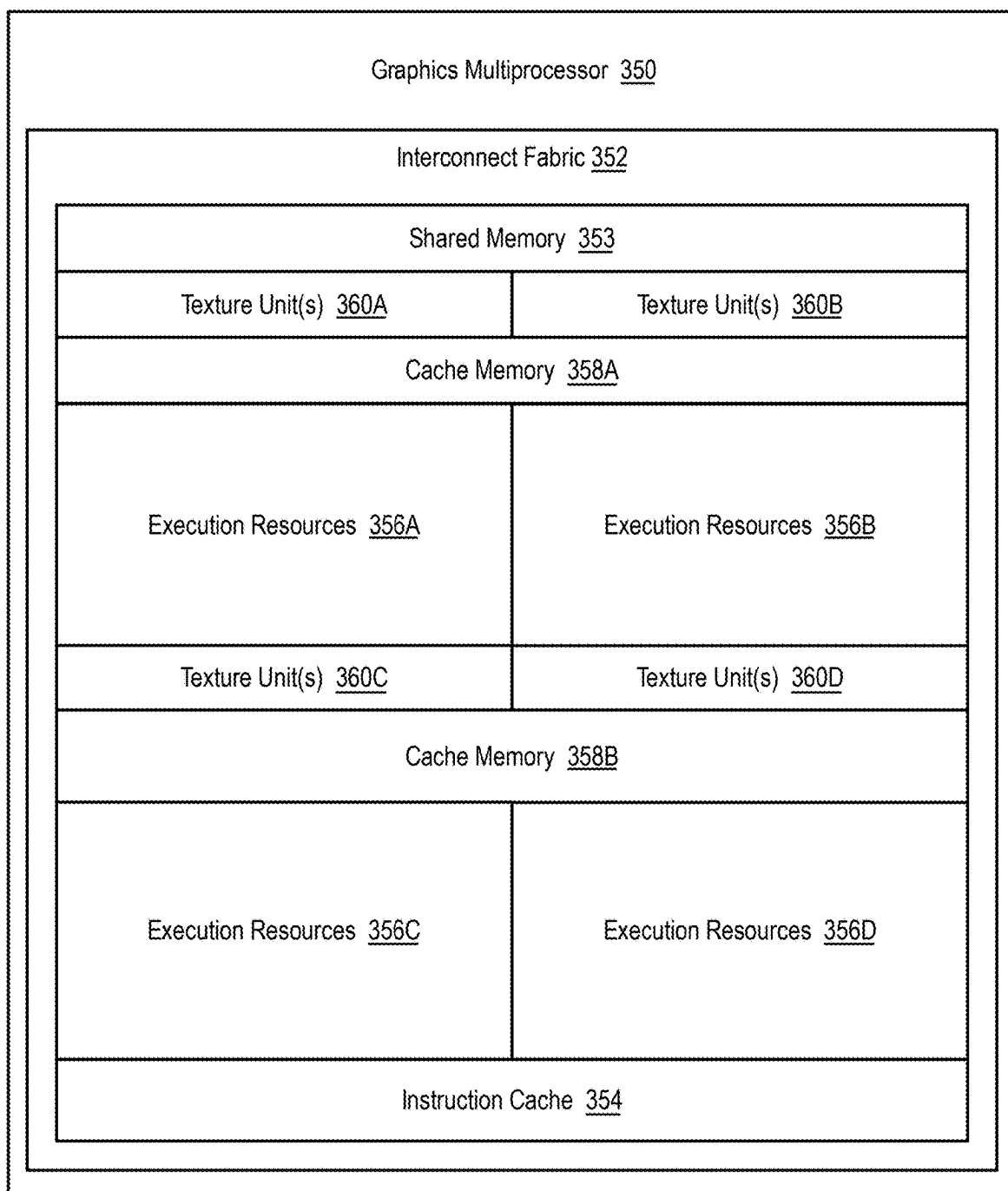


FIG. 2D

**FIG. 3A**

**FIG. 3B**

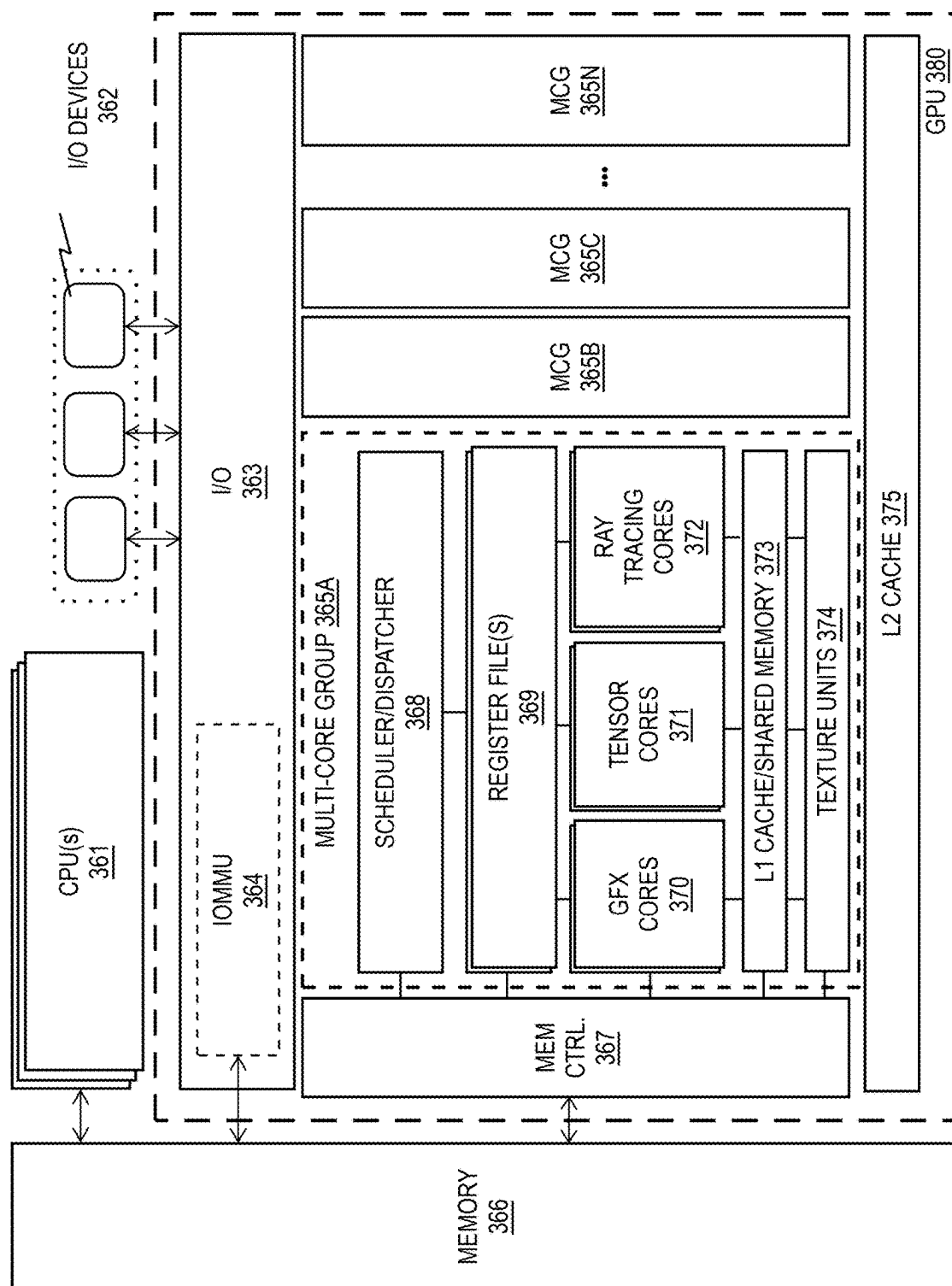


FIG. 3C

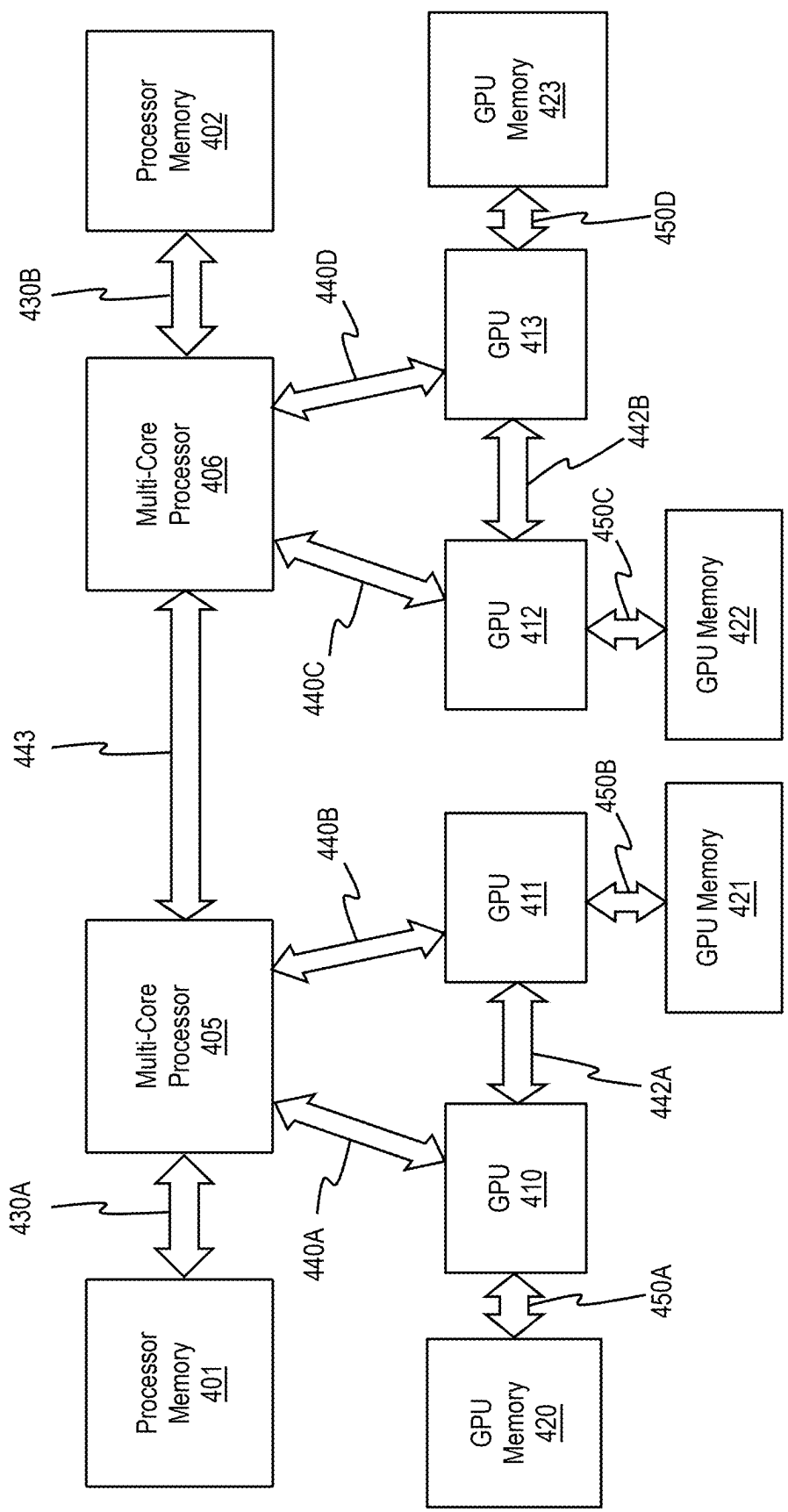


FIG. 4A

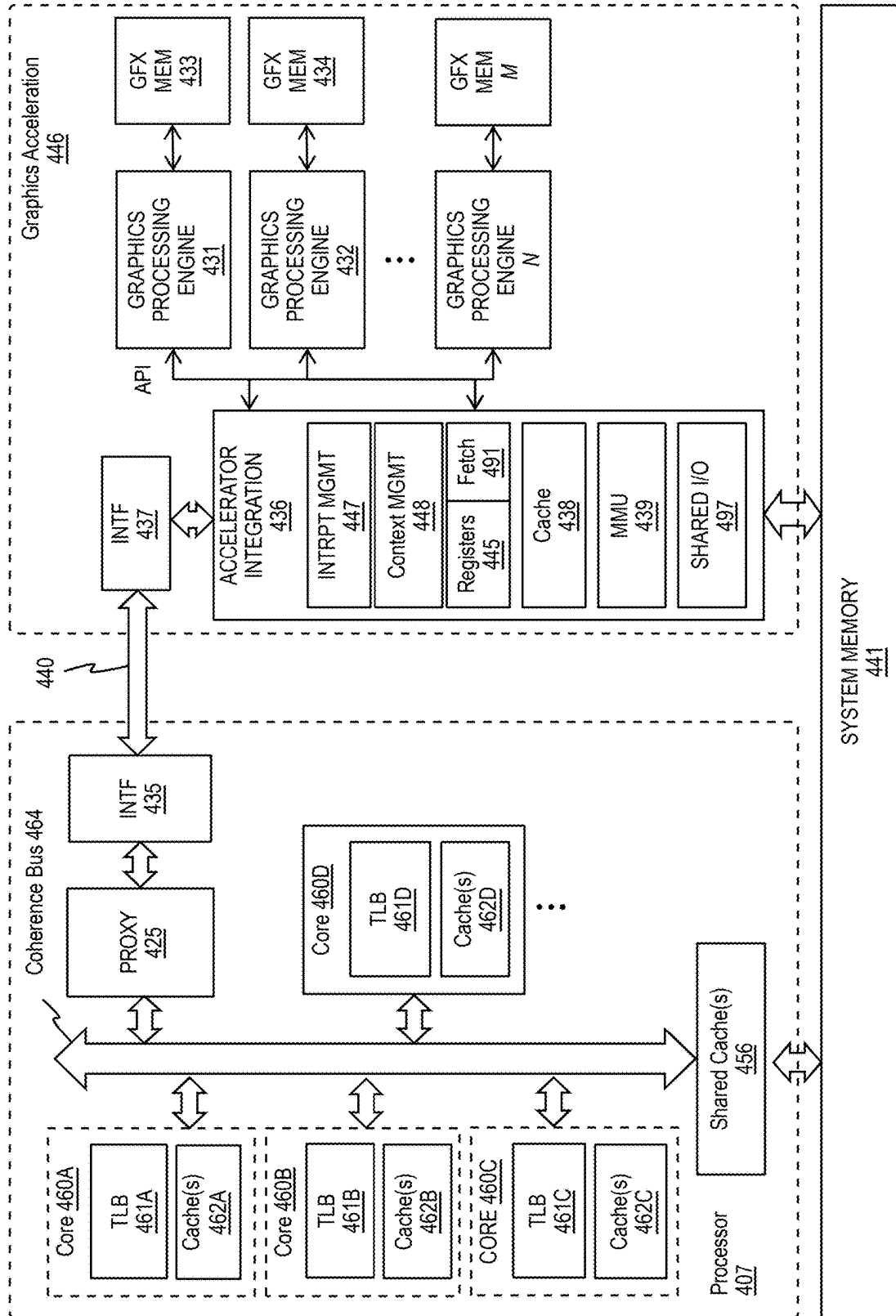


FIG. 4B

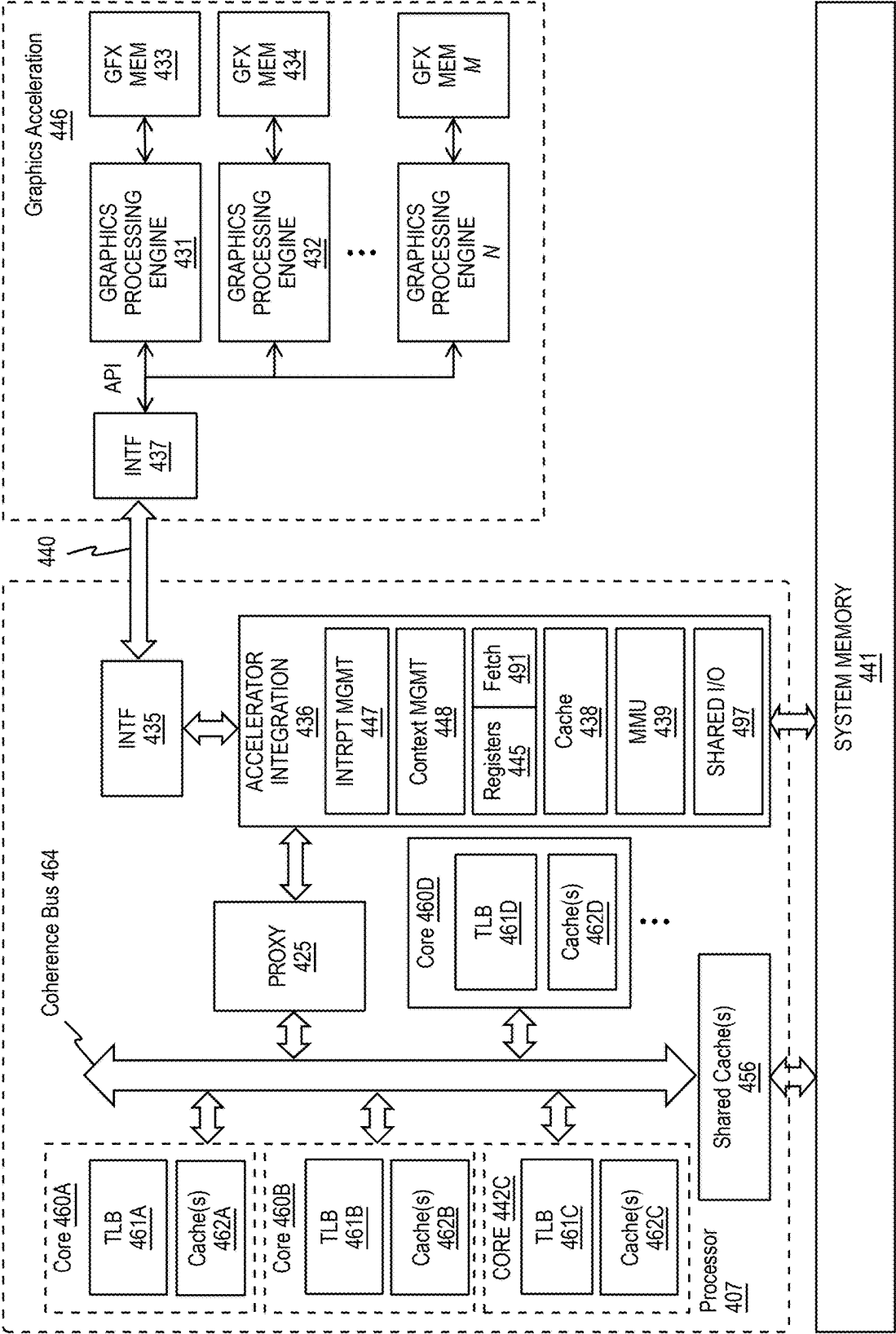


FIG. 4C

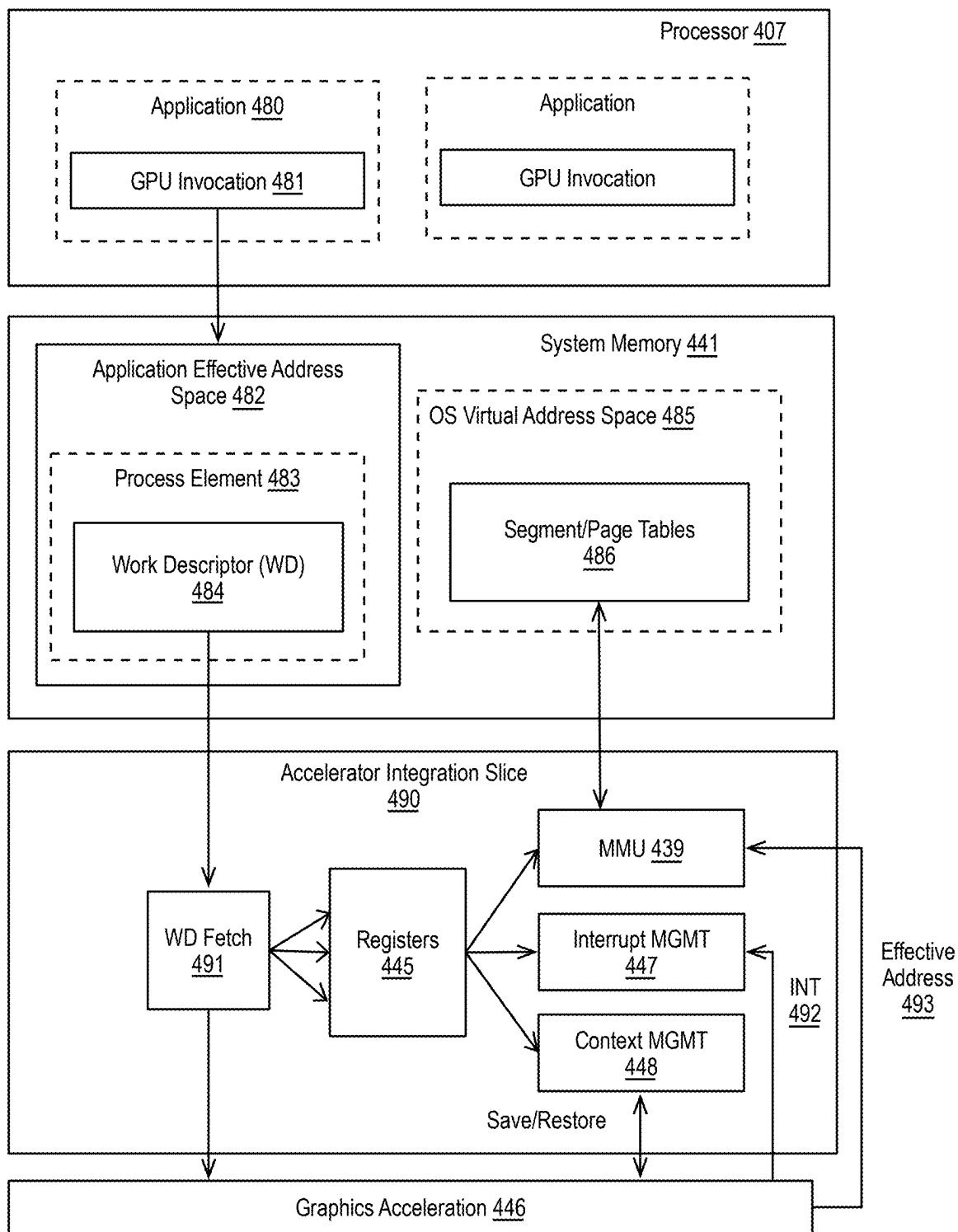


FIG. 4D

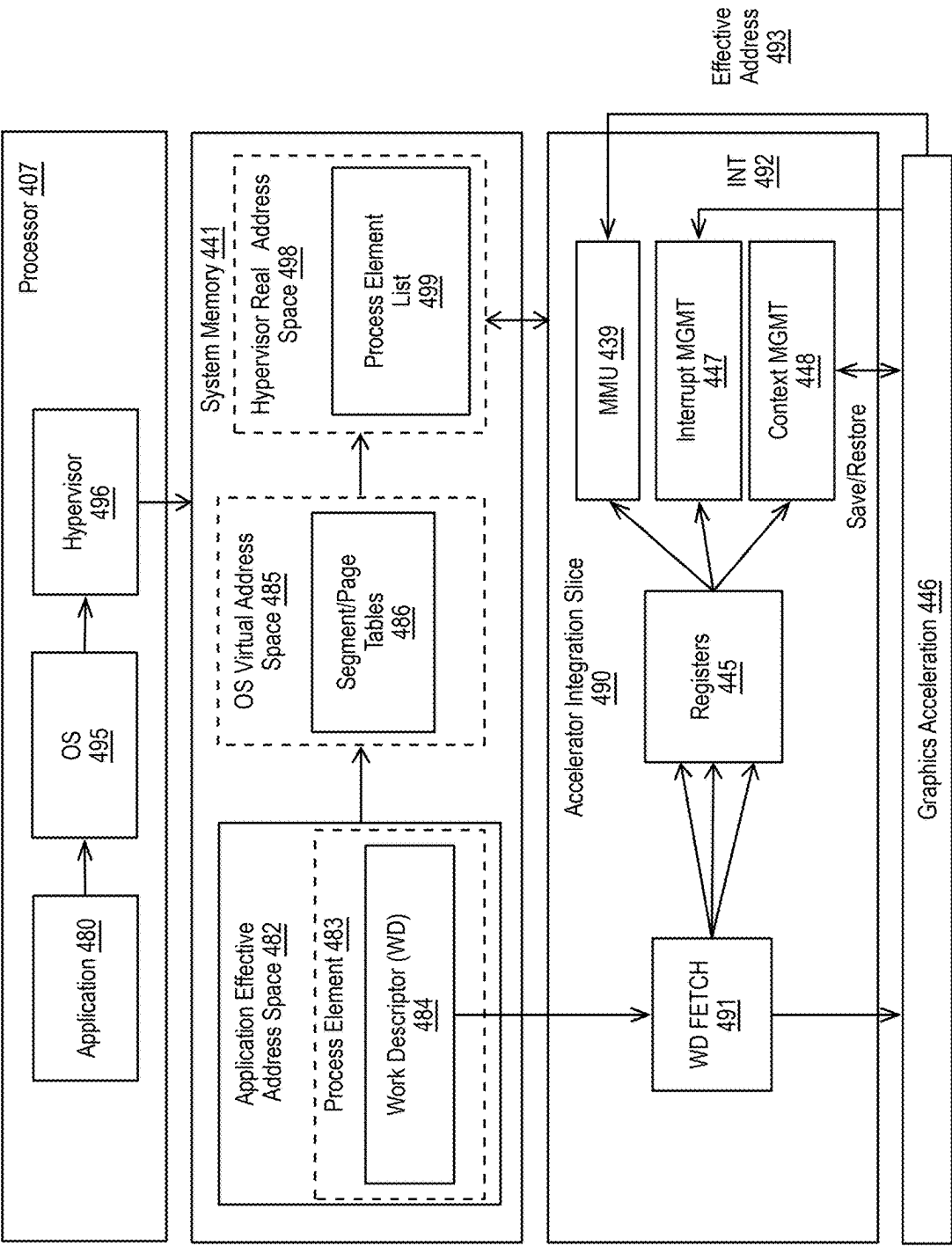


FIG. 4E

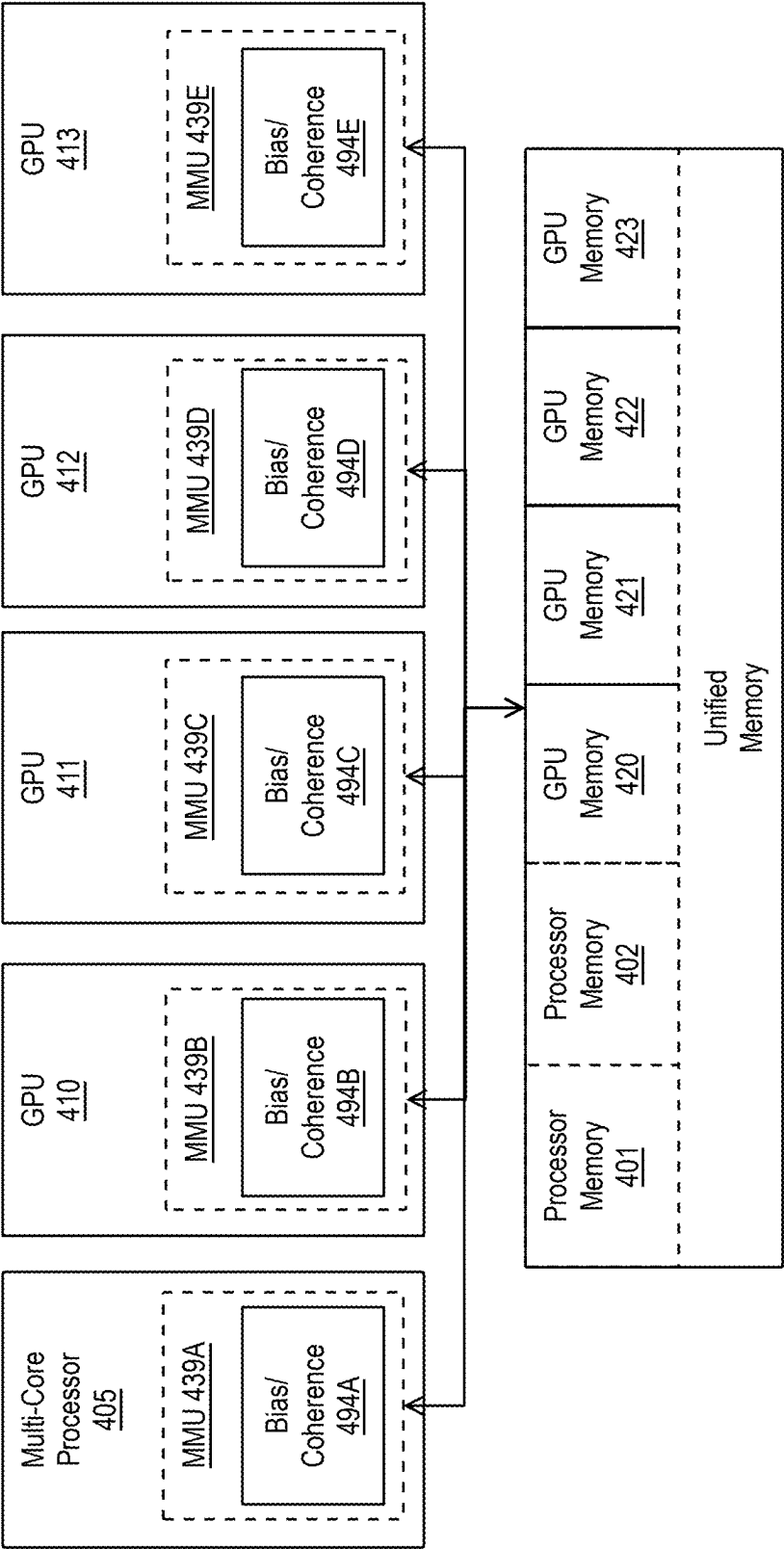
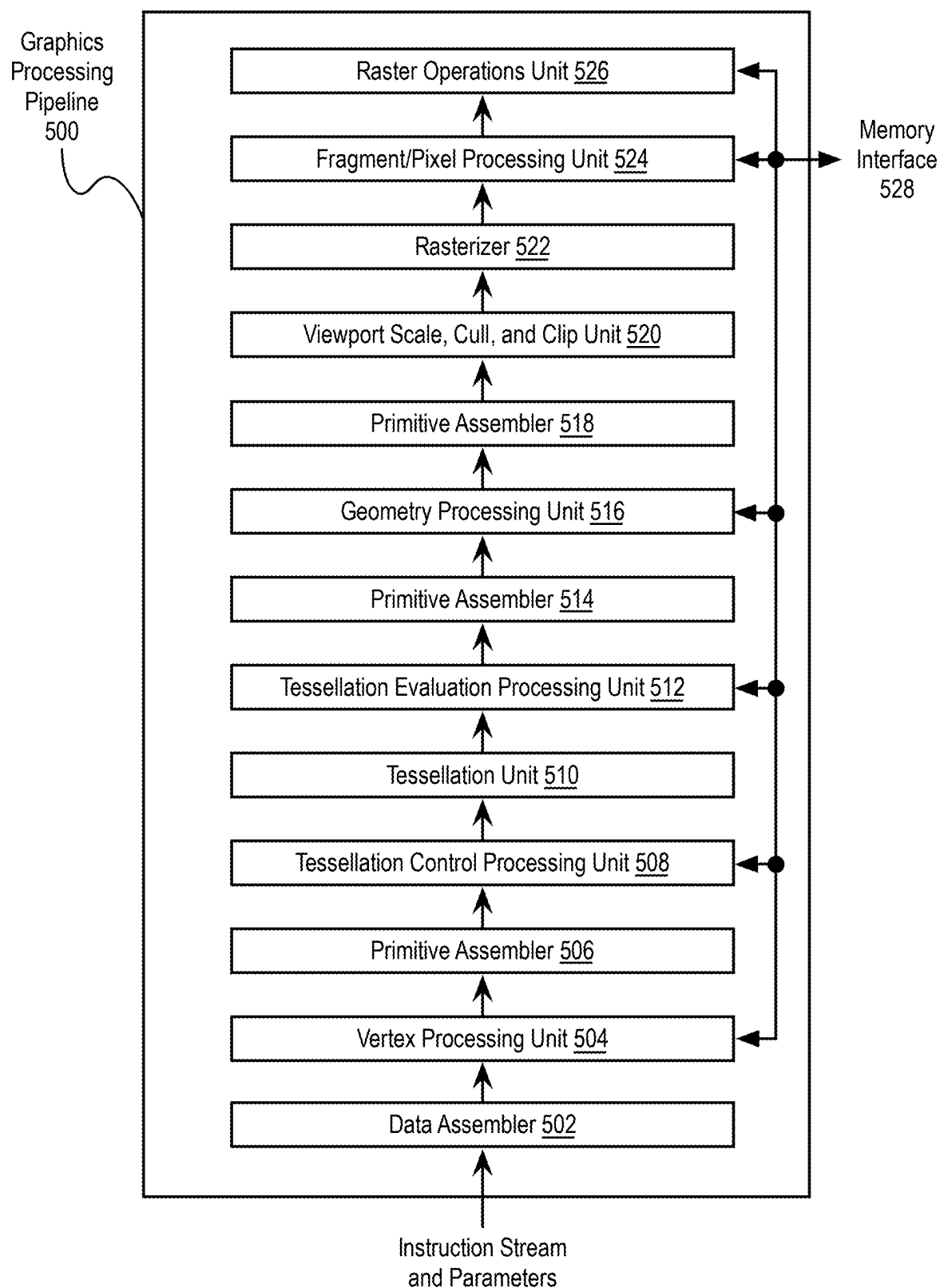


FIG. 4F

**FIG. 5**

600

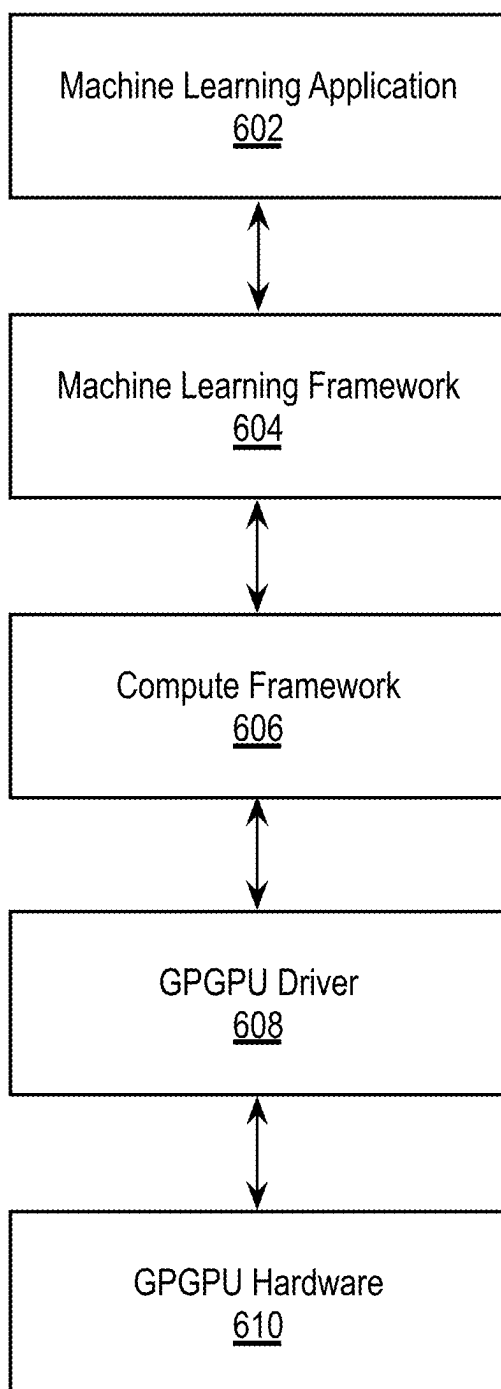


FIG. 6

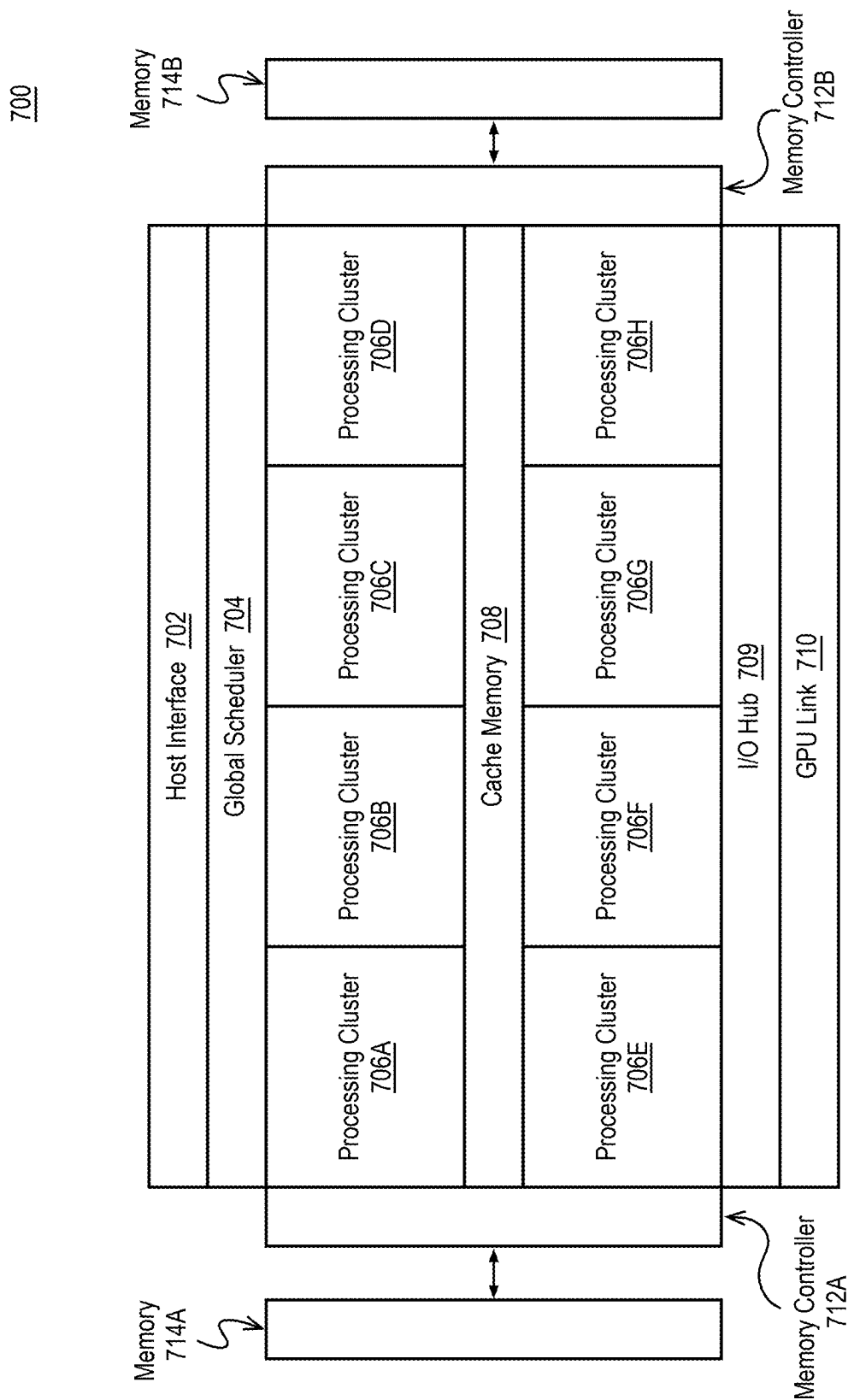
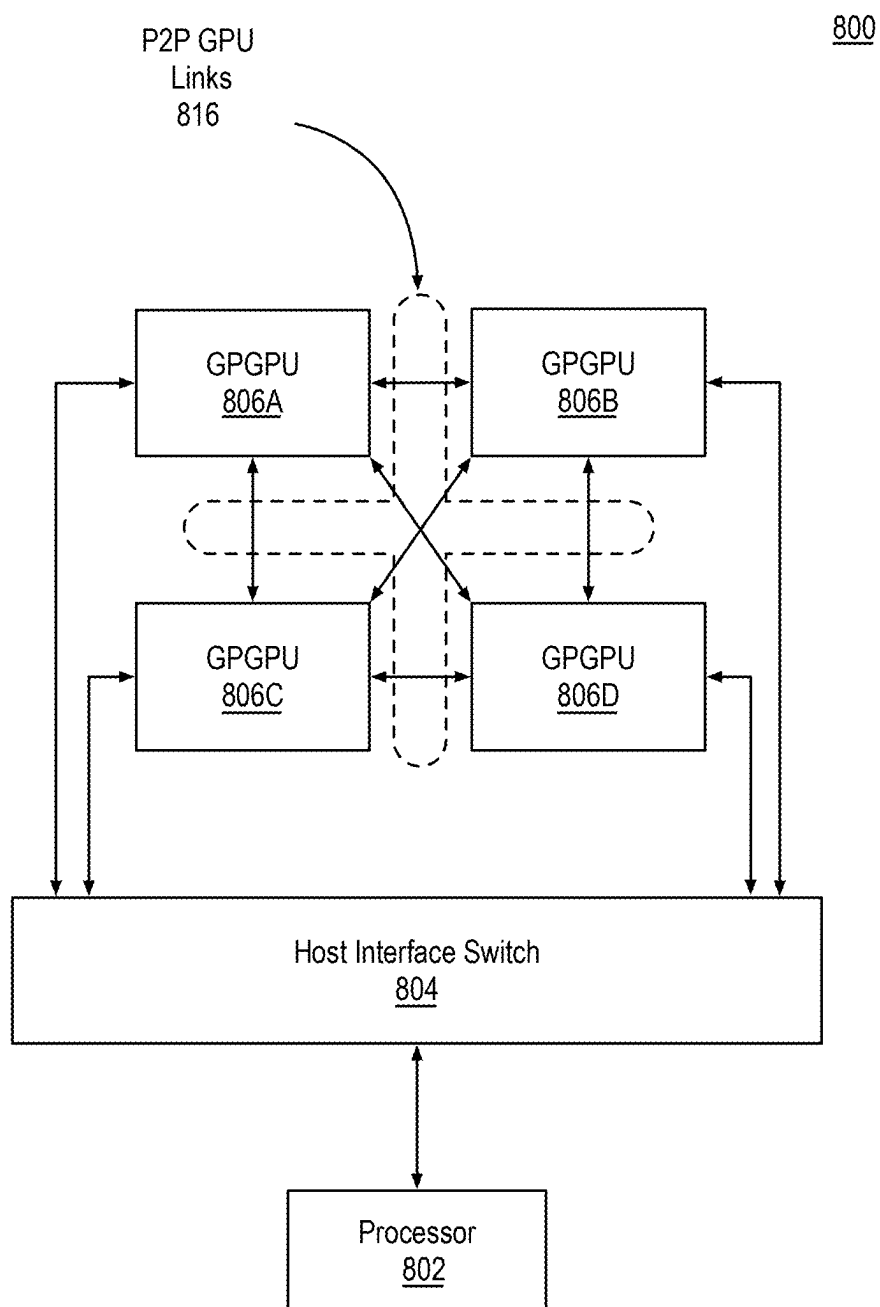


FIG. 7

**FIG. 8**

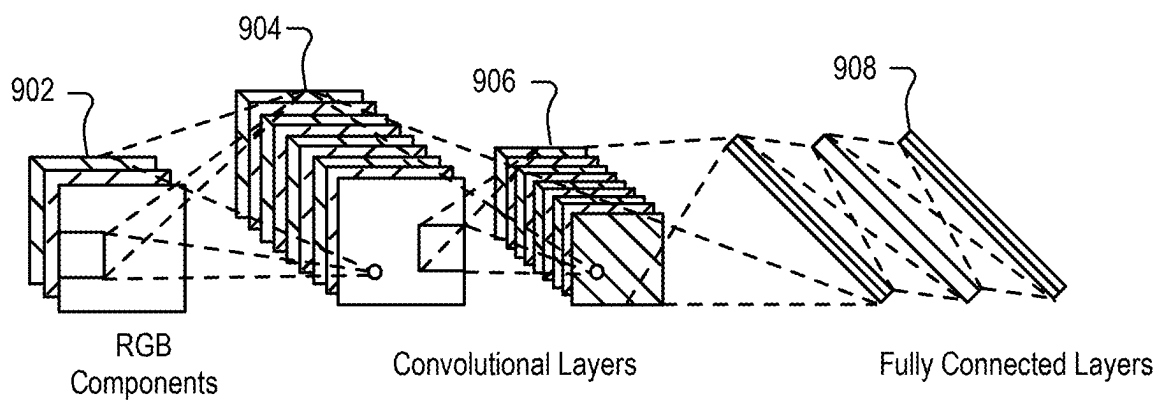


FIG. 9A

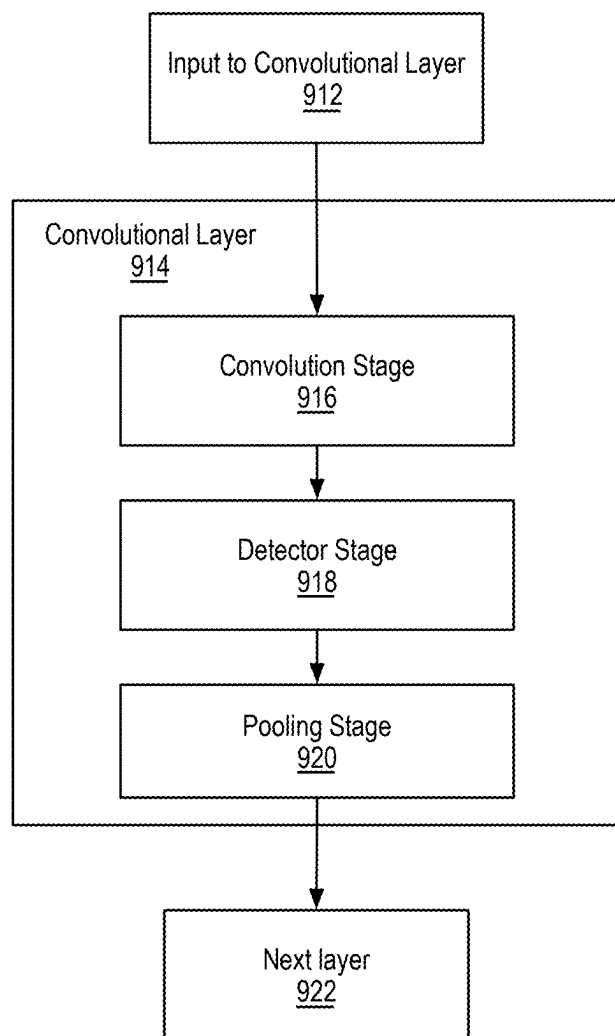


FIG. 9B

1000

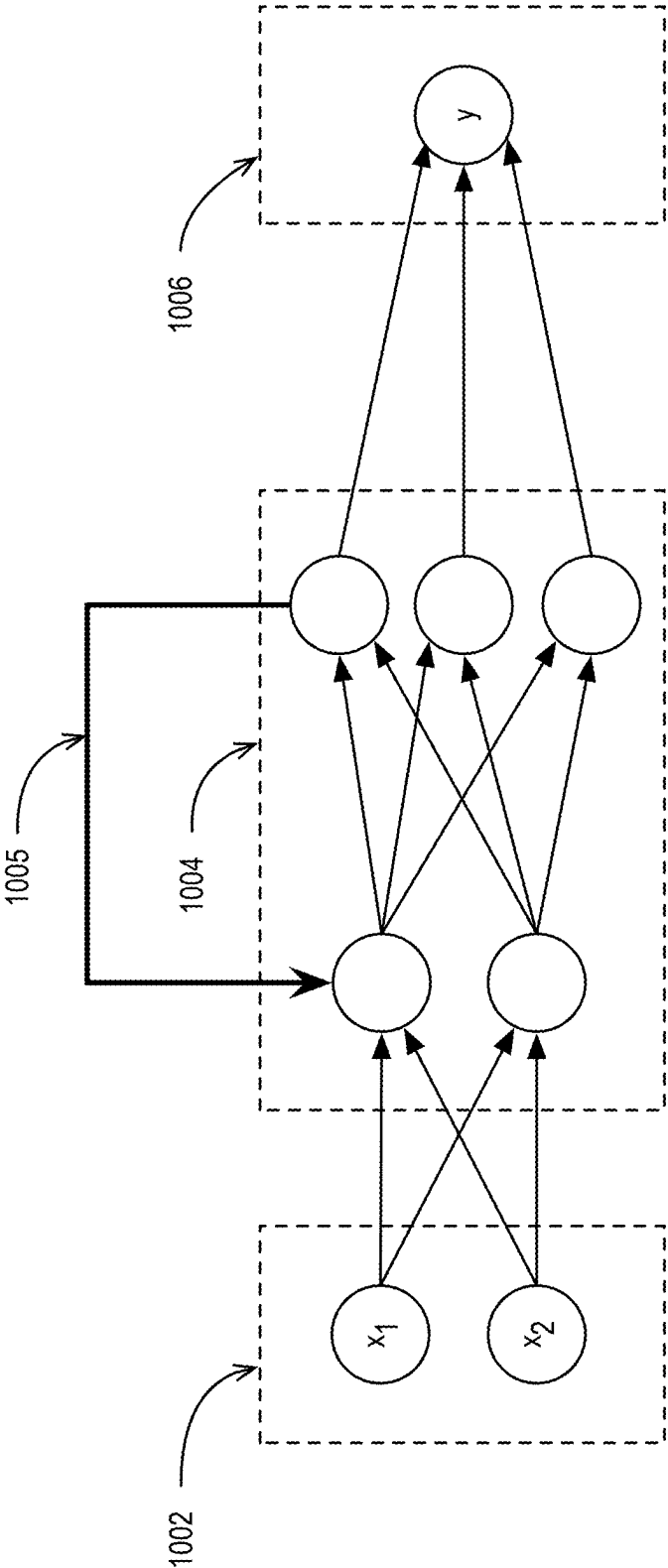


FIG. 10

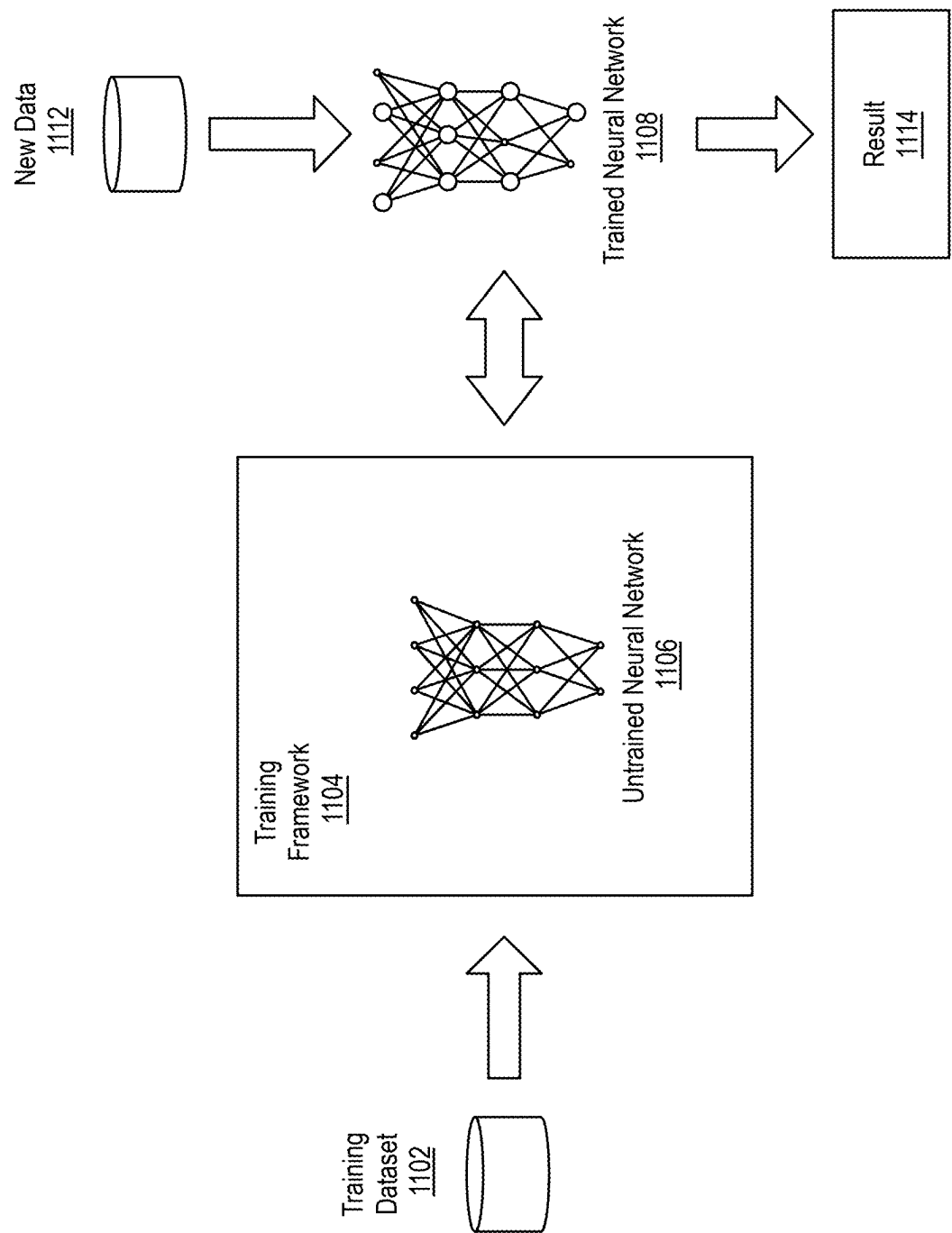


FIG. 11

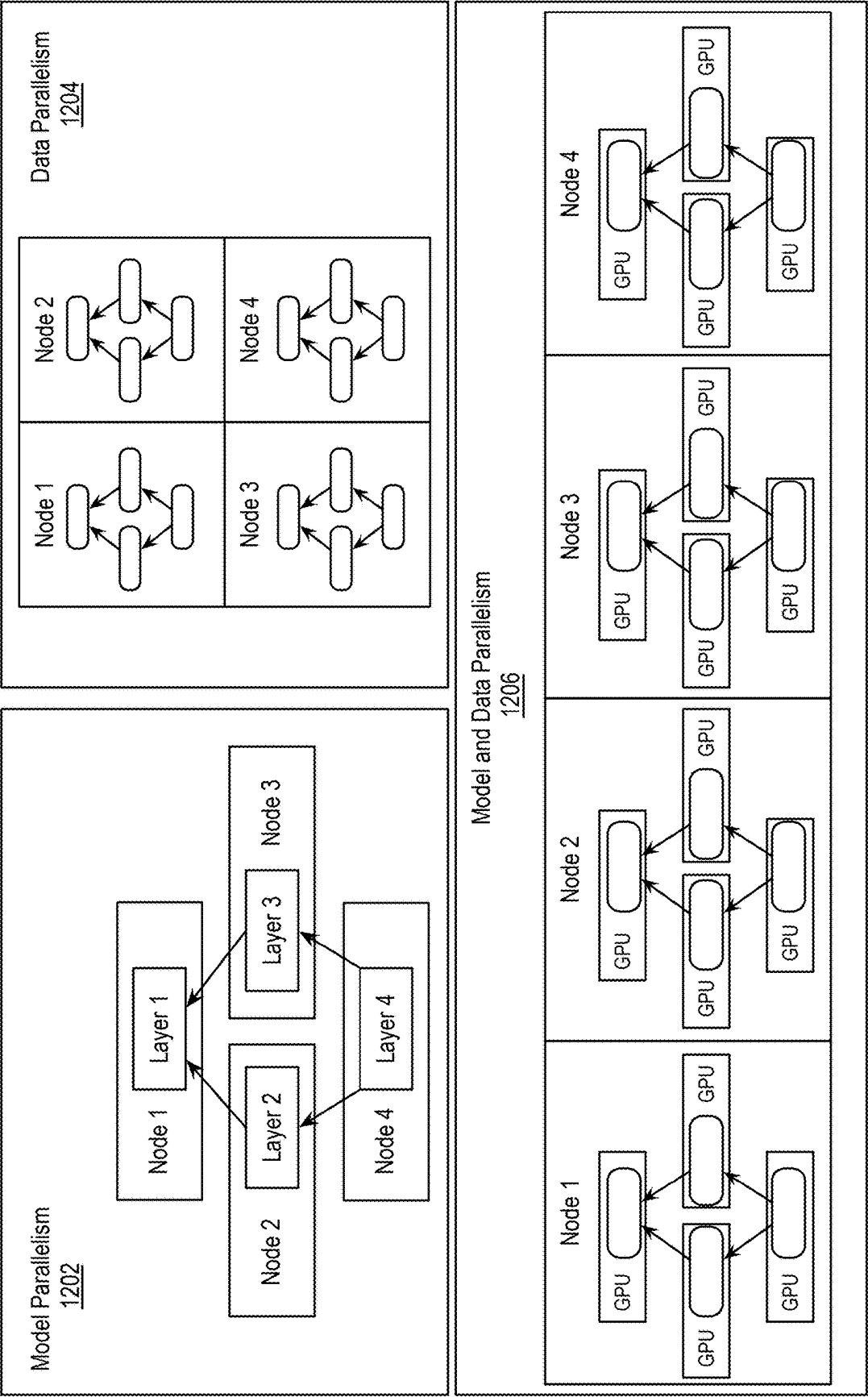


FIG. 12A

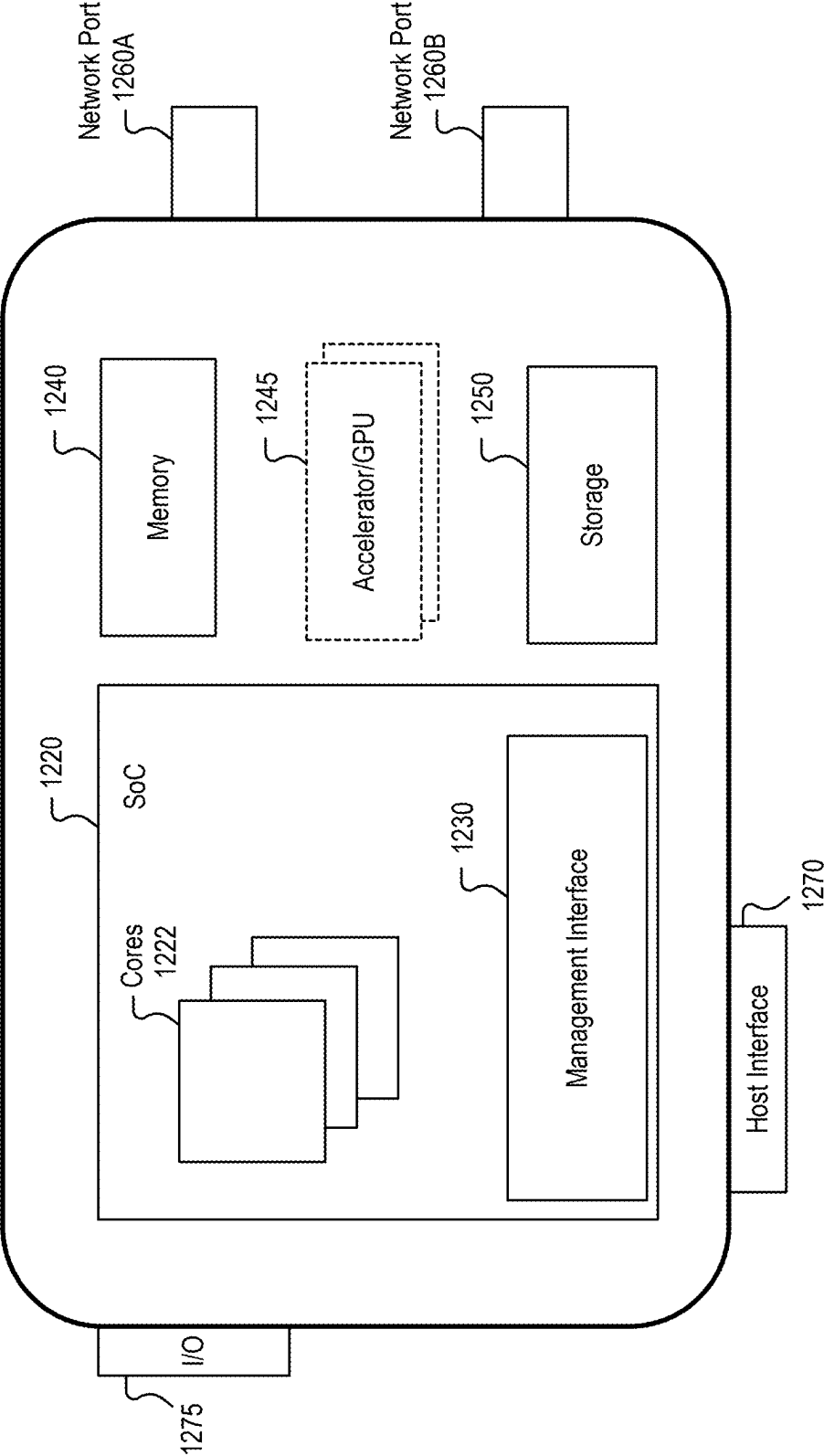


FIG. 12B

1300

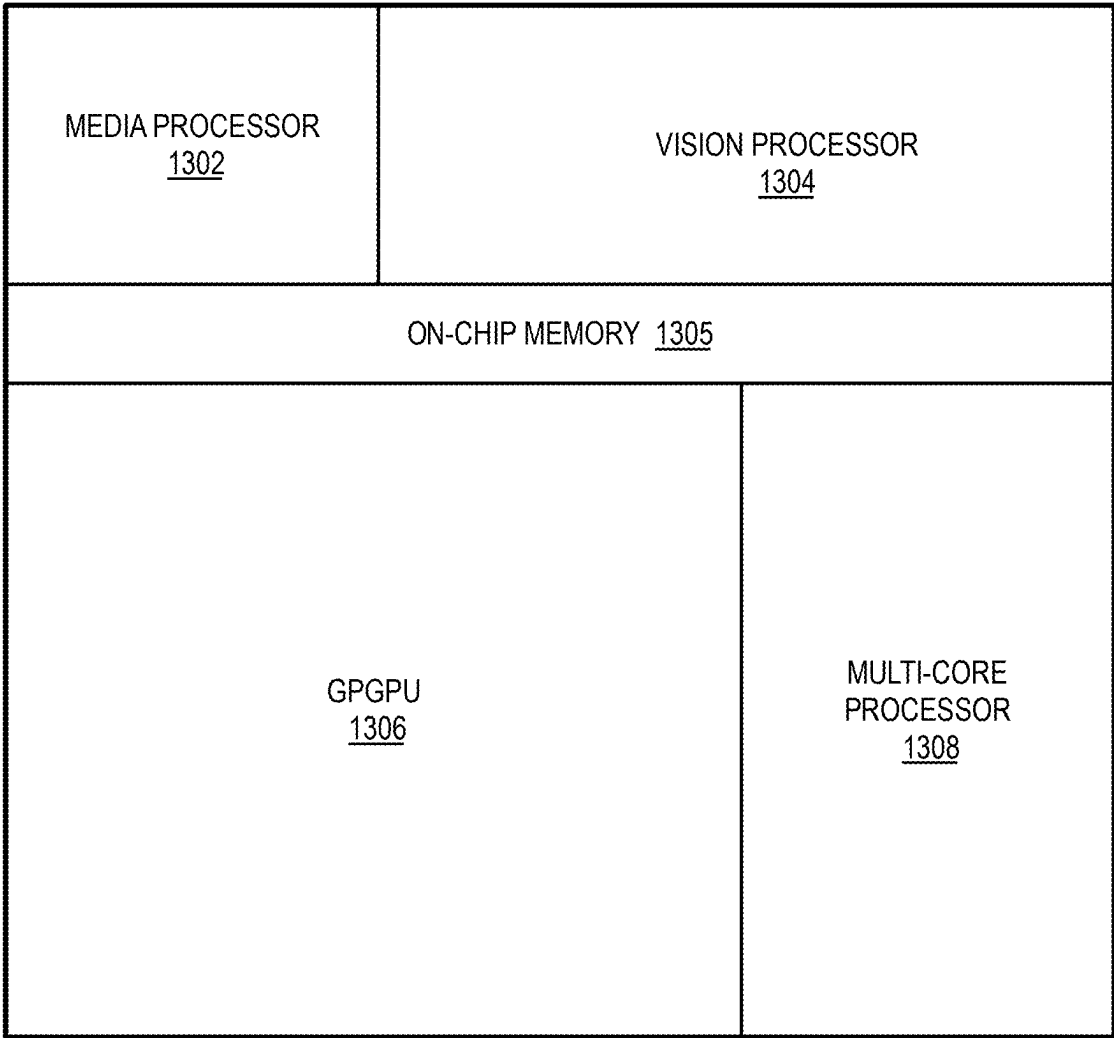


FIG. 13

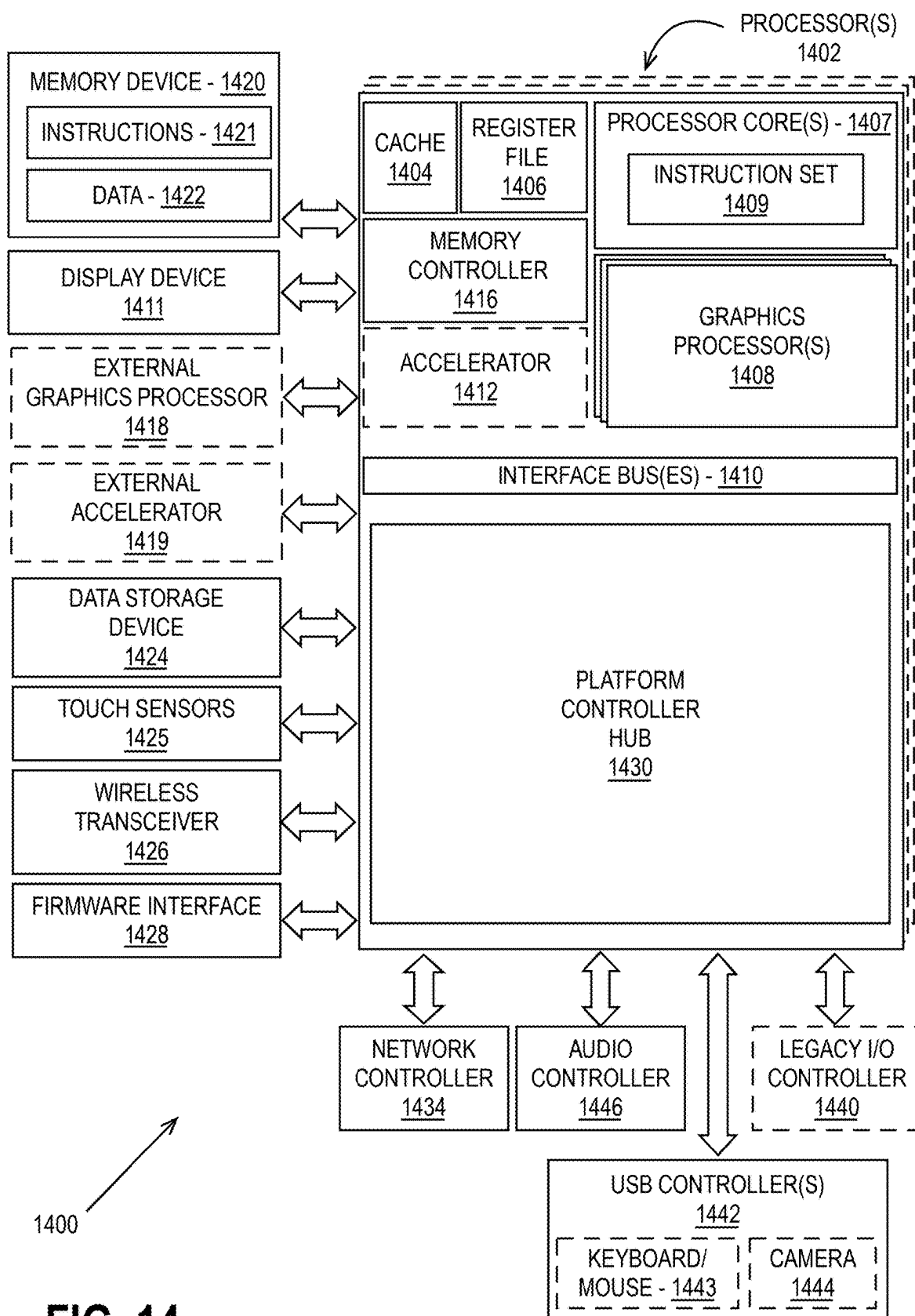


FIG. 14

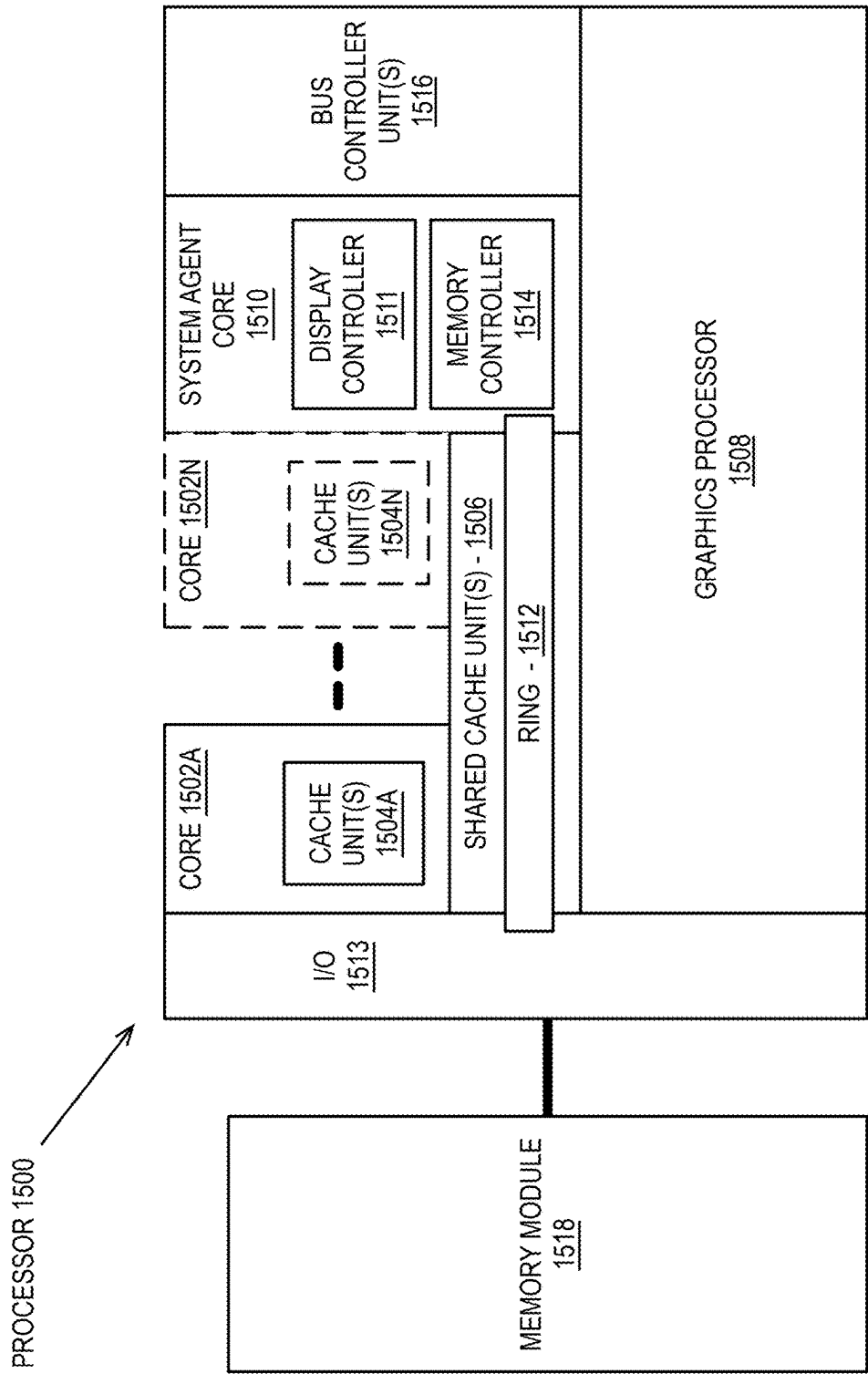


FIG. 15A

1519

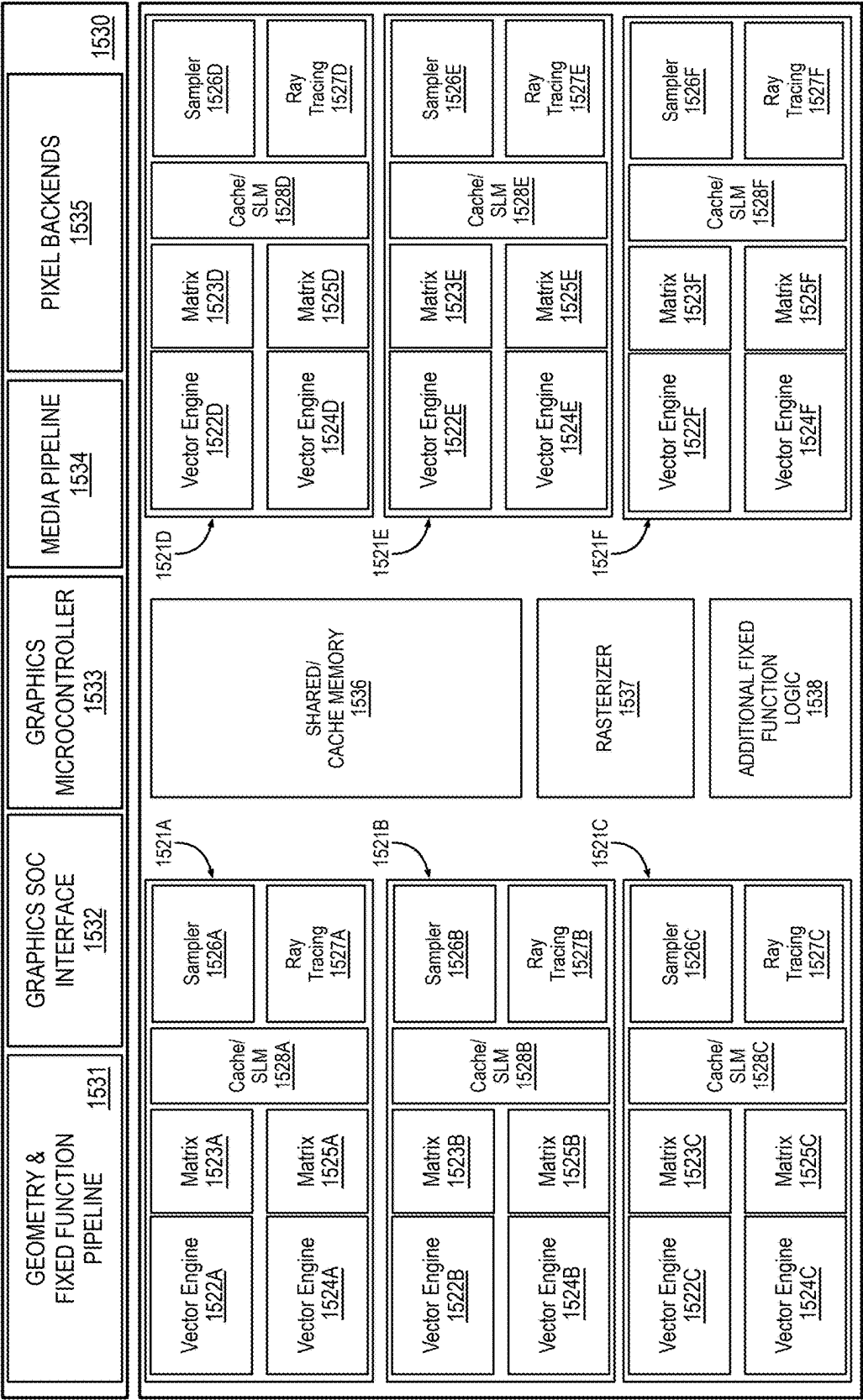


FIG. 15B

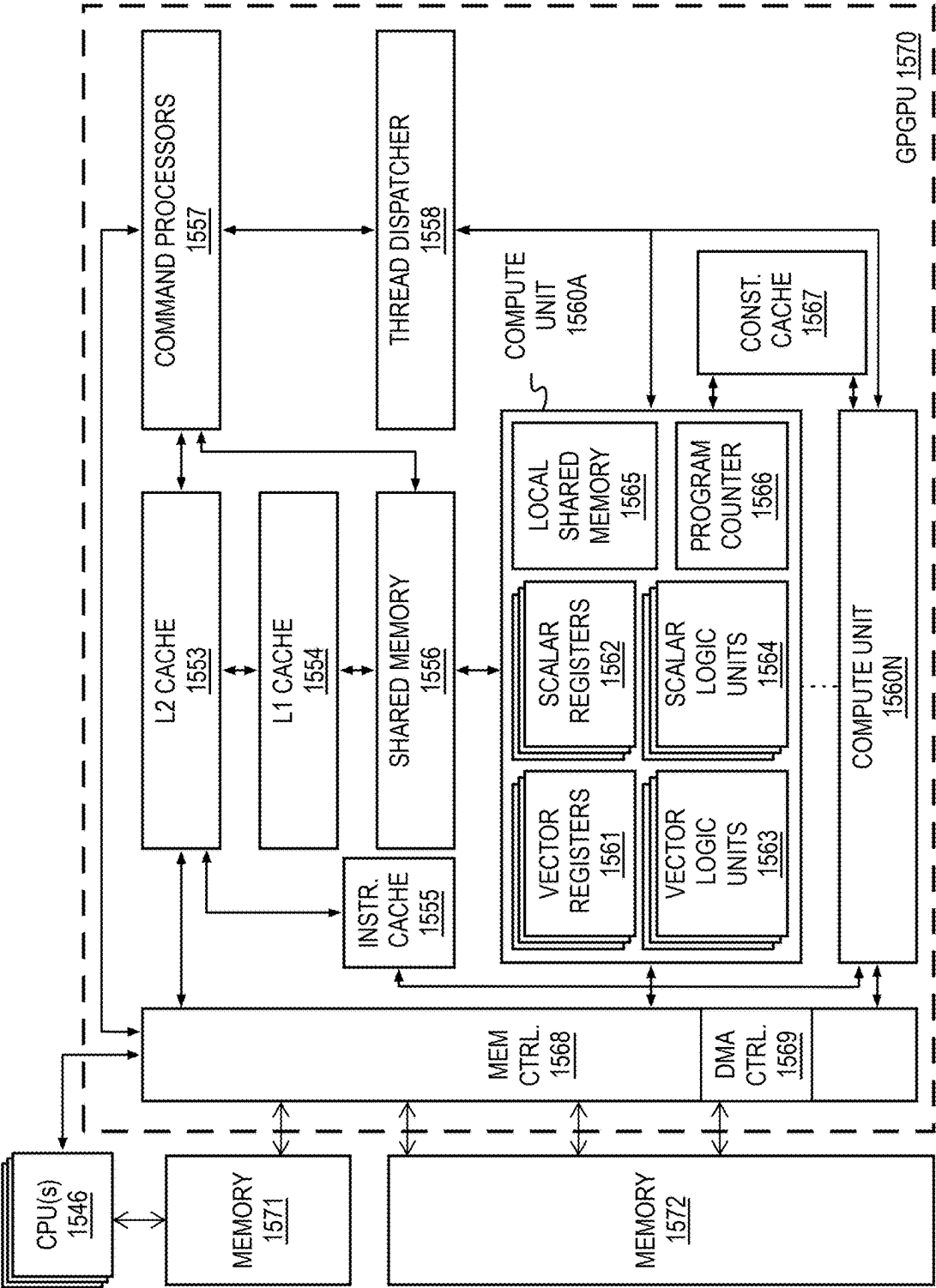


FIG. 15C

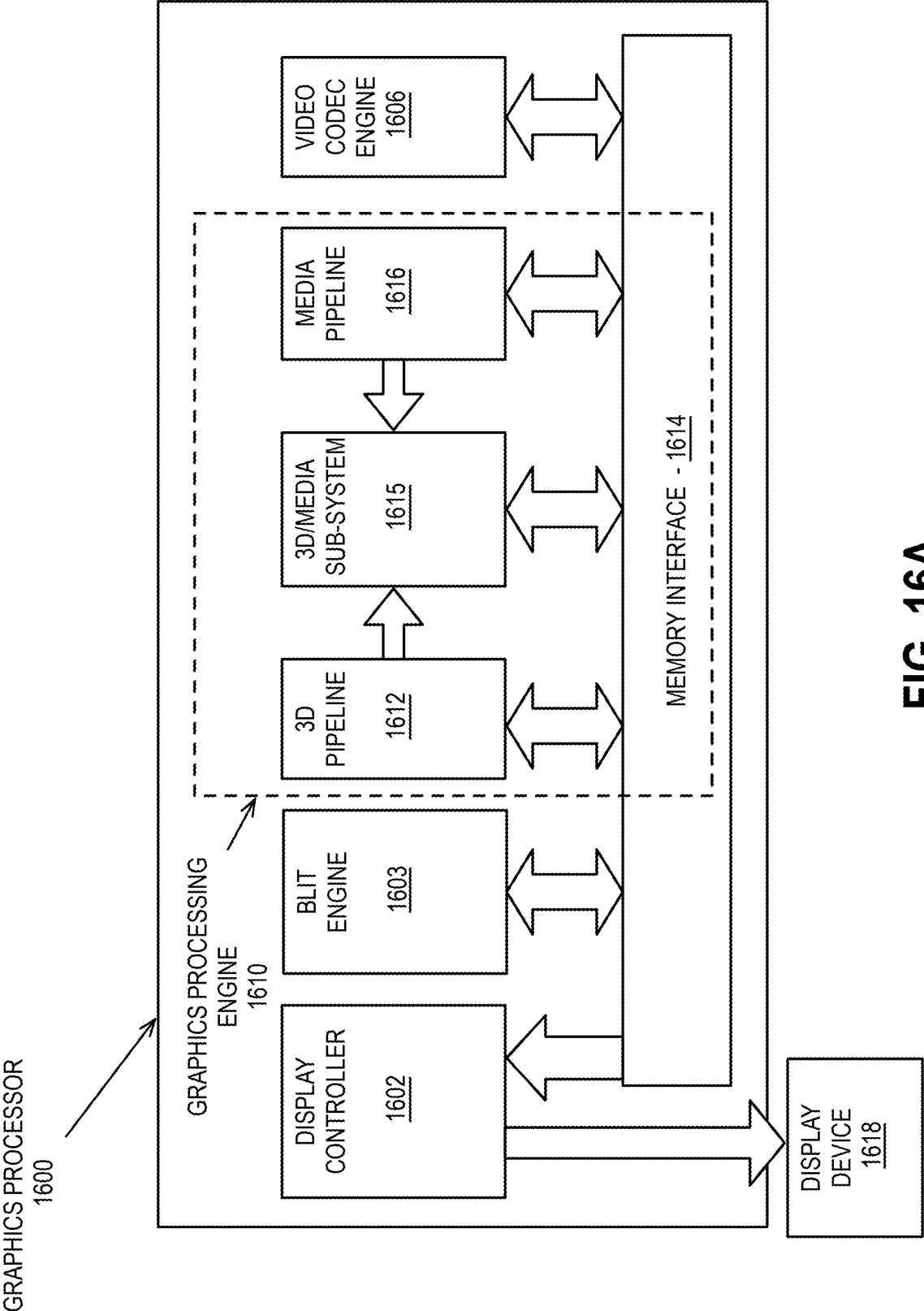


FIG. 16A

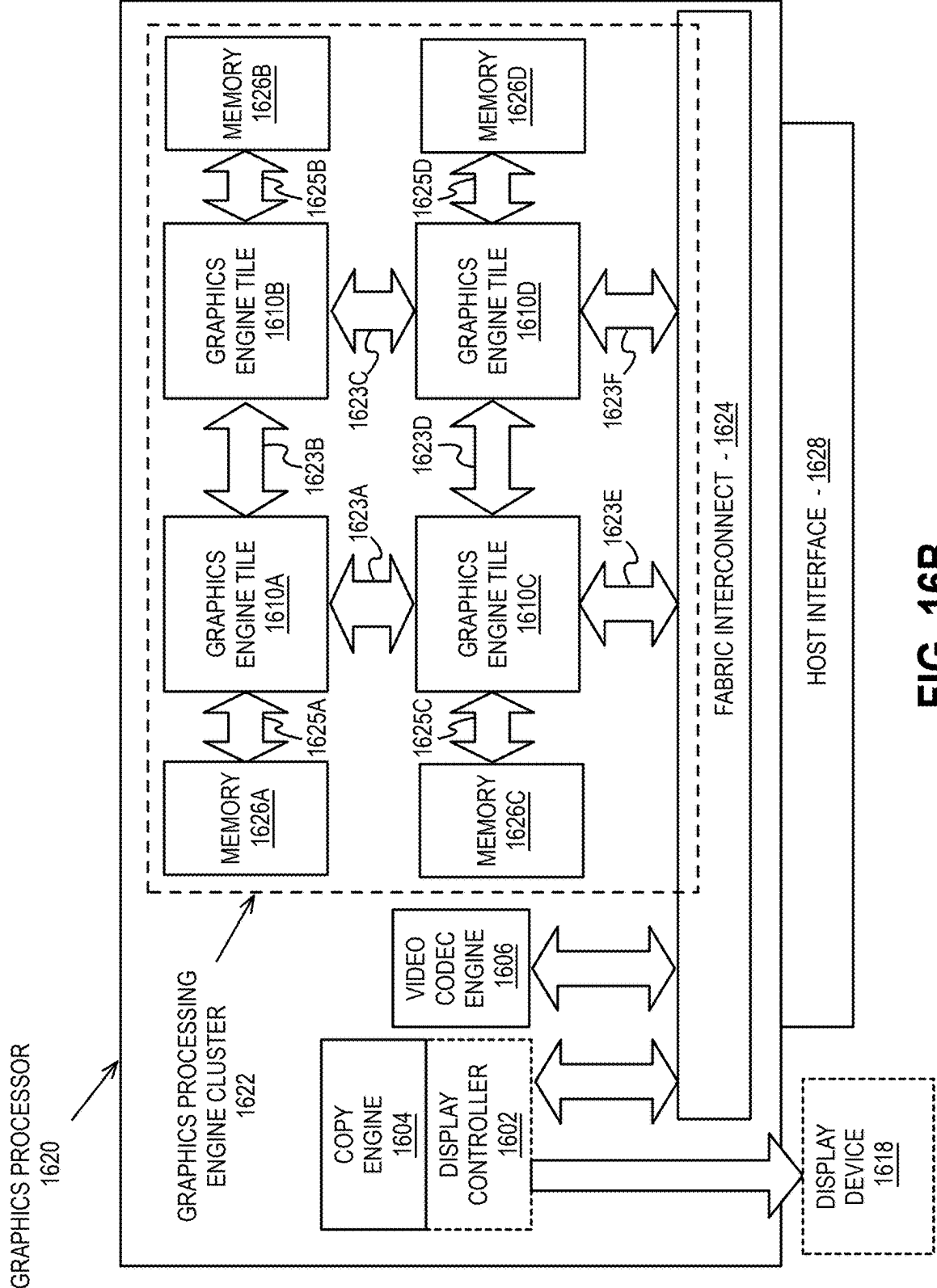


FIG. 16B

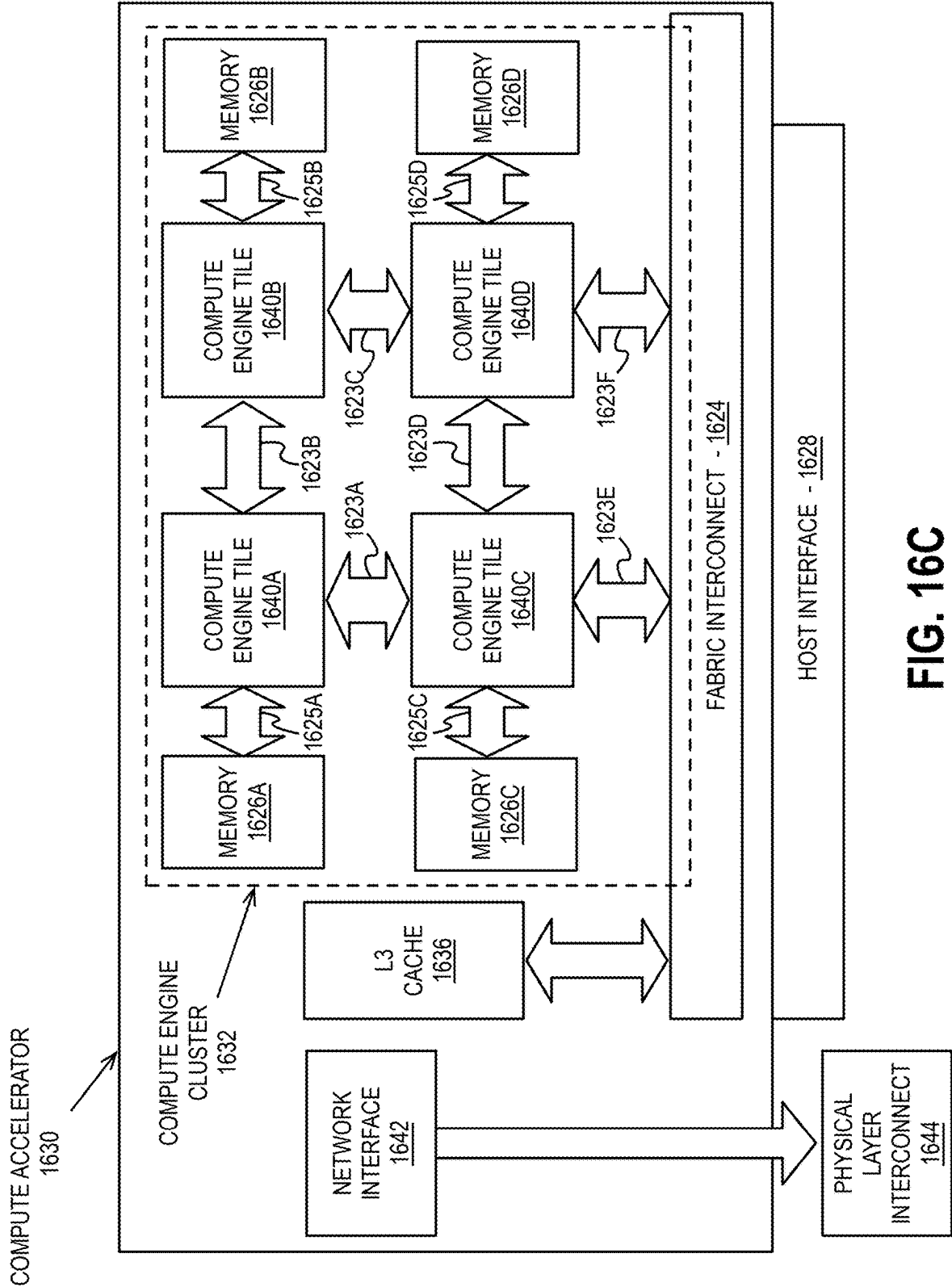


FIG. 16C

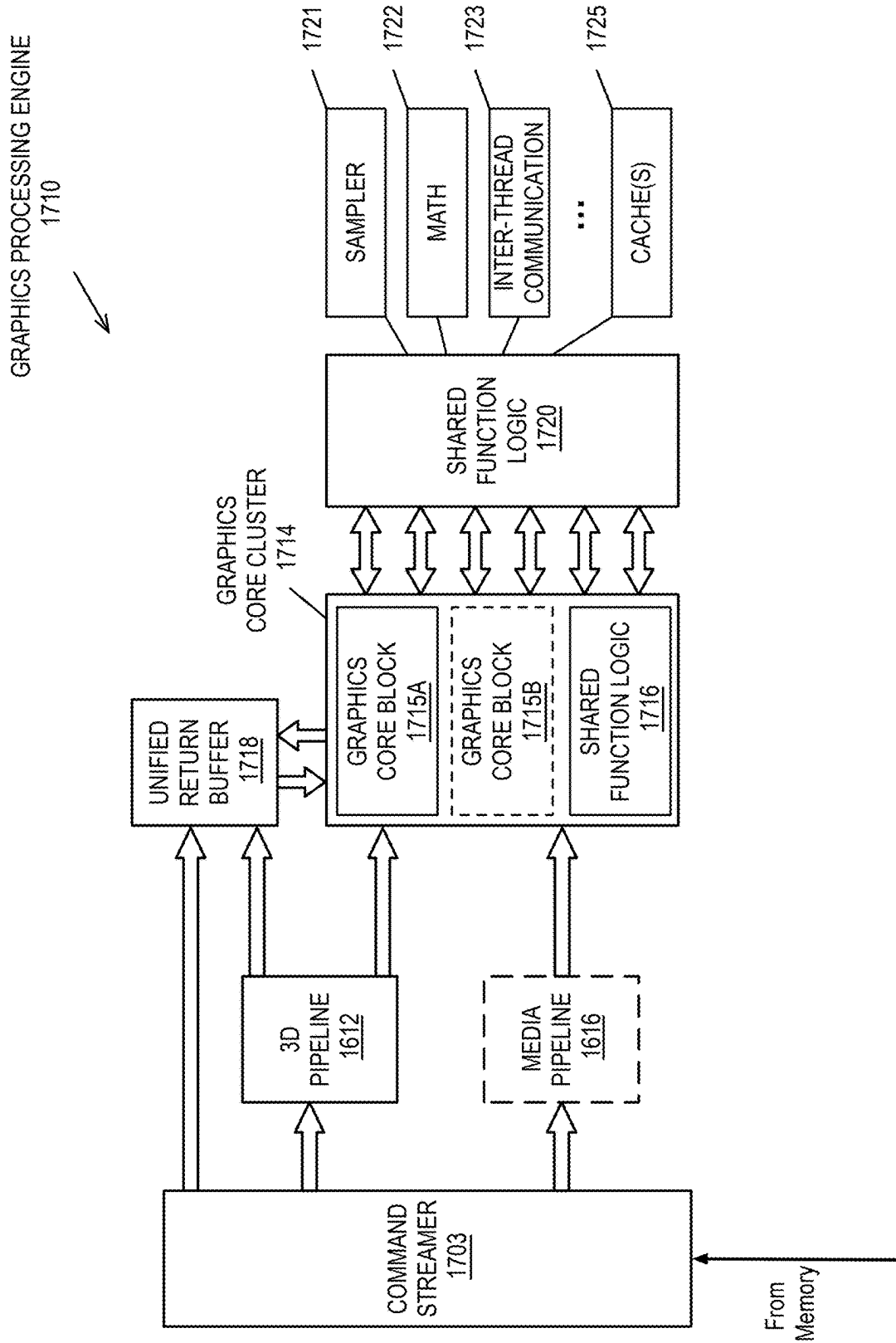


FIG. 17

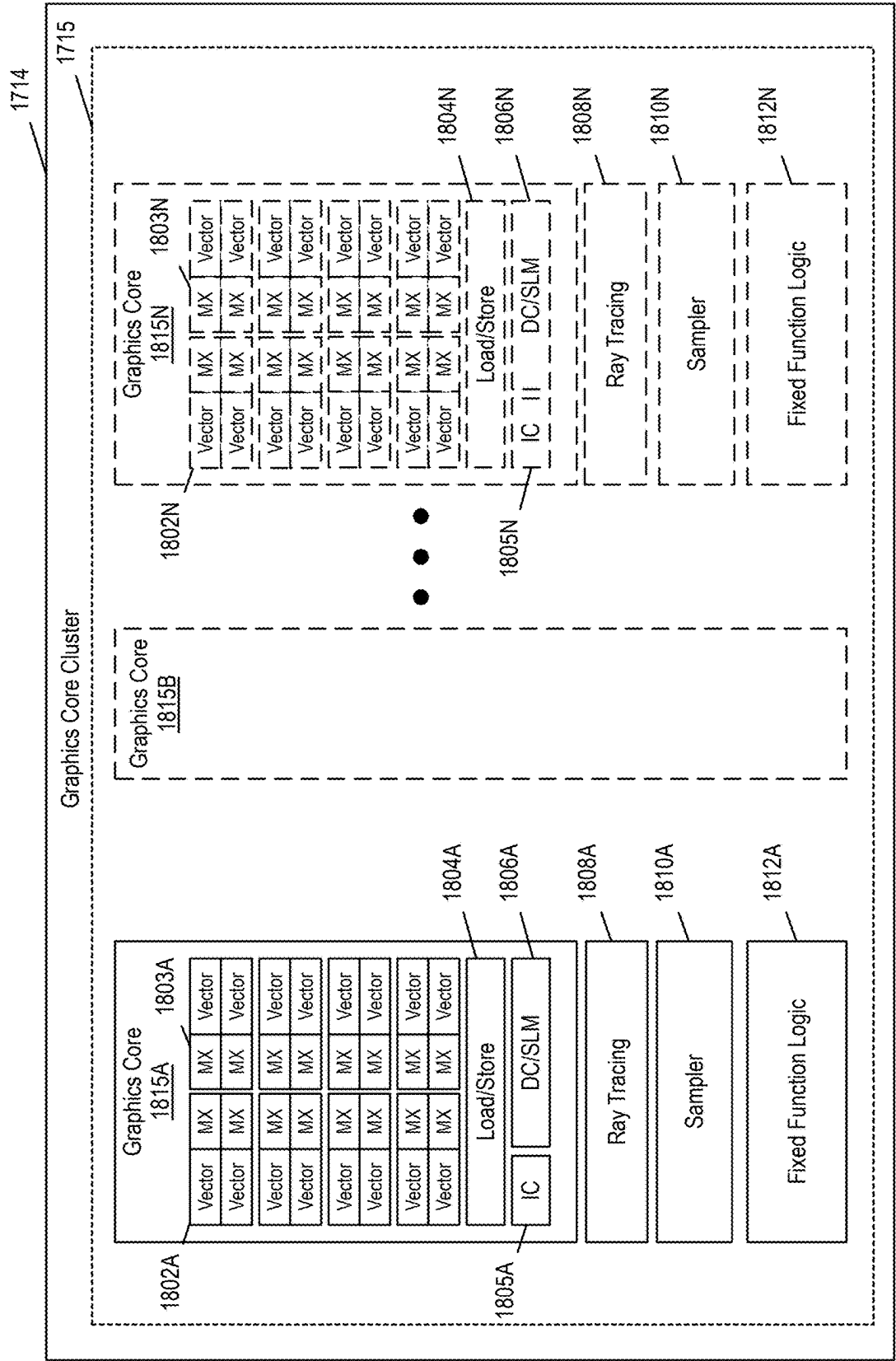


FIG. 18A

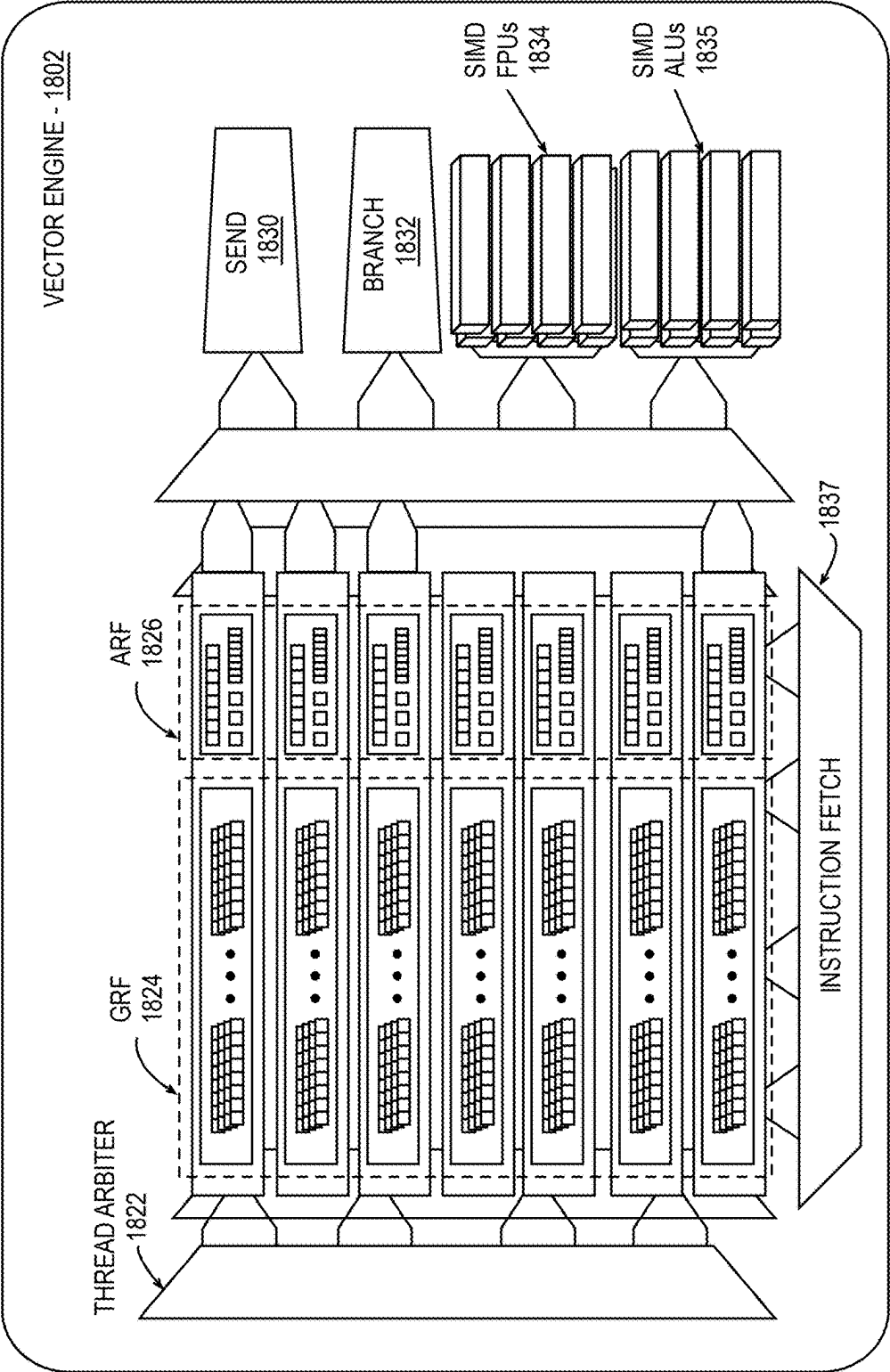


FIG. 18B

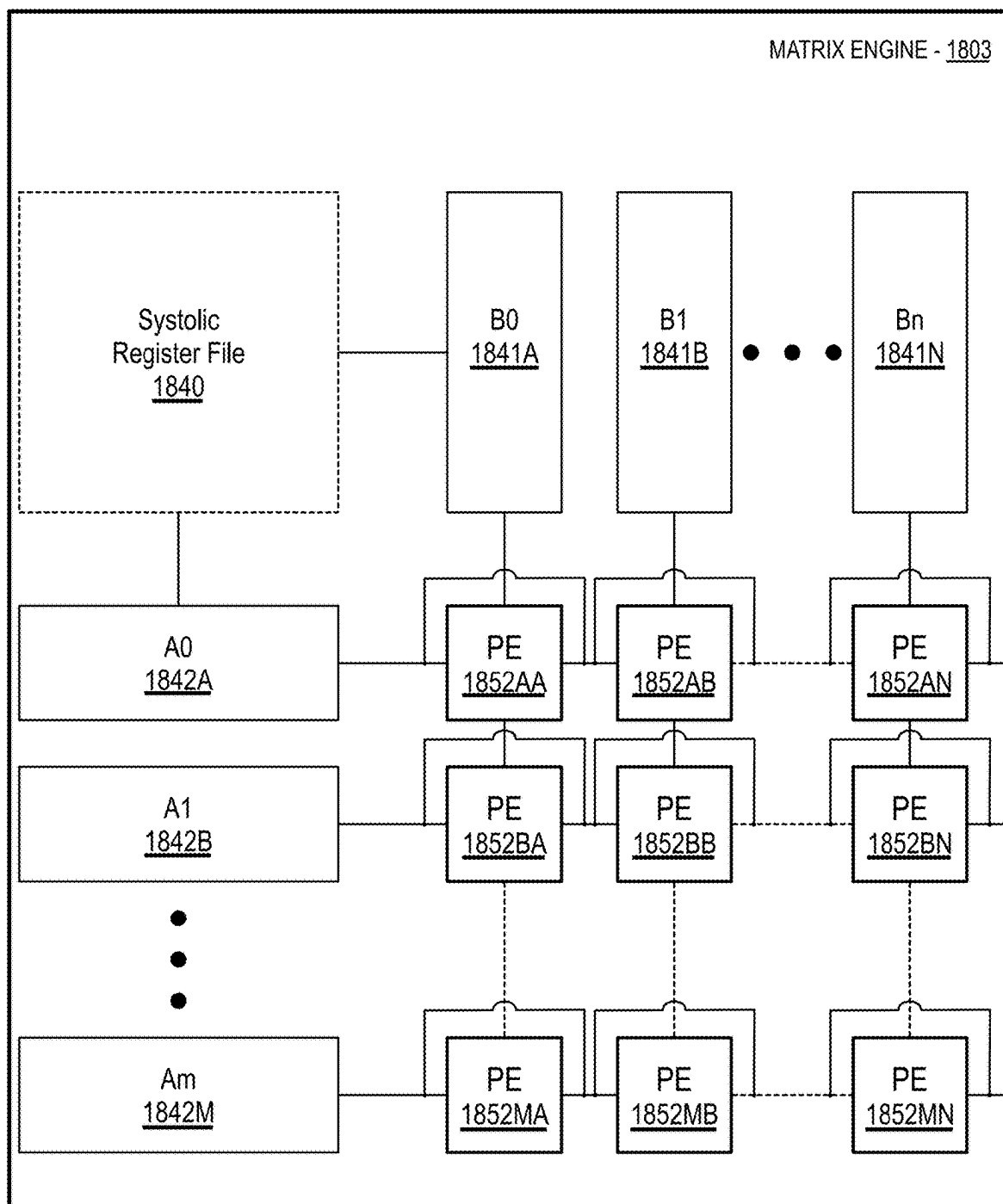


FIG. 18C

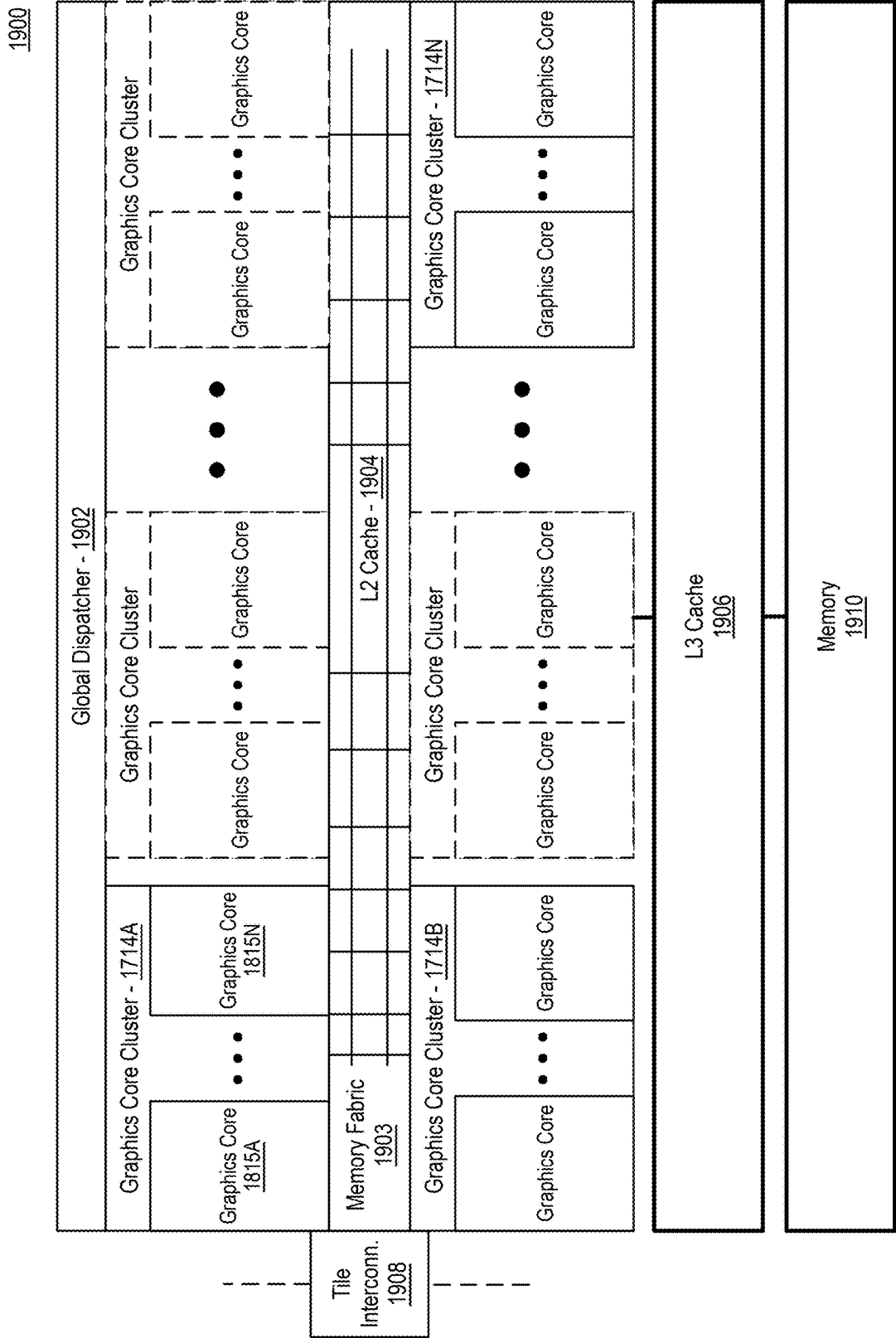


FIG. 19

GRAPHICS PROCESSOR INSTRUCTION FORMATS

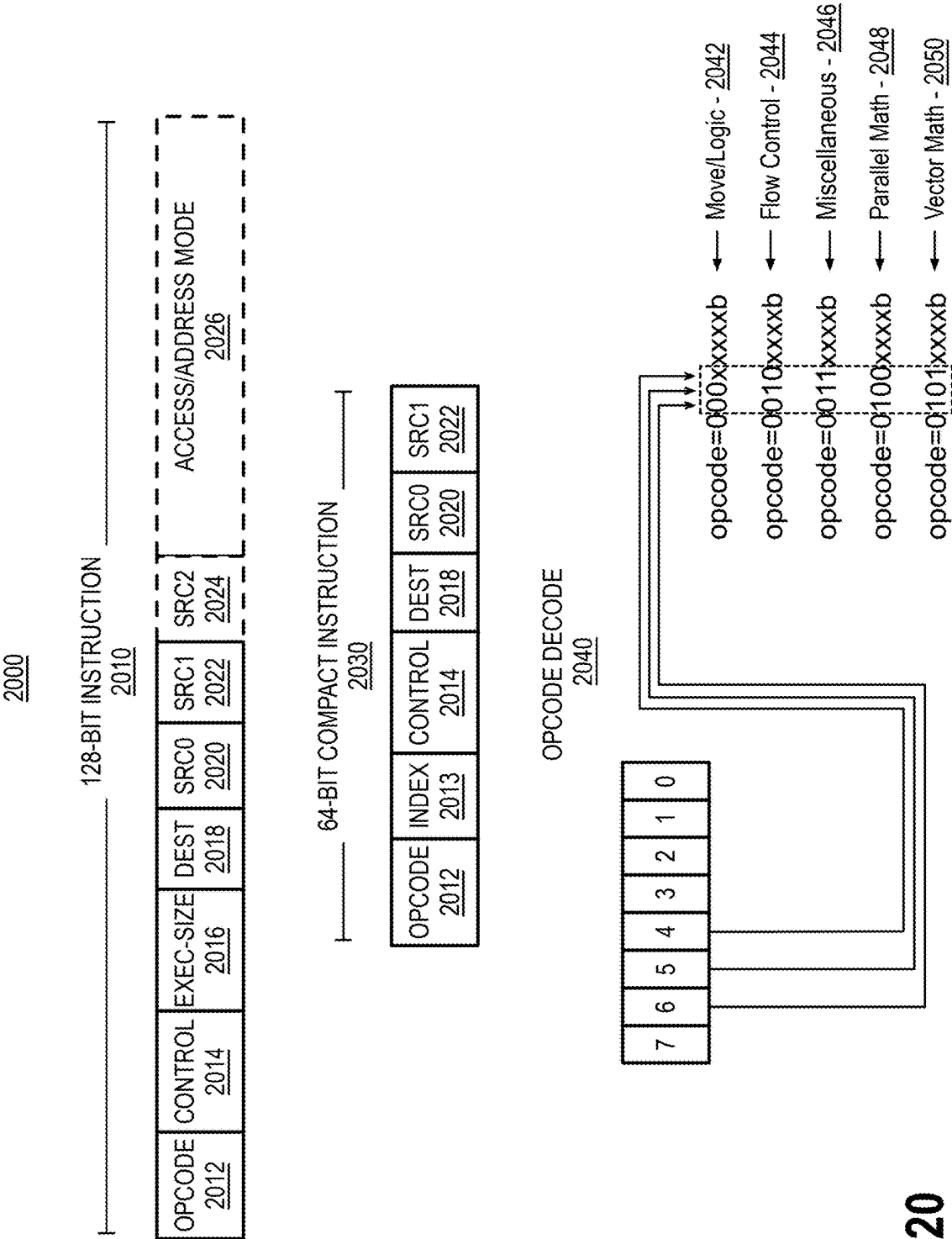


FIG. 20

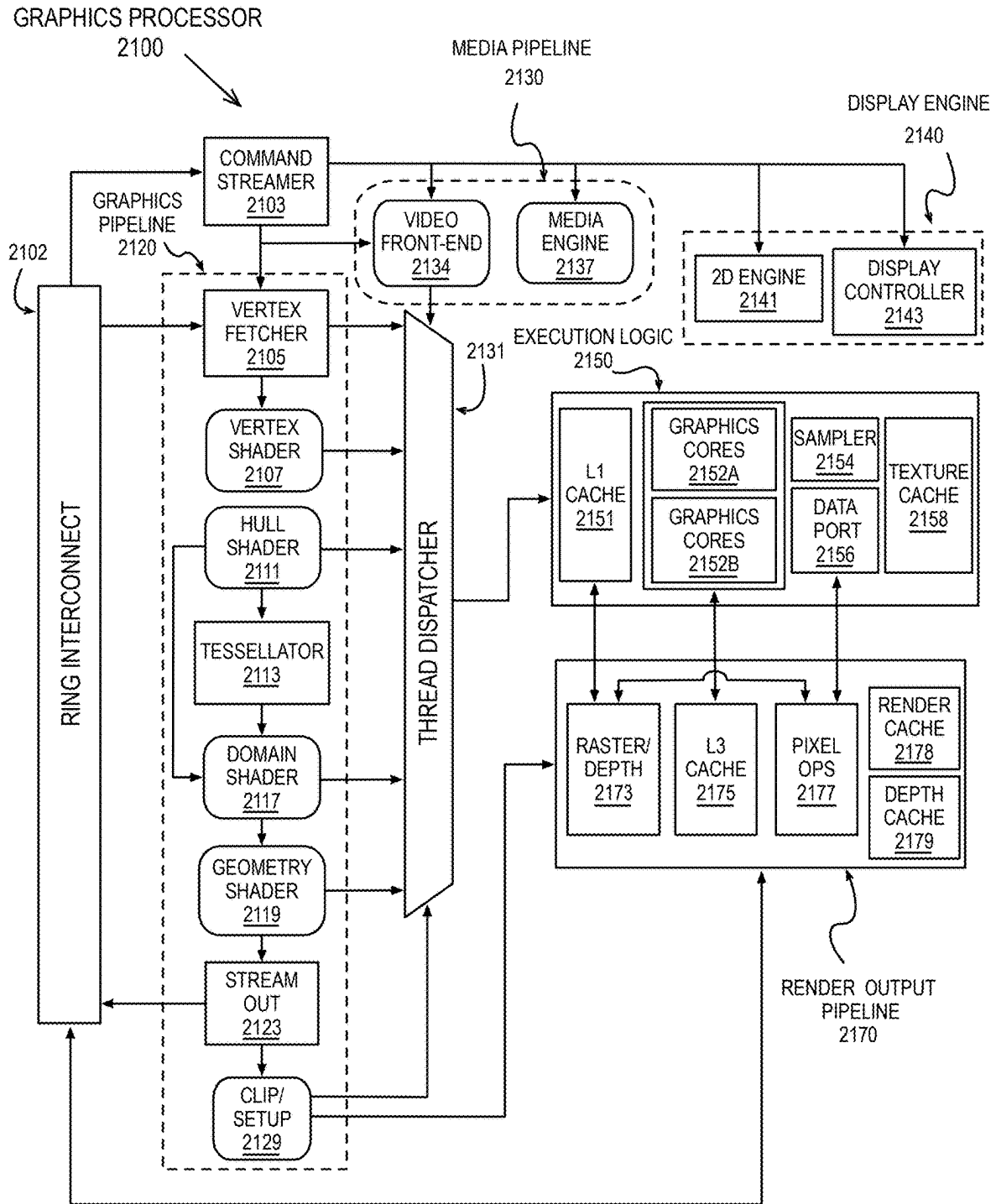


FIG. 21

FIG. 22A

GRAPHICS PROCESSOR COMMAND FORMAT

2200

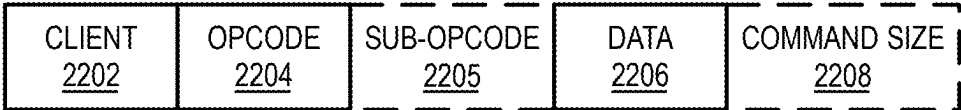
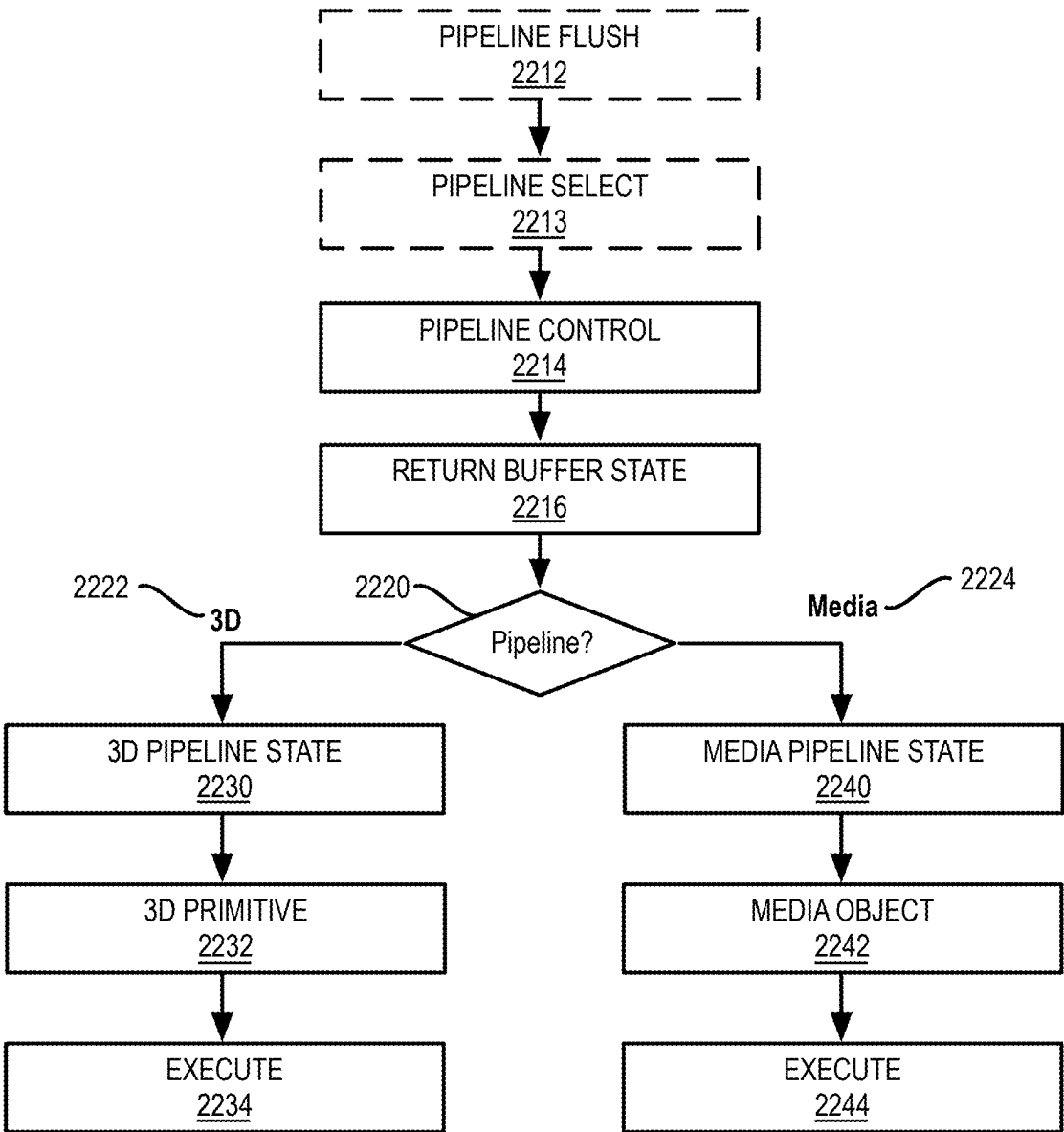


FIG. 22B

GRAPHICS PROCESSOR COMMAND SEQUENCE

2210



DATA PROCESSING SYSTEM - 2300

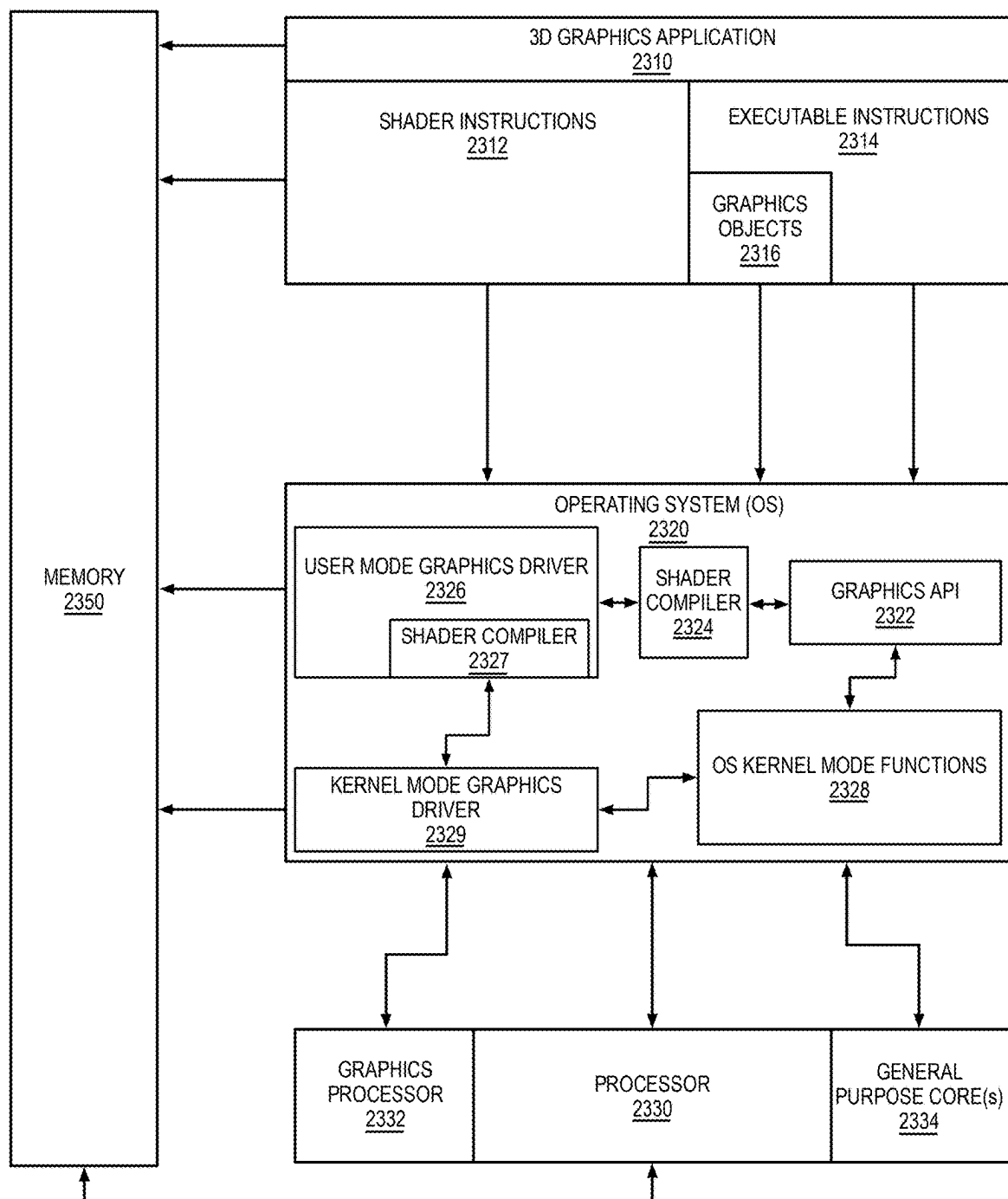


FIG. 23

IP CORE DEVELOPMENT - 2400

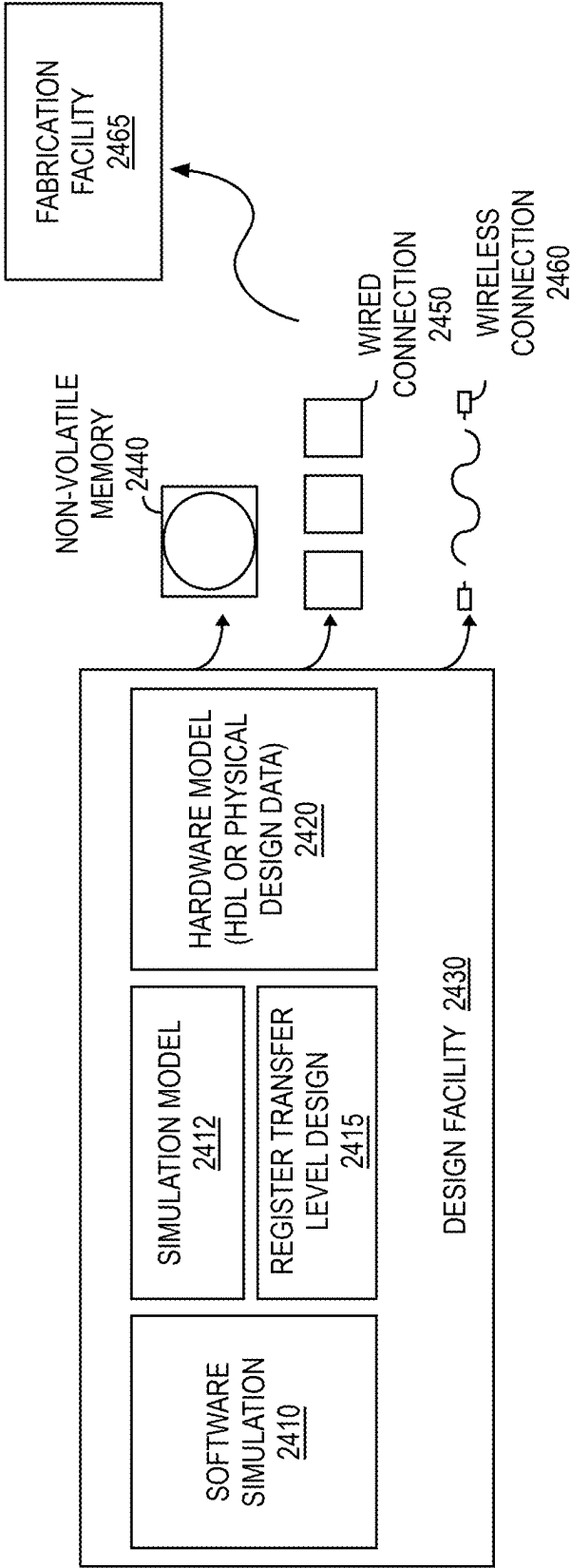


FIG. 24A

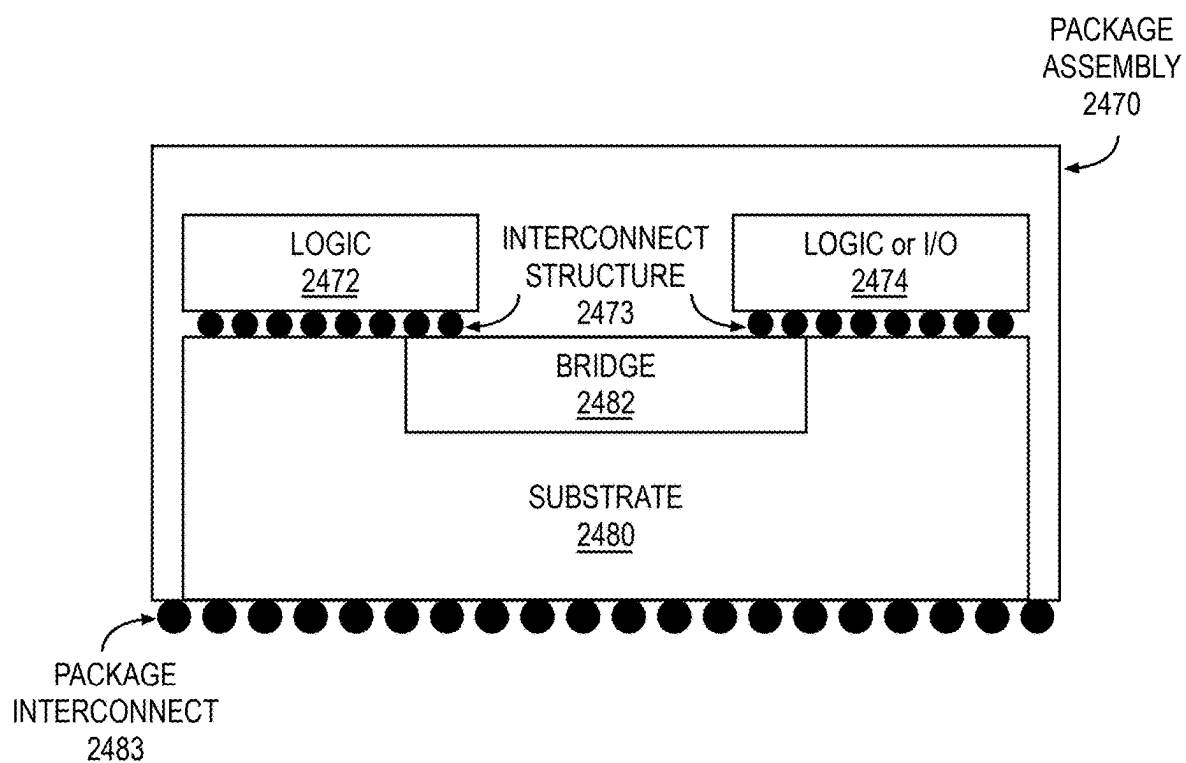


FIG. 24B

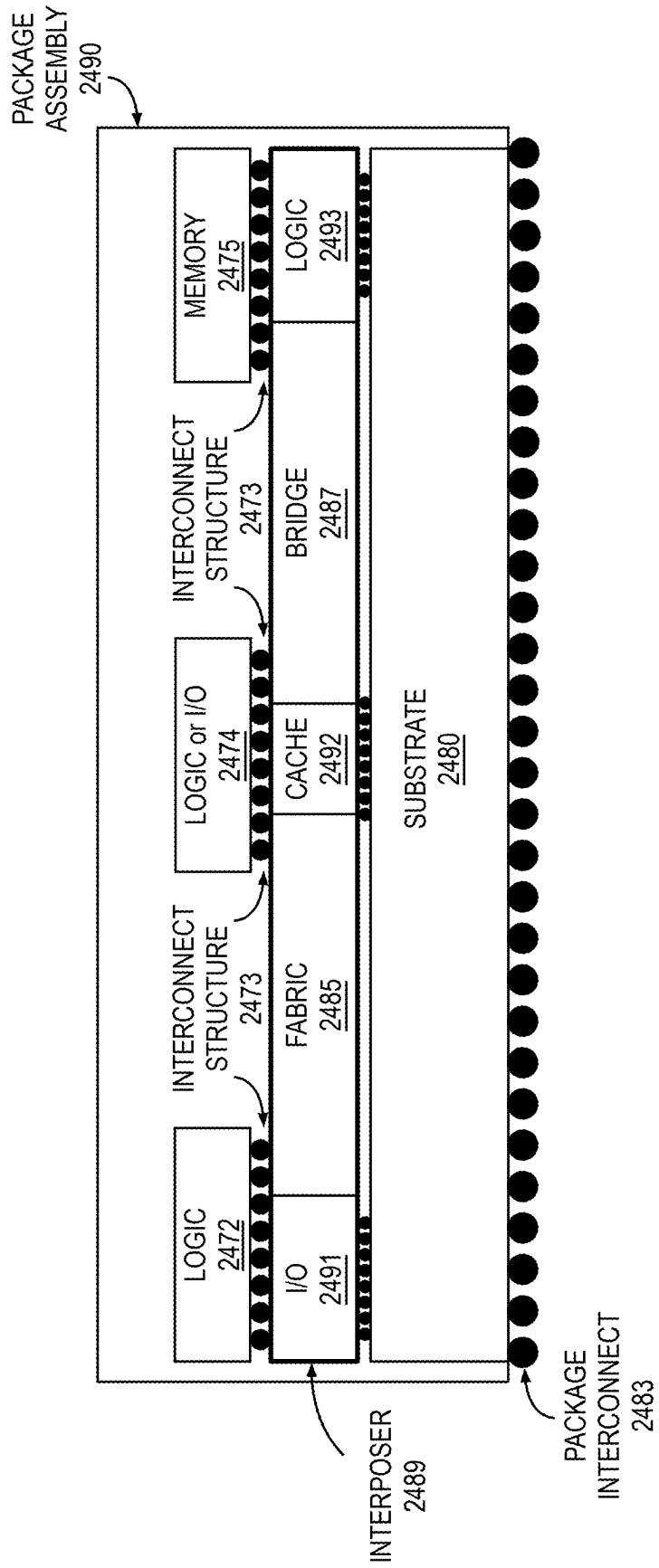


FIG. 24C

2494

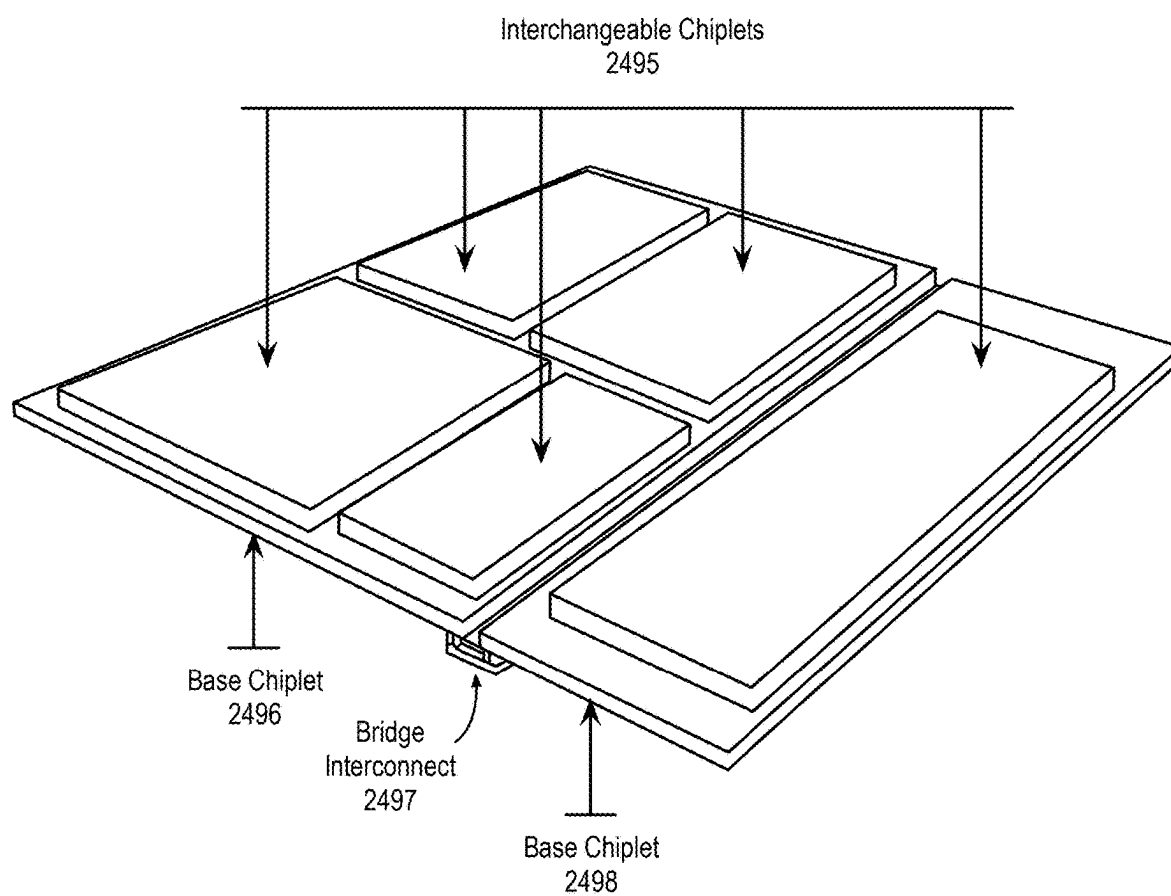


FIG. 24D

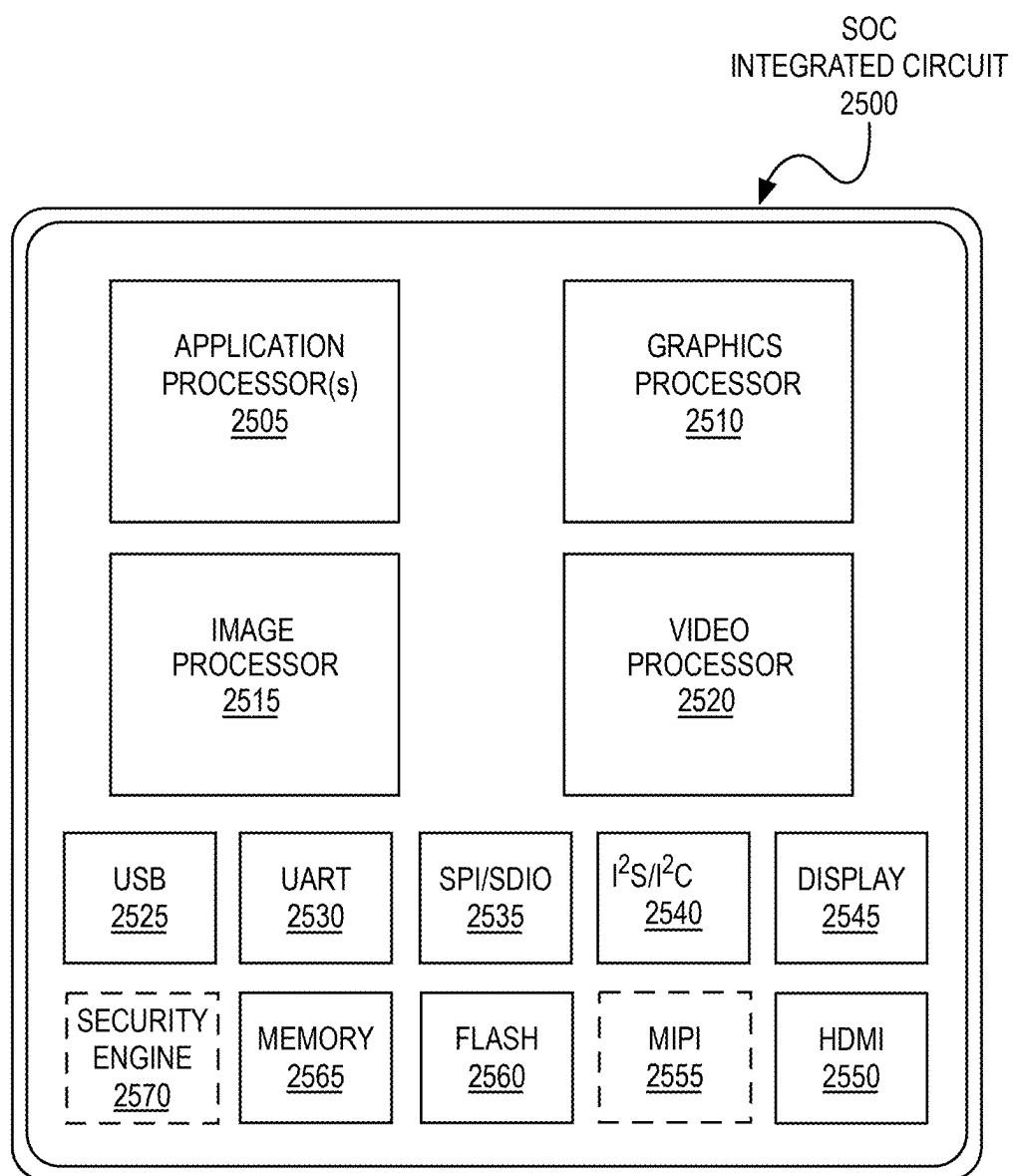


FIG. 25

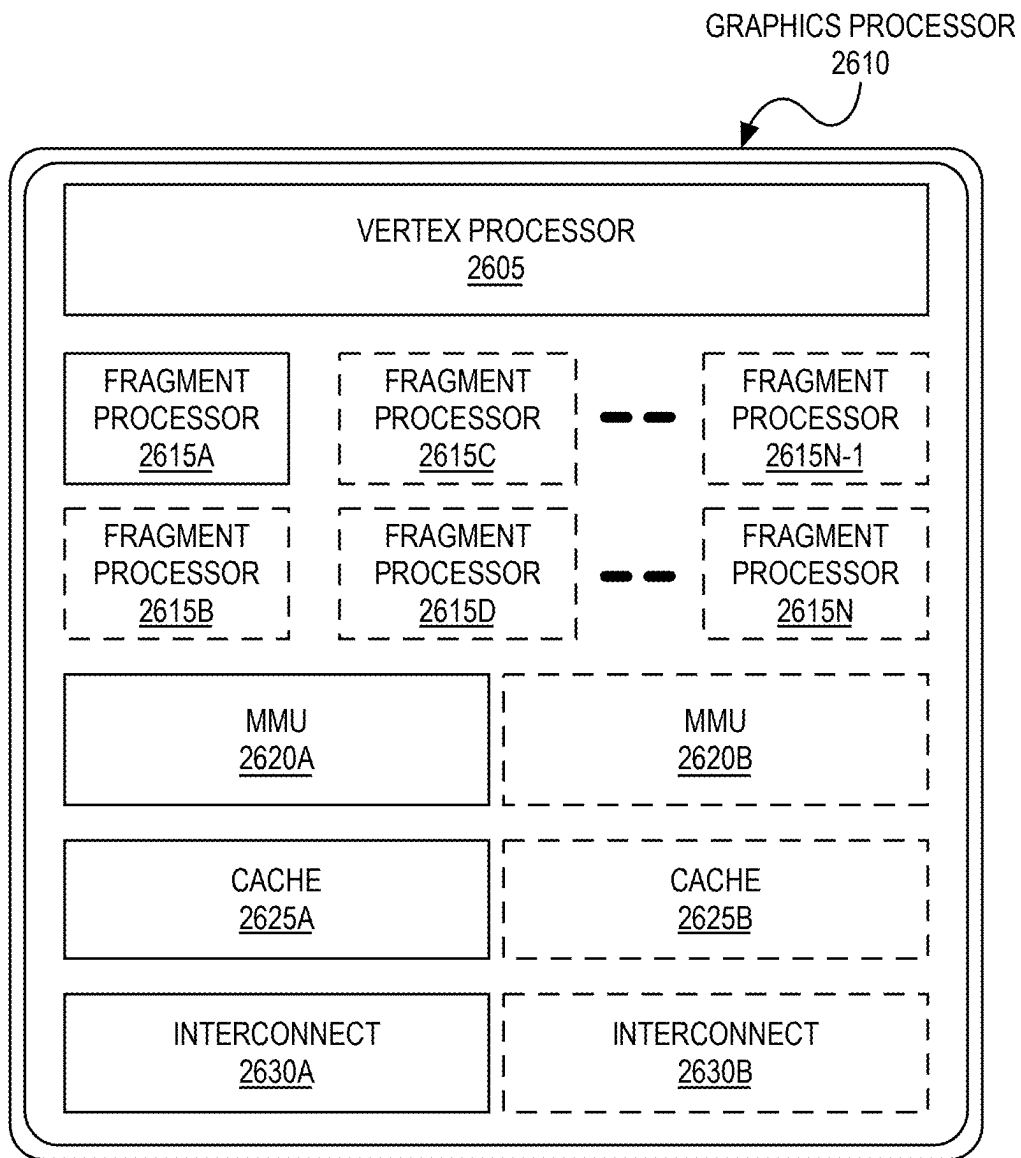


FIG. 26A

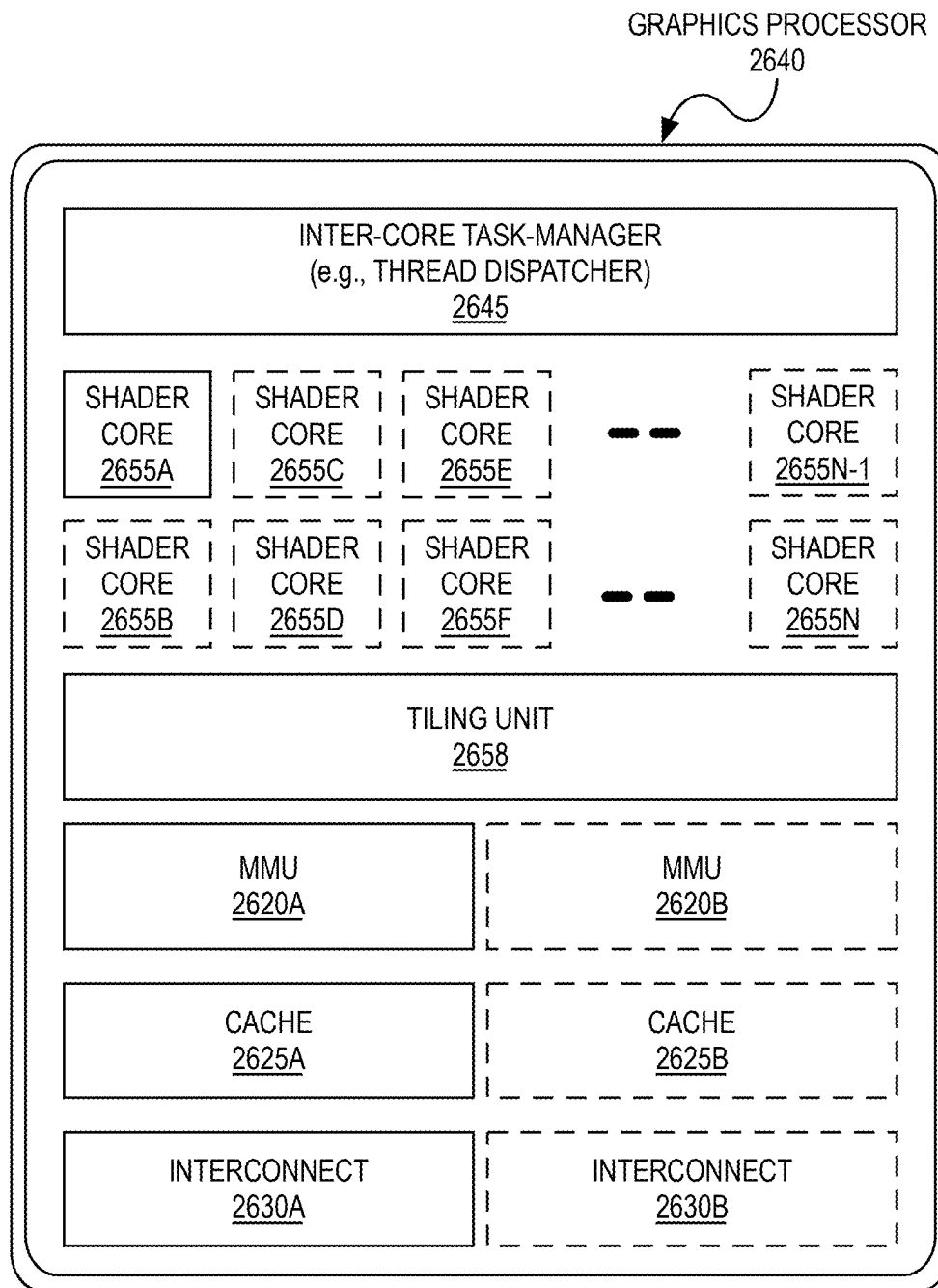


FIG. 26B

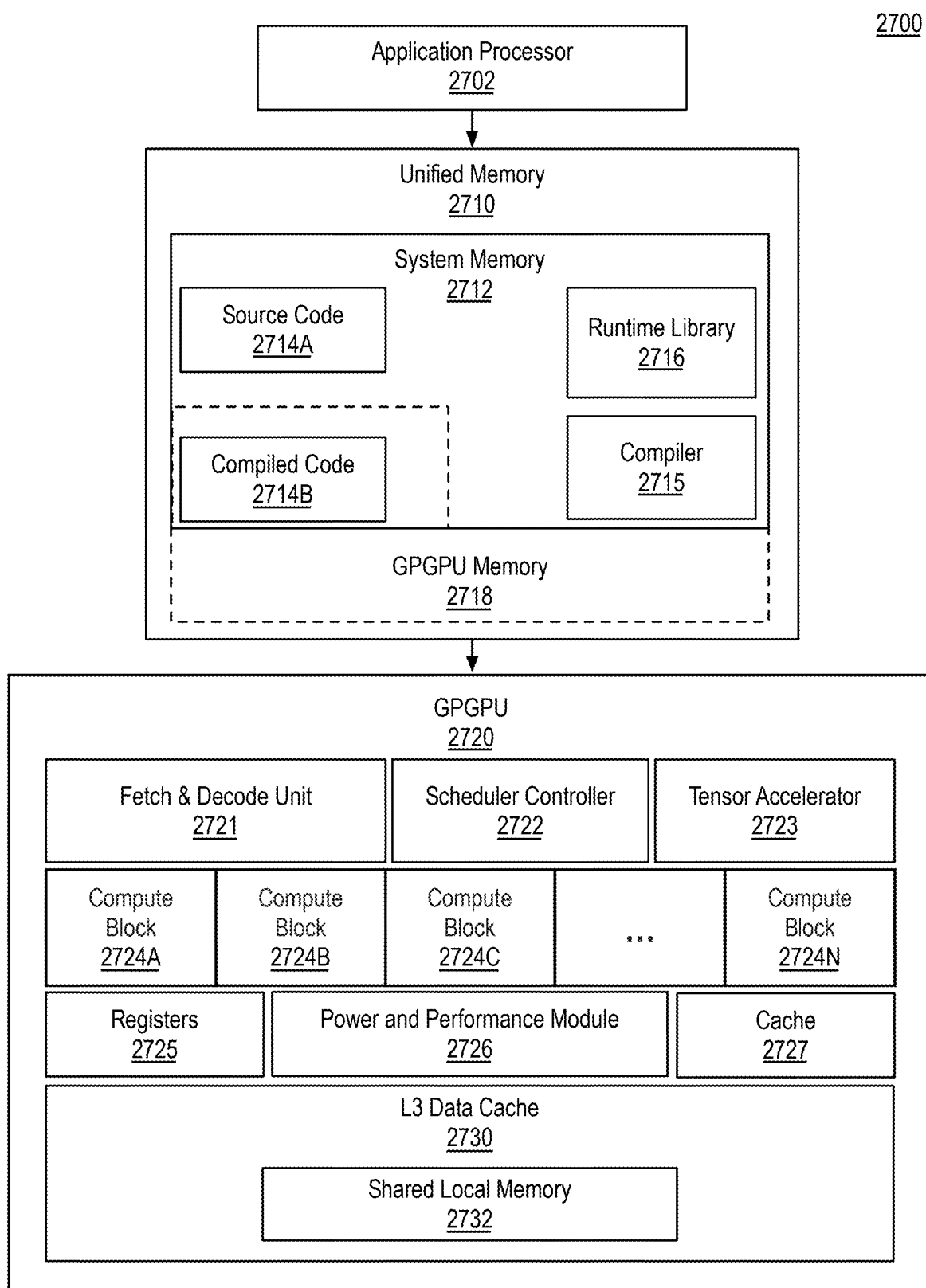


FIG. 27

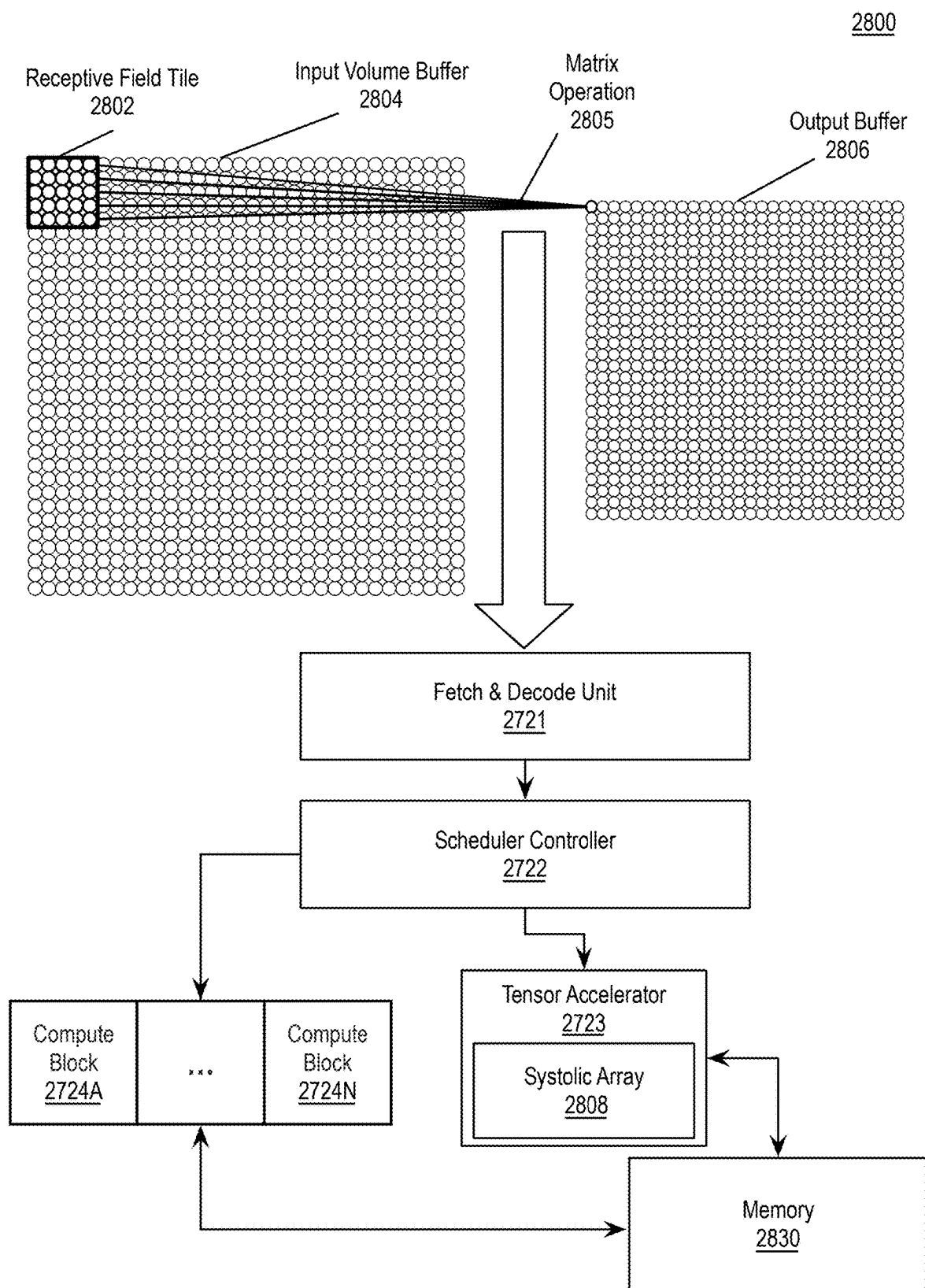


FIG. 28A

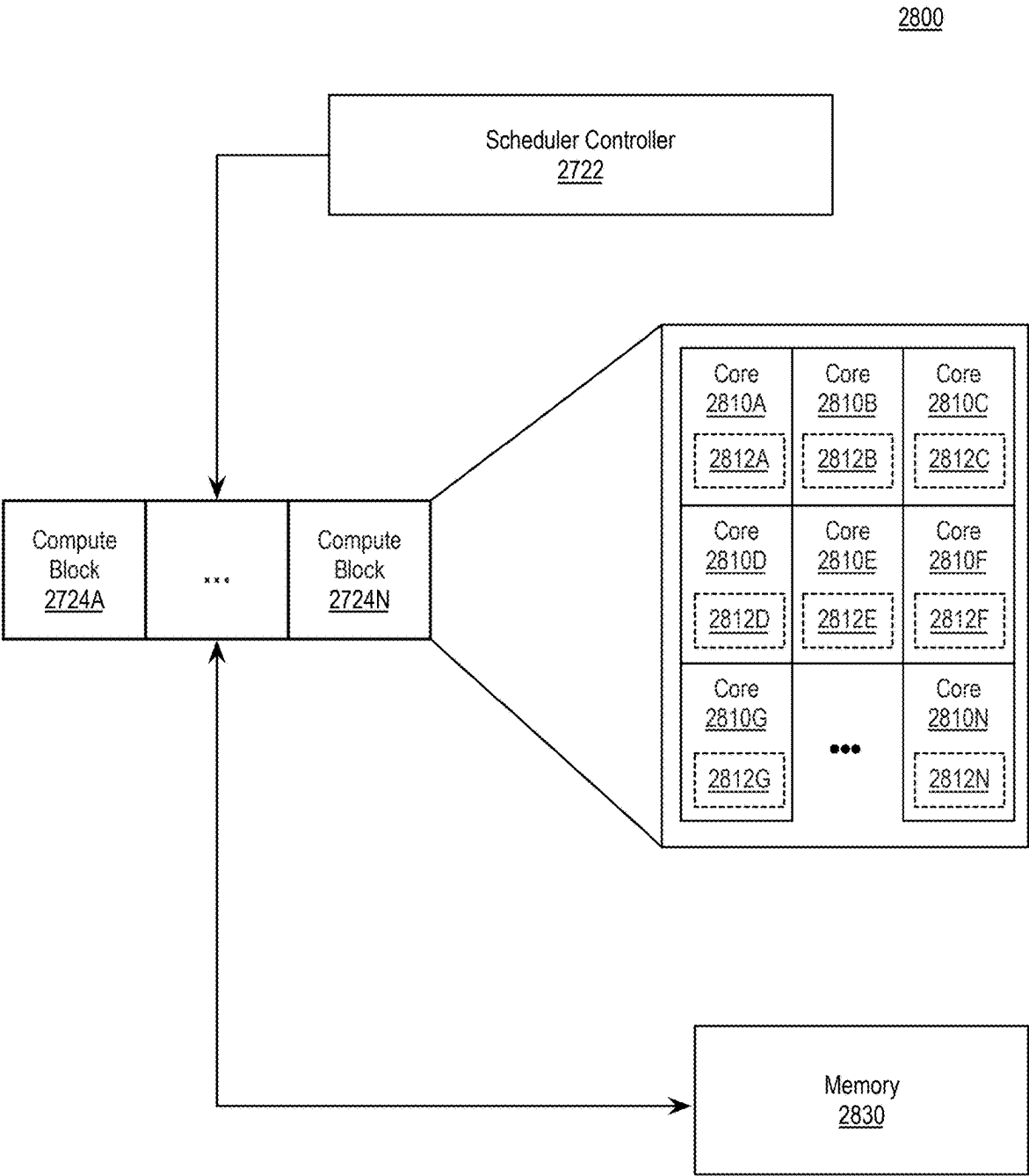


FIG. 28B

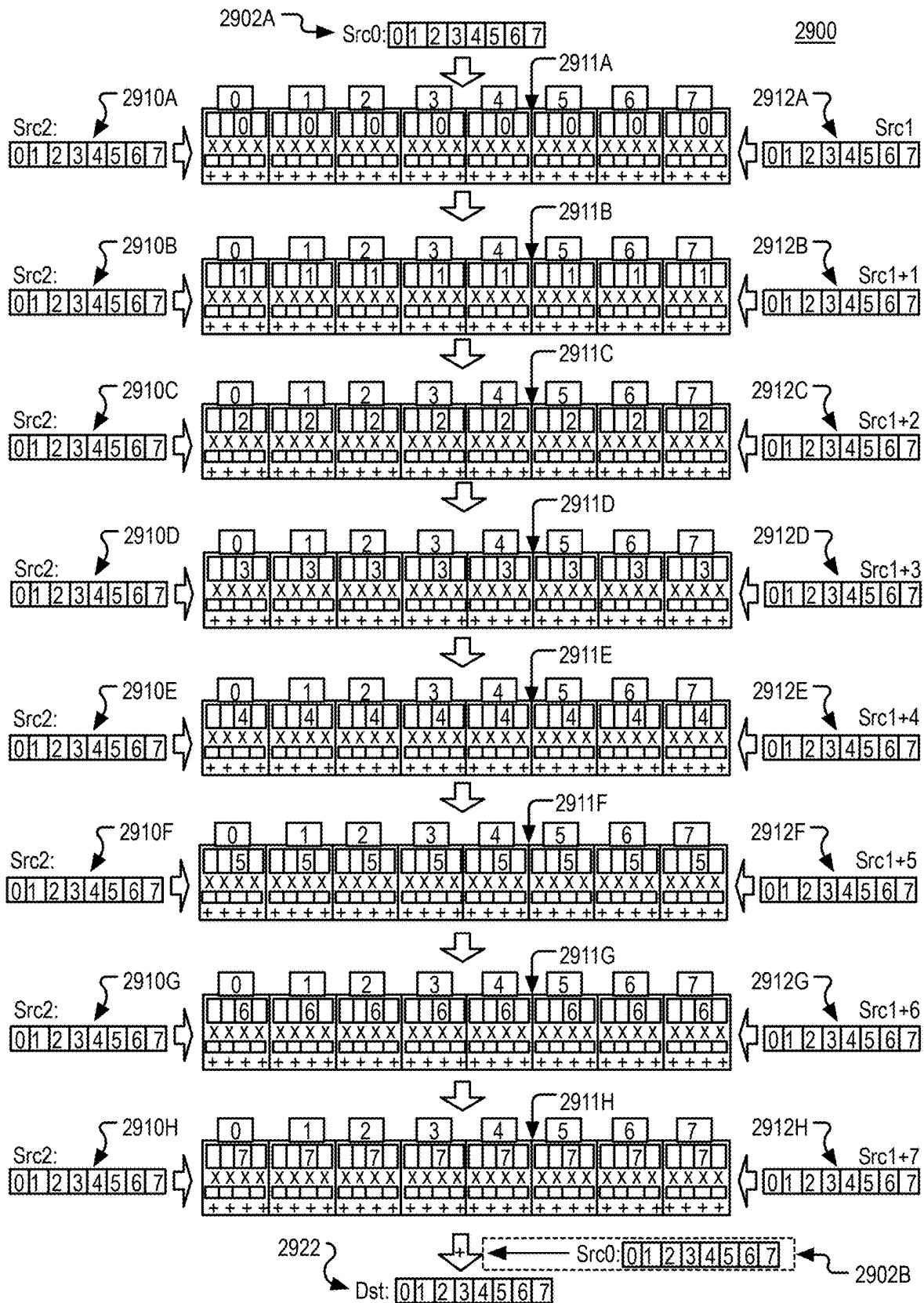


FIG. 29

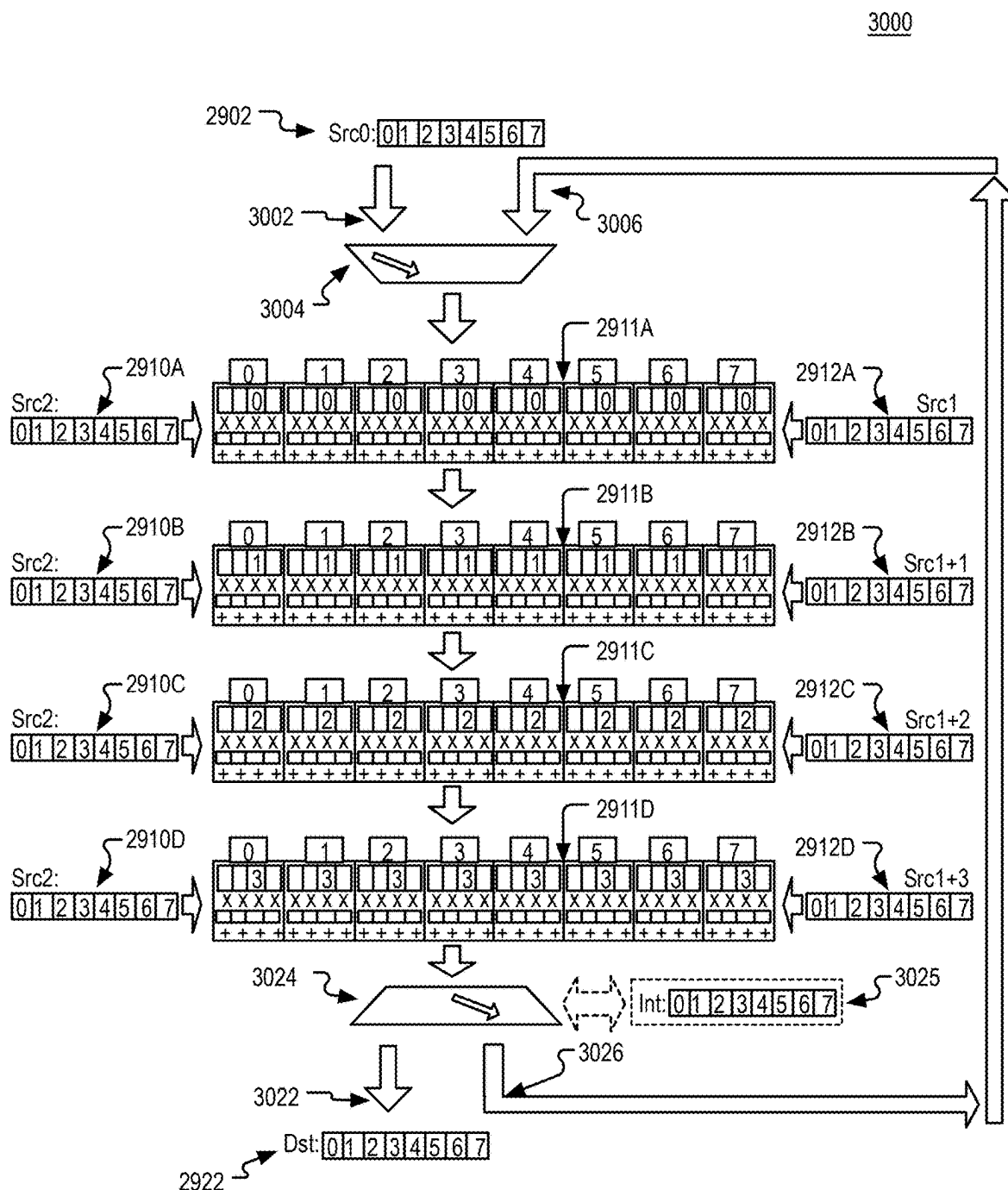


FIG. 30A

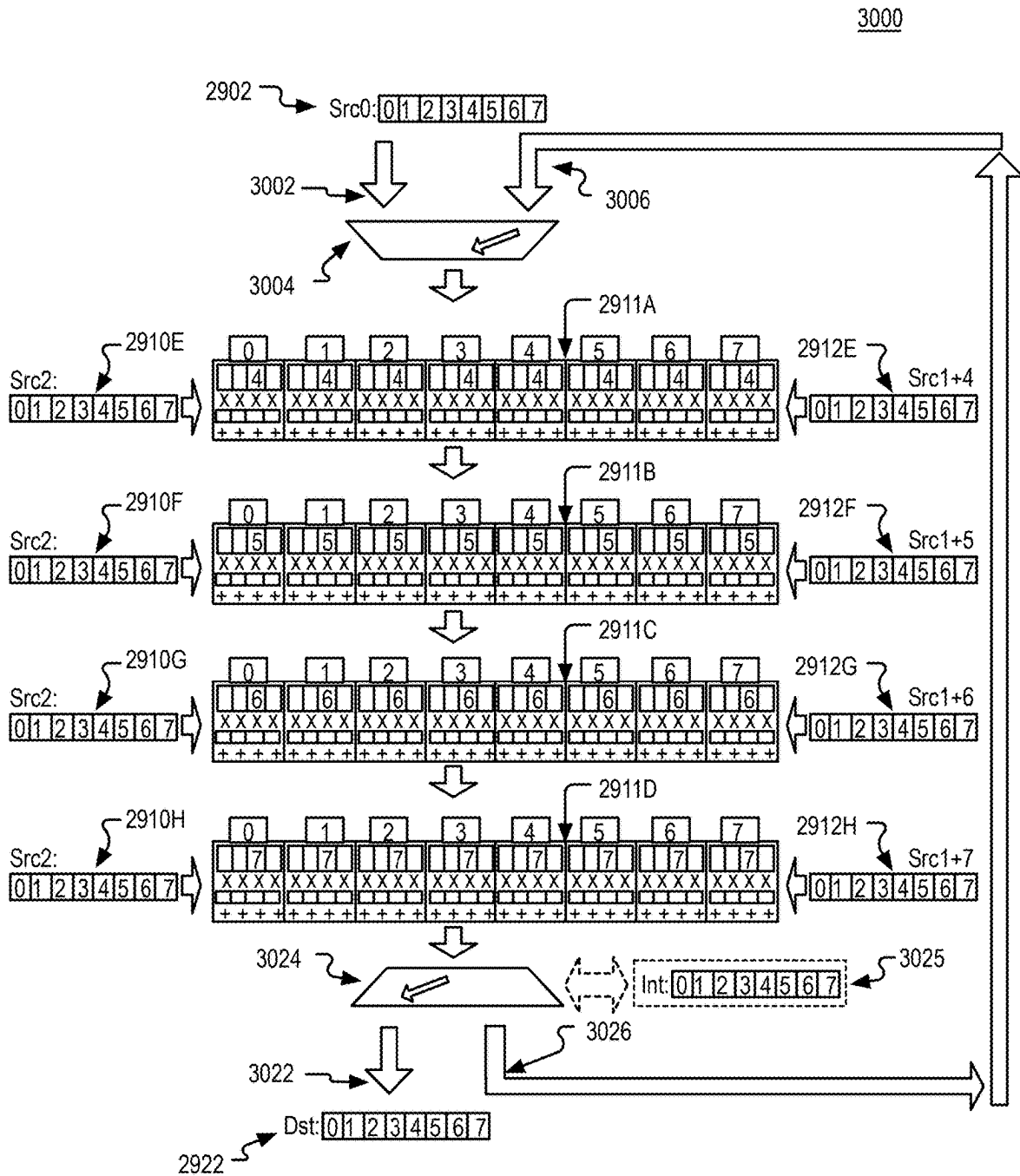


FIG. 30B

3100

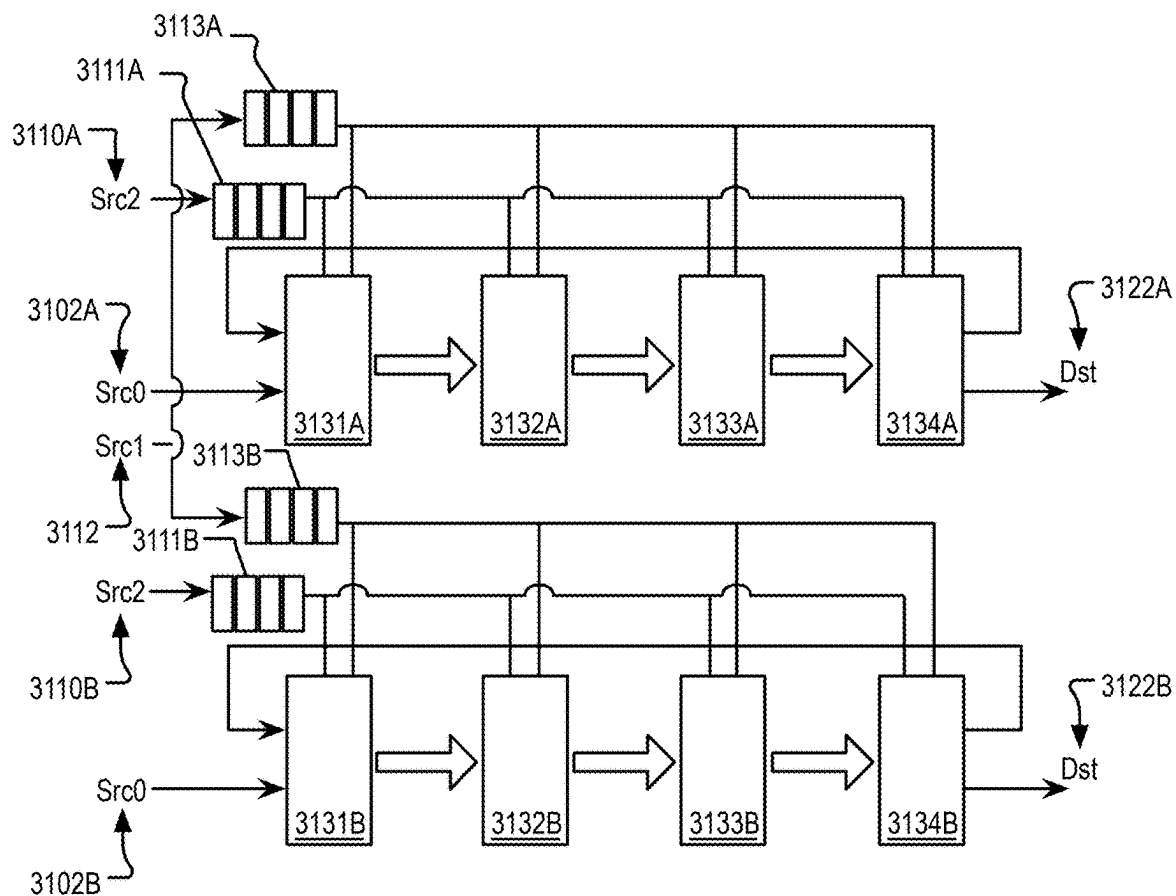


FIG. 31

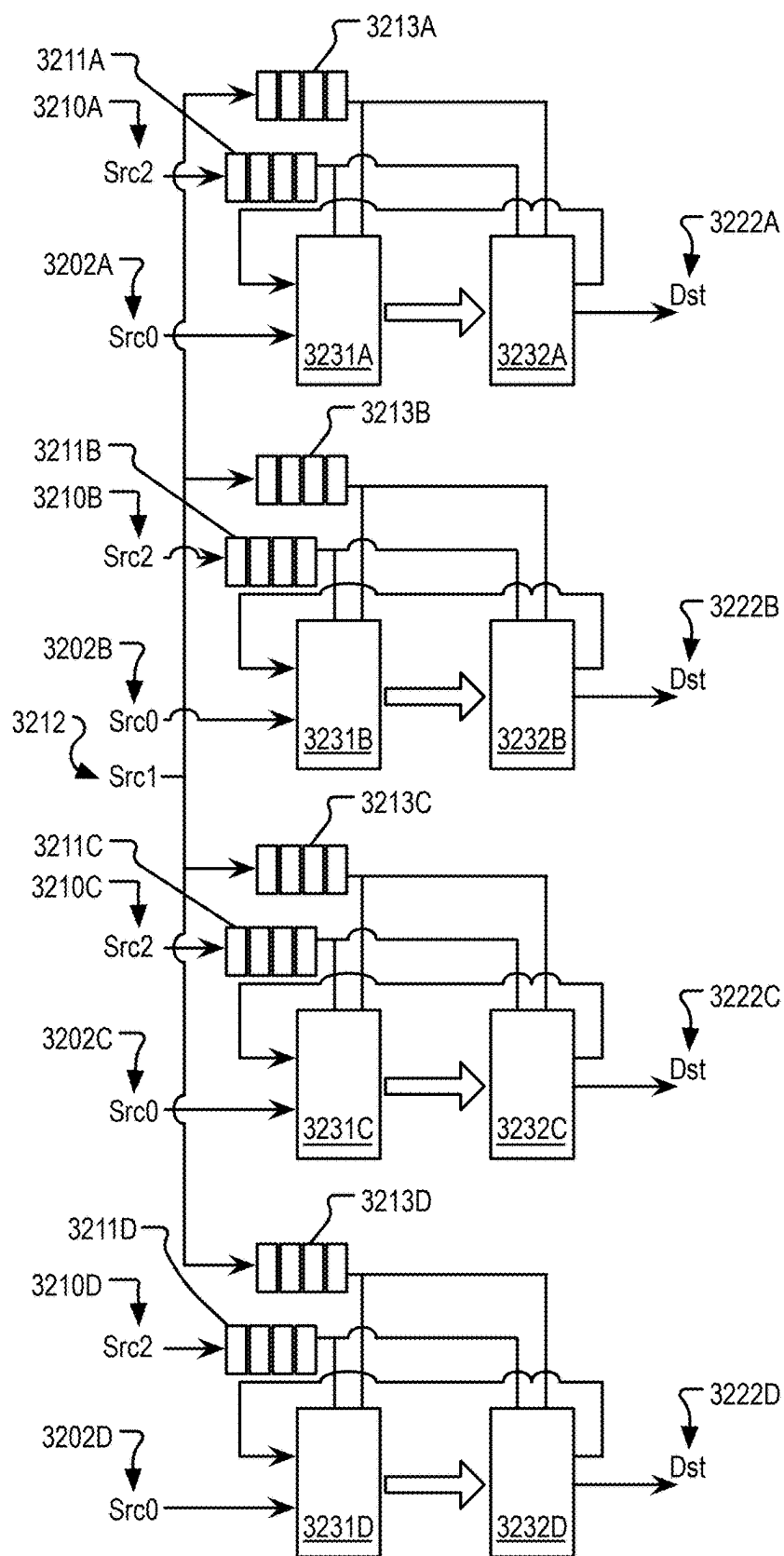


FIG. 32

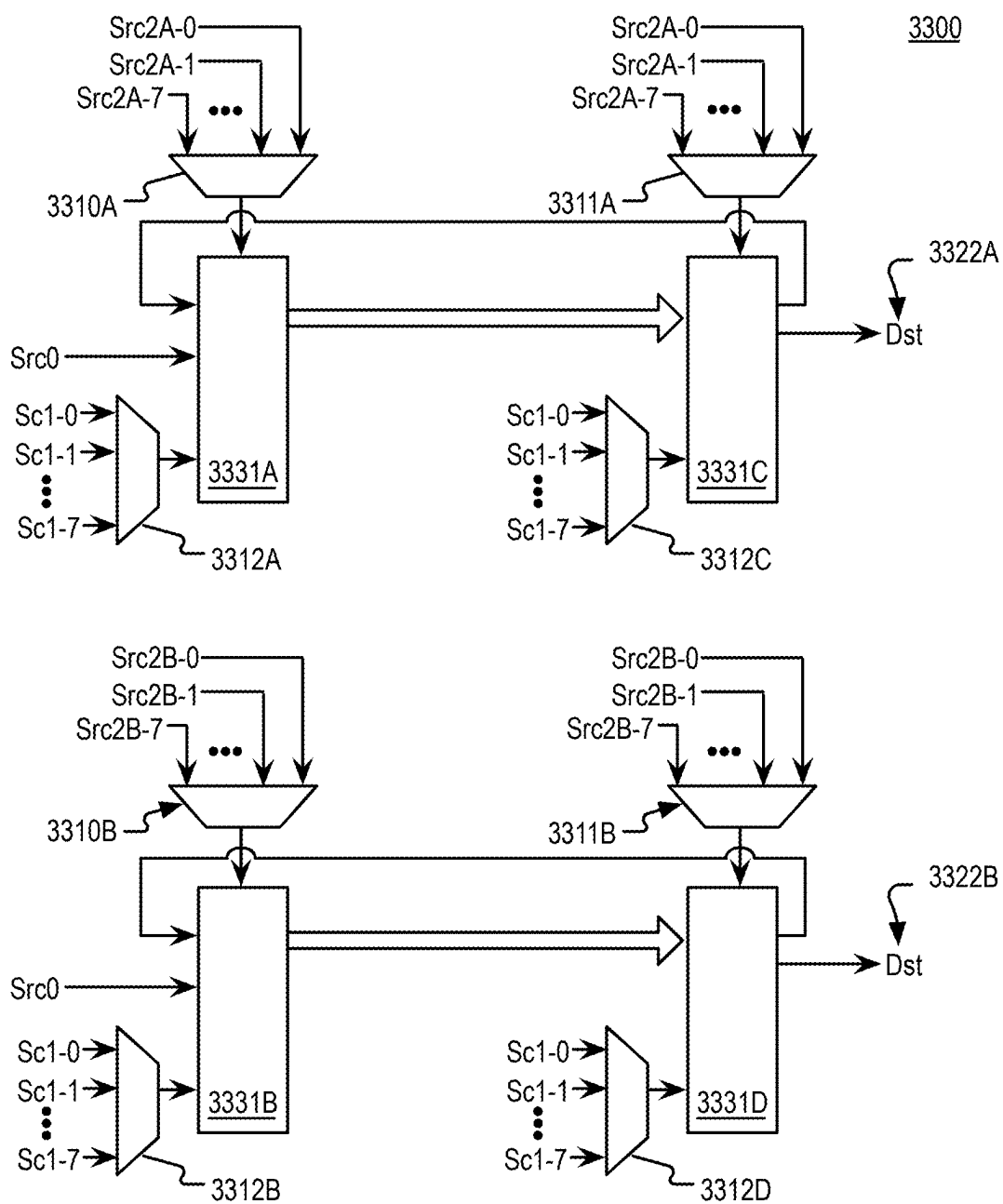


FIG. 33

3400

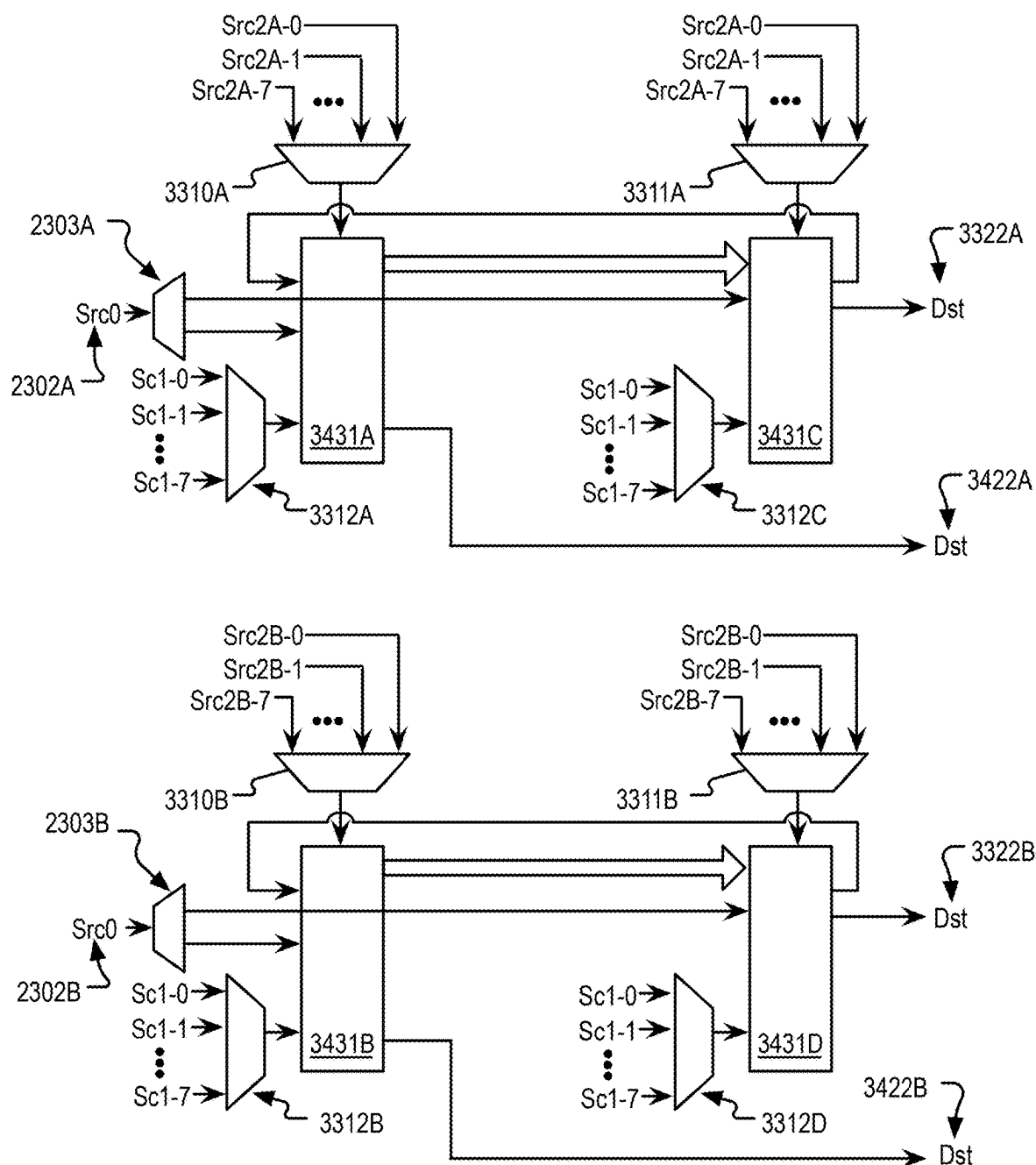


FIG. 34

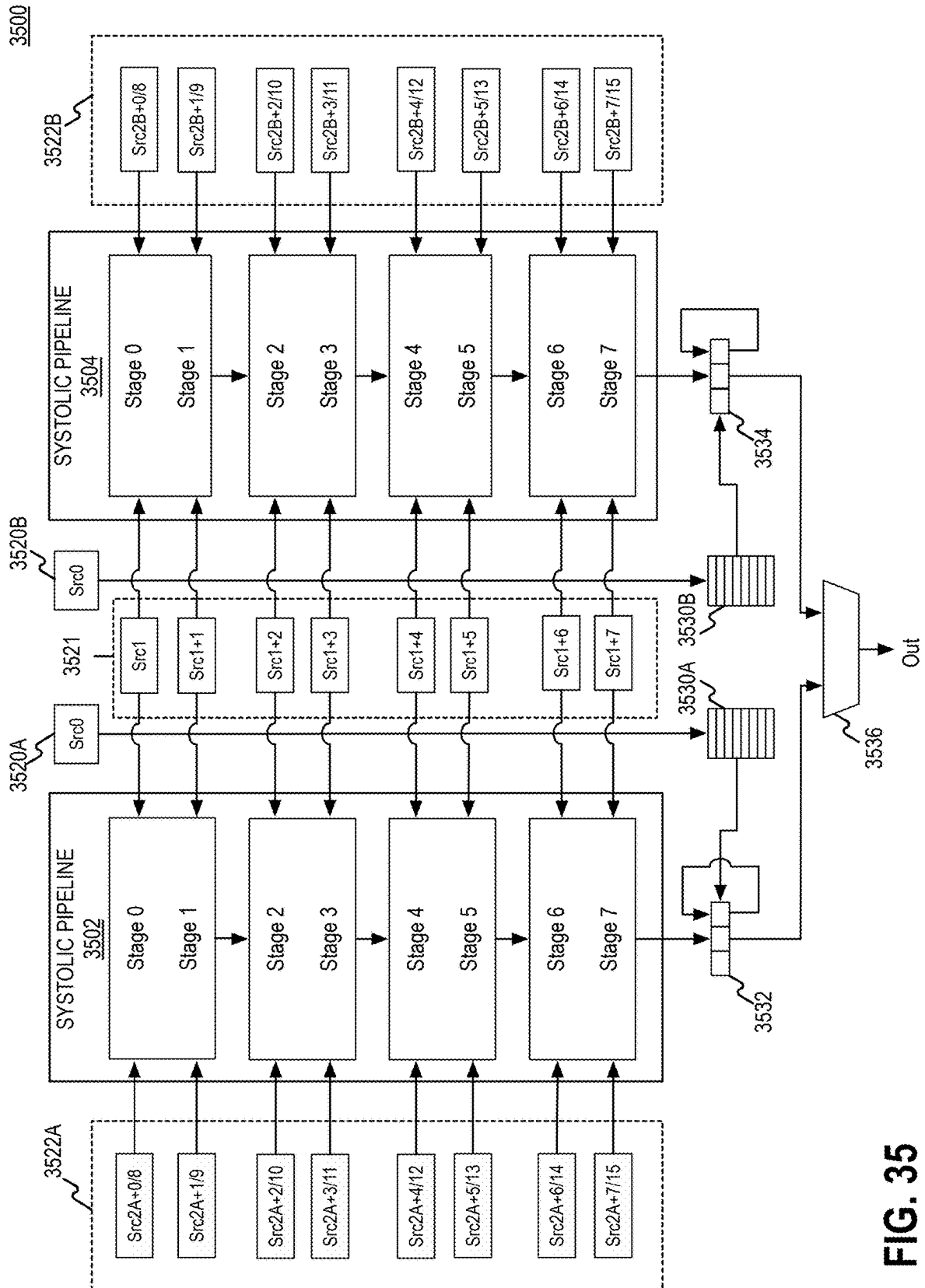


FIG. 35

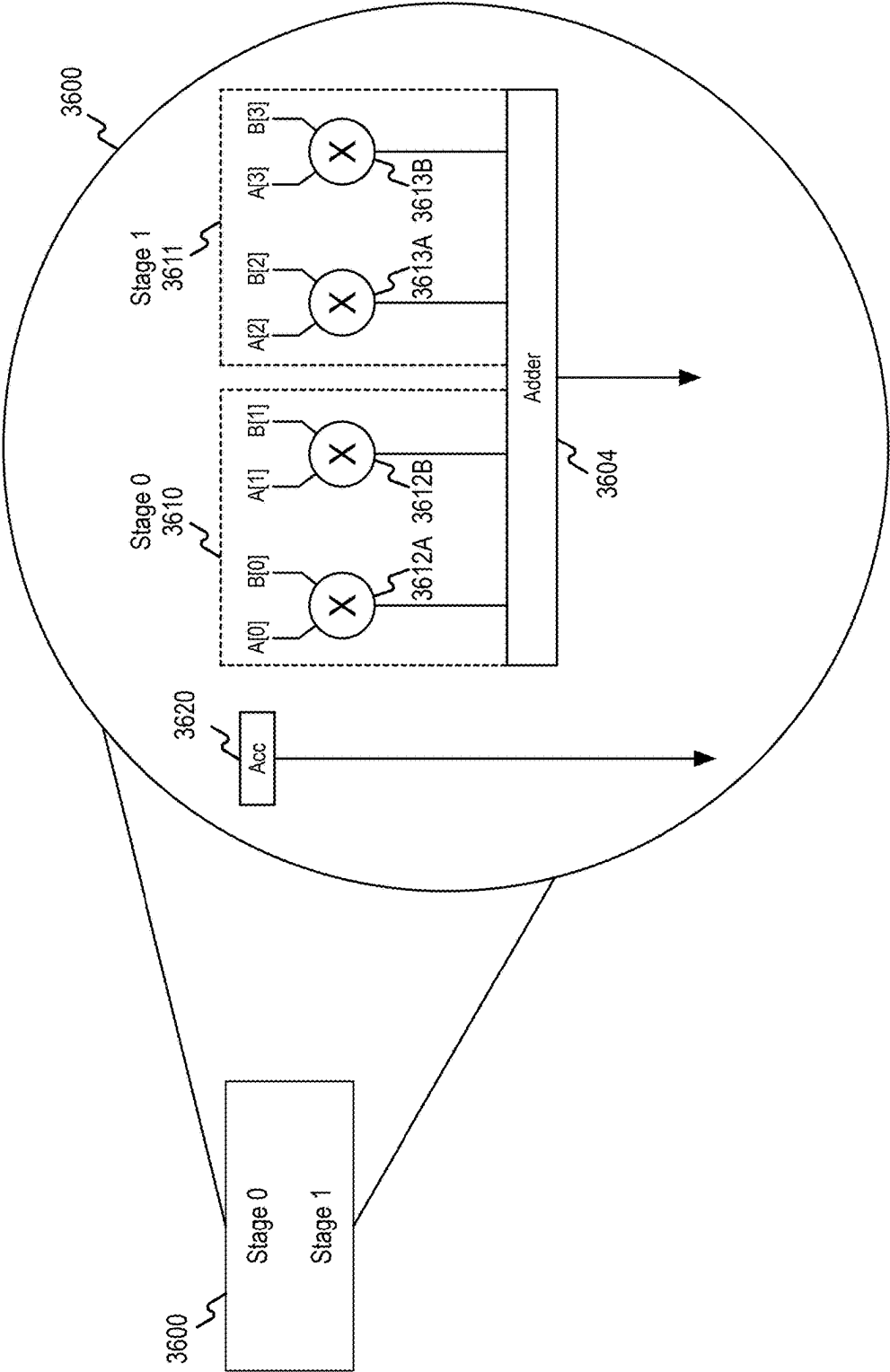


FIG. 36

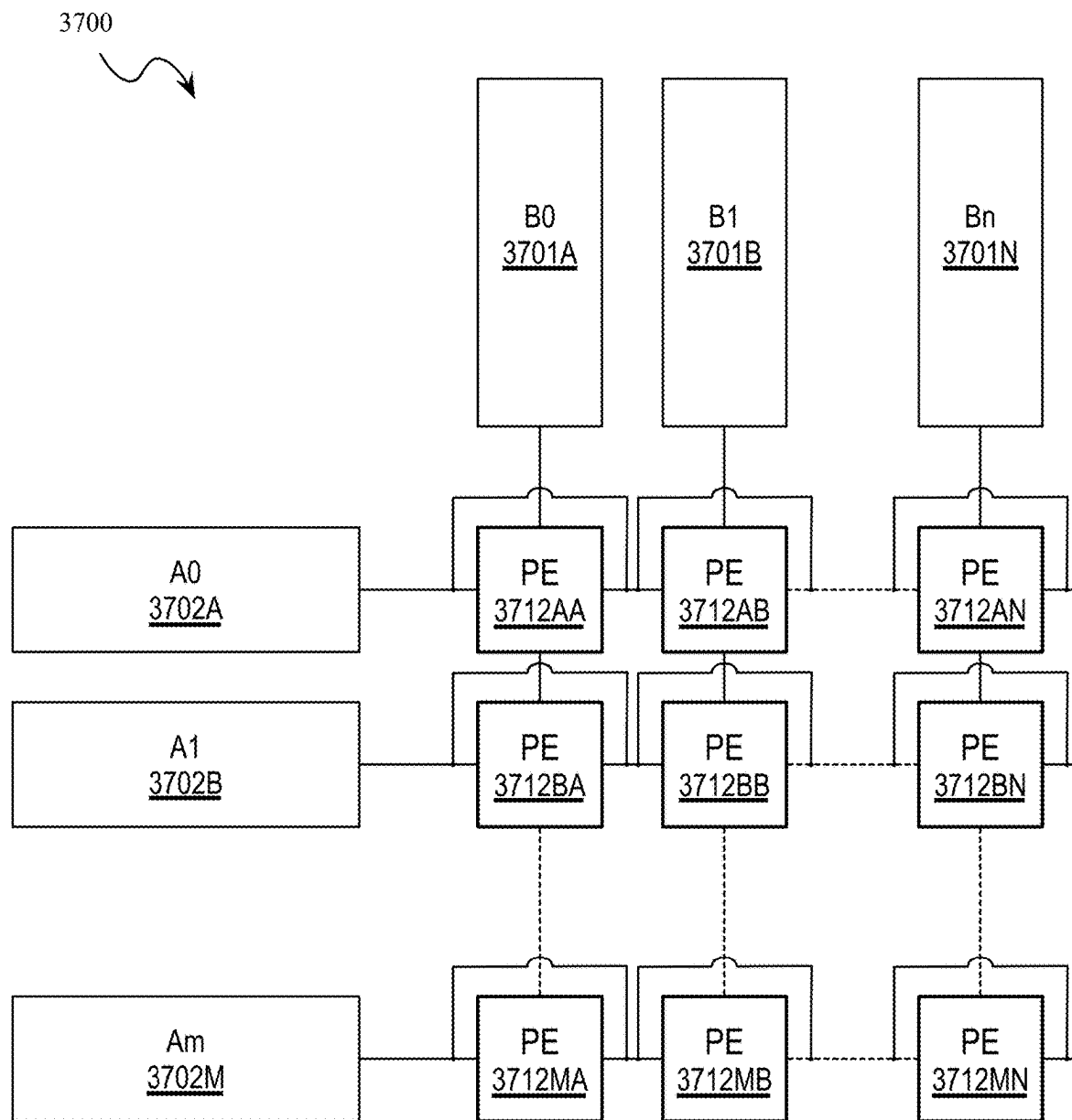


FIG. 37

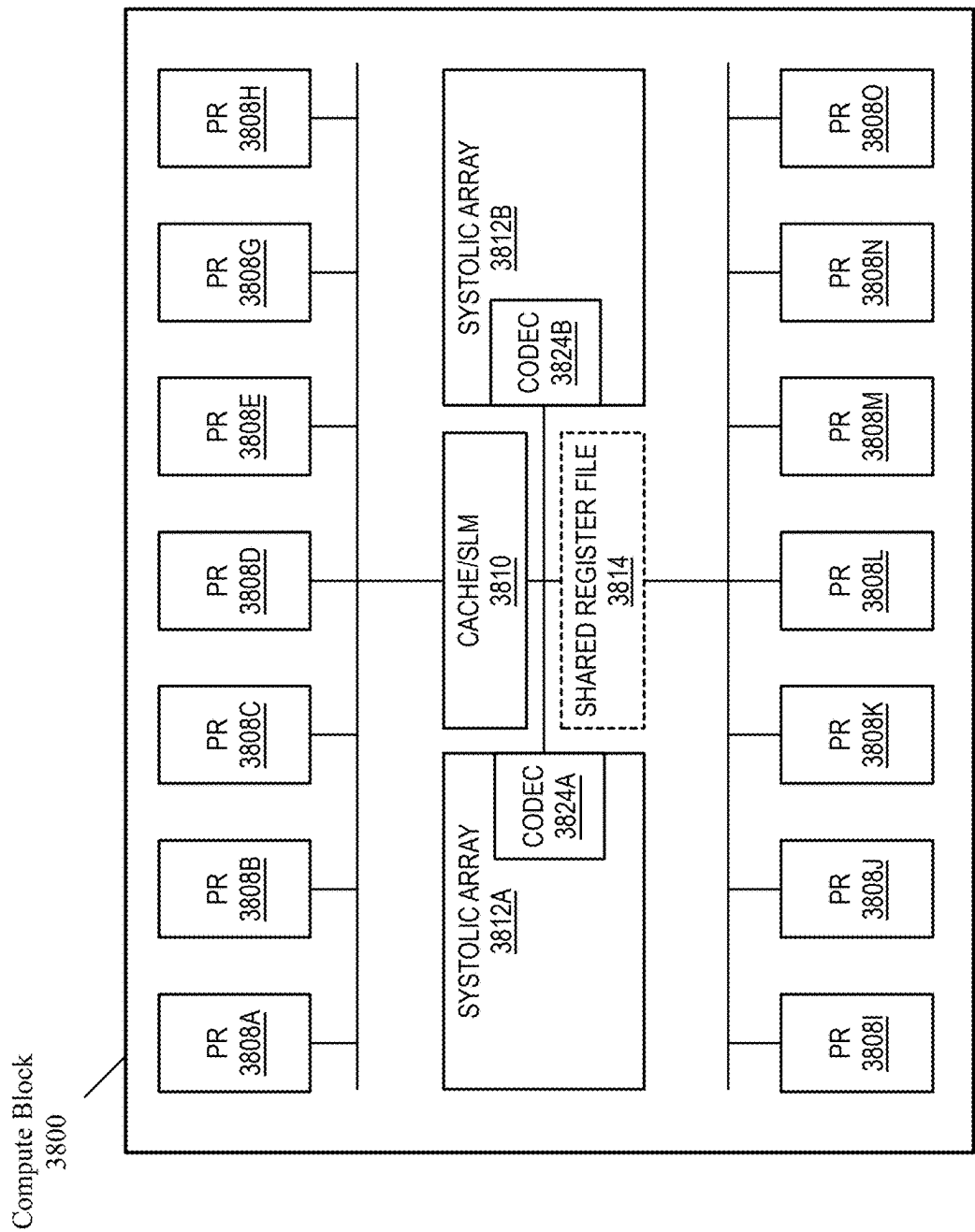
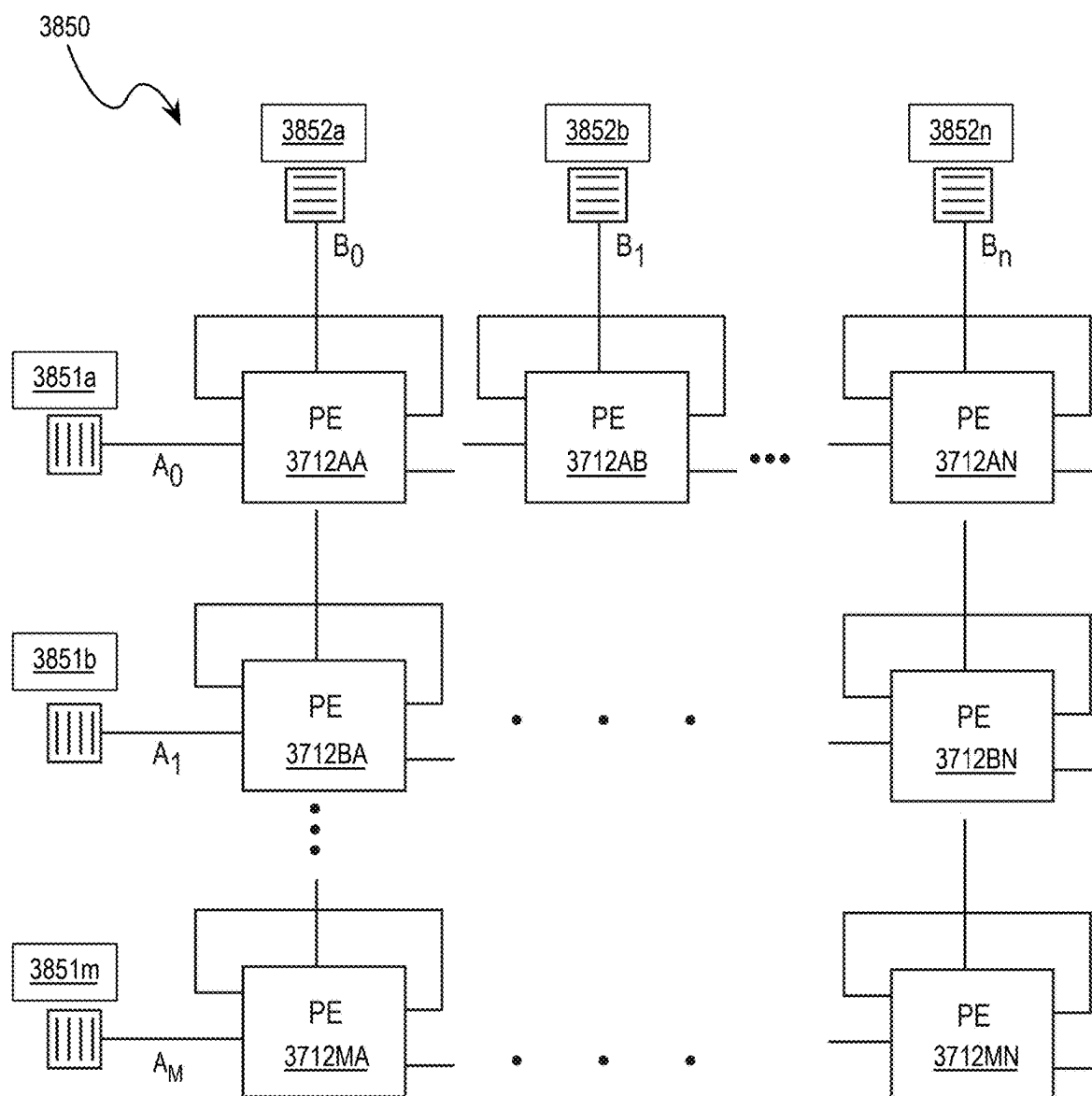


FIG. 38A

**FIG. 38B**

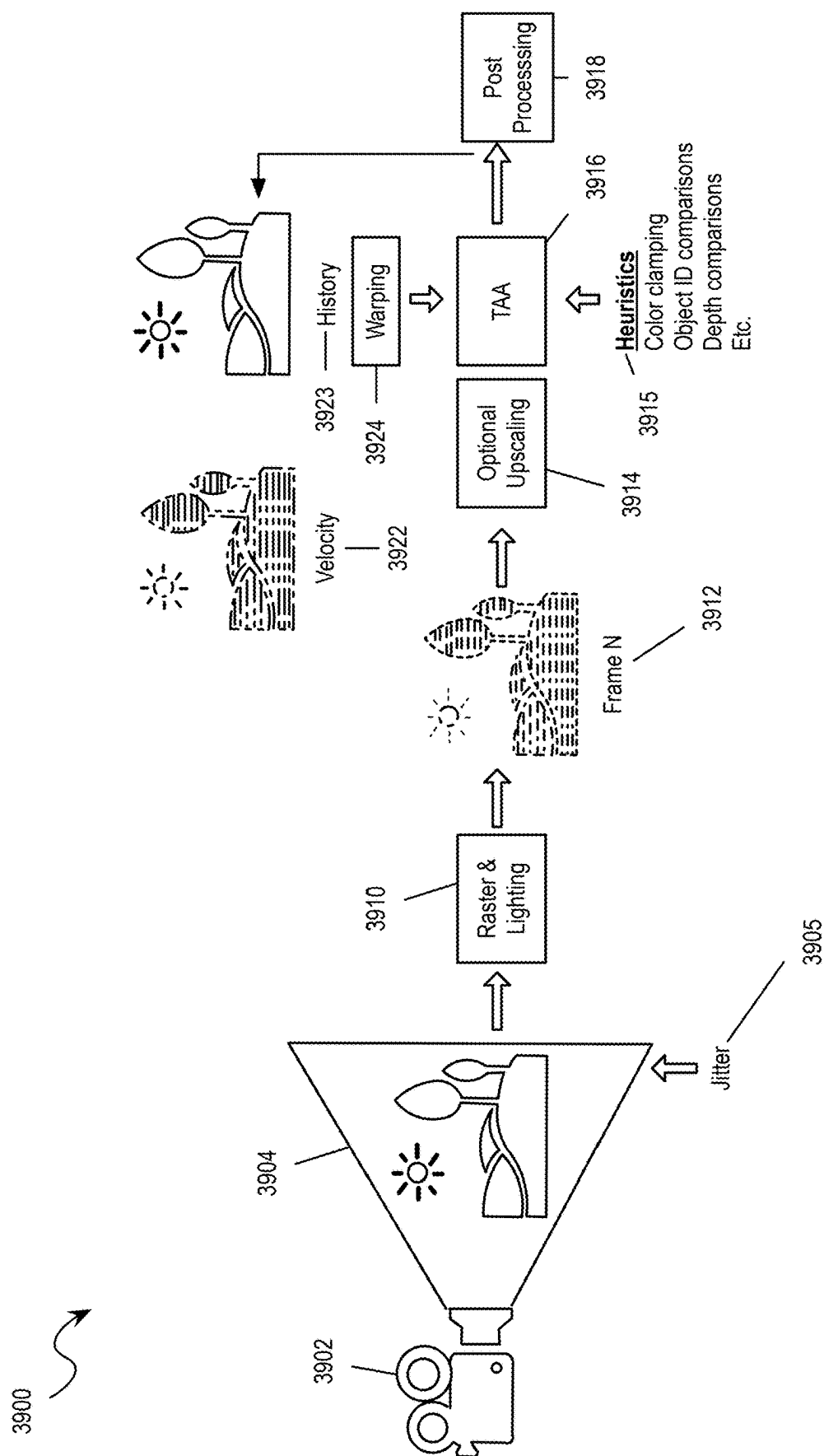


FIG. 39

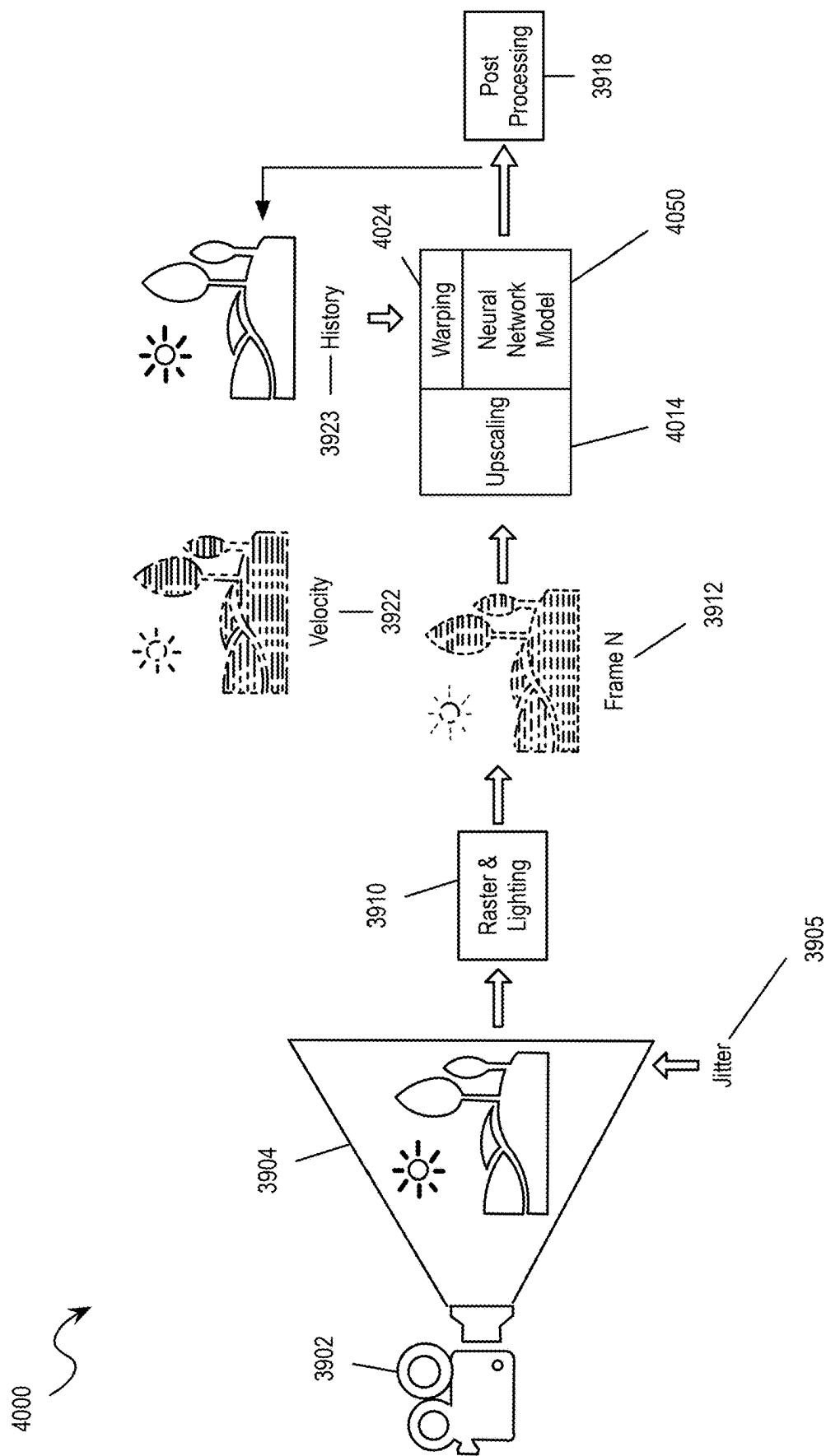


FIG. 40

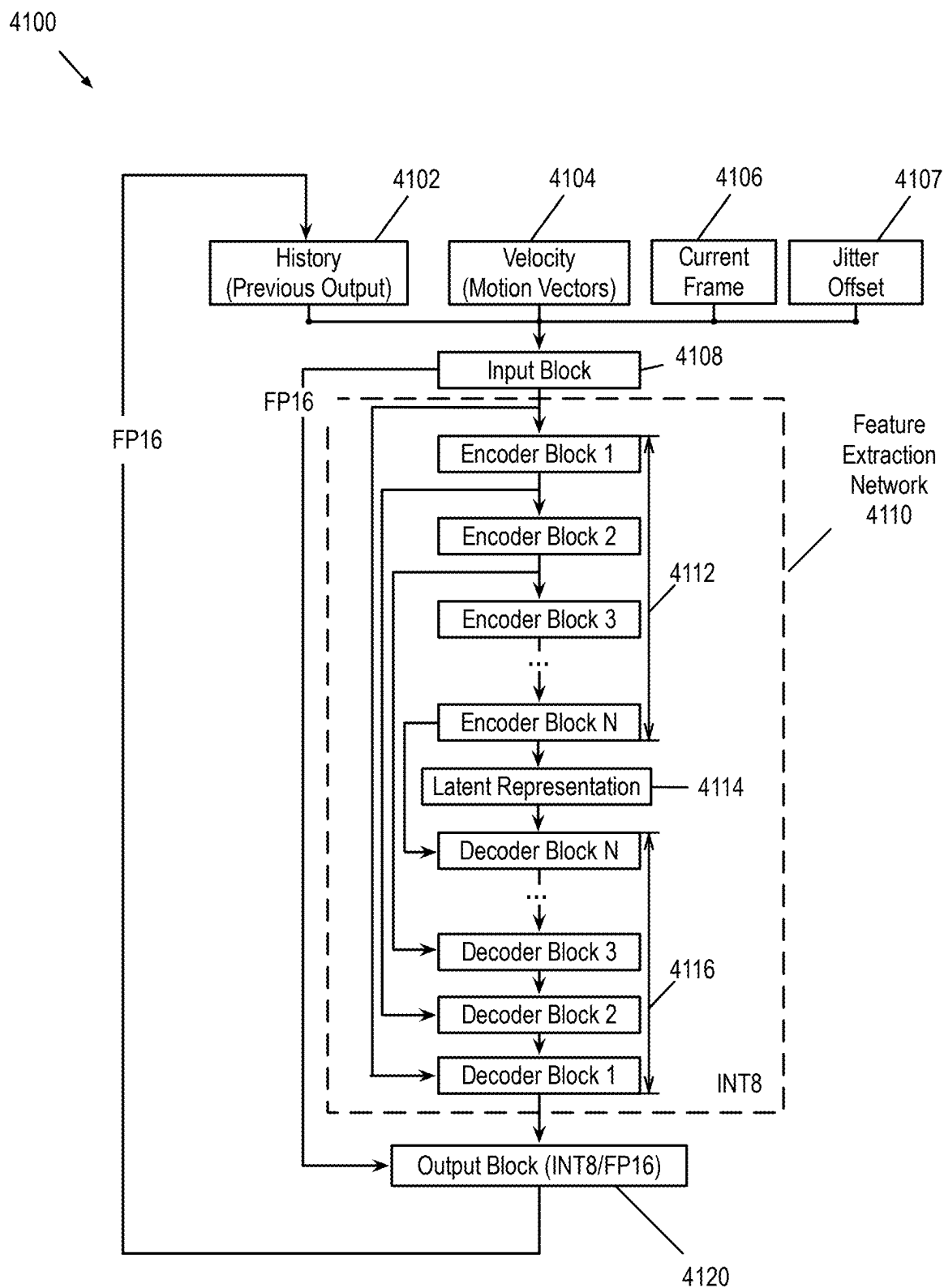


FIG. 41

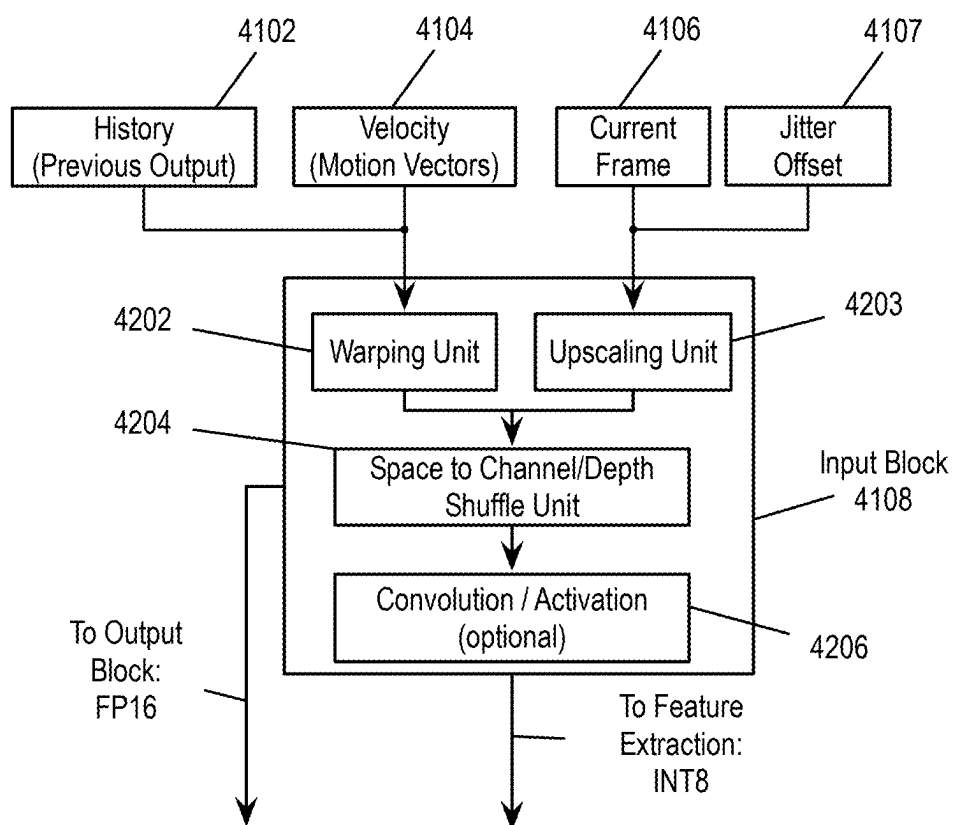


FIG. 42

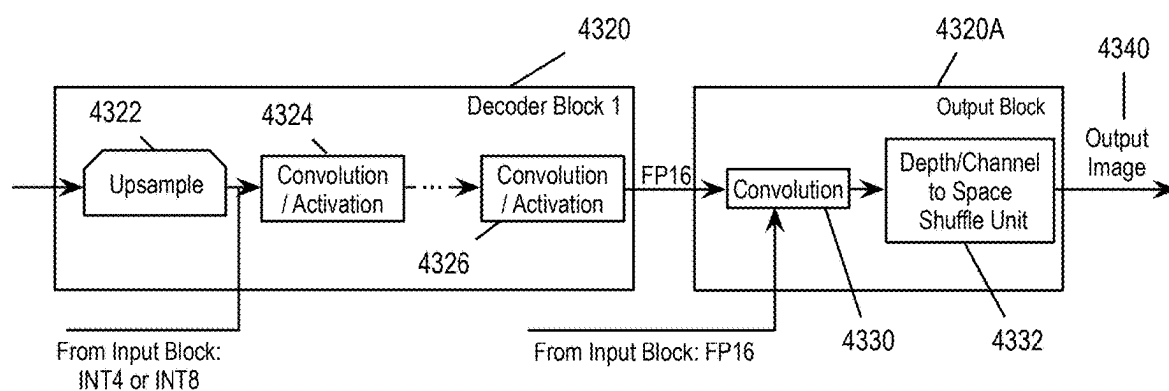


FIG. 43A

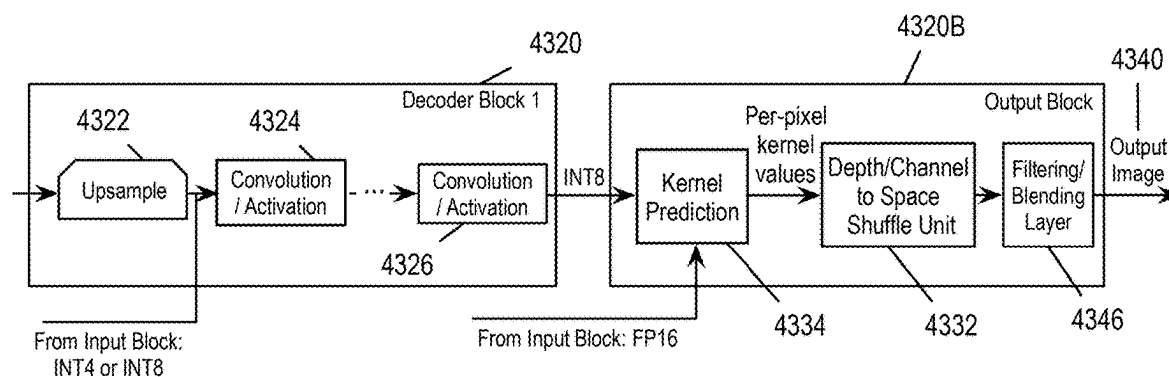


FIG. 43B

4400

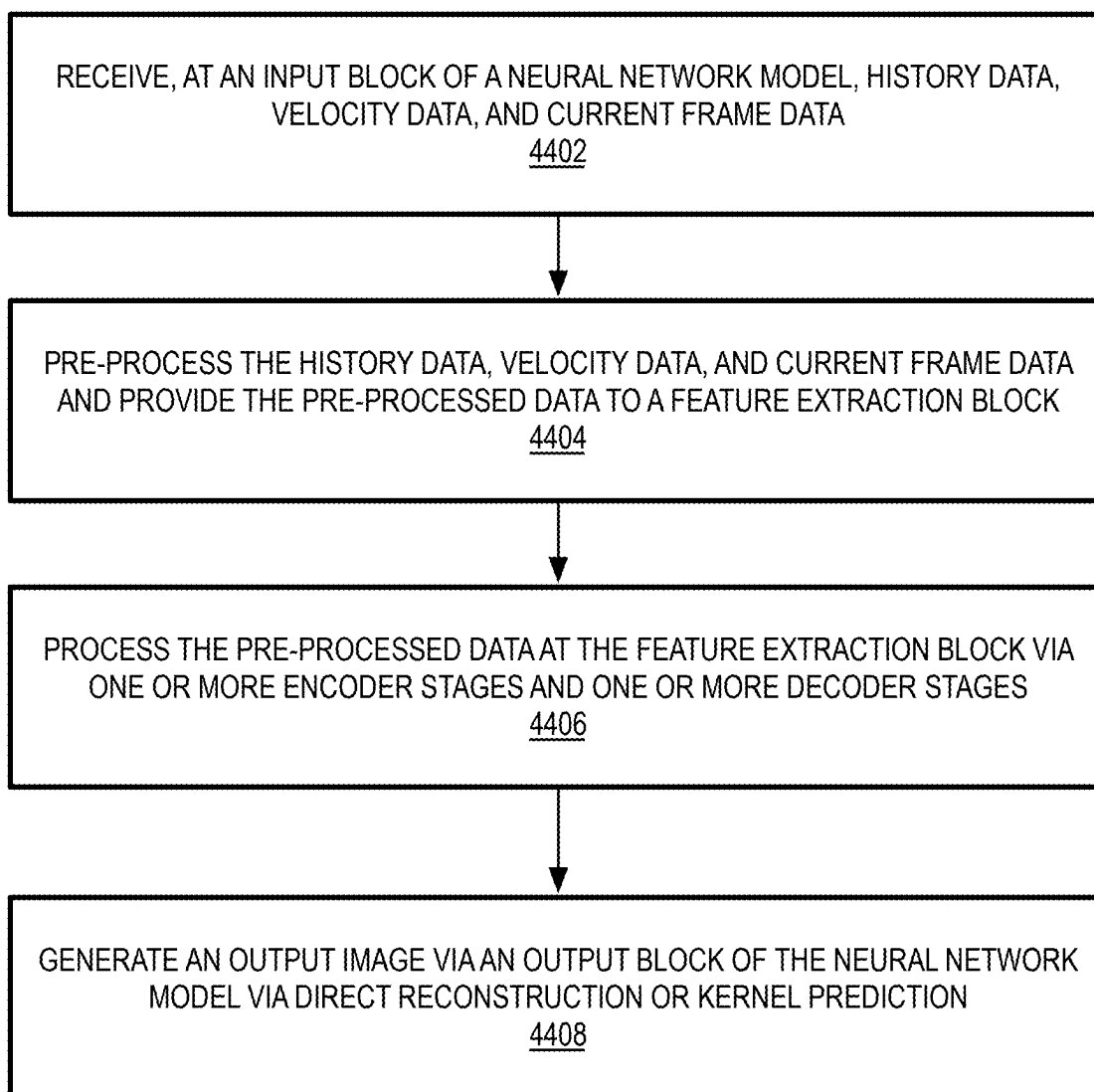
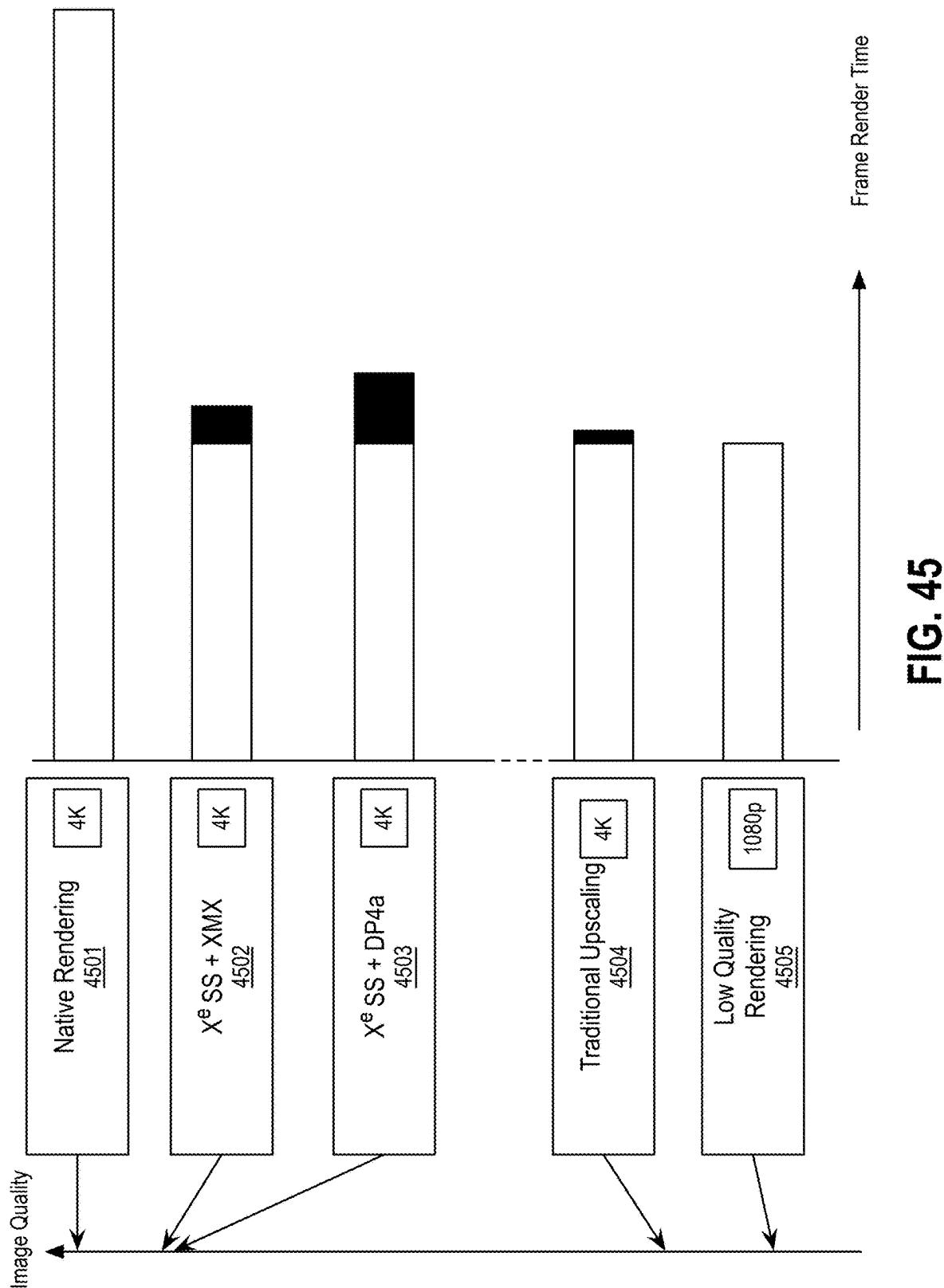


FIG. 44



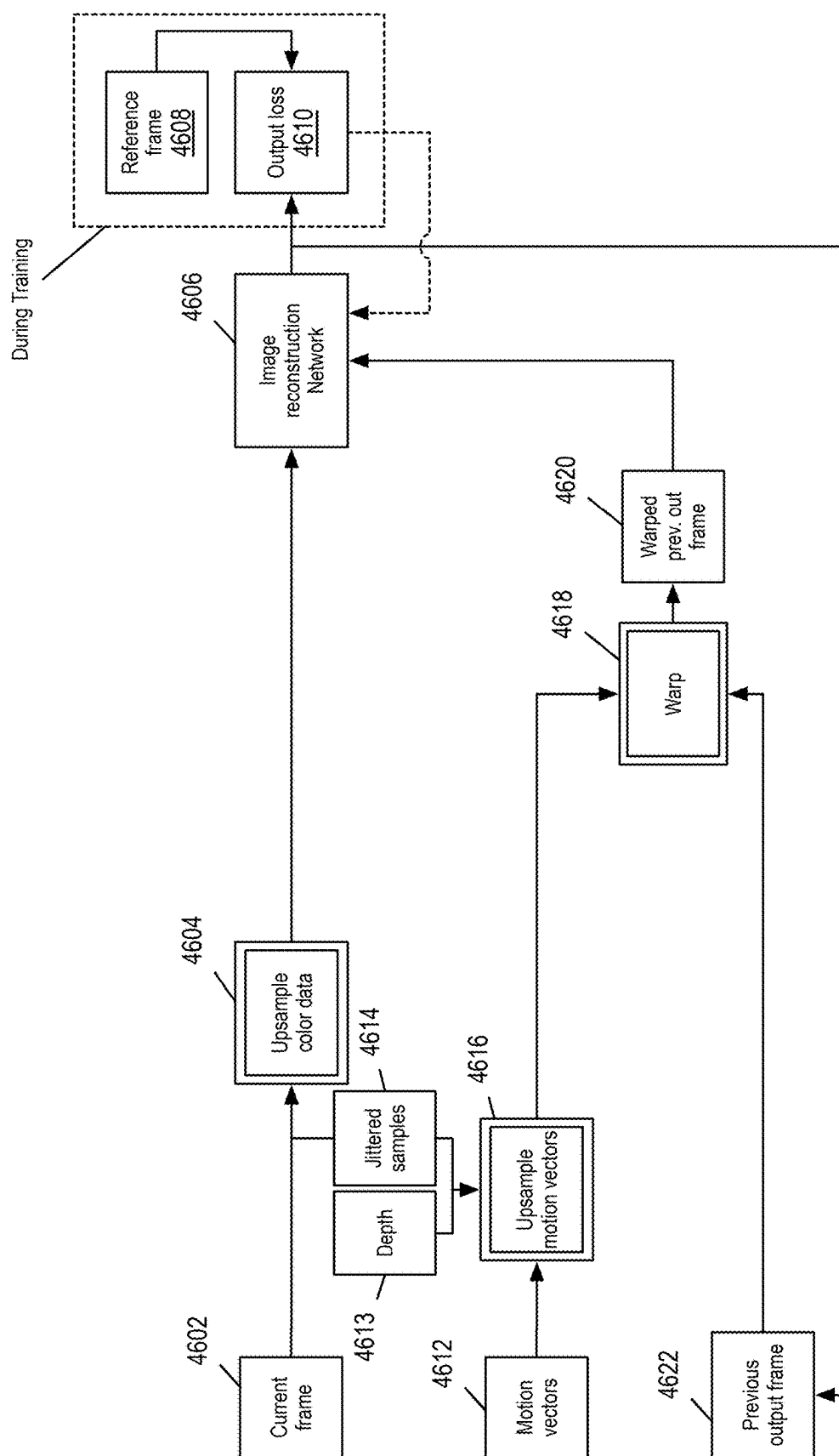


FIG. 46

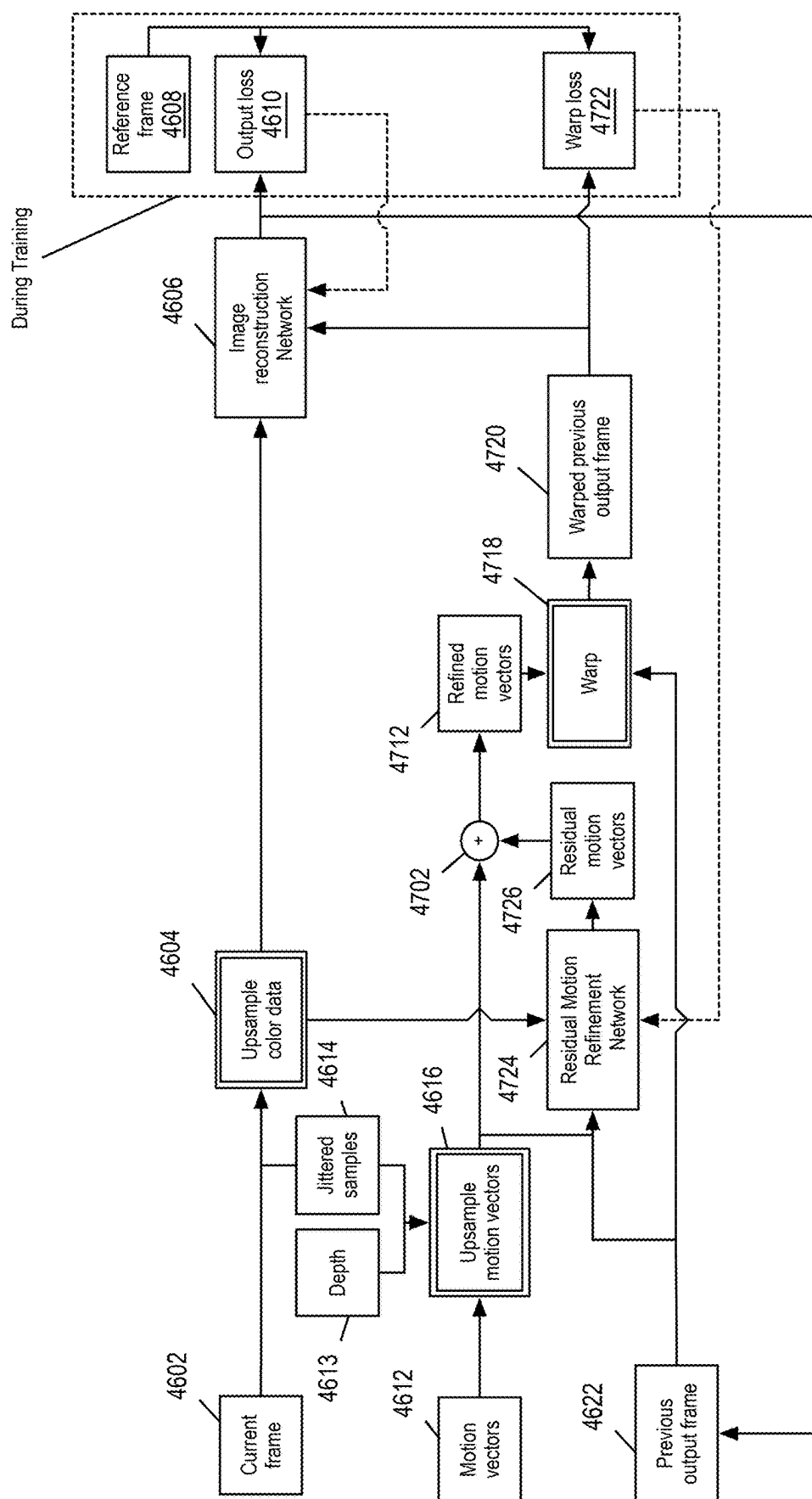


FIG. 47

4800

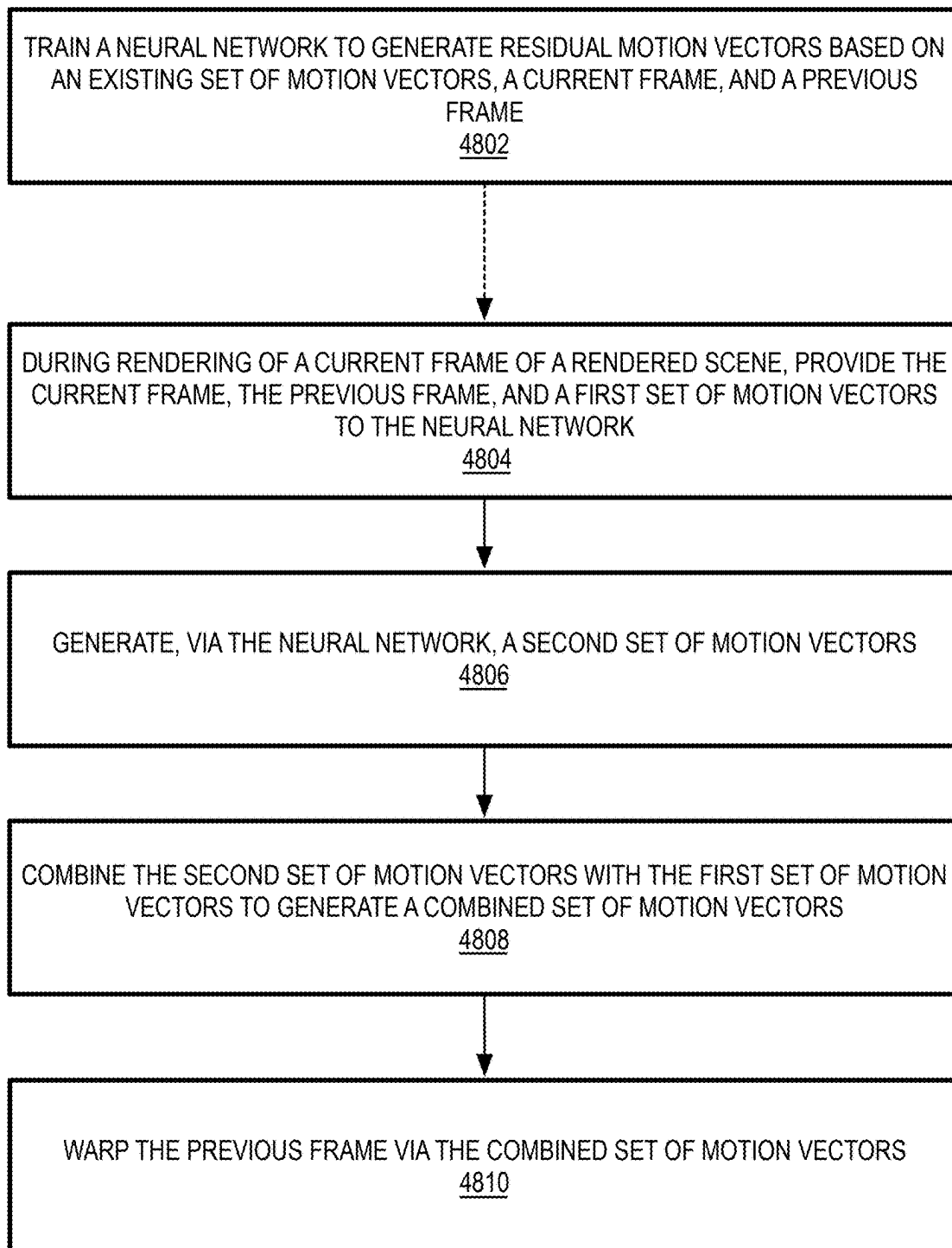


FIG. 48

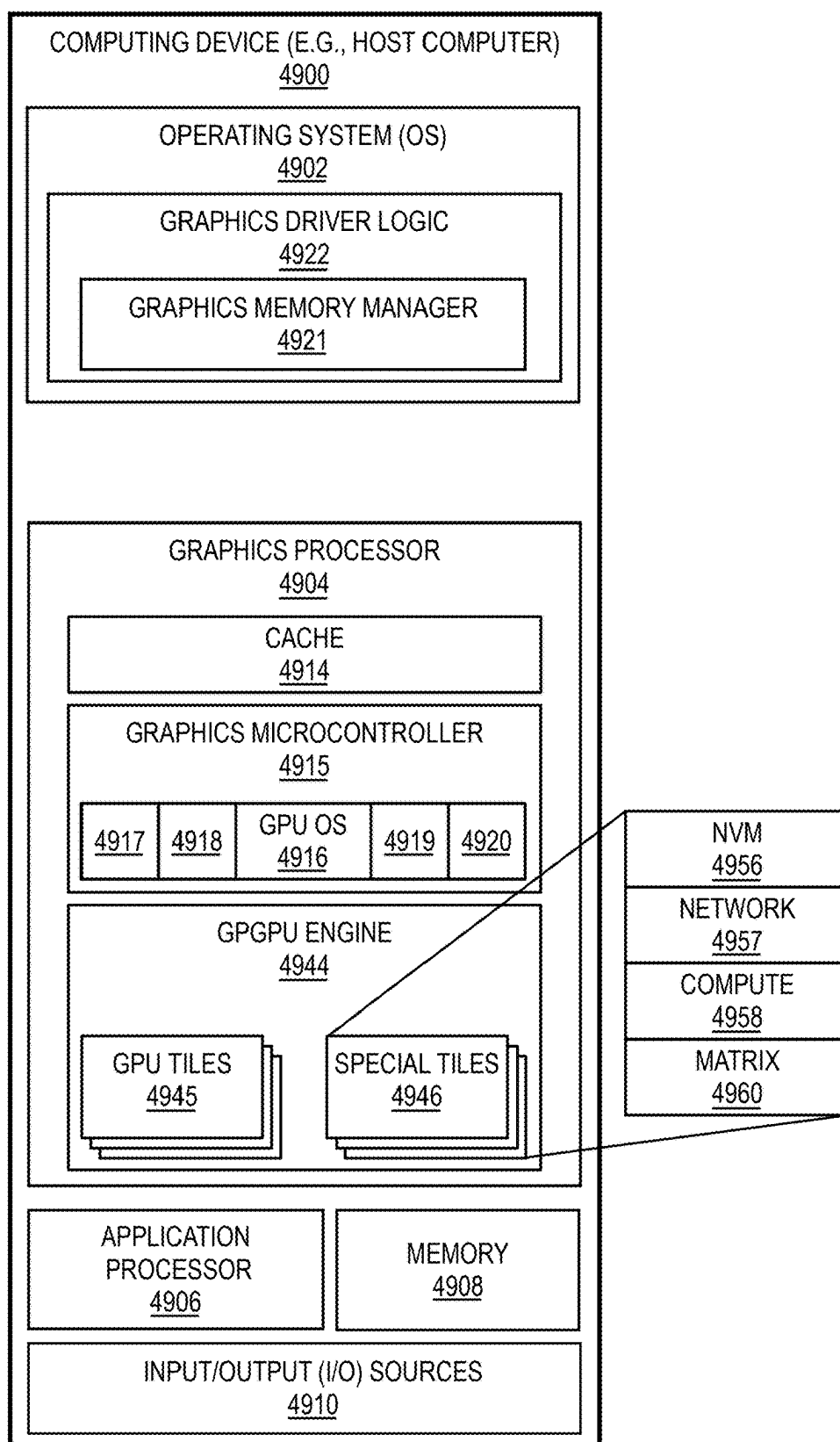


FIG. 49

MOTION VECTOR REFINEMENT FOR TEMPORALLY AMORTIZED SUPERSAMPLING

CROSS-REFERENCE

[0001] This application is a continuation of and claims the benefit of and priority to U.S. application Ser. No. 17/888,253, entitled MOTION VECTOR REFINEMENT FOR TEMPORALLY AMORTIZED SUPERSAMPLING, by SungYe Kim, et al., filed Aug. 15, 2022, now allowed, which claims the benefit of and priority to from U.S. Provisional Application No. 63/276,168, filed Nov. 5, 2021, which is hereby incorporated herein by reference.

FIELD

[0002] This disclosure relates generally to graphics anti-aliasing via neural network operations performed via a matrix accelerator of a graphics processing unit.

BACKGROUND OF THE DISCLOSURE

[0003] Temporal Anti-aliasing (TAA) is an anti-aliasing technique in which the renderer jitters the camera every frame to sample different coordinates in screen space. The TAA stage accumulates these samples temporally to produce a supersampled image. The previously accumulated frame is warped using renderer generated velocity/motion vectors to align it with the current frame before accumulation. Although TAA is a widely used technique to generate temporally stable anti-aliased image, the warped sample history can be mismatched to the current pixel due to frame-to-frame changes in visibility and shading or errors in the motion vectors. This typically results in ghosting artifacts around moving object boundary.

BRIEF DESCRIPTION OF THE DRAWINGS

[0004] The present invention is illustrated by way of example and not limitation in the figures of the accompanying drawings in which like references indicate similar elements, and in which:

[0005] FIG. 1 is a block diagram illustrating a computer system configured to implement one or more aspects of the embodiments described herein;

[0006] FIG. 2A-2D illustrate parallel processor components;

[0007] FIG. 3A-3C are block diagrams of graphics multiprocessors and multiprocessor-based GPUs;

[0008] FIG. 4A-4F illustrate an exemplary architecture in which a plurality of GPUs is communicatively coupled to a plurality of multi-core processors;

[0009] FIG. 5 illustrates a graphics processing pipeline;

[0010] FIG. 6 illustrates a machine learning software stack;

[0011] FIG. 7 illustrates a general-purpose graphics processing unit;

[0012] FIG. 8 illustrates a multi-GPU computing system;

[0013] FIG. 9A-9B illustrate layers of exemplary deep neural networks;

[0014] FIG. 10 illustrates an exemplary recurrent neural network;

[0015] FIG. 11 illustrates training and deployment of a deep neural network;

[0016] FIG. 12A is a block diagram illustrating distributed learning;

[0017] FIG. 12B is a block diagram illustrating a programmable network interface and data processing unit;

[0018] FIG. 13 illustrates an exemplary inferencing system on a chip (SOC) suitable for performing inferencing using a trained model;

[0019] FIG. 14 is a block diagram of a processing system;

[0020] FIG. 15A-15C illustrate computing systems and graphics processors;

[0021] FIG. 16A-16C illustrate block diagrams of additional graphics processor and compute accelerator architectures;

[0022] FIG. 17 is a block diagram of a graphics processing engine of a graphics processor;

[0023] FIG. 18A-18C illustrate thread execution logic including an array of processing elements employed in a graphics processor core;

[0024] FIG. 19 illustrates a tile of a multi-tile processor, according to an embodiment;

[0025] FIG. 20 is a block diagram illustrating a graphics processor instruction formats;

[0026] FIG. 21 is a block diagram of an additional graphics processor architecture;

[0027] FIG. 22A-22B illustrate a graphics processor command format and command sequence;

[0028] FIG. 23 illustrates exemplary graphics software architecture for a data processing system;

[0029] FIG. 24A is a block diagram illustrating an IP core development system;

[0030] FIG. 24B illustrates a cross-section side view of an integrated circuit package assembly;

[0031] FIG. 24C illustrates a package assembly that includes multiple units of hardware logic chiplets connected to a substrate (e.g., base die);

[0032] FIG. 24D illustrates a package assembly including interchangeable chiplets;

[0033] FIG. 25 is a block diagram illustrating an exemplary system on a chip integrated circuit;

[0034] FIG. 26A-26B are block diagrams illustrating exemplary graphics processors for use within an SoC;

[0035] FIG. 27 is a block diagram of a data processing system, according to an embodiment;

[0036] FIG. 28A-28B illustrate a matrix operation performed by an instruction pipeline, according to an embodiment;

[0037] FIG. 29 illustrates a systolic array including multiplier and adder circuits organized in a pipelined fashion;

[0038] FIG. 30A-30B illustrates the use of a systolic array that can be configured to execute operations at an arbitrary systolic depth;

[0039] FIG. 31 illustrates a two-path matrix multiply accelerator in which each path has a depth of four stages;

[0040] FIG. 32 illustrates a four-path matrix multiply accelerator in which each path has a depth of two stages;

[0041] FIG. 33 illustrates a scalable sparse matrix multiply accelerator using systolic arrays with feedback inputs;

[0042] FIG. 34 shows a scalable sparse matrix multiply accelerator using systolic arrays with feedback inputs and outputs on each stage;

[0043] FIG. 35 illustrates a dual pipeline parallel systolic array for a matrix accelerator, according to an embodiment;

[0044] FIG. 36 illustrates a stage pair for a channel of a systolic array;

[0045] FIG. 37 illustrates a systolic array including partial sum loopback and circuitry to accelerate sparse matrix multiply;

[0046] FIG. 38A-38B illustrate matrix acceleration circuitry including codecs to enable the reading of sparse data in a compressed format;

[0047] FIG. 39 illustrates a conventional renderer with Temporal Anti-aliasing (TAA);

[0048] FIG. 40 illustrates a renderer that replaces the TAA stage with a temporally amortized supersampling stage;

[0049] FIG. 41 illustrate components of the neural network model, according to an embodiment;

[0050] FIG. 42 illustrates the input block of the neural network model, according to an embodiment;

[0051] FIG. 43A-43B illustrates output block variants for the neural network model, according to embodiments;

[0052] FIG. 44 illustrates a method to perform temporally amortized supersampling;

[0053] FIG. 45 illustrates exemplary rendering performance comparisons for multiple rendering techniques described herein;

[0054] FIG. 46 illustrates data flow for image reconstruction and upscaling, according to an embodiment;

[0055] FIG. 47 illustrates an extended flow that includes a residual motion refinement network and warp loss optimization, according to an embodiment;

[0056] FIG. 48 illustrates a method of motion vector refinement via a residual motion refinement network, according to an embodiment; and

[0057] FIG. 49 is a block diagram of a computing device including a graphics processor, according to an embodiment.

DETAILED DESCRIPTION

[0058] A graphics processing unit (GPU) is communicatively coupled to host/processor cores to accelerate, for example, graphics operations, machine-learning operations, pattern analysis operations, and/or various general-purpose GPU (GPGPU) functions. The GPU may be communicatively coupled to the host processor/cores over a bus or another interconnect (e.g., a high-speed interconnect such as PCIe or NV Link). Alternatively, the GPU may be integrated on the same package or chip as the cores and communicatively coupled to the cores over an internal processor bus/interconnect (i.e., internal to the package or chip). Regardless of the manner in which the GPU is connected, the processor cores may allocate work to the GPU in the form of sequences of commands/instructions contained in a work descriptor. The GPU then uses dedicated circuitry/logic for efficiently processing these commands/instructions.

[0059] Current parallel graphics data processing includes systems and methods developed to perform specific operations on graphics data such as, for example, linear interpolation, tessellation, rasterization, texture mapping, depth testing, etc. Traditionally, graphics processors used fixed function computational units to process graphics data. However, more recently, portions of graphics processors have been made programmable, enabling such processors to support a wider variety of operations for processing vertex and fragment data.

[0060] To further increase performance, graphics processors typically implement processing techniques such as pipelining that attempt to process, in parallel, as much graphics data as possible throughout the different parts of the graphics pipeline. Parallel graphics processors with single

instruction, multiple thread (SIMT) architectures are designed to maximize the amount of parallel processing in the graphics pipeline. In a SIMT architecture, groups of parallel threads attempt to execute program instructions synchronously together as often as possible to increase processing efficiency. A general overview of software and hardware for SIMT architectures can be found in Shane Cook, *CUDA Programming* Chapter 3, pages 37-51 (2013).

[0061] Temporal upsampling can be combined with TAA to upscale spatial resolution at the same time so that frames are rendered at lower spatial resolution to save render time. Post-process stages after the temporal anti-aliasing upsampling can then run at target display resolution. This allows the creation of sharper images than can be created using spatial-only upscaling techniques and effectively reduces render time than when rendering frames at native display resolution. However, such temporal anti-aliasing upsampling quality is much lower than using TAA for native resolution rendered frames. Described herein is a technique to use a mixed low precision convolutional neural network for temporally amortized supersampling to achieve a performance boost from rendering at lower resolution while also generating high quality images.

[0062] Additionally described herein is a technique to generate residual motion vectors for portions of a scene for which motion vectors are not generated by the renderer. Network inputs are typically warped by motion vectors generated analytically for directly visible surfaces only. Motion vectors are not generated for pixels associated with animated textures, mirrored/reflected objects, and/or moving objects without motion information. Embodiments described herein provide a residual motion refinement network that can be trained to generate residual motion vectors based on renderer generated motion vectors and a comparison between a previous and current frame.

[0063] In the following description, numerous specific details are set forth to provide a more thorough understanding. However, it will be apparent to one of skill in the art that the embodiments described herein may be practiced without one or more of these specific details. In other instances, well-known features have not been described to avoid obscuring the details of the present embodiments.

System Overview

[0064] FIG. 1 is a block diagram illustrating a computing system 100 configured to implement one or more aspects of the embodiments described herein. The computing system 100 includes a processing subsystem 101 having one or more processor(s) 102 and a system memory 104 communicating via an interconnection path that may include a memory hub 105. The memory hub 105 may be a separate component within a chipset component or may be integrated within the one or more processor(s) 102. The memory hub 105 couples with an I/O subsystem 111 via a communication link 106. The I/O subsystem 111 includes an I/O hub 107 that can enable the computing system 100 to receive input from one or more input device(s) 108. Additionally, the I/O hub 107 can enable a display controller, which may be included in the one or more processor(s) 102, to provide outputs to one or more display device(s) 110A. In one embodiment the one or more display device(s) 110A coupled with the I/O hub 107 can include a local, internal, or embedded display device.

[0065] The processing subsystem 101, for example, includes one or more parallel processor(s) 112 coupled to memory hub 105 via a bus or other communication link 113. The communication link 113 may be one of any number of standards-based communication link technologies or protocols, such as, but not limited to PCI Express, or may be a vendor specific communications interface or communications fabric. The one or more parallel processor(s) 112 may form a computationally focused parallel or vector processing system that can include a large number of processing cores and/or processing clusters, such as a many integrated core (MIC) processor. For example, the one or more parallel processor(s) 112 form a graphics processing subsystem that can output pixels to one of the one or more display device(s) 110A coupled via the I/O hub 107. The one or more parallel processor(s) 112 can also include a display controller and display interface (not shown) to enable a direct connection to one or more display device(s) 110B.

[0066] Within the I/O subsystem 111, a system storage unit 114 can connect to the I/O hub 107 to provide a storage mechanism for the computing system 100. An I/O switch 116 can be used to provide an interface mechanism to enable connections between the I/O hub 107 and other components, such as a network adapter 118 and/or wireless network adapter 119 that may be integrated into the platform, and various other devices that can be added via one or more add-in device(s) 120. The add-in device(s) 120 may also include, for example, one or more external graphics processor devices, graphics cards, and/or compute accelerators. The network adapter 118 can be an Ethernet adapter or another wired network adapter. The wireless network adapter 119 can include one or more of a Wi-Fi, Bluetooth, near field communication (NFC), or other network device that includes one or more wireless radios.

[0067] The computing system 100 can include other components not explicitly shown, including USB or other port connections, optical storage drives, video capture devices, and the like, which may also be connected to the I/O hub 107. Communication paths interconnecting the various components in FIG. 1 may be implemented using any suitable protocols, such as PCI (Peripheral Component Interconnect) based protocols (e.g., PCI-Express), or any other bus or point-to-point communication interfaces and/or protocol(s), such as the NV Link high-speed interconnect, Compute Express Link™ (CXL™) (e.g., CXL.mem), Infinity Fabric (IF), Ethernet (IEEE 802.3), remote direct memory access (RDMA), InfiniBand, Internet Wide Area RDMA Protocol (iWARP), Transmission Control Protocol (TCP), User Datagram Protocol (UDP), quick UDP Internet Connections (QUIC), RDMA over Converged Ethernet (RoCE), Intel QuickPath Interconnect (QPI), Intel Ultra Path Interconnect (UPI), Intel On-Chip System Fabric (IOSF), OmniPath, HyperTransport, Advanced Microcontroller Bus Architecture (AMBA) interconnect, OpenCAPI, Gen-Z, Cache Coherent Interconnect for Accelerators (CCIX), 3GPP Long Term Evolution (LTE) (4G), 3GPP 5G, and variations thereof, or wired or wireless interconnect protocols known in the art. In some examples, data can be copied or stored to virtualized storage nodes using a protocol such as non-volatile memory express (NVM e) over Fabrics (NVMe-oF) or NVMe.

[0068] The one or more parallel processor(s) 112 may incorporate circuitry optimized for graphics and video processing, including, for example, video output circuitry, and

constitutes a graphics processing unit (GPU). Alternatively or additionally, the one or more parallel processor(s) 112 can incorporate circuitry optimized for general purpose processing, while preserving the underlying computational architecture, described in greater detail herein. Components of the computing system 100 may be integrated with one or more other system elements on a single integrated circuit. For example, the one or more parallel processor(s) 112, memory hub 105, processor(s) 102, and I/O hub 107 can be integrated into a system on chip (SoC) integrated circuit. Alternatively, the components of the computing system 100 can be integrated into a single package to form a system in package (SIP) configuration. In one embodiment at least a portion of the components of the computing system 100 can be integrated into a multi-chip module (MCM), which can be interconnected with other multi-chip modules into a modular computing system.

[0069] It will be appreciated that the computing system 100 shown herein is illustrative and that variations and modifications are possible. The connection topology, including the number and arrangement of bridges, the number of processor(s) 102, and the number of parallel processor(s) 112, may be modified as desired. For instance, system memory 104 can be connected to the processor(s) 102 directly rather than through a bridge, while other devices communicate with system memory 104 via the memory hub 105 and the processor(s) 102. In other alternative topologies, the parallel processor(s) 112 are connected to the I/O hub 107 or directly to one of the one or more processor(s) 102, rather than to the memory hub 105. In other embodiments, the I/O hub 107 and memory hub 105 may be integrated into a single chip. It is also possible that two or more sets of processor(s) 102 are attached via multiple sockets, which can couple with two or more instances of the parallel processor(s) 112.

[0070] Some of the particular components shown herein are optional and may not be included in all implementations of the computing system 100. For example, any number of add-in cards or peripherals may be supported, or some components may be eliminated. Furthermore, some architectures may use different terminology for components similar to those illustrated in FIG. 1. For example, the memory hub 105 may be referred to as a Northbridge in some architectures, while the I/O hub 107 may be referred to as a Southbridge.

[0071] FIG. 2A illustrates a parallel processor 200. The parallel processor 200 may be a GPU, GPGPU or the like as described herein. The various components of the parallel processor 200 may be implemented using one or more integrated circuit devices, such as programmable processors, application specific integrated circuits (ASICs), or field programmable gate arrays (FPGA). The illustrated parallel processor 200 may be one or more of the parallel processor(s) 112 shown in FIG. 1.

[0072] The parallel processor 200 includes a parallel processing unit 202. The parallel processing unit includes an I/O unit 204 that enables communication with other devices, including other instances of the parallel processing unit 202. The I/O unit 204 may be directly connected to other devices. For instance, the I/O unit 204 connects with other devices via the use of a hub or switch interface, such as memory hub 105. The connections between the memory hub 105 and the I/O unit 204 form a communication link 113. Within the parallel processing unit 202, the I/O unit 204 connects with

a host interface **206** and a memory crossbar **216**, where the host interface **206** receives commands directed to performing processing operations and the memory crossbar **216** receives commands directed to performing memory operations.

[0073] When the host interface **206** receives a command buffer via the I/O unit **204**, the host interface **206** can direct work operations to perform those commands to a front end **208**. In one embodiment the front end **208** couples with a scheduler **210**, which is configured to distribute commands or other work items to a processing cluster array **212**. The scheduler **210** ensures that the processing cluster array **212** is properly configured and in a valid state before tasks are distributed to the processing clusters of the processing cluster array **212**. The scheduler **210** may be implemented via firmware logic executing on a microcontroller. The microcontroller implemented scheduler **210** is configurable to perform complex scheduling and work distribution operations at coarse and fine granularity, enabling rapid preemption and context switching of threads executing on the processing cluster array **212**. Preferably, the host software can prove workloads for scheduling on the processing cluster array **212** via one of multiple graphics processing doorbells. In other examples, polling for new workloads or interrupts can be used to identify or indicate availability of work to perform. The workloads can then be automatically distributed across the processing cluster array **212** by the scheduler **210** logic within the scheduler microcontroller.

[0074] The processing cluster array **212** can include up to “N” processing clusters (e.g., cluster **214A**, cluster **214B**, through cluster **214N**). Each cluster **214A-214N** of the processing cluster array **212** can execute a large number of concurrent threads. The scheduler **210** can allocate work to the clusters **214A-214N** of the processing cluster array **212** using various scheduling and/or work distribution algorithms, which may vary depending on the workload arising for each type of program or computation. The scheduling can be handled dynamically by the scheduler **210** or can be assisted in part by compiler logic during compilation of program logic configured for execution by the processing cluster array **212**. Optionally, different clusters **214A-214N** of the processing cluster array **212** can be allocated for processing different types of programs or for performing different types of computations.

[0075] The processing cluster array **212** can be configured to perform various types of parallel processing operations. For example, the processing cluster array **212** is configured to perform general-purpose parallel compute operations. For example, the processing cluster array **212** can include logic to execute processing tasks including filtering of video and/or audio data, performing modeling operations, including physics operations, and performing data transformations.

[0076] The processing cluster array **212** is configured to perform parallel graphics processing operations. In such embodiments in which the parallel processor **200** is configured to perform graphics processing operations, the processing cluster array **212** can include additional logic to support the execution of such graphics processing operations, including, but not limited to texture sampling logic and other vertex processing logic. Additionally, the processing cluster array **212** can be configured to execute graphics processing related shader programs such as, but not limited

to vertex shaders, tessellation shaders, geometry shaders, and pixel shaders. The parallel processing unit **202** can transfer data from system memory via the I/O unit **204** for processing. During processing the transferred data can be stored to on-chip memory (e.g., parallel processor memory **222**) during processing, then written back to system memory.

[0077] In embodiments in which the parallel processing unit **202** is used to perform graphics processing, the scheduler **210** may be configured to divide the processing workload into approximately equal sized tasks, to better enable distribution of the graphics processing operations to multiple clusters **214A-214N** of the processing cluster array **212**. In some of these embodiments, portions of the processing cluster array **212** can be configured to perform different types of processing. For example, a first portion may be configured to perform vertex shading and topology generation, a second portion may be configured to perform tessellation and geometry shading, and a third portion may be configured to perform pixel shading or other screen space operations, to produce a rendered image for display. Intermediate data produced by one or more of the clusters **214A-214N** may be stored in buffers to allow the intermediate data to be transmitted between clusters **214A-214N** for further processing.

[0078] During operation, the processing cluster array **212** can receive processing tasks to be executed via the scheduler **210**, which receives commands defining processing tasks from front end **208**. For graphics processing operations, processing tasks can include indices of data to be processed, e.g., surface (patch) data, primitive data, vertex data, and/or pixel data, as well as state parameters and commands defining how the data is to be processed (e.g., what program is to be executed). The scheduler **210** may be configured to fetch the indices corresponding to the tasks or may receive the indices from the front end **208**. The front end **208** can be configured to ensure the processing cluster array **212** is configured to a valid state before the workload specified by incoming command buffers (e.g., batch-buffers, push buffers, etc.) is initiated.

[0079] Each of the one or more instances of the parallel processing unit **202** can couple with parallel processor memory **222**. The parallel processor memory **222** can be accessed via the memory crossbar **216**, which can receive memory requests from the processing cluster array **212** as well as the I/O unit **204**. The memory crossbar **216** can access the parallel processor memory **222** via a memory interface **218**. The memory interface **218** can include multiple partition units (e.g., partition unit **220A**, partition unit **220B**, through partition unit **220N**) that can each couple to a portion (e.g., memory unit) of parallel processor memory **222**. The number of partition units **220A-220N** may be configured to be equal to the number of memory units, such that a first partition unit **220A** has a corresponding first memory unit **224A**, a second partition unit **220B** has a corresponding second memory unit **224B**, and an Nth partition unit **220N** has a corresponding Nth memory unit **224N**. In other embodiments, the number of partition units **220A-220N** may not be equal to the number of memory devices.

[0080] The memory units **224A-224N** can include various types of memory devices, including dynamic random-access memory (DRAM) or graphics random access memory, such as synchronous graphics random access memory (SGRAM),

including graphics double data rate (GDDR) memory. Optionally, the memory units 224A-224N may also include 3D stacked memory, including but not limited to high bandwidth memory (HBM). Persons skilled in the art will appreciate that the specific implementation of the memory units 224A-224N can vary and can be selected from one of various conventional designs. Render targets, such as frame buffers or texture maps may be stored across the memory units 224A-224N, allowing partition units 220A-220N to write portions of each render target in parallel to efficiently use the available bandwidth of parallel processor memory 222. In some embodiments, a local instance of the parallel processor memory 222 may be excluded in favor of a unified memory design that utilizes system memory in conjunction with local cache memory.

[0081] Optionally, any one of the clusters 214A-214N of the processing cluster array 212 has the ability to process data that will be written to any of the memory units 224A-224N within parallel processor memory 222. The memory crossbar 216 can be configured to transfer the output of each cluster 214A-214N to any partition unit 220A-220N or to another cluster 214A-214N, which can perform additional processing operations on the output. Each cluster 214A-214N can communicate with the memory interface 218 through the memory crossbar 216 to read from or write to various external memory devices. In one of the embodiments with the memory crossbar 216 the memory crossbar 216 has a connection to the memory interface 218 to communicate with the I/O unit 204, as well as a connection to a local instance of the parallel processor memory 222, enabling the processing units within the different processing clusters 214A-214N to communicate with system memory or other memory that is not local to the parallel processing unit 202. Generally, the memory crossbar 216 may, for example, be able to use virtual channels to separate traffic streams between the clusters 214A-214N and the partition units 220A-220N.

[0082] While a single instance of the parallel processing unit 202 is illustrated within the parallel processor 200, any number of instances of the parallel processing unit 202 can be included. For example, multiple instances of the parallel processing unit 202 can be provided on a single add-in card, or multiple add-in cards can be interconnected. For example, the parallel processor 200 can be an add-in device, such as add-in device 120 of FIG. 1, which may be a graphics card such as a discrete graphics card that includes one or more GPUs, one or more memory devices, and device-to-device or network or fabric interfaces. The different instances of the parallel processing unit 202 can be configured to interoperate even if the different instances have different numbers of processing cores, different amounts of local parallel processor memory, and/or other configuration differences. Optionally, some instances of the parallel processing unit 202 can include higher precision floating point units relative to other instances. Systems incorporating one or more instances of the parallel processing unit 202 or the parallel processor 200 can be implemented in a variety of configurations and form factors, including but not limited to desktop, laptop, or handheld personal computers, servers, workstations, game consoles, and/or embedded systems. An orchestrator can form composite nodes for workload performance using one or more of: disaggregated processor resources, cache resources, memory resources, storage resources, and networking resources.

[0083] In one embodiment, the parallel processing unit 202 can be partitioned into multiple instances. Those multiple instances can be configured to execute workloads associated with different clients in an isolated manner, enabling a pre-determined quality of service to be provided for each client. For example, each cluster 214A-214N can be compartmentalized and isolated from other clusters, allowing the processing cluster array 212 to be divided into multiple compute partitions or instances. In such configuration, workloads that execute on an isolated partition are protected from faults or errors associated with a different workload that executes on a different partition. The partition units 220A-220N can be configured to enable a dedicated and/or isolated path to memory for the clusters 214A-214N associated with the respective compute partitions. This data-path isolation enables the compute resources within a partition can communicate with one or more assigned memory units 224A-224N without being subjected to inference by the activities of other partitions.

[0084] FIG. 2B is a block diagram of a partition unit 220. The partition unit 220 may be an instance of one of the partition units 220A-220N of FIG. 2A. As illustrated, the partition unit 220 includes an L2 cache 221, a frame buffer interface 225, and a ROP 226 (raster operations unit). The L2 cache 221 is a read/write cache that is configured to perform load and store operations received from the memory crossbar 216 and ROP 226. Read misses and urgent write-back requests are output by L2 cache 221 to frame buffer interface 225 for processing. Updates can also be sent to the frame buffer via the frame buffer interface 225 for processing. In one embodiment the frame buffer interface 225 interfaces with one of the memory units in parallel processor memory, such as the memory units 224A-224N of FIG. 2A (e.g., within parallel processor memory 222). The partition unit 220 may additionally or alternatively also interface with one of the memory units in parallel processor memory via a memory controller (not shown).

[0085] In graphics applications, the ROP 226 is a processing unit that performs raster operations such as stencil, z test, blending, and the like. The ROP 226 then outputs processed graphics data that is stored in graphics memory. In some embodiments the ROP 226 includes or couples with a CODEC 227 that includes compression logic to compress depth or color data that is written to memory or the L2 cache 221 and decompress depth or color data that is read from memory or the L2 cache 221. The compression logic can be lossless compression logic that makes use of one or more of multiple compression algorithms. The type of compression that is performed by the CODEC 227 can vary based on the statistical characteristics of the data to be compressed. For example, in one embodiment, delta color compression is performed on depth and color data on a per-tile basis. In one embodiment the CODEC 227 includes compression and decompression logic that can compress and decompress compute data associated with machine learning operations. The CODEC 227 can, for example, compress sparse matrix data for sparse machine learning operations. The CODEC 227 can also compress sparse matrix data that is encoded in a sparse matrix format (e.g., coordinate list encoding (COO), compressed sparse row (CSR), compress sparse column (CSC), etc.) to generate compressed and encoded sparse matrix data. The compressed and encoded sparse matrix data can be decompressed and/or decoded before being processed by processing elements or the processing elements can be

configured to consume compressed, encoded, or compressed and encoded data for processing.

[0086] The ROP 226 may be included within each processing cluster (e.g., cluster 214A-214N of FIG. 2A) instead of within the partition unit 220. In such embodiment, read and write requests for pixel data are transmitted over the memory crossbar 216 instead of pixel fragment data. The processed graphics data may be displayed on a display device, such as one of the one or more display device(s) 110A-110B of FIG. 1, routed for further processing by the processor(s) 102, or routed for further processing by one of the processing entities within the parallel processor 200 of FIG. 2A.

[0087] FIG. 2C is a block diagram of a processing cluster 214 within a parallel processing unit. For example, the processing cluster is an instance of one of the processing clusters 214A-214N of FIG. 2A. The processing cluster 214 can be configured to execute many threads in parallel, where the term “thread” refers to an instance of a particular program executing on a particular set of input data. Optionally, single-instruction, multiple-data (SIMD) instruction issue techniques may be used to support parallel execution of a large number of threads without providing multiple independent instruction units. Alternatively, single-instruction, multiple-thread (SIMT) techniques may be used to support parallel execution of a large number of generally synchronized threads, using a common instruction unit configured to issue instructions to a set of processing engines within each one of the processing clusters. Unlike a SIMD execution regime, where all processing engines typically execute identical instructions, SIMT execution allows different threads to more readily follow divergent execution paths through a given thread program. Persons skilled in the art will understand that a SIMD processing regime represents a functional subset of a SIMT processing regime.

[0088] Operation of the processing cluster 214 can be controlled via a pipeline manager 232 that distributes processing tasks to SIMT parallel processors. The pipeline manager 232 receives instructions from the scheduler 210 of FIG. 2A and manages execution of those instructions via a graphics multiprocessor 234 and/or a texture unit 236. The illustrated graphics multiprocessor 234 is an exemplary instance of a SIMT parallel processor. However, various types of SIMT parallel processors of differing architectures may be included within the processing cluster 214. One or more instances of the graphics multiprocessor 234 can be included within a processing cluster 214. The graphics multiprocessor 234 can process data and a data crossbar 240 can be used to distribute the processed data to one of multiple possible destinations, including other shader units. The pipeline manager 232 can facilitate the distribution of processed data by specifying destinations for processed data to be distributed via the data crossbar 240.

[0089] Each graphics multiprocessor 234 within the processing cluster 214 can include an identical set of functional execution logic (e.g., arithmetic logic units, load-store units, etc.). The functional execution logic can be configured in a pipelined manner in which new instructions can be issued before previous instructions are complete. The functional execution logic supports a variety of operations including integer and floating-point arithmetic, comparison operations, Boolean operations, bit-shifting, and computation of various algebraic functions. The same functional-unit hard-

ware could be leveraged to perform different operations and any combination of functional units may be present.

[0090] The instructions transmitted to the processing cluster 214 constitute a thread. A set of threads executing across the set of parallel processing engines is a thread group. A thread group executes the same program on different input data. Each thread within a thread group can be assigned to a different processing engine within a graphics multiprocessor 234. A thread group may include fewer threads than the number of processing engines within the graphics multiprocessor 234. When a thread group includes fewer threads than the number of processing engines, one or more of the processing engines may be idle during cycles in which that thread group is being processed. A thread group may also include more threads than the number of processing engines within the graphics multiprocessor 234. When the thread group includes more threads than the number of processing engines within the graphics multiprocessor 234, processing can be performed over consecutive clock cycles. Optionally, multiple thread groups can be executed concurrently on the graphics multiprocessor 234.

[0091] The graphics multiprocessor 234 may include an internal cache memory to perform load and store operations. Optionally, the graphics multiprocessor 234 can forego an internal cache and use a cache memory (e.g., level 1 (L1) cache 248) within the processing cluster 214. Each graphics multiprocessor 234 also has access to level 2 (L2) caches within the partition units (e.g., partition units 220A-220N of FIG. 2A) that are shared among all processing clusters 214 and may be used to transfer data between threads. The graphics multiprocessor 234 may also access off-chip global memory, which can include one or more of local parallel processor memory and/or system memory. Any memory external to the parallel processing unit 202 may be used as global memory. Embodiments in which the processing cluster 214 includes multiple instances of the graphics multiprocessor 234 can share common instructions and data, which may be stored in the L1 cache 248.

[0092] Each processing cluster 214 may include an MMU 245 (memory management unit) that is configured to map virtual addresses into physical addresses. In other embodiments, one or more instances of the MMU 245 may reside within the memory interface 218 of FIG. 2A. The MMU 245 includes a set of page table entries (PTEs) used to map a virtual address to a physical address of a tile and optionally a cache line index. The MMU 245 may include address translation lookaside buffers (TLB) or caches that may reside within the graphics multiprocessor 234 or the L1 cache 248 of processing cluster 214. The physical address is processed to distribute surface data access locality to allow efficient request interleaving among partition units. The cache line index may be used to determine whether a request for a cache line is a hit or miss.

[0093] In graphics and computing applications, a processing cluster 214 may be configured such that each graphics multiprocessor 234 is coupled to a texture unit 236 for performing texture mapping operations, e.g., determining texture sample positions, reading texture data, and filtering the texture data. Texture data is read from an internal texture L1 cache (not shown) or in some embodiments from the L1 cache within graphics multiprocessor 234 and is fetched from an L2 cache, local parallel processor memory, or system memory, as needed. Each graphics multiprocessor 234 outputs processed tasks to the data crossbar 240 to

provide the processed task to another processing cluster **214** for further processing or to store the processed task in an L2 cache, local parallel processor memory, or system memory via the memory crossbar **216**. A preROP **242** (pre-raster operations unit) is configured to receive data from graphics multiprocessor **234**, direct data to ROP units, which may be located with partition units as described herein (e.g., partition units **220A-220N** of FIG. 2A). The preROP **242** unit can perform optimizations for color blending, organize pixel color data, and perform address translations.

[0094] It will be appreciated that the core architecture described herein is illustrative and that variations and modifications are possible. Any number of processing units, e.g., graphics multiprocessor **234**, texture units **236**, preROPs **242**, etc., may be included within a processing cluster **214**. Further, while only one processing cluster **214** is shown, a parallel processing unit as described herein may include any number of instances of the processing cluster **214**. Optionally, each processing cluster **214** can be configured to operate independently of other processing clusters **214** using separate and distinct processing units, L1 caches, L2 caches, etc.

[0095] FIG. 2D shows an example of the graphics multiprocessor **234** in which the graphics multiprocessor **234** couples with the pipeline manager **232** of the processing cluster **214**. The graphics multiprocessor **234** has an execution pipeline including but not limited to an instruction cache **252**, an instruction unit **254**, an address mapping unit **256**, a register file **258**, one or more general purpose graphics processing unit (GPGPU) cores **262**, and one or more load/store units **266**. The GPGPU cores **262** and load/store units **266** are coupled with cache memory **272** and shared memory **270** via a memory and cache interconnect **268**. The graphics multiprocessor **234** may additionally include tensor and/or ray-tracing cores **263** that include hardware logic to accelerate matrix and/or ray-tracing operations.

[0096] The instruction cache **252** may receive a stream of instructions to execute from the pipeline manager **232**. The instructions are cached in the instruction cache **252** and dispatched for execution by the instruction unit **254**. The instruction unit **254** can dispatch instructions as thread groups (e.g., warps), with each thread of the thread group assigned to a different execution unit within GPGPU core **262**. An instruction can access any of a local, shared, or global address space by specifying an address within a unified address space. The address mapping unit **256** can be used to translate addresses in the unified address space into a distinct memory address that can be accessed by the load/store units **266**.

[0097] The register file **258** provides a set of registers for the functional units of the graphics multiprocessor **234**. The register file **258** provides temporary storage for operands connected to the data paths of the functional units (e.g., GPGPU cores **262**, load/store units **266**) of the graphics multiprocessor **234**. The register file **258** may be divided between each of the functional units such that each functional unit is allocated a dedicated portion of the register file **258**. For example, the register file **258** may be divided between the different warps being executed by the graphics multiprocessor **234**.

[0098] The GPGPU cores **262** can each include floating point units (FPUs) and/or integer arithmetic logic units (ALUs) that are used to execute instructions of the graphics

multiprocessor **234**. In some implementations, the GPGPU cores **262** can include hardware logic that may otherwise reside within the tensor and/or ray-tracing cores **263**. The GPGPU cores **262** can be similar in architecture or can differ in architecture. For example and in one embodiment, a first portion of the GPGPU cores **262** include a single precision FPU and an integer ALU while a second portion of the GPGPU cores include a double precision FPU. Optionally, the FPUs can implement the IEEE 754-2008 standard for floating point arithmetic or enable variable precision floating point arithmetic. The graphics multiprocessor **234** can additionally include one or more fixed function or special function units to perform specific functions such as copy rectangle or pixel blending operations. One or more of the GPGPU cores can also include fixed or special function logic.

[0099] The GPGPU cores **262** may include SIMD logic capable of performing a single instruction on multiple sets of data. Optionally, GPGPU cores **262** can physically execute SIMD4, SIMD8, and SIMD16 instructions and logically execute SIMD1, SIMD2, and SIMD32 instructions. The SIMD instructions for the GPGPU cores can be generated at compile time by a shader compiler or automatically generated when executing programs written and compiled for single program multiple data (SPMD) or SIMT architectures. Multiple threads of a program configured for the SIMT execution model can be executed via a single SIMD instruction. For example and in one embodiment, eight SIMT threads that perform the same or similar operations can be executed in parallel via a single SIMD8 logic unit.

[0100] The memory and cache interconnect **268** is an interconnect network that connects each of the functional units of the graphics multiprocessor **234** to the register file **258** and to the shared memory **270**. For example, the memory and cache interconnect **268** is a crossbar interconnect that allows the load/store unit **266** to implement load and store operations between the shared memory **270** and the register file **258**. The register file **258** can operate at the same frequency as the GPGPU cores **262**, thus data transfer between the GPGPU cores **262** and the register file **258** is very low latency. The shared memory **270** can be used to enable communication between threads that execute on the functional units within the graphics multiprocessor **234**. The cache memory **272** can be used as a data cache for example, to cache texture data communicated between the functional units and the texture unit **236**. The shared memory **270** can also be used as a program managed cache. The shared memory **270** and the cache memory **272** can couple with the data crossbar **240** to enable communication with other components of the processing cluster. Threads executing on the GPGPU cores **262** can programmatically store data within the shared memory in addition to the automatically cached data that is stored within the cache memory **272**.

[0101] FIG. 3A-3C illustrate additional graphics multiprocessors, according to embodiments. FIG. 3A-3B illustrate graphics multiprocessors **325**, **350**, which are related to the graphics multiprocessor **234** of FIG. 2C and may be used in place of one of those. Therefore, the disclosure of any features in combination with the graphics multiprocessor **234** herein also discloses a corresponding combination with the graphics multiprocessor(s) **325**, **350**, but is not limited to such. FIG. 3C illustrates a graphics processing unit (GPU) **380** which includes dedicated sets of graphics processing resources arranged into multi-core groups **365A-365N**,

which correspond to the graphics multiprocessors **325**, **350**. The illustrated graphics multiprocessors **325**, **350** and the multi-core groups **365A-365N** can be streaming multiprocessors (SM) capable of simultaneous execution of a large number of execution threads.

[0102] The graphics multiprocessor **325** of FIG. 3A includes multiple additional instances of execution resource units relative to the graphics multiprocessor **234** of FIG. 2D. For example, the graphics multiprocessor **325** can include multiple instances of the instruction unit **332A-332B**, register file **334A-334B**, and texture unit(s) **344A-344B**. The graphics multiprocessor **325** also includes multiple sets of graphics or compute execution units (e.g., GPGPU core **336A-336B**, tensor core **337A-337B**, ray-tracing core **338A-338B**) and multiple sets of load/store units **340A-340B**. The execution resource units have a common instruction cache **330**, texture and/or data cache memory **342**, and shared memory **346**.

[0103] The various components can communicate via an interconnect fabric **327**. The interconnect fabric **327** may include one or more crossbar switches to enable communication between the various components of the graphics multiprocessor **325**. The interconnect fabric **327** may be a separate, high-speed network fabric layer upon which each component of the graphics multiprocessor **325** is stacked. The components of the graphics multiprocessor **325** communicate with remote components via the interconnect fabric **327**. For example, the cores **336A-336B**, **337A-337B**, and **338A-338B** can each communicate with shared memory **346** via the interconnect fabric **327**. The interconnect fabric **327** can arbitrate communication within the graphics multiprocessor **325** to ensure a fair bandwidth allocation between components.

[0104] The graphics multiprocessor **350** of FIG. 3B includes multiple sets of execution resources **356A-356D**, where each set of execution resource includes multiple instruction units, register files, GPGPU cores, and load store units, as illustrated in FIG. 2D and FIG. 3A. The execution resources **356A-356D** can work in concert with texture unit(s) **360A-360D** for texture operations, while sharing an instruction cache **354**, and shared memory **353**. For example, the execution resources **356A-356D** can share an instruction cache **354** and shared memory **353**, as well as multiple instances of a texture and/or data cache memory **358A-358B**. The various components can communicate via an interconnect fabric **352** similar to the interconnect fabric **327** of FIG. 3A.

[0105] Persons skilled in the art will understand that the architecture described in FIGS. 1, 2A-2D, and 3A-3B are descriptive and not limiting as to the scope of the present embodiments. Thus, the techniques described herein may be implemented on any properly configured processing unit, including, without limitation, one or more mobile application processors, one or more desktop or server central processing units (CPUs) including multi-core CPUs, one or more parallel processing units, such as the parallel processing unit **202** of FIG. 2A, as well as one or more graphics processors or special purpose processing units, without departure from the scope of the embodiments described herein.

[0106] The parallel processor or GPGPU as described herein may be communicatively coupled to host/processor cores to accelerate graphics operations, machine-learning operations, pattern analysis operations, and various general-

purpose GPU (GPGPU) functions. The GPU may be communicatively coupled to the host processor/cores over a bus or other interconnect (e.g., a high-speed interconnect such as PCIe, NV Link, or other known protocols, standardized protocols, or proprietary protocols). In other embodiments, the GPU may be integrated on the same package or chip as the cores and communicatively coupled to the cores over an internal processor bus/interconnect (i.e., internal to the package or chip). Regardless of the manner in which the GPU is connected, the processor cores may allocate work to the GPU in the form of sequences of commands/instructions contained in a work descriptor. The GPU then uses dedicated circuitry/logic for efficiently processing these commands/instructions.

[0107] FIG. 3C illustrates a graphics processing unit (GPU) **380** which includes dedicated sets of graphics processing resources arranged into multi-core groups **365A-365N**. While the details of only a single multi-core group **365A** are provided, it will be appreciated that the other multi-core groups **365B-365N** may be equipped with the same or similar sets of graphics processing resources. Details described with respect to the multi-core groups **365A-365N** may also apply to any graphics multiprocessor **234**, **325**, **350** described herein.

[0108] As illustrated, a multi-core group **365A** may include a set of graphics cores **370**, a set of tensor cores **371**, and a set of ray tracing cores **372**. A scheduler/dispatcher **368** schedules and dispatches the graphics threads for execution on the various cores **370**, **371**, **372**. A set of register files **369** store operand values used by the cores **370**, **371**, **372** when executing the graphics threads. These may include, for example, integer registers for storing integer values, floating point registers for storing floating point values, vector registers for storing packed data elements (integer and/or floating-point data elements) and tile registers for storing tensor/matrix values. The tile registers may be implemented as combined sets of vector registers.

[0109] One or more combined level 1 (L1) caches and shared memory units **373** store graphics data such as texture data, vertex data, pixel data, ray data, bounding volume data, etc., locally within each multi-core group **365A**. One or more texture units **374** can also be used to perform texturing operations, such as texture mapping and sampling. A Level 2 (L2) cache **375** shared by all or a subset of the multi-core groups **365A-365N** stores graphics data and/or instructions for multiple concurrent graphics threads. As illustrated, the L2 cache **375** may be shared across a plurality of multi-core groups **365A-365N**. One or more memory controllers **367** couple the GPU **380** to a memory **366** which may be a system memory (e.g., DRAM) and/or a dedicated graphics memory (e.g., GDDR6 memory).

[0110] Input/output (I/O) circuitry **363** couples the GPU **380** to one or more I/O devices **362** such as digital signal processors (DSPs), network controllers, or user input devices. An on-chip interconnect may be used to couple the I/O devices **362** to the GPU **380** and memory **366**. One or more I/O memory management units (IOMMUs) **364** of the I/O circuitry **363** couple the I/O devices **362** directly to the system memory **366**. Optionally, the IOMMU **364** manages multiple sets of page tables to map virtual addresses to physical addresses in system memory **366**. The I/O devices **362**, CPU(s) **361**, and GPU(s) **380** may then share the same virtual address space.

[0111] In one implementation of the IOMMU 364, the IOMMU 364 supports virtualization. In this case, it may manage a first set of page tables to map guest/graphics virtual addresses to guest/graphics physical addresses and a second set of page tables to map the guest/graphics physical addresses to system/host physical addresses (e.g., within system memory 366). The base addresses of each of the first and second sets of page tables may be stored in control registers and swapped out on a context switch (e.g., so that the new context is provided with access to the relevant set of page tables). While not illustrated in FIG. 3C, each of the cores 370, 371, 372 and/or multi-core groups 365A-365N may include translation lookaside buffers (TLBs) to cache guest virtual to guest physical translations, guest physical to host physical translations, and guest virtual to host physical translations.

[0112] The CPU(s) 361, GPUs 380, and I/O devices 362 may be integrated on a single semiconductor chip and/or chip package. The illustrated memory 366 may be integrated on the same chip or may be coupled to the memory controllers 367 via an off-chip interface. In one implementation, the memory 366 comprises GDDR6 memory which shares the same virtual address space as other physical system-level memories, although the underlying principles described herein are not limited to this specific implementation.

[0113] The tensor cores 371 may include a plurality of execution resources specifically designed to perform matrix operations, which are the fundamental compute operation used to perform deep learning operations. For example, simultaneous matrix multiplication operations may be used for neural network training and inferencing. The tensor cores 371 may perform matrix processing using a variety of operand precisions including single precision floating-point (e.g., 32 bits), half-precision floating point (e.g., 16 bits), integer words (16 bits), bytes (8 bits), and half-bytes (4 bits). For example, a neural network implementation extracts features of each rendered scene, potentially combining details from multiple frames, to construct a high-quality final image.

[0114] In deep learning implementations, parallel matrix multiplication work may be scheduled for execution on the tensor cores 371. The training of neural networks, in particular, requires a significant number of matrix dot product operations. In order to process an inner-product formulation of an $N \times N \times N$ matrix multiply, the tensor cores 371 may include at least N dot-product processing elements. Before the matrix multiply begins, one entire matrix is loaded into tile registers and at least one column of a second matrix is loaded each cycle for N cycles. Each cycle, there are N dot products that are processed.

[0115] Matrix elements may be stored at different precisions depending on the particular implementation, including 16-bit words, 8-bit bytes (e.g., INT8) and 4-bit half-bytes (e.g., INT4). Different precision modes may be specified for the tensor cores 371 to ensure that the most efficient precision is used for different workloads (e.g., such as inferencing workloads which can tolerate quantization to bytes and half-bytes). Supported formats additionally include 64-bit floating point (FP64) and non-IEEE floating point formats such as the bfloat16 format (e.g., Brain floating point), a 16-bit floating point format with one sign bit, eight exponent bits, and eight significand bits, of which seven are explicitly stored. One embodiment includes support for a reduced precision tensor-float (TF32) mode, which performs com-

putations using the range of FP32 (8-bits) and the precision of FP16 (10-bits). Reduced precision TF32 operations can be performed on FP32 inputs and produce FP32 outputs at higher performance relative to FP32 and increased precision relative to FP16. In one embodiment, one or more 8-bit floating point formats (FP8) are supported.

[0116] In one embodiment the tensor cores 371 support a sparse mode of operation for matrices in which the vast majority of values are zero. The tensor cores 371 include support for sparse input matrices that are encoded in a sparse matrix representation (e.g., coordinate list encoding (COO), compressed sparse row (CSR), compressed sparse column (CSC), etc.). The tensor cores 371 also include support for compressed sparse matrix representations in the event that the sparse matrix representation may be further compressed. Compressed, encoded, and/or compressed and encoded matrix data, along with associated compression and/or encoding metadata, can be read by the tensor cores 371 and the non-zero values can be extracted. For example, for a given input matrix A, a non-zero value can be loaded from the compressed and/or encoded representation of at least a portion of matrix A. Based on the location in matrix A for the non-zero value, which may be determined from index or coordinate metadata associated with the non-zero value, a corresponding value in input matrix B may be loaded. Depending on the operation to be performed (e.g., multiply), the load of the value from input matrix B may be bypassed if the corresponding value is a zero value. In one embodiment, the pairings of values for certain operations, such as multiply operations, may be pre-scanned by scheduler logic and only operations between non-zero inputs are scheduled. Depending on the dimensions of matrix A and matrix B and the operation to be performed, output matrix C may be dense or sparse. Where output matrix C is sparse and depending on the configuration of the tensor cores 371, output matrix C may be output in a compressed format, a sparse encoding, or a compressed sparse encoding.

[0117] The ray tracing cores 372 may accelerate ray tracing operations for both real-time ray tracing and non-real-time ray tracing implementations. In particular, the ray tracing cores 372 may include ray traversal/intersection circuitry for performing ray traversal using bounding volume hierarchies (BVHs) and identifying intersections between rays and primitives enclosed within the BVH volumes. The ray tracing cores 372 may also include circuitry for performing depth testing and culling (e.g., using a Z buffer or similar arrangement). In one implementation, the ray tracing cores 372 perform traversal and intersection operations in concert with the image denoising techniques described herein, at least a portion of which may be executed on the tensor cores 371. For example, the tensor cores 371 may implement a deep learning neural network to perform denoising of frames generated by the ray tracing cores 372. However, the CPU(s) 361, graphics cores 370, and/or ray tracing cores 372 may also implement all or a portion of the denoising and/or deep learning algorithms.

[0118] In addition, as described above, a distributed approach to denoising may be employed in which the GPU 380 is in a computing device coupled to other computing devices over a network or high-speed interconnect. In this distributed approach, the interconnected computing devices may share neural network learning/training data to improve

the speed with which the overall system learns to perform denoising for different types of image frames and/or different graphics applications.

[0119] The ray tracing cores 372 may process all BVH traversal and/or ray-primitive intersections, saving the graphics cores 370 from being overloaded with thousands of instructions per ray. For example, each ray tracing core 372 includes a first set of specialized circuitry for performing bounding box tests (e.g., for traversal operations) and/or a second set of specialized circuitry for performing the ray-triangle intersection tests (e.g., intersecting rays which have been traversed). Thus, for example, the multi-core group 365A can simply launch a ray probe, and the ray tracing cores 372 independently perform ray traversal and intersection and return hit data (e.g., a hit, no hit, multiple hits, etc.) to the thread context. The other cores 370, 371 are freed to perform other graphics or compute work while the ray tracing cores 372 perform the traversal and intersection operations.

[0120] Optionally, each ray tracing core 372 may include a traversal unit to perform BVH testing operations and/or an intersection unit which performs ray-primitive intersection tests. The intersection unit generates a “hit,” “no hit,” or “multiple hit” response, which it provides to the appropriate thread. During the traversal and intersection operations, the execution resources of the other cores (e.g., graphics cores 370 and tensor cores 371) are freed to perform other forms of graphics work.

[0121] In one optional embodiment described below, a hybrid rasterization/ray tracing approach is used in which work is distributed between the graphics cores 370 and ray tracing cores 372.

[0122] The ray tracing cores 372 (and/or other cores 370, 371) may include hardware support for a ray tracing instruction set such as Microsoft’s DirectX Ray Tracing (DXR) which includes a DispatchRays command, as well as ray-generation, closest-hit, any-hit, and miss shaders, which enable the assignment of unique sets of shaders and textures for each object. Another ray tracing platform which may be supported by the ray tracing cores 372, graphics cores 370 and tensor cores 371 is Vulkan API (e.g., Vulkan version 1.1.85 and later). Note, however, that the underlying principles described herein are not limited to any particular ray tracing ISA.

[0123] In general, the various cores 372, 371, 370 may support a ray tracing instruction set that includes instructions/functions for one or more of ray generation, closest hit, any hit, ray-primitive intersection, per-primitive and hierarchical bounding box construction, miss, visit, and exceptions. More specifically, a preferred embodiment includes ray tracing instructions to perform one or more of the following functions:

[0124] Ray Generation—Ray generation instructions may be executed for each pixel, sample, or other user-defined work assignment.

[0125] Closest Hit—A closest hit instruction may be executed to locate the closest intersection point of a ray with primitives within a scene.

[0126] Any Hit—An any hit instruction identifies multiple intersections between a ray and primitives within a scene, potentially to identify a new closest intersection point.

[0127] Intersection—An intersection instruction performs a ray-primitive intersection test and outputs a result.

[0128] Per-primitive Bounding box Construction—This instruction builds a bounding box around a given primitive or group of primitives (e.g., when building a new BVH or other acceleration data structure).

[0129] Miss—Indicates that a ray misses all geometry within a scene, or specified region of a scene.

[0130] Visit—Indicates the child volumes a ray will traverse.

[0131] Exceptions—Includes various types of exception handlers (e.g., invoked for various error conditions).

[0132] In one embodiment the ray tracing cores 372 may be adapted to accelerate general-purpose compute operations that can be accelerated using computational techniques that are analogous to ray intersection tests. A compute framework can be provided that enables shader programs to be compiled into low level instructions and/or primitives that perform general-purpose compute operations via the ray tracing cores. Exemplary computational problems that can benefit from compute operations performed on the ray tracing cores 372 include computations involving beam, wave, ray, or particle propagation within a coordinate space. Interactions associated with that propagation can be computed relative to a geometry or mesh within the coordinate space. For example, computations associated with electromagnetic signal propagation through an environment can be accelerated via the use of instructions or primitives that are executed via the ray tracing cores. Diffraction and reflection of the signals by objects in the environment can be computed as direct ray-tracing analogies.

[0133] Ray tracing cores 372 can also be used to perform computations that are not directly analogous to ray tracing. For example, mesh projection, mesh refinement, and volume sampling computations can be accelerated using the ray tracing cores 372. Generic coordinate space calculations, such as nearest neighbor calculations can also be performed. For example, the set of points near a given point can be discovered by defining a bounding box in the coordinate space around the point. BVH and ray probe logic within the ray tracing cores 372 can then be used to determine the set of point intersections within the bounding box. The intersections constitute the origin point and the nearest neighbors to that origin point. Computations that are performed using the ray tracing cores 372 can be performed in parallel with computations performed on the graphics cores 372 and tensor cores 371. A shader compiler can be configured to compile a compute shader or other general-purpose graphics processing program into low level primitives that can be parallelized across the graphics cores 370, tensor cores 371, and ray tracing cores 372.

Techniques for GPU to Host Processor Interconnection

[0134] FIG. 4A illustrates an exemplary architecture in which a plurality of GPUs 410-413, e.g., such as the parallel processors 200 shown in FIG. 2A, are communicatively coupled to a plurality of multi-core processors 405-406 over high-speed links 440A-440D (e.g., buses, point-to-point interconnects, etc.). The high-speed links 440A-440D may support a communication throughput of 4 GB/s, 30 GB/s, 80 GB/s or higher, depending on the implementation. Various interconnect protocols may be used including, but not lim-

ited to, PCIe 4.0 or 5.0 and NV Link 2.0. However, the underlying principles described herein are not limited to any particular communication protocol or throughput.

[0135] Two or more of the GPUs **410-413** may be interconnected over high-speed links **442A-442B**, which may be implemented using the same or different protocols/links than those used for high-speed links **440A-440D**. Similarly, two or more of the multi-core processors **405-406** may be connected over high-speed link **443** which may be symmetric multi-processor (SMP) buses operating at 20 GB/s, 30 GB/s, 120 GB/s or lower or higher speeds. Alternatively, all communication between the various system components shown in FIG. 4A may be accomplished using the same protocols/links (e.g., over a common interconnection fabric). As mentioned, however, the underlying principles described herein are not limited to any particular type of interconnect technology.

[0136] Each of multi-core processor **405** and multi-core processor **406** may be communicatively coupled to a processor memory **401-402**, via memory interconnects **430A-430B**, respectively, and each GPU **410-413** is communicatively coupled to GPU memory **420-423** over GPU memory interconnects **450A-450D**, respectively. The memory interconnects **430A-430B** and **450A-450D** may utilize the same or different memory access technologies. By way of example, and not limitation, the processor memories **401-402** and GPU memories **420-423** may be volatile memories such as dynamic random-access memories (DRAMs) (including stacked DRAMs), Graphics DDR SDRAM (GDDR) (e.g., GDDR5, GDDR6), or High Bandwidth Memory (HBM) and/or may be non-volatile memories such as 3D X Point/Optane or Nano-Ram. For example, some portion of the memories may be volatile memory and another portion may be non-volatile memory (e.g., using a two-level memory (2L M) hierarchy). A memory subsystem as described herein may be compatible with a number of memory technologies, such as Double Data Rate versions released by JEDEC (Joint Electronic Device Engineering Council).

[0137] As described below, although the various processors **405-406** and GPUs **410-413** may be physically coupled to a particular memory **401-402**, **420-423**, respectively, a unified memory architecture may be implemented in which the same virtual system address space (also referred to as the “effective address” space) is distributed among all of the various physical memories. For example, processor memories **401-402** may each comprise 64 GB of the system memory address space and GPU memories **420-423** may each comprise 32 GB of the system memory address space (resulting in a total of 256 GB addressable memory in this example).

[0138] FIG. 4B illustrates additional optional details for an interconnection between a multi-core processor **407** and a graphics acceleration module **446**. The graphics acceleration module **446** may include one or more GPU chips integrated on a line card which is coupled to the processor **407** via the high-speed link **440**. Alternatively, the graphics acceleration module **446** may be integrated on the same package or chip as the processor **407**.

[0139] The illustrated processor **407** includes a plurality of cores **460A-460D**, each with a translation lookaside buffer **461A-461D** and one or more caches **462A-462D**. The cores may include various other components for executing instructions and processing data which are not illustrated to avoid

obscuring the underlying principles of the components described herein (e.g., instruction fetch units, branch prediction units, decoders, execution units, reorder buffers, etc.). The caches **462A-462D** may comprise level 1 (L1) and level 2 (L2) caches. In addition, one or more shared caches **456** may be included in the caching hierarchy and shared by sets of the cores **460A-460D**. For example, one embodiment of the processor **407** includes 24 cores, each with its own L1 cache, twelve shared L2 caches, and twelve shared L3 caches. In this embodiment, one of the L2 and L3 caches are shared by two adjacent cores. The processor **407** and the graphics accelerator integration module **446** connect with system memory **441**, which may include processor memories **401-402**.

[0140] Coherency is maintained for data and instructions stored in the various caches **462A-462D**, **456** and system memory **441** via inter-core communication over a coherence bus **464**. For example, each cache may have cache coherency logic/circuitry associated therewith to communicate to over the coherence bus **464** in response to detected reads or writes to particular cache lines. In one implementation, a cache snooping protocol is implemented over the coherence bus **464** to snoop cache accesses. Cache snooping/coherency techniques are well understood by those of skill in the art and will not be described in detail here to avoid obscuring the underlying principles described herein.

[0141] A proxy circuit **425** may be provided that communicatively couples the graphics acceleration module **446** to the coherence bus **464**, allowing the graphics acceleration module **446** to participate in the cache coherence protocol as a peer of the cores. In particular, an interface **435** provides connectivity to the proxy circuit **425** over high-speed link **440** (e.g., a PCIe bus, NV Link, etc.) and an interface **437** connects the graphics acceleration module **446** to the high-speed link **440**.

[0142] In one implementation, an accelerator integration circuit **436** provides cache management, memory access, context management, and interrupt management services on behalf of a plurality of graphics processing engines **431**, **432**, N of the graphics acceleration module **446**. The graphics processing engines **431**, **432**, N may each comprise a separate graphics processing unit (GPU). Alternatively, the graphics processing engines **431**, **432**, N may comprise different types of graphics processing engines within a GPU such as graphics execution units, media processing engines (e.g., video encoders/decoders), samplers, and blit engines. In other words, the graphics acceleration module may be a GPU with a plurality of graphics processing engines **431-432**, N or the graphics processing engines **431-432**, N may be individual GPUs integrated on a common package, line card, or chip. The graphics processing engines **431-432**, N may be configured with any graphics processor or compute accelerator architecture described herein.

[0143] The accelerator integration circuit **436** may include a memory management unit (MMU) **439** for performing various memory management functions such as virtual-to-physical memory translations (also referred to as effective-to-real memory translations) and memory access protocols for accessing system memory **441**. The MMU **439** may also include a translation lookaside buffer (TLB) (not shown) for caching the virtual/effective to physical/real address translations. In one implementation, a cache **438** stores commands and data for efficient access by the graphics processing engines **431**, **432**, N. The data stored in cache **438** and

graphics memories **433-434**, M may be kept coherent with the core caches **462A-462D**, **456** and system memory **441**. As mentioned, this may be accomplished via proxy circuit **425** which takes part in the cache coherency mechanism on behalf of cache **438** and memories **433-434**, M (e.g., sending updates to the cache **438** related to modifications/accesses of cache lines on processor caches **462A-462D**, **456** and receiving updates from the cache **438**).

[0144] A set of registers **445** store context data for threads executed by the graphics processing engines **431-432**, N and a context management circuit **448** manages the thread contexts. For example, the context management circuit **448** may perform save and restore operations to save and restore contexts of the various threads during contexts switches (e.g., where a first thread is saved and a second thread is restored so that the second thread can be execute by a graphics processing engine). For example, on a context switch, the context management circuit **448** may store current register values to a designated region in memory (e.g., identified by a context pointer). It may then restore the register values when returning to the context. An interrupt management circuit **447**, for example, may receive and processes interrupts received from system devices.

[0145] In one implementation, virtual/effective addresses from a graphics processing engine **431** are translated to real/physical addresses in system memory **441** by the MMU **439**. Optionally, the accelerator integration circuit **436** supports multiple (e.g., 4, 8, 16) graphics accelerator modules **446** and/or other accelerator devices. The graphics accelerator module **446** may be dedicated to a single application executed on the processor **407** or may be shared between multiple applications. Optionally, a virtualized graphics execution environment is provided in which the resources of the graphics processing engines **431-432**, N are shared with multiple applications, virtual machines (VMs), or containers. The resources may be subdivided into “slices” which are allocated to different VMs and/or applications based on the processing requirements and priorities associated with the VMs and/or applications, or based on a pre-determined partitioning profile for a graphics accelerator module **446**. VMs and containers can be used interchangeably herein.

[0146] A virtual machine (VM) can be software that runs an operating system and one or more applications. A VM can be defined by specification, configuration files, virtual disk file, non-volatile random-access memory (NVRAM) setting file, and the log file and is backed by the physical resources of a host computing platform. A VM can include an operating system (OS) or application environment that is installed on software, which imitates dedicated hardware. The end user has the same experience on a virtual machine as they would have on dedicated hardware. Specialized software, called a hypervisor, emulates the PC client or server’s CPU, memory, hard disk, network, and other hardware resources completely, enabling virtual machines to share the resources. The hypervisor can emulate multiple virtual hardware platforms that are isolated from each other, allowing virtual machines to run Linux®, Windows® Server, VM ware ESXi, and other operating systems on the same underlying physical host.

[0147] A container can be a software package of applications, configurations, and dependencies so the applications run reliably on one computing environment to another. Containers can share an operating system installed on the server platform and run as isolated processes. A container

can be a software package that contains everything the software needs to run such as system tools, libraries, and settings. Containers are not installed like traditional software programs, which allows them to be isolated from the other software and the operating system itself. The isolated nature of containers provides several benefits. First, the software in a container will run the same in different environments. For example, a container that includes PHP and MySQL can run identically on both a Linux® computer and a Windows® machine. Second, containers provide added security since the software will not affect the host operating system. While an installed application may alter system settings and modify resources, such as the Windows registry, a container can only modify settings within the container.

[0148] Thus, the accelerator integration circuit **436** acts as a bridge to the system for the graphics acceleration module **446** and provides address translation and system memory cache services. In one embodiment, to facilitate the bridging functionality, the accelerator integration circuit **436** may also include shared I/O **497** (e.g., PCIe, USB, or others) and hardware to enable system control of voltage, clocking, performance, thermals, and security. The shared I/O **497** may utilize separate physical connections or may traverse the high-speed link **440**. In addition, the accelerator integration circuit **436** may provide virtualization facilities for the host processor to manage virtualization of the graphics processing engines, interrupts, and memory management.

[0149] Because hardware resources of the graphics processing engines **431-432**, N are mapped explicitly to the real address space seen by the host processor **407**, any host processor can address these resources directly using an effective address value. One optional function of the accelerator integration circuit **436** is the physical separation of the graphics processing engines **431-432**, N so that they appear to the system as independent units.

[0150] One or more graphics memories **433-434**, M may be coupled to each of the graphics processing engines **431-432**, N, respectively. The graphics memories **433-434**, M store instructions and data being processed by each of the graphics processing engines **431-432**, N. The graphics memories **433-434**, M may be volatile memories such as DRAMs (including stacked DRAMs), GDDR memory (e.g., GDDR5, GDDR6), or HBM, and/or may be non-volatile memories such as 3D X Point/Optane, Samsung Z-NAND, or Nano-Ram.

[0151] To reduce data traffic over the high-speed link **440**, biasing techniques may be used to ensure that the data stored in graphics memories **433-434**, M is data which will be used most frequently by the graphics processing engines **431-432**, N and preferably not used by the cores **460A-460D** (at least not frequently). Similarly, the biasing mechanism attempts to keep data needed by the cores (and preferably not the graphics processing engines **431-432**, N) within the caches **462A-462D**, **456** of the cores and system memory **441**.

[0152] According to a variant shown in FIG. 4C the accelerator integration circuit **436** is integrated within the processor **407**. The graphics processing engines **431-432**, N communicate directly over the high-speed link **440** to the accelerator integration circuit **436** via interface **437** and interface **435** (which, again, may be utilize any form of bus or interface protocol). The accelerator integration circuit **436** may perform the same operations as those described with

respect to FIG. 4B, but potentially at a higher throughput given its close proximity to the coherence bus 464 and caches 462A-462D, 456.

[0153] The embodiments described may support different programming models including a dedicated-process programming model (no graphics acceleration module virtualization) and shared programming models (with virtualization). The latter may include programming models which are controlled by the accelerator integration circuit 436 and programming models which are controlled by the graphics acceleration module 446.

[0154] In the embodiments of the dedicated process model, graphics processing engines 431, 432, . . . N may be dedicated to a single application or process under a single operating system. The single application can funnel other application requests to the graphics engines 431, 432, . . . N, providing virtualization within a VM/partition.

[0155] In the dedicated-process programming models, the graphics processing engines 431, 432, N, may be shared by multiple VM/application partitions. The shared models require a system hypervisor to virtualize the graphics processing engines 431-432, N to allow access by each operating system. For single-partition systems without a hypervisor, the graphics processing engines 431-432, N are owned by the operating system. In both cases, the operating system can virtualize the graphics processing engines 431-432, N to provide access to each process or application.

[0156] For the shared programming model, the graphics acceleration module 446 or an individual graphics processing engine 431-432, N selects a process element using a process handle. The process elements may be stored in system memory 441 and be addressable using the effective address to real address translation techniques described herein. The process handle may be an implementation-specific value provided to the host process when registering its context with the graphics processing engine 431-432, N (that is, calling system software to add the process element to the process element linked list). The lower 16-bits of the process handle may be the offset of the process element within the process element linked list.

[0157] FIG. 4D illustrates an exemplary accelerator integration slice 490. As used herein, a “slice” comprises a specified portion of the processing resources of the accelerator integration circuit 436. Application effective address space 482 within system memory 441 stores process elements 483. The process elements 483 may be stored in response to GPU invocations 481 from applications 480 executed on the processor 407. A process element 483 contains the process state for the corresponding application 480. A work descriptor (WD) 484 contained in the process element 483 can be a single job requested by an application or may contain a pointer to a queue of jobs. In the latter case, the WD 484 is a pointer to the job request queue in the application’s address space 482.

[0158] The graphics acceleration module 446 and/or the individual graphics processing engines 431-432, N can be shared by all or a subset of the processes in the system. For example, the technologies described herein may include an infrastructure for setting up the process state and sending a WD 484 to a graphics acceleration module 446 to start a job in a virtualized environment.

[0159] In one implementation, the dedicated-process programming model is implementation-specific. In this model, a single process owns the graphics acceleration module 446

or an individual graphics processing engine 431. Because the graphics acceleration module 446 is owned by a single process, the hypervisor initializes the accelerator integration circuit 436 for the owning partition and the operating system initializes the accelerator integration circuit 436 for the owning process at the time when the graphics acceleration module 446 is assigned.

[0160] In operation, a WD fetch unit 491 in the accelerator integration slice 490 fetches the next WD 484 which includes an indication of the work to be done by one of the graphics processing engines of the graphics acceleration module 446. Data from the WD 484 may be stored in registers 445 and used by the MMU 439, interrupt management circuit 447 and/or context management circuit 448 as illustrated. For example, the MMU 439 may include segment/page walk circuitry for accessing segment/page tables 486 within the OS virtual address space 485. The interrupt management circuit 447 may process interrupt events 492 received from the graphics acceleration module 446. When performing graphics operations, an effective address 493 generated by a graphics processing engine 431-432, N is translated to a real address by the MMU 439.

[0161] The same set of registers 445 may be duplicated for each graphics processing engine 431-432, N and/or graphics acceleration module 446 and may be initialized by the hypervisor or operating system. Each of these duplicated registers may be included in an accelerator integration slice 490. In one embodiment, each graphics processing engine 431-432, N may be presented to the hypervisor 496 as a distinct graphics processor device. QoS settings can be configured for clients of a specific graphics processing engine 431-432, N and data isolation between the clients of each engine can be enabled. Exemplary registers that may be initialized by the hypervisor are shown in Table 1.

TABLE 1

Hypervisor Initialized Registers	
1	Slice Control Register
2	Real Address (RA) Scheduled Processes Area Pointer
3	Authority Mask Override Register
4	Interrupt Vector Table Entry Offset
5	Interrupt Vector Table Entry Limit
6	State Register
7	Logical Partition ID
8	Real address (RA) Hypervisor Accelerator Utilization Record Pointer
9	Storage Description Register

[0162] Exemplary registers that may be initialized by the operating system are shown in Table 2.

TABLE 2

Operating System Initialized Registers	
1	Process and Thread Identification
2	Effective Address (EA)
	Context Save/Restore Pointer
3	Virtual Address (VA) Accelerator Utilization Record Pointer
4	Virtual Address (VA) Storage Segment Table Pointer
5	Authority Mask
6	Work descriptor

[0163] Each WD **484** may be specific to a particular graphics acceleration module **446** and/or graphics processing engine **431-432**, N. It contains all the information a graphics processing engine **431-432**, N requires to do its work, or it can be a pointer to a memory location where the application has set up a command queue of work to be completed.

[0164] FIG. 4E illustrates additional optional details of a shared model. It includes a hypervisor real address space **498** in which a process element list **499** is stored. The hypervisor real address space **498** is accessible via a hypervisor **496** which virtualizes the graphics acceleration module engines for the operating system **495**.

[0165] The shared programming models allow for all or a subset of processes from all or a subset of partitions in the system to use a graphics acceleration module **446**. There are two programming models where the graphics acceleration module **446** is shared by multiple processes and partitions: time-sliced shared and graphics directed shared.

[0166] In this model, the system hypervisor **496** owns the graphics acceleration module **446** and makes its function available to all operating systems **495**. For a graphics acceleration module **446** to support virtualization by the system hypervisor **496**, the graphics acceleration module **446** may adhere to the following requirements: 1) An application's job request must be autonomous (that is, the state does not need to be maintained between jobs), or the graphics acceleration module **446** must provide a context save and restore mechanism. 2) An application's job request is guaranteed by the graphics acceleration module **446** to complete in a specified amount of time, including any translation faults, or the graphics acceleration module **446** provides the ability to preempt the processing of the job. 3) The graphics acceleration module **446** must be guaranteed fairness between processes when operating in the directed shared programming model.

[0167] For the directed shared model, the application **480** may be required to make an operating system **495** system call with a graphics acceleration module **446** type, a work descriptor (WD), an authority mask register (AMR) value, and a context save/restore area pointer (CSRP). The graphics acceleration module **446** type describes the targeted acceleration function for the system call. The graphics acceleration module **446** type may be a system-specific value. The WD is formatted specifically for the graphics acceleration module **446** and can be in the form of a graphics acceleration module **446** command, an effective address pointer to a user-defined structure, an effective address pointer to a queue of commands, or any other data structure to describe the work to be done by the graphics acceleration module **446**. In one embodiment, the AMR value is the AMR state to use for the current process. The value passed to the operating system is similar to an application setting the AMR. If the accelerator integration circuit **436** and graphics acceleration module **446** implementations do not support a User Authority Mask Override Register (UAMOR), the operating system may apply the current UAMOR value to the AMR value before passing the AMR in the hypervisor call. The hypervisor **496** may optionally apply the current Authority Mask Override Register (AMOR) value before placing the AMR into the process element **483**. The CSRP may be one of the registers **445** containing the effective address of an area in the application's address space **482** for the graphics acceleration module **446** to save and restore the

context state. This pointer is optional if no state is required to be saved between jobs or when a job is preempted. The context save/restore area may be pinned system memory.

[0168] Upon receiving the system call, the operating system **495** may verify that the application **480** has registered and been given the authority to use the graphics acceleration module **446**. The operating system **495** then calls the hypervisor **496** with the information shown in Table 3.

TABLE 3

OS to Hypervisor Call Parameters	
1	A work descriptor (WD)
2	An Authority Mask Register (AMR) value (potentially masked).
3	An effective address (EA) Context Save/Restore Area Pointer (CSRP)
4	A process ID (PID) and optional thread ID (TID)
5	A virtual address (VA) accelerator utilization record pointer (AURP)
6	The virtual address of the storage segment table pointer (SSTP)
7	A logical interrupt service number (LISN)

[0169] Upon receiving the hypervisor call, the hypervisor **496** verifies that the operating system **495** has registered and been given the authority to use the graphics acceleration module **446**. The hypervisor **496** then puts the process element **483** into the process element linked list for the corresponding graphics acceleration module **446** type. The process element may include the information shown in Table 4.

TABLE 4

Process Element Information	
1	A work descriptor (WD)
2	An Authority Mask Register (AMR) value (potentially masked).
3	An effective address (EA) Context Save/Restore Area Pointer (CSRP)
4	A process ID (PID) and optional thread ID (TID)
5	A virtual address (VA) accelerator utilization record pointer (AURP)
6	The virtual address of the storage segment table pointer (SSTP)
7	A logical interrupt service number (LISN)
8	Interrupt vector table, derived from the hypervisor call parameters.
9	A state register (SR) value
10	A logical partition ID (LPID)
11	A real address (RA) hypervisor accelerator utilization record pointer
12	The Storage Descriptor Register (SDR)

[0170] The hypervisor may initialize a plurality of accelerator integration slice **490** registers **445**.

[0171] As illustrated in FIG. 4F, in one optional implementation a unified memory addressable via a common virtual memory address space used to access the physical processor memories **401-402** and GPU memories **420-423** is employed. In this implementation, operations executed on the GPUs **410-413** utilize the same virtual/effective memory address space to access the processors memories **401-402** and vice versa, thereby simplifying programmability. A first portion of the virtual/effective address space may be allocated to the processor memory **401**, a second portion to the second processor memory **402**, a third portion to the GPU

memory 420, and so on. The entire virtual/effective memory space (sometimes referred to as the effective address space) may thereby be distributed across each of the processor memories 401-402 and GPU memories 420-423, allowing any processor or GPU to access any physical memory with a virtual address mapped to that memory.

[0172] Bias/coherence management circuitry 494A-494E within one or more of the MMUs 439A-439E may be provided that ensures cache coherence between the caches of the host processors (e.g., 405) and the GPUs 410-413 and implements biasing techniques indicating the physical memories in which certain types of data should be stored. While multiple instances of bias/coherence management circuitry 494A-494E are illustrated in FIG. 4F, the bias/coherence circuitry may be implemented within the MMU of one or more host processors 405 and/or within the accelerator integration circuit 436.

[0173] The GPU-attached memory 420-423 may be mapped as part of system memory and accessed using shared virtual memory (SVM) technology, but without suffering the typical performance drawbacks associated with full system cache coherence. The ability to GPU-attached memory 420-423 to be accessed as system memory without onerous cache coherence overhead provides a beneficial operating environment for GPU offload. This arrangement allows the host processor 405 software to setup operands and access computation results, without the overhead of tradition I/O DMA data copies. Such traditional copies involve driver calls, interrupts and memory mapped I/O (MMIO) accesses that are all inefficient relative to simple memory accesses. At the same time, the ability to access GPU attached memory 420-423 without cache coherence overheads can be critical to the execution time of an off-loaded computation. In cases with substantial streaming write memory traffic, for example, cache coherence overhead can significantly reduce the effective write bandwidth seen by a GPU 410-413. The efficiency of operand setup, the efficiency of results access, and the efficiency of GPU computation all play a role in determining the effectiveness of GPU offload.

[0174] A selection between GPU bias and host processor bias may be driven by a bias tracker data structure. A bias table may be used, for example, which may be a page-granular structure (i.e., controlled at the granularity of a memory page) that includes 1 or 2 bits per GPU-attached memory page. The bias table may be implemented in a stolen memory range of one or more GPU-attached memories 420-423, with or without a bias cache in the GPU 410-413 (e.g., to cache frequently/recently used entries of the bias table). Alternatively, the entire bias table may be maintained within the GPU.

[0175] In one implementation, the bias table entry associated with each access to the GPU-attached memory 420-423 is accessed prior the actual access to the GPU memory, causing the following operations. First, local requests from the GPU 410-413 that find their page in GPU bias are forwarded directly to a corresponding GPU memory 420-423. Local requests from the GPU that find their page in host bias are forwarded to the processor 405 (e.g., over a high-speed link as discussed above). Optionally, requests from the processor 405 that find the requested page in host processor bias complete the request like a normal memory read. Alternatively, requests directed to a GPU-biased page may

be forwarded to the GPU 410-413. The GPU may then transition the page to a host processor bias if it is not currently using the page.

[0176] The bias state of a page can be changed either by a software-based mechanism, a hardware-assisted software-based mechanism, or, for a limited set of cases, a purely hardware-based mechanism.

[0177] One mechanism for changing the bias state employs an API call (e.g., OpenCL), which, in turn, calls the GPU's device driver which, in turn, sends a message (or enqueues a command descriptor) to the GPU directing it to change the bias state and, for some transitions, perform a cache flushing operation in the host. The cache flushing operation is required for a transition from host processor 405 bias to GPU bias but is not required for the opposite transition.

[0178] Cache coherency may be maintained by temporarily rendering GPU-biased pages uncacheable by the host processor 405. To access these pages, the processor 405 may request access from the GPU 410 which may or may not grant access right away, depending on the implementation. Thus, to reduce communication between the host processor 405 and GPU 410 it is beneficial to ensure that GPU-biased pages are those which are required by the GPU but not the host processor 405 and vice versa.

Graphics Processing Pipeline

[0179] FIG. 5 illustrates a graphics processing pipeline 500. A graphics multiprocessor, such as graphics multiprocessor 234 as in FIG. 2D, graphics multiprocessor 325 of FIG. 3A, graphics multiprocessor 350 of FIG. 3B can implement the illustrated graphics processing pipeline 500. The graphics multiprocessor can be included within the parallel processing subsystems as described herein, such as the parallel processor 200 of FIG. 2A, which may be related to the parallel processor(s) 112 of FIG. 1 and may be used in place of one of those. The various parallel processing systems can implement the graphics processing pipeline 500 via one or more instances of the parallel processing unit (e.g., parallel processing unit 202 of FIG. 2A) as described herein. For example, a shader unit (e.g., graphics multiprocessor 234 of FIG. 2C) may be configured to perform the functions of one or more of a vertex processing unit 504, a tessellation control processing unit 508, a tessellation evaluation processing unit 512, a geometry processing unit 516, and a fragment/pixel processing unit 524. The functions of data assembler 502, primitive assemblers 506, 514, 518, tessellation unit 510, rasterizer 522, and raster operations unit 526 may also be performed by other processing engines within a processing cluster (e.g., processing cluster 214 of FIG. 2A) and a corresponding partition unit (e.g., partition unit 220A-220N of FIG. 2A). The graphics processing pipeline 500 may also be implemented using dedicated processing units for one or more functions. It is also possible that one or more portions of the graphics processing pipeline 500 are performed by parallel processing logic within a general-purpose processor (e.g., CPU). Optionally, one or more portions of the graphics processing pipeline 500 can access on-chip memory (e.g., parallel processor memory 222 as in FIG. 2A) via a memory interface 528, which may be an instance of the memory interface 218 of FIG. 2A. The graphics processor pipeline 500 may also be implemented via a multi-core group 365A as in FIG. 3C.

[0180] The data assembler **502** is a processing unit that may collect vertex data for surfaces and primitives. The data assembler **502** then outputs the vertex data, including the vertex attributes, to the vertex processing unit **504**. The vertex processing unit **504** is a programmable execution unit that executes vertex shader programs, lighting and transforming vertex data as specified by the vertex shader programs. The vertex processing unit **504** reads data that is stored in cache, local or system memory for use in processing the vertex data and may be programmed to transform the vertex data from an object-based coordinate representation to a world space coordinate space or a normalized device coordinate space.

[0181] A first instance of a primitive assembler **506** receives vertex attributes from the vertex processing unit **504**. The primitive assembler **506** readings stored vertex attributes as needed and constructs graphics primitives for processing by tessellation control processing unit **508**. The graphics primitives include triangles, line segments, points, patches, and so forth, as supported by various graphics processing application programming interfaces (APIs).

[0182] The tessellation control processing unit **508** treats the input vertices as control points for a geometric patch. The control points are transformed from an input representation from the patch (e.g., the patch's bases) to a representation that is suitable for use in surface evaluation by the tessellation evaluation processing unit **512**. The tessellation control processing unit **508** can also compute tessellation factors for edges of geometric patches. A tessellation factor applies to a single edge and quantifies a view-dependent level of detail associated with the edge. A tessellation unit **510** is configured to receive the tessellation factors for edges of a patch and to tessellate the patch into multiple geometric primitives such as line, triangle, or quadrilateral primitives, which are transmitted to a tessellation evaluation processing unit **512**. The tessellation evaluation processing unit **512** operates on parameterized coordinates of the subdivided patch to generate a surface representation and vertex attributes for each vertex associated with the geometric primitives.

[0183] A second instance of a primitive assembler **514** receives vertex attributes from the tessellation evaluation processing unit **512**, reading stored vertex attributes as needed, and constructs graphics primitives for processing by the geometry processing unit **516**. The geometry processing unit **516** is a programmable execution unit that executes geometry shader programs to transform graphics primitives received from primitive assembler **514** as specified by the geometry shader programs. The geometry processing unit **516** may be programmed to subdivide the graphics primitives into one or more new graphics primitives and calculate parameters used to rasterize the new graphics primitives.

[0184] The geometry processing unit **516** may be able to add or delete elements in the geometry stream. The geometry processing unit **516** outputs the parameters and vertices specifying new graphics primitives to primitive assembler **518**. The primitive assembler **518** receives the parameters and vertices from the geometry processing unit **516** and constructs graphics primitives for processing by a viewport scale, cull, and clip unit **520**. The geometry processing unit **516** reads data that is stored in parallel processor memory or system memory for use in processing the geometry data. The viewport scale, cull, and clip unit **520** performs clipping,

culling, and viewport scaling and outputs processed graphics primitives to a rasterizer **522**.

[0185] The rasterizer **522** can perform depth culling and other depth-based optimizations. The rasterizer **522** also performs scan conversion on the new graphics primitives to generate fragments and output those fragments and associated coverage data to the fragment/pixel processing unit **524**. The fragment/pixel processing unit **524** is a programmable execution unit that is configured to execute fragment shader programs or pixel shader programs. The fragment/pixel processing unit **524** transforming fragments or pixels received from rasterizer **522**, as specified by the fragment or pixel shader programs. For example, the fragment/pixel processing unit **524** may be programmed to perform operations included but not limited to texture mapping, shading, blending, texture correction and perspective correction to produce shaded fragments or pixels that are output to a raster operations unit **526**. The fragment/pixel processing unit **524** can read data that is stored in either the parallel processor memory or the system memory for use when processing the fragment data. Fragment or pixel shader programs may be configured to shade at sample, pixel, tile, or other granularities depending on the sampling rate configured for the processing units.

[0186] The raster operations unit **526** is a processing unit that performs raster operations including, but not limited to stencil, z-test, blending, and the like, and outputs pixel data as processed graphics data to be stored in graphics memory (e.g., parallel processor memory **222** as in FIG. 2A, and/or system memory **104** as in FIG. 1), to be displayed on the one or more display device(s) **110A-110B** or for further processing by one of the one or more processor(s) **102** or parallel processor(s) **112**. The raster operations unit **526** may be configured to compress z or color data that is written to memory and decompress z or color data that is read from memory.

Machine Learning Overview

[0187] The architecture described above can be applied to perform training and inference operations using machine learning models. Machine learning has been successful at solving many kinds of tasks. The computations that arise when training and using machine learning algorithms (e.g., neural networks) lend themselves naturally to efficient parallel implementations. Accordingly, parallel processors such as general-purpose graphics processing units (GPGPUs) have played a significant role in the practical implementation of deep neural networks. Parallel graphics processors with single instruction, multiple thread (SIMT) architectures are designed to maximize the amount of parallel processing in the graphics pipeline. In an SIMT architecture, groups of parallel threads attempt to execute program instructions synchronously together as often as possible to increase processing efficiency. The efficiency provided by parallel machine learning algorithm implementations allows the use of high-capacity networks and enables those networks to be trained on larger datasets.

[0188] A machine learning algorithm is an algorithm that can learn based on a set of data. For example, machine learning algorithms can be designed to model high-level abstractions within a data set. For example, image recognition algorithms can be used to determine which of several categories to which a given input belong; regression algorithms can output a numerical value given an input; and

pattern recognition algorithms can be used to generate translated text or perform text to speech and/or speech recognition.

[0189] An exemplary type of machine learning algorithm is a neural network. There are many types of neural networks; a simple type of neural network is a feedforward network. A feedforward network may be implemented as an acyclic graph in which the nodes are arranged in layers. Typically, a feedforward network topology includes an input layer and an output layer that are separated by at least one hidden layer. The hidden layer transforms input received by the input layer into a representation that is useful for generating output in the output layer. The network nodes are fully connected via edges to the nodes in adjacent layers, but there are no edges between nodes within each layer. Data received at the nodes of an input layer of a feedforward network are propagated (i.e., “fed forward”) to the nodes of the output layer via an activation function that calculates the states of the nodes of each successive layer in the network based on coefficients (“weights”) respectively associated with each of the edges connecting the layers. Depending on the specific model being represented by the algorithm being executed, the output from the neural network algorithm can take various forms.

[0190] Before a machine learning algorithm can be used to model a particular problem, the algorithm is trained using a training data set. Training a neural network involves selecting a network topology, using a set of training data representing a problem being modeled by the network, and adjusting the weights until the network model performs with a minimal error for all instances of the training data set. For example, during a supervised learning training process for a neural network, the output produced by the network in response to the input representing an instance in a training data set is compared to the “correct” labeled output for that instance, an error signal representing the difference between the output and the labeled output is calculated, and the weights associated with the connections are adjusted to minimize that error as the error signal is backward propagated through the layers of the network. The network is considered “trained” when the errors for each of the outputs generated from the instances of the training data set are minimized.

[0191] The accuracy of a machine learning algorithm can be affected significantly by the quality of the data set used to train the algorithm. The training process can be computationally intensive and may require a significant amount of time on a conventional general-purpose processor. Accordingly, parallel processing hardware is used to train many types of machine learning algorithms. This is particularly useful for optimizing the training of neural networks, as the computations performed in adjusting the coefficients in neural networks lend themselves naturally to parallel implementations. Specifically, many machine learning algorithms and software applications have been adapted to make use of the parallel processing hardware within general-purpose graphics processing devices.

[0192] FIG. 6 is a generalized diagram of a machine learning software stack 600. A machine learning application 602 is any logic that can be configured to train a neural network using a training dataset or to use a trained deep neural network to implement machine intelligence. The machine learning application 602 can include training and inference functionality for a neural network and/or special-

ized software that can be used to train a neural network before deployment. The machine learning application 602 can implement any type of machine intelligence including but not limited to image recognition, mapping and localization, autonomous navigation, speech synthesis, medical imaging, or language translation. Example machine learning applications 602 include, but are not limited to, voice-based virtual assistants, image or facial recognition algorithms, autonomous navigation, and the software tools that are used to train the machine learning models used by the machine learning applications 602.

[0193] Hardware acceleration for the machine learning application 602 can be enabled via a machine learning framework 604. The machine learning framework 604 can provide a library of machine learning primitives. Machine learning primitives are basic operations that are commonly performed by machine learning algorithms. Without the machine learning framework 604, developers of machine learning algorithms would be required to create and optimize the main computational logic associated with the machine learning algorithm, then re-optimize the computational logic as new parallel processors are developed. Instead, the machine learning application can be configured to perform the necessary computations using the primitives provided by the machine learning framework 604. Exemplary primitives include tensor convolutions, activation functions, and pooling, which are computational operations that are performed while training a convolutional neural network (CNN). The machine learning framework 604 can also provide primitives to implement basic linear algebra subprograms performed by many machine-learning algorithms, such as matrix and vector operations. Examples of a machine learning framework 604 include, but are not limited to, TensorFlow, TensorRT, PyTorch, MXNet, Caffe, and other high-level machine learning frameworks.

[0194] The machine learning framework 604 can process input data received from the machine learning application 602 and generate the appropriate input to a compute framework 606. The compute framework 606 can abstract the underlying instructions provided to the GPGPU driver 608 to enable the machine learning framework 604 to take advantage of hardware acceleration via the GPGPU hardware 610 without requiring the machine learning framework 604 to have intimate knowledge of the architecture of the GPGPU hardware 610. Additionally, the compute framework 606 can enable hardware acceleration for the machine learning framework 604 across a variety of types and generations of the GPGPU hardware 610. Exemplary compute frameworks 606 include the CUDA compute framework and associated machine learning libraries, such as the CUDA Deep Neural Network (cuDNN) library. The machine learning software stack 600 can also include communication libraries or frameworks to facilitate multi-GPU and multi-node compute.

GPGPU Machine Learning Acceleration

[0195] FIG. 7 illustrates a general-purpose graphics processing unit, which may be the parallel processor 200 of FIG. 2A or the parallel processor(s) 112 of FIG. 1. The general-purpose processing unit (GPGPU 700) may be configured to provide support for hardware acceleration of primitives provided by a machine learning framework to accelerate the processing the type of computational workloads associated with training deep neural networks. Addi-

tionally, the GPGPU 700 can be linked directly to other instances of the GPGPU to create a multi-GPU cluster to improve training speed for particularly deep neural networks. Primitives are also supported to accelerate inference operations for deployed neural networks.

[0196] The GPGPU 700 includes a host interface 702 to enable a connection with a host processor. The host interface 702 may be a PCI Express interface. However, the host interface can also be a vendor specific communications interface or communications fabric. The GPGPU 700 receives commands from the host processor and uses a global scheduler 704 to distribute execution threads associated with those commands to a set of processing clusters 706A-706H. The processing clusters 706A-706H share a cache memory 708. The cache memory 708 can serve as a higher-level cache for cache memories within the processing clusters 706A-706H. The illustrated processing clusters 706A-706H may correspond with processing clusters 214A-214N as in FIG. 2A.

[0197] The GPGPU 700 includes memory 714A-714B coupled with the processing clusters 706A-706H via a set of memory controllers 712A-712B. The memory 714A-714B can include various types of memory devices including dynamic random-access memory (DRAM) or graphics random access memory (SGRAM), including graphics double data rate (GDDR) memory. The memory 714A-714B may also include 3D stacked memory, including but not limited to high bandwidth memory (HBM).

[0198] Each of the processing clusters 706A-706H may include a set of graphics multiprocessors, such as the graphics multiprocessor 234 of FIG. 2D, graphics multiprocessor 325 of FIG. 3A, graphics multiprocessor 350 of FIG. 3B, or may include a multi-core group 365A-365N as in FIG. 3C. The graphics multiprocessors of the compute cluster include multiple types of integer and floating-point logic units that can perform computational operations at a range of precisions including suited for machine learning computations. For example, at least a subset of the floating-point units in each of the processing clusters 706A-706H can be configured to perform 16-bit or 32-bit floating point operations, while a different subset of the floating-point units can be configured to perform 64-bit floating point operations.

[0199] Multiple instances of the GPGPU 700 can be configured to operate as a compute cluster. The communication mechanism used by the compute cluster for synchronization and data exchange varies across embodiments. For example, the multiple instances of the GPGPU 700 communicate over the host interface 702. In one embodiment the GPGPU 700 includes an I/O hub 709 that couples the GPGPU 700 with a GPU link 710 that enables a direct connection to other instances of the GPGPU. The GPU link 710 may be coupled to a dedicated GPU-to-GPU bridge that enables communication and synchronization between multiple instances of the GPGPU 700. Optionally, the GPU link 710 couples with a high-speed interconnect to transmit and receive data to other GPGPUs or parallel processors. The multiple instances of the GPGPU 700 may be located in separate data processing systems and communicate via a network device that is accessible via the host interface 702. The GPU link 710 may be configured to enable a connection to a host processor in addition to or as an alternative to the host interface 702.

[0200] While the illustrated configuration of the GPGPU 700 can be configured to train neural networks, an alternate configuration of the GPGPU 700 can be configured for deployment within a high performance or low power inferencing platform. In an inferencing configuration, the GPGPU 700 includes fewer of the processing clusters 706A-706H relative to the training configuration. Additionally, memory technology associated with the memory 714A-714B may differ between inferencing and training configurations. In one embodiment, the inferencing configuration of the GPGPU 700 can support inferencing specific instructions. For example, an inferencing configuration can provide support for one or more 8-bit integer or floating-point dot product instructions, which are commonly used during inferencing operations for deployed neural networks.

[0201] FIG. 8 illustrates a multi-GPU computing system 800. The multi-GPU computing system 800 can include a processor 802 coupled to multiple GPGPUs 806A-806D via a host interface switch 804. The host interface switch 804 may be a PCI express switch device that couples the processor 802 to a PCI express bus over which the processor 802 can communicate with the set of GPGPUs 806A-806D. Each of the multiple GPGPUs 806A-806D can be an instance of the GPGPU 700 of FIG. 7. The GPGPUs 806A-806D can interconnect via a set of high-speed point to point GPU to GPU links 816. The high-speed GPU to GPU links can connect to each of the GPGPUs 806A-806D via a dedicated GPU link, such as the GPU link 710 as in FIG. 7. The P2P GPU links 816 enable direct communication between each of the GPGPUs 806A-806D without requiring communication over the host interface bus to which the processor 802 is connected. With GPU-to-GPU traffic directed to the P2P GPU links, the host interface bus remains available for system memory access or to communicate with other instances of the multi-GPU computing system 800, for example, via one or more network devices. While in FIG. 8 the GPGPUs 806A-806D connect to the processor 802 via the host interface switch 804, the processor 802 may alternatively include direct support for the P2P GPU links 816 and connect directly to the GPGPUs 806A-806D. In one embodiment the P2P GPU link 816 enable the multi-GPU computing system 800 to operate as a single logical GPU.

Machine Learning Neural Network Implementations

[0202] The computing architecture described herein can be configured to perform the types of parallel processing that is particularly suited for training and deploying neural networks for machine learning. A neural network can be generalized as a network of functions having a graph relationship. As is well-known in the art, there are a variety of types of neural network implementations used in machine learning. One exemplary type of neural network is the feedforward network, as previously described.

[0203] A second exemplary type of neural network is the Convolutional Neural Network (CNN). A CNN is a specialized feedforward neural network for processing data having a known, grid-like topology, such as image data. Accordingly, CNNs are commonly used for compute vision and image recognition applications, but they also may be used for other types of pattern recognition such as speech and language processing. The nodes in the CNN input layer are organized into a set of “filters” (feature detectors inspired by the receptive fields found in the retina), and the output of each set of filters is propagated to nodes in successive layers

of the network. The computations for a CNN include applying the convolution mathematical operation to each filter to produce the output of that filter. Convolution is a specialized kind of mathematical operation performed by two functions to produce a third function that is a modified version of one of the two original functions. In convolutional network terminology, the first function to the convolution can be referred to as the input, while the second function can be referred to as the convolution kernel. The output may be referred to as the feature map. For example, the input to a convolution layer can be a multidimensional array of data that defines the various color components of an input image. The convolution kernel can be a multidimensional array of parameters, where the parameters are adapted by the training process for the neural network.

[0204] Recurrent neural networks (RNNs) are a family of feedforward neural networks that include feedback connections between layers. RNNs enable modeling of sequential data by sharing parameter data across different parts of the neural network. The architecture for an RNN includes cycles. The cycles represent the influence of a present value of a variable on its own value at a future time, as at least a portion of the output data from the RNN is used as feedback for processing subsequent input in a sequence. This feature makes RNNs particularly useful for language processing due to the variable nature in which language data can be composed.

[0205] The figures described below present exemplary feedforward, CNN, and RNN networks, as well as describe a general process for respectively training and deploying each of those types of networks. It will be understood that these descriptions are exemplary and non-limiting as to any specific embodiment described herein and the concepts illustrated can be applied generally to deep neural networks and machine learning techniques in general.

[0206] The exemplary neural networks described above can be used to perform deep learning. Deep learning is machine learning using deep neural networks. The deep neural networks used in deep learning are artificial neural networks composed of multiple hidden layers, as opposed to shallow neural networks that include only a single hidden layer. Deeper neural networks are generally more computationally intensive to train. However, the additional hidden layers of the network enable multistep pattern recognition that results in reduced output error relative to shallow machine learning techniques.

[0207] Deep neural networks used in deep learning typically include a front-end network to perform feature recognition coupled to a back-end network which represents a mathematical model that can perform operations (e.g., object classification, speech recognition, etc.) based on the feature representation provided to the model. Deep learning enables machine learning to be performed without requiring hand crafted feature engineering to be performed for the model. Instead, deep neural networks can learn features based on statistical structure or correlation within the input data. The learned features can be provided to a mathematical model that can map detected features to an output. The mathematical model used by the network is generally specialized for the specific task to be performed, and different models will be used to perform different task.

[0208] Once the neural network is structured, a learning model can be applied to the network to train the network to perform specific tasks. The learning model describes how to

adjust the weights within the model to reduce the output error of the network. Backpropagation of errors is a common method used to train neural networks. An input vector is presented to the network for processing. The output of the network is compared to the desired output using a loss function and an error value is calculated for each of the neurons in the output layer. The error values are then propagated backwards until each neuron has an associated error value which roughly represents its contribution to the original output. The network can then learn from those errors using an algorithm, such as the stochastic gradient descent algorithm, to update the weights of the of the neural network.

[0209] FIG. 9A-9B illustrate an exemplary convolutional neural network. FIG. 9A illustrates various layers within a CNN. As shown in FIG. 9A, an exemplary CNN used to model image processing can receive input **902** describing the red, green, and blue (RGB) components of an input image. The input **902** can be processed by multiple convolutional layers (e.g., convolutional layer **904**, convolutional layer **906**). The output from the multiple convolutional layers may optionally be processed by a set of fully connected layers **908**. Neurons in a fully connected layer have full connections to all activations in the previous layer, as previously described for a feedforward network. The output from the fully connected layers **908** can be used to generate an output result from the network. The activations within the fully connected layers **908** can be computed using matrix multiplication instead of convolution. Not all CNN implementations make use of fully connected layers **908**. For example, in some implementations the convolutional layer **906** can generate output for the CNN.

[0210] The convolutional layers are sparsely connected, which differs from traditional neural network configuration found in the fully connected layers **908**. Traditional neural network layers are fully connected, such that every output unit interacts with every input unit. However, the convolutional layers are sparsely connected because the output of the convolution of a field is input (instead of the respective state value of each of the nodes in the field) to the nodes of the subsequent layer, as illustrated. The kernels associated with the convolutional layers perform convolution operations, the output of which is sent to the next layer. The dimensionality reduction performed within the convolutional layers is one aspect that enables the CNN to scale to process large images.

[0211] FIG. 9B illustrates exemplary computation stages within a convolutional layer of a CNN. Input to a convolutional layer **912** of a CNN can be processed in three stages of a convolutional layer **914**. The three stages can include a convolution stage **916**, a detector stage **918**, and a pooling stage **920**. The convolutional layer **914** can then output data to a successive convolutional layer. The final convolutional layer of the network can generate output feature map data or provide input to a fully connected layer, for example, to generate a classification value for the input to the CNN.

[0212] In the convolution stage **916** performs several convolutions in parallel to produce a set of linear activations. The convolution stage **916** can include an affine transformation, which is any transformation that can be specified as a linear transformation plus a translation. Affine transformations include rotations, translations, scaling, and combinations of these transformations. The convolution stage computes the output of functions (e.g., neurons) that are

connected to specific regions in the input, which can be determined as the local region associated with the neuron. The neurons compute a dot product between the weights of the neurons and the region in the local input to which the neurons are connected. The output from the convolution stage **916** defines a set of linear activations that are processed by successive stages of the convolutional layer **914**.

[0213] The linear activations can be processed by a detector stage **918**. In the detector stage **918**, each linear activation is processed by a non-linear activation function. The non-linear activation function increases the nonlinear properties of the overall network without affecting the receptive fields of the convolution layer. Several types of non-linear activation functions may be used. One particular type is the rectified linear unit (ReLU), which uses an activation function defined as $f(x)=\max(0, x)$, such that the activation is thresholded at zero.

[0214] The pooling stage **920** uses a pooling function that replaces the output of the convolutional layer **906** with a summary statistic of the nearby outputs. The pooling function can be used to introduce translation invariance into the neural network, such that small translations to the input do not change the pooled outputs. Invariance to local translation can be useful in scenarios where the presence of a feature in the input data is more important than the precise location of the feature. Various types of pooling functions can be used during the pooling stage **920**, including max pooling, average pooling, and 12-norm pooling. Additionally, some CNN implementations do not include a pooling stage. Instead, such implementations substitute and additional convolution stage having an increased stride relative to previous convolution stages.

[0215] The output from the convolutional layer **914** can then be processed by the next layer **922**. The next layer **922** can be an additional convolutional layer or one of the fully connected layers **908**. For example, the first convolutional layer **904** of FIG. 9A can output to the second convolutional layer **906**, while the second convolutional layer can output to a first layer of the fully connected layers **908**.

[0216] FIG. 10 illustrates an exemplary recurrent neural network. In a recurrent neural network (RNN), the previous state of the network influences the output of the current state of the network. RNNs can be built in a variety of ways using a variety of functions. The use of RNNs generally revolves around using mathematical models to predict the future based on a prior sequence of inputs. For example, an RNN may be used to perform statistical language modeling to predict an upcoming word given a previous sequence of words. The RNN **1000** of FIG. 10 can be described as having an input layer **1002** that receives an input vector, hidden layers **1004** to implement a recurrent function, a feedback mechanism **1005** to enable a 'memory' of previous states, and an output layer **1006** to output a result. The RNN **1000** operates based on time-steps. The state of the RNN at a given time step is influenced based on the previous time step via the feedback mechanism **1005**. For a given time step, the state of the hidden layers **1004** is defined by the previous state and the input at the current time step. An initial input (x_1) at a first-time step can be processed by the hidden layer **1004**. A second input (x_2) can be processed by the hidden layer **1004** using state information that is determined during the processing of the initial input (x_1). A given state can be computed as $s_t=f(Ux_t+Ws_{t-1})$, where U and W are parameter matrices. The function f is generally a non-

linearity, such as the hyperbolic tangent function (Tanh) or a variant of the rectifier function $f(x)=\max(0,x)$. However, the specific mathematical function used in the hidden layers **1004** can vary depending on the specific implementation details of the RNN **1000**.

[0217] In addition to the basic CNN and RNN networks described, acceleration for variations on those networks may be enabled. One example RNN variant is the long short term memory (LSTM) RNN. LSTM RNNs are capable of learning long-term dependencies that may be necessary for processing longer sequences of language. A variant on the CNN is a convolutional deep belief network, which has a structure similar to a CNN and is trained in a manner similar to a deep belief network. A deep belief network (DBN) is a generative neural network that is composed of multiple layers of stochastic (random) variables. DBNs can be trained layer-by-layer using greedy unsupervised learning. The learned weights of the DBN can then be used to provide pre-train neural networks by determining an optimal initial set of weights for the neural network. In further embodiments, acceleration for reinforcement learning is enabled. In reinforcement learning, an artificial agent learns by interacting with its environment. The agent is configured to optimize certain objectives to maximize cumulative rewards.

[0218] FIG. 11 illustrates training and deployment of a deep neural network. Once a given network has been structured for a task the neural network is trained using a training dataset **1102**. Various training frameworks **1104** have been developed to enable hardware acceleration of the training process. For example, the machine learning framework **604** of FIG. 6 may be configured as a training framework **1104**. The training framework **1104** can hook into an untrained neural network **1106** and enable the untrained neural net to be trained using the parallel processing resources described herein to generate a trained neural network **1108**.

[0219] To start the training process the initial weights may be chosen randomly or by pre-training using a deep belief network. The training cycle then be performed in either a supervised or unsupervised manner.

[0220] Supervised learning is a learning method in which training is performed as a mediated operation, such as when the training dataset **1102** includes input paired with the desired output for the input, or where the training dataset includes input having known output and the output of the neural network is manually graded. The network processes the inputs and compares the resulting outputs against a set of expected or desired outputs. Errors are then propagated back through the system. The training framework **1104** can adjust to adjust the weights that control the untrained neural network **1106**. The training framework **1104** can provide tools to monitor how well the untrained neural network **1106** is converging towards a model suitable to generating correct answers based on known input data. The training process occurs repeatedly as the weights of the network are adjusted to refine the output generated by the neural network. The training process can continue until the neural network reaches a statistically desired accuracy associated with a trained neural network **1108**. The trained neural network **1108** can then be deployed to implement any number of machine learning operations to generate an inference result **1114** based on input of new data **1112**.

[0221] Unsupervised learning is a learning method in which the network attempts to train itself using unlabeled

data. Thus, for unsupervised learning the training dataset **1102** will include input data without any associated output data. The untrained neural network **1106** can learn groupings within the unlabeled input and can determine how individual inputs are related to the overall dataset. Unsupervised training can be used to generate a self-organizing map, which is a type of trained neural network **1108** capable of performing operations useful in reducing the dimensionality of data. Unsupervised training can also be used to perform anomaly detection, which allows the identification of data points in an input dataset that deviate from the normal patterns of the data.

[0222] Variations on supervised and unsupervised training may also be employed. Semi-supervised learning is a technique in which in the training dataset **1102** includes a mix of labeled and unlabeled data of the same distribution. Incremental learning is a variant of supervised learning in which input data is continuously used to further train the model. Incremental learning enables the trained neural network **1108** to adapt to the new data **1112** without forgetting the knowledge instilled within the network during initial training.

[0223] Whether supervised or unsupervised, the training process for particularly deep neural networks may be too computationally intensive for a single compute node. Instead of using a single compute node, a distributed network of computational nodes can be used to accelerate the training process.

[0224] FIG. 12A is a block diagram illustrating distributed learning. Distributed learning is a training model that uses multiple distributed computing nodes to perform supervised or unsupervised training of a neural network. The distributed computational nodes can each include one or more host processors and one or more of the general-purpose processing nodes, such as the GPGPU **700** as in FIG. 7. As illustrated, distributed learning can be performed with model parallelism **1202**, data parallelism **1204**, or a combination of model and data parallelism **1206**.

[0225] In model parallelism **1202**, different computational nodes in a distributed system can perform training computations for different parts of a single network. For example, each layer of a neural network can be trained by a different processing node of the distributed system. The benefits of model parallelism include the ability to scale to particularly large models. Splitting the computations associated with different layers of the neural network enables the training of very large neural networks in which the weights of all layers would not fit into the memory of a single computational node. In some instances, model parallelism can be particularly useful in performing unsupervised training of large neural networks.

[0226] In data parallelism **1204**, the different nodes of the distributed network have a complete instance of the model and each node receives a different portion of the data. The results from the different nodes are then combined. While different approaches to data parallelism are possible, data parallel training approaches all require a technique of combining results and synchronizing the model parameters between each node. Exemplary approaches to combining data include parameter averaging and update-based data parallelism. Parameter averaging trains each node on a subset of the training data and sets the global parameters (e.g., weights, biases) to the average of the parameters from each node. Parameter averaging uses a central parameter

server that maintains the parameter data. Update based data parallelism is similar to parameter averaging except that instead of transferring parameters from the nodes to the parameter server, the updates to the model are transferred. Additionally, update-based data parallelism can be performed in a decentralized manner, where the updates are compressed and transferred between nodes.

[0227] Combined model and data parallelism **1206** can be implemented, for example, in a distributed system in which each computational node includes multiple GPUs. Each node can have a complete instance of the model with separate GPUs within each node are used to train different portions of the model.

[0228] Distributed training has increased overhead relative to training on a single machine. However, the parallel processors and GPGPUs described herein can each implement various techniques to reduce the overhead of distributed training, including techniques to enable high bandwidth GPU-to-GPU data transfer and accelerated remote data synchronization.

[0229] FIG. 12B is a block diagram illustrating a programmable network interface **1210** and data processing unit. The programmable network interface **1210** is a programmable network engine that can be used to accelerate network-based compute tasks within a distributed environment. The programmable network interface **1210** can couple with a host system via host interface **1270**. The programmable network interface **1210** can be used to accelerate network or storage operations for CPUs or GPUs of the host system. The host system can be, for example, a node of a distributed learning system used to perform distributed training, for example, as shown in FIG. 12A. The host system can also be a data center node within a data center.

[0230] In one embodiment, access to remote storage containing model data can be accelerated by the programmable network interface **1210**. For example, the programmable network interface **1210** can be configured to present remote storage devices as local storage devices to the host system. The programmable network interface **1210** can also accelerate remote direct memory access (RDMA) operations performed between GPUs of the host system with GPUs of remote systems. In one embodiment, the programmable network interface **1210** can enable storage functionality such as, but not limited to NVME-oF. The programmable network interface **1210** can also accelerate encryption, data integrity, compression, and other operations for remote storage on behalf of the host system, allowing remote storage to approach the latencies of storage devices that are directly attached to the host system.

[0231] The programmable network interface **1210** can also perform resource allocation and management on behalf of the host system. Storage security operations can be offloaded to the programmable network interface **1210** and performed in concert with the allocation and management of remote storage resources. Network-based operations to manage access to the remote storage that would otherwise be performed by a processor of the host system can instead be performed by the programmable network interface **1210**.

[0232] In one embodiment, network and/or data security operations can be offloaded from the host system to the programmable network interface **1210**. Data center security policies for a data center node can be handled by the programmable network interface **1210** instead of the processors of the host system. For example, the programmable

network interface **1210** can detect and mitigate against an attempted network-based attack (e.g., DDoS) on the host system, preventing the attack from compromising the availability of the host system.

[0233] The programmable network interface **1210** can include a system on a chip (SoC **1220**) that executes an operating system via multiple processor cores **1222**. The processor cores **1222** can include general-purpose processor (e.g., CPU) cores. In one embodiment the processor cores **1222** can also include one or more GPU cores. The SoC **1220** can execute instructions stored in a memory device **1240**. A storage device **1250** can store local operating system data. The storage device **1250** and memory device **1240** can also be used to cache remote data for the host system. Network ports **1260A-1260B** enable a connection to a network or fabric and facilitate network access for the SoC **1220** and, via the host interface **1270**, for the host system. The programmable network interface **1210** can also include an I/O interface **1275**, such as a USB interface. The I/O interface **1275** can be used to couple external devices to the programmable network interface **1210** or as a debug interface. The programmable network interface **1210** also includes a management interface **1230** that enables software on the host device to manage and configure the programmable network interface **1210** and/or SoC **1220**. In one embodiment the programmable network interface **1210** may also include one or more accelerators or GPUs **1245** to accept offload of parallel compute tasks from the SoC **1220**, host system, or remote systems coupled via the network ports **1260A-1260B**.

Exemplary Machine Learning Applications

[0234] Machine learning can be applied to solve a variety of technological problems, including but not limited to computer vision, autonomous driving and navigation, speech recognition, and language processing. Computer vision has traditionally been one of the most active research areas for machine learning applications. Applications of computer vision range from reproducing human visual abilities, such as recognizing faces, to creating new categories of visual abilities. For example, computer vision applications can be configured to recognize sound waves from the vibrations induced in objects visible in a video. Parallel processor accelerated machine learning enables computer vision applications to be trained using significantly larger training dataset than previously feasible and enables inferencing systems to be deployed using low power parallel processors.

[0235] Parallel processor accelerated machine learning has autonomous driving applications including lane and road sign recognition, obstacle avoidance, navigation, and driving control. Accelerated machine learning techniques can be used to train driving models based on datasets that define the appropriate responses to specific training input. The parallel processors described herein can enable rapid training of the increasingly complex neural networks used for autonomous driving solutions and enables the deployment of low power inferencing processors in a mobile platform suitable for integration into autonomous vehicles.

[0236] Parallel processor accelerated deep neural networks have enabled machine learning approaches to automatic speech recognition (ASR). ASR includes the creation of a function that computes the most probable linguistic sequence given an input acoustic sequence. Accelerated

machine learning using deep neural networks have enabled the replacement of the hidden Markov models (HMMs) and Gaussian mixture models (GMMs) previously used for ASR.

[0237] Parallel processor accelerated machine learning can also be used to accelerate natural language processing. Automatic learning procedures can make use of statistical inference algorithms to produce models that are robust to erroneous or unfamiliar input. Exemplary natural language processor applications include automatic machine translation between human languages.

[0238] The parallel processing platforms used for machine learning can be divided into training platforms and deployment platforms. Training platforms are generally highly parallel and include optimizations to accelerate multi-GPU single node training and multi-node, multi-GPU training. Exemplary parallel processors suited for training include the GPGPU **700** of FIG. 7 and the multi-GPU computing system **800** of FIG. 8. On the contrary, deployed machine learning platforms generally include lower power parallel processors suitable for use in products such as cameras, autonomous robots, and autonomous vehicles.

[0239] Additionally, machine learning techniques can be applied to accelerate or enhance graphics processing activities. For example, a machine learning model can be trained to recognize output generated by a GPU accelerated application and generate an upscaled version of that output. Such techniques can be applied to accelerate the generation of high-resolution images for a gaming application. Various other graphics pipeline activities can benefit from the use of machine learning. For example, machine learning models can be trained to perform tessellation operations on geometry data to increase the complexity of geometric models, allowing fine-detailed geometry to be automatically generated from geometry of relatively lower detail.

[0240] FIG. 13 illustrates an exemplary inferencing system on a chip (SOC) **1300** suitable for performing inferencing using a trained model. The SOC **1300** can integrate processing components including a media processor **1302**, a vision processor **1304**, a GPGPU **1306** and a multi-core processor **1308**. The GPGPU **1306** may be a GPGPU as described herein, such as the GPGPU **700**, and the multi-core processor **1308** may be a multi-core processor described herein, such as the multi-core processors **405-406**. The SOC **1300** can additionally include on-chip memory **1305** that can enable a shared on-chip data pool that is accessible by each of the processing components. The processing components can be optimized for low power operation to enable deployment to a variety of machine learning platforms, including autonomous vehicles and autonomous robots. For example, one implementation of the SOC **1300** can be used as a portion of the main control system for an autonomous vehicle. Where the SOC **1300** is configured for use in autonomous vehicles the SOC is designed and configured for compliance with the relevant functional safety standards of the deployment jurisdiction.

[0241] During operation, the media processor **1302** and vision processor **1304** can work in concert to accelerate computer vision operations. The media processor **1302** can enable low latency decode of multiple high-resolution (e.g., 4K, 8K) video streams. The decoded video streams can be written to a buffer in the on-chip memory **1305**. The vision processor **1304** can then parse the decoded video and perform preliminary processing operations on the frames of the decoded video in preparation of processing the frames

using a trained image recognition model. For example, the vision processor **1304** can accelerate convolution operations for a CNN that is used to perform image recognition on the high-resolution video data, while back-end model computations are performed by the GPGPU **1306**.

[0242] The multi-core processor **1308** can include control logic to assist with sequencing and synchronization of data transfers and shared memory operations performed by the media processor **1302** and the vision processor **1304**. The multi-core processor **1308** can also function as an application processor to execute software applications that can make use of the inferencing compute capability of the GPGPU **1306**. For example, at least a portion of the navigation and driving logic can be implemented in software executing on the multi-core processor **1308**. Such software can directly issue computational workloads to the GPGPU **1306** or the computational workloads can be issued to the multi-core processor **1308**, which can offload at least a portion of those operations to the GPGPU **1306**.

[0243] The GPGPU **1306** can include compute clusters such as a low power configuration of the processing clusters **706A-706H** within the GPGPU **700**. The compute clusters within the GPGPU **1306** can support instruction that are specifically optimized to perform inferencing computations on a trained neural network. For example, the GPGPU **1306** can support instructions to perform low precision computations such as 8-bit and 4-bit integer vector operations.

Additional System Overview

[0244] FIG. **14** is a block diagram of a processing system **1400**. The elements of FIG. **14** having the same or similar names as the elements of any other figure herein describe the same elements as in the other figures, can operate or function in a manner similar to that, can comprise the same components, and can be linked to other entities, as those described elsewhere herein, but are not limited to such. System **1400** may be used in a single processor desktop system, a multiprocessor workstation system, or a server system having a large number of processors **1402** or processor cores **1407**. The system **1400** may be a processing platform incorporated within a system-on-a-chip (SoC) integrated circuit for use in mobile, handheld, or embedded devices such as within Internet-of-things (IoT) devices with wired or wireless connectivity to a local or wide area network.

[0245] The system **1400** may be a processing system having components that correspond with those of FIG. **1**. For example, in different configurations, processor(s) **1402** or processor core(s) **1407** may correspond with processor(s) **102** of FIG. **1**. Graphics processor(s) **1408** may correspond with parallel processor(s) **112** of FIG. **1**. External graphics processor **1418** may be one of the add-in device(s) **120** of FIG. **1**.

[0246] The system **1400** can include, couple with, or be integrated within: a server-based gaming platform; a game console, including a game and media console; a mobile gaming console, a handheld game console, or an online game console. The system **1400** may be part of a mobile phone, smart phone, tablet computing device or mobile Internet-connected device such as a laptop with low internal storage capacity. Processing system **1400** can also include, couple with, or be integrated within: a wearable device, such as a smart watch wearable device; smart eyewear or clothing enhanced with augmented reality (AR) or virtual reality (VR) features to provide visual, audio or tactile outputs to

supplement real world visual, audio or tactile experiences or otherwise provide text, audio, graphics, video, holographic images or video, or tactile feedback; other augmented reality (AR) device; or other virtual reality (VR) device. The processing system **1400** may include or be part of a television or set top box device. The system **1400** can include, couple with, or be integrated within a self-driving vehicle such as a bus, tractor trailer, car, motor or electric power cycle, plane or glider (or any combination thereof). The self-driving vehicle may use system **1400** to process the environment sensed around the vehicle.

[0247] The one or more processors **1402** may include one or more processor cores **1407** to process instructions which, when executed, perform operations for system or user software. The least one of the one or more processor cores **1407** may be configured to process a specific instruction set **1409**. The instruction set **1409** may facilitate Complex Instruction Set Computing (CISC), Reduced Instruction Set Computing (RISC), or computing via a Very Long Instruction Word (VLIW). One or more processor cores **1407** may process a different instruction set **1409**, which may include instructions to facilitate the emulation of other instruction sets. Processor core **1407** may also include other processing devices, such as a Digital Signal Processor (DSP).

[0248] The processor **1402** may include cache memory **1404**. Depending on the architecture, the processor **1402** can have a single internal cache or multiple levels of internal cache. In some embodiments, the cache memory is shared among various components of the processor **1402**. In some embodiments, the processor **1402** also uses an external cache (e.g., a Level-3 (L3) cache or Last Level Cache (LLC)) (not shown), which may be shared among processor cores **1407** using known cache coherency techniques. A register file **1406** can be additionally included in processor **1402** and may include different types of registers for storing different types of data (e.g., integer registers, floating point registers, status registers, and an instruction pointer register). Some registers may be general-purpose registers, while other registers may be specific to the design of the processor **1402**.

[0249] The one or more processor(s) **1402** may be coupled with one or more interface bus(es) **1410** to transmit communication signals such as address, data, or control signals between processor **1402** and other components in the system **1400**. The interface bus **1410**, in one of these embodiments, can be a processor bus, such as a version of the Direct Media Interface (DMI) bus. However, processor busses are not limited to the DMI bus, and may include one or more Peripheral Component Interconnect buses (e.g., PCI, PCI express), memory busses, or other types of interface busses. For example, the processor(s) **1402** may include an integrated memory controller **1416** and a platform controller hub **1430**. The memory controller **1416** facilitates communication between a memory device and other components of the system **1400**, while the platform controller hub (PCH) **1430** provides connections to I/O devices via a local I/O bus.

[0250] The memory device **1420** can be a dynamic random-access memory (DRAM) device, a static random-access memory (SRAM) device, flash memory device, phase-change memory device, or some other memory device having suitable performance to serve as process memory. The memory device **1420** can, for example, operate as system memory for the system **1400**, to store data **1422** and instructions **1421** for use when the one or more processors

1402 executes an application or process. Memory controller **1416** also couples with an optional external graphics processor **1418**, which may communicate with the one or more graphics processors **1408** in processors **1402** to perform graphics and media operations. In some embodiments, graphics, media, and or compute operations may be assisted by an accelerator **1412** which is a coprocessor that can be configured to perform a specialized set of graphics, media, or compute operations. For example, the accelerator **1412** may be a matrix multiplication accelerator used to optimize machine learning or compute operations. The accelerator **1412** can be a ray-tracing accelerator that can be used to perform ray-tracing operations in concert with the graphics processor **1408**. In one embodiment, an external accelerator **1419** may be used in place of or in concert with the accelerator **1412**.

[0251] A display device **1411** may be provided that can connect to the processor(s) **1402**. The display device **1411** can be one or more of an internal display device, as in a mobile electronic device or a laptop device or an external display device attached via a display interface (e.g., DisplayPort, etc.). The display device **1411** can be a head mounted display (HMD) such as a stereoscopic display device for use in virtual reality (VR) applications or augmented reality (AR) applications.

[0252] The platform controller hub **1430** may enable peripherals to connect to memory device **1420** and processor **1402** via a high-speed I/O bus. The I/O peripherals include, but are not limited to, an audio controller **1446**, a network controller **1434**, a firmware interface **1428**, a wireless transceiver **1426**, touch sensors **1425**, a data storage device **1424** (e.g., non-volatile memory, volatile memory, hard disk drive, flash memory, NAND, 3D NAND, 3D X Point/Optane, etc.). The data storage device **1424** can connect via a storage interface (e.g., SATA) or via a peripheral bus, such as a Peripheral Component Interconnect bus (e.g., PCI, PCI express). The touch sensors **1425** can include touch screen sensors, pressure sensors, or fingerprint sensors. The wireless transceiver **1426** can be a Wi-Fi transceiver, a Bluetooth transceiver, or a mobile network transceiver such as a 3G, 4G, 5G, or Long-Term Evolution (LTE) transceiver. The firmware interface **1428** enables communication with system firmware, and can be, for example, a unified extensible firmware interface (UEFI). The network controller **1434** can enable a network connection to a wired network. In some embodiments, a high-performance network controller (not shown) couples with the interface bus **1410**. The audio controller **1446** may be a multi-channel high-definition audio controller. In some of these embodiments the system **1400** includes an optional legacy I/O controller **1440** for coupling legacy (e.g., Personal System 2 (PS/2)) devices to the system. The platform controller hub **1430** can also connect to one or more Universal Serial Bus (USB) controllers **1442** connect input devices, such as keyboard and mouse **1443** combinations, a camera **1444**, or other USB input devices.

[0253] It will be appreciated that the system **1400** shown is exemplary and not limiting, as other types of data processing systems that are differently configured may also be used. For example, an instance of the memory controller **1416** and platform controller hub **1430** may be integrated into a discrete external graphics processor, such as the external graphics processor **1418**. The platform controller hub **1430** and/or memory controller **1416** may be external to

the one or more processor(s) **1402**. For example, the system **1400** can include an external memory controller **1416** and platform controller hub **1430**, which may be configured as a memory controller hub and peripheral controller hub within a system chipset that is in communication with the processor (s) **1402**.

[0254] For example, circuit boards (“sleds”) can be used on which components such as CPUs, memory, and other components are placed are designed for increased thermal performance. Processing components such as the processors may be located on a top side of a sled while near memory, such as DIM Ms, are located on a bottom side of the sled. As a result of the enhanced airflow provided by this design, the components may operate at higher frequencies and power levels than in typical systems, thereby increasing performance. Furthermore, the sleds are configured to blindly mate with power and data communication cables in a rack, thereby enhancing their ability to be quickly removed, upgraded, reinstalled, and/or replaced. Similarly, individual components located on the sleds, such as processors, accelerators, memory, and data storage drives, are configured to be easily upgraded due to their increased spacing from each other. In the illustrative embodiment, the components additionally include hardware attestation features to prove their authenticity.

[0255] A data center can utilize a single network architecture (“fabric”) that supports multiple other network architectures including Ethernet and Omni-Path. The sleds can be coupled to switches via optical fibers, which provide higher bandwidth and lower latency than typical twisted pair cabling (e.g., Category 5, Category 5e, Category 6, etc.). Due to the high bandwidth, low latency interconnections and network architecture, the data center may, in use, pool resources, such as memory, accelerators (e.g., GPUs, graphics accelerators, FPGAs, ASICs, neural network and/or artificial intelligence accelerators, etc.), and data storage drives that are physically disaggregated, and provide them to compute resources (e.g., processors) on an as needed basis, enabling the compute resources to access the pooled resources as if they were local.

[0256] A power supply or source can provide voltage and/or current to system **1400** or any component or system described herein. In one example, the power supply includes an AC to DC (alternating current to direct current) adapter to plug into a wall outlet. Such AC power can be renewable energy (e.g., solar power) power source. In one example, the power source includes a DC power source, such as an external AC to DC converter. A power source or power supply may also include wireless charging hardware to charge via proximity to a charging field. The power source can include an internal battery, alternating current supply, motion-based power supply, solar power supply, or fuel cell source.

[0257] FIG. 15A-15C illustrate computing systems and graphics processors. The elements of FIG. 15A-15C having the same or similar names as the elements of any other figure herein describe the same elements as in the other figures, can operate or function in a manner similar to that, can comprise the same components, and can be linked to other entities, as those described elsewhere herein, but are not limited to such.

[0258] FIG. 15A is a block diagram of a processor **1500**, which may be a variant of one of the processors **1402** and may be used in place of one of those. Therefore, the disclosure of any features in combination with the processor

1500 herein also discloses a corresponding combination with the processor(s) **1402** but is not limited to such. The processor **1500** may have one or more processor cores **1502A-1502N**, an integrated memory controller **1514**, and an integrated graphics processor **1508**. Where an integrated graphics processor **1508** is excluded, the system that includes the processor will include a graphics processor device within a system chipset or coupled via a system bus. Processor **1500** can include additional cores up to and including additional core **1502N** represented by the dashed lined boxes. Each of processor cores **1502A-1502N** includes one or more internal cache units **1504A-1504N**. In some embodiments each processor core **1502A-1502N** also has access to one or more shared cache units **1506**. The internal cache units **1504A-1504N** and shared cache units **1506** represent a cache memory hierarchy within the processor **1500**. The cache memory hierarchy may include at least one level of instruction and data cache within each processor core and one or more levels of shared mid-level cache, such as a Level 2 (L2), Level 3 (L3), Level 4 (L4), or other levels of cache, where the highest level of cache before external memory is classified as the LLC. In some embodiments, cache coherency logic maintains coherency between the various cache units **1506** and **1504A-1504N**.

[0259] The processor **1500** may also include a set of one or more bus controller units **1516** and a system agent core **1510**. The one or more bus controller units **1516** manage a set of peripheral buses, such as one or more PCI or PCI express busses. System agent core **1510** provides management functionality for the various processor components. The system agent core **1510** may include one or more integrated memory controllers **1514** to manage access to various external memory devices (not shown).

[0260] For example, one or more of the processor cores **1502A-1502N** may include support for simultaneous multi-threading. The system agent core **1510** includes components for coordinating and operating cores **1502A-1502N** during multi-threaded processing. System agent core **1510** may additionally include a power control unit (PCU), which includes logic and components to regulate the power state of processor cores **1502A-1502N** and graphics processor **1508**.

[0261] The processor **1500** may additionally include graphics processor **1508** to execute graphics processing operations. In some of these embodiments, the graphics processor **1508** couples with the set of shared cache units **1506**, and the system agent core **1510**, including the one or more integrated memory controllers **1514**. The system agent core **1510** may also include a display controller **1511** to drive graphics processor output to one or more coupled displays. The display controller **1511** may also be a separate module coupled with the graphics processor via at least one interconnect or may be integrated within the graphics processor **1508**.

[0262] A ring-based interconnect **1512** may be used to couple the internal components of the processor **1500**. However, an alternative interconnect unit may be used, such as a point-to-point interconnect, a switched interconnect, or other techniques, including techniques well known in the art. In some of these embodiments with a ring-based interconnect **1512**, the graphics processor **1508** couples with the ring-based interconnect **1512** via an I/O link **1513**.

[0263] The exemplary I/O link **1513** represents at least one of multiple varieties of I/O interconnects, including an on package I/O interconnect which facilitates communication

between various processor components and a high-performance memory module **1518**, such as an eDRAM module or a high-bandwidth memory (HBM) module. Optionally, each of the processor cores **1502A-1502N** and graphics processor **1508** can use the high-performance memory module **1518** as a shared Last Level Cache.

[0264] The processor cores **1502A-1502N** may, for example, be homogenous cores executing the same instruction set architecture. Alternatively, the processor cores **1502A-1502N** are heterogeneous in terms of instruction set architecture (ISA), where one or more of processor cores **1502A-1502N** execute a first instruction set, while at least one of the other cores executes a subset of the first instruction set or a different instruction set. The processor cores **1502A-1502N** may be heterogeneous in terms of micro-architecture, where one or more cores having a relatively higher power consumption couple with one or more power cores having a lower power consumption. As another example, the processor cores **1502A-1502N** are heterogeneous in terms of computational capability. Additionally, processor **1500** can be implemented on one or more chips or as an SoC integrated circuit having the illustrated components, in addition to other components.

[0265] FIG. 15B is a block diagram of hardware logic of a graphics processor core block **1519**, according to some embodiments described herein. In some embodiments, elements of FIG. 15B having the same reference numbers (or names) as the elements of any other figure herein may operate or function in a manner similar to that described elsewhere herein. In one embodiment, the graphics processor core block **1519** is exemplary of one partition of a graphics processor. The graphics processor core block **1519** can be included within the integrated graphics processor **1508** of FIG. 15A or a discrete graphics processor, parallel processor, and/or compute accelerator. A graphics processor as described herein may include multiple graphics core blocks based on target power and performance envelopes. Each graphics processor core block **1519** can include a function block **1530** coupled with multiple graphics cores **1521A-1521F** that include modular blocks of fixed function logic and general-purpose programmable logic. The graphics processor core block **1519** also includes shared/cache memory **1536** that is accessible by all graphics cores **1521A-1521F**, rasterizer logic **1537**, and additional fixed function logic **1538**.

[0266] In some embodiments, the function block **1530** includes a geometry/fixed function pipeline **1531** that can be shared by all graphics cores in the graphics processor core block **1519**. In various embodiments, the geometry/fixed function pipeline **1531** includes a 3D geometry pipeline a video front-end unit, a thread spawner and global thread dispatcher, and a unified return buffer manager, which manages unified return buffers. In one embodiment the function block **1530** also includes a graphics SoC interface **1532**, a graphics microcontroller **1533**, and a media pipeline **1534**. The graphics SoC interface **1532** provides an interface between the graphics processor core block **1519** and other core blocks within a graphics processor or compute accelerator SoC. The graphics microcontroller **1533** is a programmable sub-processor that is configurable to manage various functions of the graphics processor core block **1519**, including thread dispatch, scheduling, and pre-emption. The media pipeline **1534** includes logic to facilitate the decoding, encoding, pre-processing, and/or post-processing of multi-

media data, including image and video data. The media pipeline **1534** implement media operations via requests to compute or sampling logic within the graphics cores **1521-1521F**. One or more pixel backends **1535** can also be included within the function block **1530**. The pixel backends **1535** include a cache memory to store pixel color values and can perform blend operations and lossless color compression of rendered pixel data.

[0267] In one embodiment the graphics SoC interface **1532** enables the graphics processor core block **1519** to communicate with general-purpose application processor cores (e.g., CPUs) and/or other components within an SoC or a system host CPU that is coupled with the SoC via a peripheral interface. The graphics SoC interface **1532** also enables communication with off-chip memory hierarchy elements such as a shared last level cache memory, system RAM, and/or embedded on-chip or on-package DRAM. The SoC interface **1532** can also enable communication with fixed function devices within the SoC, such as camera imaging pipelines, and enables the use of and/or implements global memory atomics that may be shared between the graphics processor core block **1519** and CPUs within the SoC. The graphics SoC interface **1532** can also implement power management controls for the graphics processor core block **1519** and enable an interface between a clock domain of the graphics processor core block **1519** and other clock domains within the SoC. In one embodiment the graphics SoC interface **1532** enables receipt of command buffers from a command streamer and global thread dispatcher that are configured to provide commands and instructions to each of one or more graphics cores within a graphics processor. The commands and instructions can be dispatched to the media pipeline **1534** when media operations are to be performed, the geometry and fixed function pipeline **1531** when graphics processing operations are to be performed. When compute operations are to be performed, compute dispatch logic can dispatch the commands to the graphics cores **1521A-1521F**, bypassing the geometry and media pipelines.

[0268] The graphics microcontroller **1533** can be configured to perform various scheduling and management tasks for the graphics processor core block **1519**. In one embodiment the graphics microcontroller **1533** can perform graphics and/or compute workload scheduling on the various vector engines **1522A-1522F**, **1524A-1524F** and matrix engines **1523A-1523F**, **1525A-1525F** within the graphics cores **1521A-1521F**. In this scheduling model, host software executing on a CPU core of an SoC including the graphics processor core block **1519** can submit workloads one of multiple graphics processor doorbells, which invokes a scheduling operation on the appropriate graphics engine. Scheduling operations include determining which workload to run next, submitting a workload to a command streamer, pre-empting existing workloads running on an engine, monitoring progress of a workload, and notifying host software when a workload is complete. In one embodiment the graphics microcontroller **1533** can also facilitate low-power or idle states for the graphics processor core block **1519**, providing the graphics processor core block **1519** with the ability to save and restore registers within the graphics processor core block **1519** across low-power state transitions independently from the operating system and/or graphics driver software on the system.

[0269] The graphics processor core block **1519** may have greater than or fewer than the illustrated graphics cores **1521A-1521F**, up to N modular graphics cores. For each set of N graphics cores, the graphics processor core block **1519** can also include shared/cache memory **1536**, which can be configured as shared memory or cache memory, rasterizer logic **1537**, and additional fixed function logic **1538** to accelerate various graphics and compute processing operations.

[0270] Within each graphics cores **1521A-1521F** is set of execution resources that may be used to perform graphics, media, and compute operations in response to requests by graphics pipeline, media pipeline, or shader programs. The graphics cores **1521A-1521F** include multiple vector engines **1522A-1522F**, **1524A-1524F**, matrix acceleration units **1523A-1523F**, **1525A-1525D**, cache/shared local memory (SLM), a sampler **1526A-1526F**, and a ray tracing unit **1527A-1527F**.

[0271] The vector engines **1522A-1522F**, **1524A-1524F** are general-purpose graphics processing units capable of performing floating-point and integer/fixed-point logic operations in service of a graphics, media, or compute operation, including graphics, media, or compute/GPGPU programs. The vector engines **1522A-1522F**, **1524A-1524F** can operate at variable vector widths using SIMD, SIMT, or SIMT+SIMD execution modes. The matrix acceleration units **1523A-1523F**, **1525A-1525D** include matrix-matrix and matrix-vector acceleration logic that improves performance on matrix operations, particularly low and mixed precision (e.g., INT8, FP16, BF16, FP8) matrix operations used for machine learning. In one embodiment, each of the matrix acceleration units **1523A-1523F**, **1525A-1525D** includes one or more systolic arrays of processing elements that can perform concurrent matrix multiply or dot product operations on matrix elements.

[0272] The sampler **1526A-1526F** can read media or texture data into memory and can sample data differently based on a configured sampler state and the texture/media format that is being read. Threads executing on the vector engines **1522A-1522F**, **1524A-1524F** or matrix acceleration units **1523A-1523F**, **1525A-1525D** can make use of the cache/SLM **1528A-1528F** within each of the graphics cores **1521A-1521F**. The cache/SLM **1528A-1528F** can be configured as cache memory or as a pool of shared memory that is local to each of the respective graphics cores **1521A-1521F**. The ray tracing units **1527A-1527F** within the graphics cores **1521A-1521F** include ray traversal/intersection circuitry for performing ray traversal using bounding volume hierarchies (BVHs) and identifying intersections between rays and primitives enclosed within the BVH volumes. In one embodiment the ray tracing units **1527A-1527F** include circuitry for performing depth testing and culling (e.g., using a depth buffer or similar arrangement). In one implementation, the ray tracing units **1527A-1527F** perform traversal and intersection operations in concert with image denoising, at least a portion of which may be performed using an associated matrix acceleration unit **1523A-1523F**, **1525A-1525D**.

[0273] FIG. 15C is a block diagram of general-purpose graphics processing unit (GPGPU) **1570** that can be configured as a graphics processor, e.g., the graphics processor **1508**, and/or compute accelerator, according to embodiments described herein. The GPGPU **1570** can interconnect with host processors (e.g., one or more CPU(s) **1546**) and

memory **1571**, **1572** via one or more system and/or memory busses. Memory **1571** may be system memory that can be shared with the one or more CPU(s) **1546**, while memory **1572** is device memory that is dedicated to the GPGPU **1570**. For example, components within the GPGPU **1570** and memory **1572** may be mapped into memory addresses that are accessible to the one or more CPU(s) **1546**. Access to memory **1571** and **1572** may be facilitated via a memory controller **1568**. The memory controller **1568** may include an internal direct memory access (DMA) controller **1569** or can include logic to perform operations that would otherwise be performed by a DMA controller.

[0274] The GPGPU **1570** includes multiple cache memories, including an L2 cache **1553**, L1 cache **1554**, an instruction cache **1555**, and shared memory **1556**, at least a portion of which may also be partitioned as a cache memory. The GPGPU **1570** also includes multiple compute units **1560A-1560N**. Each compute unit **1560A-1560N** includes a set of vector registers **1561**, scalar registers **1562**, vector logic units **1563**, and scalar logic units **1564**. The compute units **1560A-1560N** can also include local shared memory **1565** and a program counter **1566**. The compute units **1560A-1560N** can couple with a constant cache **1567**, which can be used to store constant data, which is data that will not change during the run of kernel or shader program that executes on the GPGPU **1570**. The constant cache **1567** may be a scalar data cache and cached data can be fetched directly into the scalar registers **1562**.

[0275] During operation, the one or more CPU(s) **1546** can write commands into registers or memory in the GPGPU **1570** that has been mapped into an accessible address space. The command processors **1557** can read the commands from registers or memory and determine how those commands will be processed within the GPGPU **1570**. A thread dispatcher **1558** can then be used to dispatch threads to the compute units **1560A-1560N** to perform those commands. Each compute unit **1560A-1560N** can execute threads independently of the other compute units. Additionally, each compute unit **1560A-1560N** can be independently configured for conditional computation and can conditionally output the results of computation to memory. The command processors **1557** can interrupt the one or more CPU(s) **1546** when the submitted commands are complete.

[0276] FIG. 16A-16C illustrate block diagrams of additional graphics processor and compute accelerator architectures provided by embodiments described herein, e.g., in accordance with FIG. 15A-15C. The elements of FIG. 16A-16C having the same or similar names as the elements of any other figure herein describe the same elements as in the other figures, can operate or function in a manner similar to that, can comprise the same components, and can be linked to other entities, as those described elsewhere herein, but are not limited to such.

[0277] FIG. 16A is a block diagram of a graphics processor **1600**, which may be a discrete graphics processing unit, or may be a graphics processor integrated with a plurality of processing cores, or other semiconductor devices such as, but not limited to, memory devices or network interfaces. The graphics processor **1600** may be a variant of the graphics processor **1508** and may be used in place of the graphics processor **1508**. Therefore, the disclosure of any features in combination with the graphics processor **1508** herein also discloses a corresponding combination with the graphics processor **1600** but is not limited to such. The

graphics processor may communicate via a memory mapped I/O interface to registers on the graphics processor and with commands placed into the processor memory. Graphics processor **1600** may include a memory interface **1614** to access memory. Memory interface **1614** can be an interface to local memory, one or more internal caches, one or more shared external caches, and/or to system memory.

[0278] Optionally, graphics processor **1600** also includes a display controller **1602** to drive display output data to a display device **1618**. Display controller **1602** includes hardware for one or more overlay planes for the display and composition of multiple layers of video or user interface elements. The display device **1618** can be an internal or external display device. In one embodiment the display device **1618** is a head mounted display device, such as a virtual reality (VR) display device or an augmented reality (AR) display device. Graphics processor **1600** may include a video codec engine **1606** to encode, decode, or transcode media to, from, or between one or more media encoding formats, including, but not limited to Moving Picture Experts Group (MPEG) formats such as MPEG-2, Advanced Video Coding (AVC) formats such as H.264/MPEG-4 AVC, H.265/HEVC, Alliance for Open Media (AOMedia) VP8, VP9, as well as the Society of Motion Picture & Television Engineers (SMPTE) 421M/VC-1, and Joint Photographic Experts Group (JPEG) formats such as JPEG, and Motion JPEG (MJPEG) formats.

[0279] Graphics processor **1600** may include a block image transfer (BLIT) engine **1603** to perform two-dimensional (2D) rasterizer operations including, for example, bit-boundary block transfers. However, alternatively, 2D graphics operations may be performed using one or more components of graphics processing engine (GPE **1610**). In some embodiments, GPE **1610** is a compute engine for performing graphics operations, including three-dimensional (3D) graphics operations and media operations.

[0280] GPE **1610** may include a 3D pipeline **1612** for performing 3D operations, such as rendering three-dimensional images and scenes using processing functions that act upon 3D primitive shapes (e.g., rectangle, triangle, etc.). The 3D pipeline **1612** includes programmable and fixed function elements that perform various tasks within the element and/or spawn execution threads to a 3D/Media subsystem **1615**. While 3D pipeline **1612** can be used to perform media operations, an embodiment of GPE **1610** also includes a media pipeline **1616** that is specifically used to perform media operations, such as video post-processing and image enhancement.

[0281] Media pipeline **1616** may include fixed function or programmable logic units to perform one or more specialized media operations, such as video decode acceleration, video de-interlacing, and video encode acceleration in place of, or on behalf of video codec engine **1606**. Media pipeline **1616** may additionally include a thread spawning unit to spawn threads for execution on 3D/Media subsystem **1615**. The spawned threads perform computations for the media operations on one or more graphics execution units included in 3D/Media subsystem **1615**.

[0282] The 3D/Media subsystem **1615** may include logic for executing threads spawned by 3D pipeline **1612** and media pipeline **1616**. The pipelines may send thread execution requests to 3D/Media subsystem **1615**, which includes thread dispatch logic for arbitrating and dispatching the various requests to available thread execution resources. The

execution resources include an array of graphics execution resources to process the 3D and media threads. The 3D/Media subsystem **1615** may include one or more internal caches for thread instructions and data. Additionally, the 3D/Media subsystem **1615** may also include shared memory, including registers and addressable memory, to share data between threads and to store output data.

[0283] FIG. **16B** illustrates a graphics processor **1620**, being a variant of the graphics processor **1600** and may be used in place of the graphics processor **1600** and vice versa. Therefore, the disclosure of any features in combination with the graphics processor **1600** herein also discloses a corresponding combination with the graphics processor **1620** but is not limited to such. The graphics processor **1620** has a tiled architecture, according to embodiments described herein. The graphics processor **1620** may include a graphics processing engine cluster **1622** having multiple instances of the GPE **1610** of FIG. **16A** within a graphics engine tile **1610A-1610D**. Each graphics engine tile **1610A-1610D** can be interconnected via a set of tile interconnects **1623A-1623F**. Each graphics engine tile **1610A-1610D** can also be connected to a memory module or memory device **1626A-1626D** via memory interconnects **1625A-1625D**. The memory devices **1626A-1626D** can use any graphics memory technology. For example, the memory devices **1626A-1626D** may be graphics double data rate (GDDR) memory. The memory devices **1626A-1626D** may be high-bandwidth memory (HBM) modules that can be on-die with their respective graphics engine tile **1610A-1610D**. The memory devices **1626A-1626D** may be stacked memory devices that can be stacked on top of their respective graphics engine tile **1610A-1610D**. Each graphics engine tile **1610A-1610D** and associated memory **1626A-1626D** may reside on separate chiplets, which are bonded to a base die or base substrate, as described in further detail in FIG. **24B-24D**.

[0284] The graphics processor **1620** may be configured with a non-uniform memory access (NUMA) system in which memory devices **1626A-1626D** are coupled with associated graphics engine tiles **1610A-1610D**. A given memory device may be accessed by graphics engine tiles other than the tile to which it is directly connected. However, access latency to the memory devices **1626A-1626D** may be lowest when accessing a local tile. In one embodiment, a cache coherent NUMA (ccNUMA) system is enabled that uses the tile interconnects **1623A-1623F** to enable communication between cache controllers within the graphics engine tiles **1610A-1610D** to keep a consistent memory image when more than one cache stores the same memory location.

[0285] The graphics processing engine cluster **1622** can connect with an on-chip or on-package fabric interconnect **1624**. In one embodiment the fabric interconnect **1624** includes a network processor, network on a chip (NoC), or another switching processor to enable the fabric interconnect **1624** to act as a packet switched fabric interconnect that switches data packets between components of the graphics processor **1620**. The fabric interconnect **1624** can enable communication between graphics engine tiles **1610A-1610D** and components such as the video codec engine **1606** and one or more copy engines **1604**. The copy engines **1604** can be used to move data out of, into, and between the memory devices **1626A-1626D** and memory that is external to the graphics processor **1620** (e.g., system memory). The

fabric interconnect **1624** can also be used to interconnect the graphics engine tiles **1610A-1610D**. The graphics processor **1620** may optionally include a display controller **1602** to enable a connection with an external display device **1618**. The graphics processor may also be configured as a graphics or compute accelerator. In the accelerator configuration, the display controller **1602** and display device **1618** may be omitted.

[0286] The graphics processor **1620** can connect to a host system via a host interface **1628**. The host interface **1628** can enable communication between the graphics processor **1620**, system memory, and/or other system components. The host interface **1628** can be, for example, a PCI express bus or another type of host system interface. For example, the host interface **1628** may be an NV Link or NV Switch interface. The host interface **1628** and fabric interconnect **1624** can cooperate to enable multiple instances of the graphics processor **1620** to act as single logical device. Cooperation between the host interface **1628** and fabric interconnect **1624** can also enable the individual graphics engine tiles **1610A-1610D** to be presented to the host system as distinct logical graphics devices.

[0287] FIG. **16C** illustrates a compute accelerator **1630**, according to embodiments described herein. The compute accelerator **1630** can include architectural similarities with the graphics processor **1620** of FIG. **16B** and is optimized for compute acceleration. A compute engine cluster **1632** can include a set of compute engine tiles **1640A-1640D** that include execution logic that is optimized for parallel or vector-based general-purpose compute operations. The compute engine tiles **1640A-1640D** may not include fixed function graphics processing logic, although in some embodiments one or more of the compute engine tiles **1640A-1640D** can include logic to perform media acceleration. The compute engine tiles **1640A-1640D** can connect to memory **1626A-1626D** via memory interconnects **1625A-1625D**. The memory **1626A-1626D** and memory interconnects **1625A-1625D** may be similar technology as in graphics processor **1620** or can be different. The graphics compute engine tiles **1640A-1640D** can also be interconnected via a set of tile interconnects **1623A-1623F** and may be connected with and/or interconnected by a fabric interconnect **1624**. In one embodiment the compute accelerator **1630** includes a large L3 cache **1636** that can be configured as a device-wide cache. The compute accelerator **1630** can also connect to a host processor and memory via a host interface **1628** in a similar manner as the graphics processor **1620** of FIG. **16B**.

[0288] The compute accelerator **1630** can also include an integrated network interface **1642**. In one embodiment the integrated network interface **1642** includes a network processor and controller logic that enables the compute engine cluster **1632** to communicate over a physical layer interconnect **1644** without requiring data to traverse memory of a host system. In one embodiment, one of the compute engine tiles **1640A-1640D** is replaced by network processor logic and data to be transmitted or received via the physical layer interconnect **1644** may be transmitted directly to or from memory **1626A-1626D**. Multiple instances of the compute accelerator **1630** may be joined via the physical layer interconnect **1644** into a single logical device. Alternatively, the various compute engine tiles **1640A-1640D** may be presented as distinct network accessible compute accelerator devices.

Graphics Processing Engine

[0289] FIG. 17 is a block diagram of a graphics processing engine of a graphics processor in accordance with some embodiments. The graphics processing engine (GPE 1710) may be a version of the GPE 1610 shown in FIG. 16A and may also represent a graphics engine tile 1610A-1610D of FIG. 16B. The elements of FIG. 17 having the same or similar names as the elements of any other figure herein describe the same elements as in the other figures, can operate or function in a manner similar to that, can comprise the same components, and can be linked to other entities, as those described elsewhere herein, but are not limited to such. For example, the 3D pipeline 1612 and media pipeline 1616 of FIG. 16A are also illustrated in FIG. 17. The media pipeline 1616 is optional in some embodiments of the GPE 1710 and may not be explicitly included within the GPE 1710. For example and in at least one embodiment, a separate media and/or image processor is coupled to the GPE 1710.

[0290] GPE 1710 may couple with or include a command streamer 1703, which provides a command stream to the 3D pipeline 1612 and/or media pipelines 1616. Alternatively or additionally, the command streamer 1703 may be directly coupled to a unified return buffer (URB 1718). The URB 1718 may be communicatively coupled to a graphics core cluster 1714. Optionally, the command streamer 1703 is coupled with memory, which can be system memory, or one or more of internal cache memory and shared cache memory. The command streamer 1703 may receive commands from the memory and sends the commands to 3D pipeline 1612 and/or media pipeline 1616. The commands are directives fetched from a ring buffer, which stores commands for the 3D pipeline 1612 and media pipeline 1616. The ring buffer can additionally include batch command buffers storing batches of multiple commands. The commands for the 3D pipeline 1612 can also include references to data stored in memory, such as but not limited to vertex and geometry data for the 3D pipeline 1612 and/or image data and memory objects for the media pipeline 1616. The 3D pipeline 1612 and media pipeline 1616 process the commands and data by performing operations via logic within the respective pipelines or by dispatching one or more execution threads to the graphics core cluster 1714. The graphics core cluster 1714 may include one or more blocks of graphics cores (e.g., graphics core block 1715A, graphics core block 1715B), each block including one or more graphics cores. Each graphics core includes a set of graphics execution resources that includes general-purpose and graphics specific execution logic to perform graphics and compute operations, as well as fixed function texture processing and/or machine learning and artificial intelligence acceleration logic.

[0291] In various embodiments the 3D pipeline 1612 can include fixed function and programmable logic to process one or more shader programs, such as vertex shaders, geometry shaders, pixel shaders, fragment shaders, compute shaders, or other shader programs, by processing the instructions and dispatching execution threads to the graphics core cluster 1714. The graphics core cluster 1714 provides a unified block of execution resources for use in processing these shader programs. Multi-purpose execution logic within the graphics core block 1715A-1715B of the graphics core cluster 1714 includes support for various 3D API

shader languages and can execute multiple simultaneous execution threads associated with multiple shaders.

[0292] The graphics core cluster 1714 may include execution logic to perform media functions, such as video and/or image processing. The execution resources may include general-purpose logic that is programmable to perform parallel general-purpose computational operations, in addition to graphics processing operations. The general-purpose logic can perform processing operations in parallel or in conjunction with general-purpose logic within the processor core(s) 1407 of FIG. 14 or core 1502A-1502N as in FIG. 15A.

[0293] Output data generated by threads executing on the graphics core cluster 1714 can output data to memory in the URB 1718. The URB 1718 can store data for multiple threads. The URB 1718 may be used to send data between different threads executing on the graphics core cluster 1714. The URB 1718 may additionally be used for synchronization between threads on the graphics core cluster 1714 and fixed function logic within the shared function logic 1720.

[0294] Optionally, the graphics core cluster 1714 may be scalable, such that the array includes a variable number of graphics cores, each having a variable number of execution resources based on the target power and performance level of GPE 1710. The execution resources may be dynamically scalable, such that execution resources may be enabled or disabled as needed.

[0295] The graphics core cluster 1714 couples with shared function logic 1720 that includes multiple resources that are shared between the graphics cores in the graphics core array. The shared functions within the shared function logic 1720 are hardware logic units that provide specialized supplemental functionality to the graphics core cluster 1714. In various embodiments, shared function logic 1720 includes but is not limited to sampler 1721, math 1722, and inter-thread communication (ITC) 1723 logic. Additionally, one or more cache(s) 1725 within the shared function logic 1720 may be implemented.

[0296] A shared function is implemented at least in a case where the demand for a given specialized function is insufficient for inclusion within the graphics core cluster 1714. Instead, a single instantiation of that specialized function is implemented as a stand-alone entity in the shared function logic 1720 and shared among the execution resources within the graphics core cluster 1714. The precise set of functions that are shared between the graphics core cluster 1714 and included within the graphics core cluster 1714 varies across embodiments. Specific shared functions within the shared function logic 1720 that are used extensively by the graphics core cluster 1714 may be included within shared function logic 1716 within the graphics core cluster 1714. Optionally, the shared function logic 1716 within the graphics core cluster 1714 can include some or all logic within the shared function logic 1720. All logic elements within the shared function logic 1720 may be duplicated within the shared function logic 1716 of the graphics core cluster 1714. Alternatively, the shared function logic 1720 is excluded in favor of the shared function logic 1716 within the graphics core cluster 1714.

Graphics Processing Resources

[0297] FIG. 18A-18C illustrate execution logic including an array of processing elements employed in a graphics processor, according to embodiments described herein. FIG.

18A illustrates graphics core cluster, according to an embodiment. **FIG. 18B** illustrates a vector engine of a graphics core, according to an embodiment. **FIG. 18C** illustrates a matrix engine of a graphics core, according to an embodiment. Elements of **FIG. 18A-18C** having the same reference numbers as the elements of any other figure herein may operate or function in any manner similar to that described elsewhere herein but are not limited as such. For example, the elements of **FIG. 18A-18C** can be considered in the context of the graphics processor core block **1519** of **FIG. 15B**, and/or the graphics core blocks **1715A-1715B** of **FIG. 17**. In one embodiment, the elements of **FIG. 18A-18C** have similar functionality to equivalent components of the graphics processor **1508** of **FIG. 15A** or the GPGPU **1570** of **FIG. 15C**.

[0298] As shown in **FIG. 18A**, in one embodiment the graphics core cluster **1714** includes a graphics core block **1715**, which may be graphics core block **1715A** or graphics core block **1715B** of **FIG. 17**. The graphics core block **1715** can include any number of graphics cores (e.g., graphics core **1815A**, graphics core **1815B**, through graphics core **1815N**). Multiple instances of the graphics core block **1715** may be included. In one embodiment the elements of the graphics cores **1815A-1815N** have similar or equivalent functionality as the elements of the graphics cores **1521A-1521F** of **FIG. 15B**. In such embodiment, the graphics cores **1815A-1815N** each include circuitry including but not limited to vector engines **1802A-1802N**, matrix engines **1803A-1803N**, memory load/store units **1804A-1804N**, instruction caches **1805A-1805N**, data caches/shared local memory **1806A-1806N**, ray tracing units **1808A-1808N**, samplers **1810A-1810N**. The circuitry of the graphics cores **1815A-1815N** can additionally include fixed function logic **1812A-1812N**. The number of vector engines **1802A-1802N** and matrix engines **1803A-1803N** within the graphics cores **1815A-1815N** of a design can vary based on the workload, performance, and power targets for the design.

[0299] With reference to graphics core **1815A**, the vector engine **1802A** and matrix engine **1803A** are configurable to perform parallel compute operations on data in a variety of integer and floating-point data formats based on instructions associated with shader programs. Each vector engine **1802A** and matrix engine **1803A** can act as a programmable general-purpose computational unit that is capable of executing multiple simultaneous hardware threads while processing multiple data elements in parallel for each thread. The vector engine **1802A** and matrix engine **1803A** support the processing of variable width vectors at various SIMD widths, including but not limited to SIMD8, SIMD16, and SIMD32. Input data elements can be stored as a packed data type in a register and the vector engine **1802A** and matrix engine **1803A** can process the various elements based on the data size of the elements. For example, when operating on a 256-bit wide vector, the 256 bits of the vector are stored in a register and the vector is processed as four separate 64-bit packed data elements (Quad-Word (QW) size data elements), eight separate 32-bit packed data elements (Double Word (DW) size data elements), sixteen separate 16-bit packed data elements (Word (W) size data elements), or thirty-two separate 8-bit data elements (byte (B) size data elements). However, different vector widths and register sizes are possible. In one embodiment, the vector engine

1802A and matrix engine **1803A** are also configurable for SIMT operation on warps or thread groups of various sizes (e.g., 8, 16, or 32 threads).

[0300] Continuing with graphics core **1815A**, the memory load/store unit **1804A** services memory access requests that are issued by the vector engine **1802A**, matrix engine **1803A**, and/or other components of the graphics core **1815A** that have access to memory. The memory access request can be processed by the memory load/store unit **1804A** to load or store the requested data to or from cache or memory into a register file associated with the vector engine **1802A** and/or matrix engine **1803A**. The memory load/store unit **1804A** can also perform prefetching operations. With additional reference to **FIG. 19**, in one embodiment, the memory load/store unit **1804A** is configured to provide SIMT scatter/gather prefetching or block prefetching for data stored in memory **1910**, from memory that is local to other tiles via the tile interconnect **1908**, or from system memory. Prefetching can be performed to a specific L1 cache (e.g., data cache/shared local memory **1806A**), the L2 cache **1904** or the L3 cache **1906**. In one embodiment, a prefetch to the L3 cache **1906** automatically results in the data being stored in the L2 cache **1904**.

[0301] The instruction cache **1805A** stores instructions to be executed by the graphics core **1815A**. In one embodiment, the graphics core **1815A** also includes instruction fetch and prefetch circuitry that fetches or prefetches instructions into the instruction cache **1805A**. The graphics core **1815A** also includes instruction decode logic to decode instructions within the instruction cache **1805A**. The data cache/shared local memory **1806A** can be configured as a data cache that is managed by a cache controller that implements a cache replacement policy and/or configured as explicitly managed shared memory. The ray tracing unit **1808A** includes circuitry to accelerate ray tracing operations. The sampler **1810A** provides texture sampling for 3D operations and media sampling for media operations. The fixed function logic **1812A** includes fixed function circuitry that is shared between the various instances of the vector engine **1802A** and matrix engine **1803A**. Graphics cores **1815B-1815N** can operate in a similar manner as graphics core **1815A**.

[0302] Functionality of the instruction caches **1805A-1805N**, data caches/shared local memory **1806A-1806N**, ray tracing units **1808A-1808N**, samplers **1810A-1810N**, and fixed function logic **1812A-1812N** corresponds with equivalent functionality in the graphics processor architectures described herein. For example, the instruction caches **1805A-1805N** can operate in a similar manner as instruction cache **1555** of **FIG. 15C**. The data caches/shared local memory **1806A-1806N**, ray tracing units **1808A-1808N**, and samplers **1810A-1810N** can operate in a similar manner as the cache/SLM **1528A-1528F**, ray tracing units **1527A-1527F**, and samplers **1526A-1526F** of **FIG. 15B**. The fixed function logic **1812A-1812N** can include elements of the geometry/fixed function pipeline **1531** and/or additional fixed function logic **1538** of **FIG. 15B**. In one embodiment, the ray tracing units **1808A-1808N** include circuitry to perform ray tracing acceleration operations performed by the ray tracing cores **372** of **FIG. 3C**.

[0303] As shown in **FIG. 18B**, in one embodiment the vector engine **1802** includes an instruction fetch unit **1837**, a general register file array (GRF) **1824**, an architectural register file array (ARF) **1826**, a thread arbiter **1822**, a send

unit **1830**, a branch unit **1832**, a set of SIMD floating point units (FPUs) **1834**, and in one embodiment a set of integer SIMD ALUs **1835**. The GRF **1824** and ARF **1826** includes the set of general register files and architecture register files associated with each hardware thread that may be active in the vector engine **1802**. In one embodiment, per thread architectural state is maintained in the ARF **1826**, while data used during thread execution is stored in the GRF **1824**. The execution state of each thread, including the instruction pointers for each thread, can be held in thread-specific registers in the ARF **1826**. Register renaming may be used to dynamically allocate registers to hardware threads.

[0304] In one embodiment the vector engine **1802** has an architecture that is a combination of Simultaneous Multi-Threading (SMT) and fine-grained Interleaved Multi-Threading (IMT). The architecture has a modular configuration that can be fine-tuned at design time based on a target number of simultaneous threads and number of registers per graphics core, where graphics core resources are divided across logic used to execute multiple simultaneous threads. The number of logical threads that may be executed by the vector engine **1802** is not limited to the number of hardware threads, and multiple logical threads can be assigned to each hardware thread.

[0305] In one embodiment, the vector engine **1802** can co-issue multiple instructions, which may each be different instructions. The thread arbiter **1822** can dispatch the instructions to one of the send unit **1830**, branch unit **1832**, or SIMD FPU(s) **1834** for execution. Each execution thread can access **128** general-purpose registers within the GRF **1824**, where each register can store 32 bytes, accessible as a variable width vector of 32-bit data elements. In one embodiment, each thread has access to 4 K bytes within the GRF **1824**, although embodiments are not so limited, and greater or fewer register resources may be provided in other embodiments. In one embodiment the vector engine **1802** is partitioned into seven hardware threads that can independently perform computational operations, although the number of threads per vector engine **1802** can also vary according to embodiments. For example, in one embodiment up to 16 hardware threads are supported. In an embodiment in which seven threads may access 4 K bytes, the GRF **1824** can store a total of 28 K bytes. Where 16 threads may access 4K bytes, the GRF **1824** can store a total of 64K bytes. Flexible addressing modes can permit registers to be addressed together to build effectively wider registers or to represent strided rectangular block data structures.

[0306] In one embodiment, memory operations, sampler operations, and other longer-latency system communications are dispatched via “send” instructions that are executed by the message passing send unit **1830**. In one embodiment, branch instructions are dispatched to a dedicated branch unit **1832** to facilitate SIMD divergence and eventual convergence.

[0307] In one embodiment the vector engine **1802** includes one or more SIMD floating point units (FPU(s)) **1834** to perform floating-point operations. In one embodiment, the FPU(s) **1834** also support integer computation. In one embodiment the FPU(s) **1834** can execute up to M number of 32-bit floating-point (or integer) operations, or execute up to 2M 16-bit integer or 16-bit floating-point operations. In one embodiment, at least one of the FPU(s) provides extended math capability to support high-throughput transcendental math functions and double precision

64-bit floating-point. In some embodiments, a set of 8-bit integer SIMD ALUs **1835** are also present and may be specifically optimized to perform operations associated with machine learning computations. In one embodiment, the SIMD ALUs are replaced by an additional set of SIMD FPUs **1834** that are configurable to perform integer and floating-point operations. In one embodiment, the SIMD FPUs **1834** and SIMD ALUs **1835** are configurable to execute SIMT programs. In one embodiment, combined SIMD+SIMT operation is supported.

[0308] In one embodiment, arrays of multiple instances of the vector engine **1802** can be instantiated in a graphics core. For scalability, product architects can choose the exact number of vector engines per graphics core grouping. In one embodiment the vector engine **1802** can execute instructions across a plurality of execution channels. In a further embodiment, each thread executed on the vector engine **1802** is executed on a different channel.

[0309] As shown in FIG. **18C**, in one embodiment the matrix engine **1803** includes an array of processing elements that are configured to perform tensor operations including vector/matrix and matrix/matrix operations, such as but not limited to matrix multiply and/or dot product operations. The matrix engine **1803** is configured with M rows and N columns of processing elements (PE **1852AA**-PE **1852MN**) that include multiplier and adder circuits organized in a pipelined fashion. In one embodiment, the processing elements **1852AA**-PE **1852MN** make up the physical pipeline stages of an N wide and M deep systolic array that can be used to perform vector/matrix or matrix/matrix operations in a data-parallel manner, including matrix multiply, fused multiply-add, dot product or other general matrix-matrix multiplication (GEMM) operations. In one embodiment the matrix engine **1803** supports 16-bit and 8-bit floating point operations, as well as 8-bit, 4-bit, 2-bit, and binary integer operations. The matrix engine **1803** can also be configured to accelerate specific machine learning operations. In such embodiments, the matrix engine **1803** can be configured with support for the bfloat (brain floating point) 16-bit floating point format or a tensor float 32-bit floating point format (TF32) that have different numbers of mantissa and exponent bits relative to Institute of Electrical and Electronics Engineers (IEEE) 754 formats.

[0310] In one embodiment, during each cycle, each stage can add the result of operations performed at that stage to the output of the previous stage. In other embodiments, the pattern of data movement between the processing elements **1852AA**-**1852MN** after a set of computational cycles can vary based on the instruction or macro-operation being performed. For example, in one embodiment partial sum loopback is enabled and the processing elements may instead add the output of a current cycle with output generated in the previous cycle. In one embodiment, the final stage of the systolic array can be configured with a loopback to the initial stage of the systolic array. In such embodiment, the number of physical pipeline stages may be decoupled from the number of logical pipeline stages that are supported by the matrix engine **1803**. For example, where the processing elements **1852AA**-**1852MN** are configured as a systolic array of M physical stages, a loopback from stage M to the initial pipeline stage can enable the processing elements **1852AA**-PE **552MN** to operate as a systolic array of, for example, 2M, 3M, 4M, etc., logical pipeline stages.

[0311] In one embodiment, the matrix engine 1803 includes memory 1841A-1841N, 1842A-1842M to store input data in the form of row and column data for input matrices. Memory 1842A-1842M is configurable to store row elements (A0-Am) of a first input matrix and memory 1841A-1841N is configurable to store column elements (B0-Bn) of a second input matrix. The row and column elements are provided as input to the processing elements 1852AA-1852MN for processing. In one embodiment, row and column elements of the input matrices can be stored in a systolic register file 1840 within the matrix engine 1803 before those elements are provided to the memory 1841A-1841N, 1842A-1842M. In one embodiment, the systolic register file 1840 is excluded and the memory 1841A-1841N, 1842A-1842M is loaded from registers in an associated vector engine (e.g., GRF 1824 of vector engine 1802 of FIG. 18B) or other memory of the graphics core that includes the matrix engine 1803 (e.g., data cache/shared local memory 1806A for matrix engine 1803A of FIG. 18A). Results generated by the processing elements 1852AA-1852MN are then output to an output buffer and/or written to a register file (e.g., systolic register file 1840, GRF 1824, data cache/shared local memory 1806A-1806N) for further processing by other functional units of the graphics processor or for output to memory.

[0312] In some embodiments, the matrix engine 1803 is configured with support for input sparsity, where multiplication operations for sparse regions of input data can be bypassed by skipping multiply operations that have a zero-value operand. In one embodiment, the processing elements 1852AA-1852MN are configured to skip the performance of certain operations that have zero value input. In one embodiment, sparsity within input matrices can be detected and operations having known zero output values can be bypassed before being submitted to the processing elements 1852AA-1852MN. The loading of zero value operands into the processing elements can be bypassed and the processing elements 1852AA-1852MN can be configured to perform multiplications on the non-zero value input elements. The matrix engine 1803 can also be configured with support for output sparsity, such that operations with results that are pre-determined to be zero are bypassed. For input sparsity and/or output sparsity, in one embodiment, metadata is provided to the processing elements 1852AA-1852MN to indicate, for a processing cycle, which processing elements and/or data channels are to be active during that cycle.

[0313] In one embodiment, the matrix engine 1803 includes hardware to enable operations on sparse data having a compressed representation of a sparse matrix that stores non-zero values and metadata that identifies the positions of the non-zero values within the matrix. Exemplary compressed representations include but are not limited to compressed tensor representations such as compressed sparse row (CSR), compressed sparse column (CSC), compressed sparse fiber (CSF) representations. Support for compressed representations enable operations to be performed on input in a compressed tensor format without requiring the compressed representation to be decompressed or decoded. In such embodiment, operations can be performed only on non-zero input values and the resulting non-zero output values can be mapped into an output matrix. In some embodiments, hardware support is also provided for machine-specific lossless data compression formats that are used when transmitting data within hardware or across

system busses. Such data may be retained in a compressed format for sparse input data and the matrix engine 1803 can use the compression metadata for the compressed data to enable operations to be performed on only non-zero values, or to enable blocks of zero data input to be bypassed for multiply operations.

[0314] In various embodiments, input data can be provided by a programmer in a compressed tensor representation, or a codec can compress input data into the compressed tensor representation or another sparse data encoding. In addition to support for compressed tensor representations, streaming compression of sparse input data can be performed before the data is provided to the processing elements 1852AA-1852MN. In one embodiment, compression is performed on data written to a cache memory associated with the graphics core cluster 1714, with the compression being performed with an encoding that is supported by the matrix engine 1803. In one embodiment, the matrix engine 1803 includes support for input having structured sparsity in which a pre-determined level or pattern of sparsity is imposed on input data. This data may be compressed to a known compression ratio, with the compressed data being processed by the processing elements 1852AA-1852MN according to metadata associated with the compressed data.

[0315] FIG. 19 illustrates a tile 1900 of a multi-tile processor, according to an embodiment. In one embodiment, the tile 1900 is representative of one of the graphics engine tiles 1610A-1610D of FIG. 16B or compute engine tiles 1640A-1640D of FIG. 16C. The tile 1900 of the multi-tile graphics processor includes an array of graphics core clusters (e.g., graphics core cluster 1714A, graphics core cluster 1714B, through graphics core cluster 1714N), with each graphics core cluster having an array of graphics cores 515A-515N. The tile 1900 also includes a global dispatcher 1902 to dispatch threads to processing resources of the tile 1900.

[0316] The tile 1900 can include or couple with an L3 cache 1906 and memory 1910. In various embodiments, the L3 cache 1906 may be excluded or the tile 1900 can include additional levels of cache, such as an L4 cache. In one embodiment, each instance of the tile 1900 in the multi-tile graphics processor has an associated memory 1910, such as in FIG. 16B and FIG. 16C. In one embodiment, a multi-tile processor can be configured as a multi-chip module in which the L3 cache 1906 and/or memory 1910 reside on separate chiplets than the graphics core clusters 1714A-1714N. In this context, a chiplet is an at least partially packaged integrated circuit that includes distinct units of logic that can be assembled with other chiplets into a larger package. For example, the L3 cache 1906 can be included in a dedicated cache chiplet or can reside on the same chiplet as the graphics core clusters 1714A-1714N. In one embodiment, the L3 cache 1906 can be included in an active base die or active interposer, as illustrated in FIG. 24C.

[0317] A memory fabric 1903 enables communication among the graphics core clusters 1714A-1714N, L3 cache 1906, and memory 1910. An L2 cache 1904 couples with the memory fabric 1903 and is configurable to cache transactions performed via the memory fabric 1903. A tile interconnect 1908 enables communication with other tiles on the graphics processors and may be one of tile interconnects 1623A-1623F of FIGS. 16B and 16C. In embodiments in which the L3 cache 1906 is excluded from the tile 1900, the L2 cache 1904 may be configured as a combined L2/L3 cache. The memory fabric 1903 is configurable to route data

to the L3 cache **1906** or memory controllers associated with the memory **1910** based on the presence or absence of the L3 cache **1906** in a specific implementation. The L3 cache **1906** can be configured as a per-tile cache that is dedicated to processing resources of the tile **1900** or may be a partition of a GPU-wide L3 cache.

[0318] FIG. **20** is a block diagram illustrating graphics processor instruction formats **2000**. The graphics processor execution resources support an instruction set having instructions in multiple formats. The solid lined boxes illustrate the components that are generally included in an execution unit instruction, while the dashed lines include components that are optional or that are only included in a sub-set of the instructions. In some embodiments the graphics processor instruction formats **2000** described and illustrated are macro-instructions, in that they are instructions supplied to the execution unit, as opposed to micro-operations resulting from instruction decode once the instruction is processed. Thus, a single instruction may cause hardware to perform multiple micro-operations

[0319] The graphics processor execution resources as described herein may natively support instructions in a 128-bit instruction format **2010**. A 64-bit compacted instruction format **2030** is available for some instructions based on the selected instruction, instruction options, and number of operands. The native 128-bit instruction format **2010** provides access to all instruction options, while some options and operations are restricted in the 64-bit compacted instruction format **2030**. The native instructions available in the 64-bit compacted instruction format **2030** vary by embodiment. The instruction is compacted in part using a set of index values in an index field **2013**. The execution unit hardware references a set of compaction tables based on the index values and uses the compaction table outputs to reconstruct a native instruction in the 128-bit instruction format **2010**. Other sizes and formats of instruction can be used.

[0320] For each format, instruction opcode **2012** defines the operation that the execution unit is to perform. The execution resources execute each instruction in parallel across the multiple data elements of each operand. For example, in response to an add instruction the execution unit performs a simultaneous add operation across each color channel representing a texture element or picture element. By default, the execution unit performs each instruction across all data channels of the operands. Instruction control field **2014** may enable control over certain execution options, such as channels selection (e.g., predication) and data channel order (e.g., swizzle). For instructions in the 128-bit instruction format **2010** an exec-size field **2016** limits the number of data channels that will be executed in parallel. An exec-size field **2016** may not be available for use in the 64-bit compacted instruction format **2030**.

[0321] Some execution unit instructions have up to three operands including two source operands, src0 **2020**, src1 **2022**, and one destination operand (dest **2018**). Other instructions, such as, for example, data manipulation instructions, dot product instructions, multiply-add instructions, or multiply-accumulate instructions, can have a third source operand (e.g., SRC2 **2024**). The instruction opcode **2012** determines the number of source operands. An instruction's last source operand can be an immediate (e.g., hard-coded) value passed with the instruction. The execution resources may also support multiple destination instructions,

where one or more of the destinations is implied or implicit based on the instruction and/or the specified destination.

[0322] The 128-bit instruction format **2010** may include an access/address mode field **2026** specifying, for example, whether direct register addressing mode or indirect register addressing mode is used. When direct register addressing mode is used, the register address of one or more operands is directly provided by bits in the instruction.

[0323] The 128-bit instruction format **2010** may also include an access/address mode field **2026**, which specifies an address mode and/or an access mode for the instruction. The access mode may be used to define a data access alignment for the instruction. Access modes including a 16-byte aligned access mode and a 1-byte aligned access mode may be supported, where the byte alignment of the access mode determines the access alignment of the instruction operands. For example, when in a first mode, the instruction may use byte-aligned addressing for source and destination operands and when in a second mode, the instruction may use 16-byte-aligned addressing for all source and destination operands.

[0324] The address mode portion of the access/address mode field **2026** may determine whether the instruction is to use direct or indirect addressing. When direct register addressing mode is used bits in the instruction directly provide the register address of one or more operands. When indirect register addressing mode is used, the register address of one or more operands may be computed based on an address register value and an address immediate field in the instruction.

[0325] Instructions may be grouped based on opcode **2012** bit-fields to simplify Opcode decode **2040**. For an 8-bit opcode, bits **4**, **5**, and **6** allow the execution unit to determine the type of opcode. The precise opcode grouping shown is merely an example. A move and logic opcode group **2042** may include data movement and logic instructions (e.g., move (mov), compare (cmp)). Move and logic group **2042** may share the five least significant bits (LSB), where move (mov) instructions are in the form of 0000xxxxb and logic instructions are in the form of 0001xxxxb. A flow control instruction group **2044** (e.g., call, jump (jmp)) includes instructions in the form of 0010xxxxb (e.g., 0x20). A miscellaneous instruction group **2046** includes a mix of instructions, including synchronization instructions (e.g., wait, send) in the form of 0011xxxxb (e.g., 0x30). A parallel math instruction group **2048** includes component-wise arithmetic instructions (e.g., add, multiply (mul)) in the form of 0100xxxxb (e.g., 0x40). The parallel math instruction group **2048** performs the arithmetic operations in parallel across data channels. The vector math group **2050** includes arithmetic instructions (e.g., dp4) in the form of 0101xxxxb (e.g., 0x50). The vector math group performs arithmetic such as dot product calculations on vector operands. The illustrated opcode decode **2040**, in one embodiment, can be used to determine which portion of an execution unit will be used to execute a decoded instruction. For example, some instructions may be designated as systolic instructions that will be performed by a systolic array. Other instructions, such as ray-tracing instructions (not shown) can be routed to a ray-tracing core or ray-tracing logic within a slice or partition of execution logic.

Graphics Pipeline

[0326] FIG. 21 is a block diagram of graphics processor 2100, according to another embodiment. The elements of FIG. 21 having the same or similar names as the elements of any other figure herein describe the same elements as in the other figures, can operate or function in a manner similar to that, can comprise the same components, and can be linked to other entities, as those described elsewhere herein, but are not limited to such.

[0327] The graphics processor 2100 may include different types of graphics processing pipelines, such as a geometry pipeline 2120, a media pipeline 2130, a display engine 2140, thread execution logic 2150, and a render output pipeline 2170. Graphics processor 2100 may be a graphics processor within a multi-core processing system that includes one or more general-purpose processing cores. The graphics processor may be controlled by register writes to one or more control registers (not shown) or via commands issued to graphics processor 2100 via a ring interconnect 2102. Ring interconnect 2102 may couple graphics processor 2100 to other processing components, such as other graphics processors or general-purpose processors. Commands from ring interconnect 2102 are interpreted by a command streamer 2103, which supplies instructions to individual components of the geometry pipeline 2120 or the media pipeline 2130.

[0328] Command streamer 2103 may direct the operation of a vertex fetcher 2105 that reads vertex data from memory and executes vertex-processing commands provided by command streamer 2103. The vertex fetcher 2105 may provide vertex data to a vertex shader 2107, which performs coordinate space transformation and lighting operations to each vertex. Vertex fetcher 2105 and vertex shader 2107 may execute vertex-processing instructions by dispatching execution threads to graphics cores 2152A-2152B via a thread dispatcher 2131.

[0329] The graphics cores 2152A-2152B may be an array of vector processors having an instruction set for performing graphics and media operations. The graphics cores 2152A-2152B may have an attached L1 cache 2151 that is specific for each array or shared between the arrays. The cache can be configured as a data cache, an instruction cache, or a single cache that is partitioned to contain data and instructions in different partitions.

[0330] A geometry pipeline 2120 may include tessellation components to perform hardware-accelerated tessellation of 3D objects. A programmable hull shader 2111 may configure the tessellation operations. A programmable domain shader 2117 may provide back-end evaluation of tessellation output. A tessellator 2113 may operate at the direction of hull shader 2111 and contain special purpose logic to generate a set of detailed geometric objects based on a coarse geometric model that is provided as input to geometry pipeline 2120. In addition, if tessellation is not used, tessellation components (e.g., hull shader 2111, tessellator 2113, and domain shader 2117) can be bypassed. The tessellation components can operate based on data received from the vertex shader 2107.

[0331] Complete geometric objects may be processed by a geometry shader 2119 via one or more threads dispatched to graphics cores 2152A-2152B, or can proceed directly to the clipper 2129. The geometry shader may operate on entire geometric objects, rather than vertices or patches of vertices as in previous stages of the graphics pipeline. If the tessellation is disabled, the geometry shader 2119 receives input

from the vertex shader 2107. The geometry shader 2119 may be programmable by a geometry shader program to perform geometry tessellation if the tessellation units are disabled.

[0332] Before rasterization, a clipper 2129 processes vertex data. The clipper 2129 may be a fixed function clipper or a programmable clipper having clipping and geometry shader functions. A rasterizer and depth test component 2173 in the render output pipeline 2170 may dispatch pixel shaders to convert the geometric objects into per pixel representations. The pixel shader logic may be included in thread execution logic 2150. Optionally, an application can bypass the rasterizer and depth test component 2173 and access un-rasterized vertex data via a stream out unit 2123.

[0333] The graphics processor 2100 has an interconnect bus, interconnect fabric, or some other interconnect mechanism that allows data and message passing amongst the major components of the processor. In some embodiments, graphics cores 2152A-2152B and associated logic units (e.g., L1 cache 2151, sampler 2154, texture cache 2158, etc.) interconnect via a data port 2156 to perform memory access and communicate with render output pipeline components of the processor. A sampler 2154, caches 2151, 2158 and graphics cores 2152A-2152B each may have separate memory access paths. Optionally, the texture cache 2158 can also be configured as a sampler cache.

[0334] The render output pipeline 2170 may contain a rasterizer and depth test component 2173 that converts vertex-based objects into an associated pixel-based representation. The rasterizer logic may include a windower/masker unit to perform fixed function triangle and line rasterization. An associated render cache 2178 and depth cache 2179 are also available in some embodiments. A pixel operations component 2177 performs pixel-based operations on the data, though in some instances, pixel operations associated with 2D operations (e.g., bit block image transfers with blending) are performed by the 2D engine 2141 or substituted at display time by the display controller 2143 using overlay display planes. A shared L3 cache 2175 may be available to all graphics components, allowing the sharing of data without the use of main system memory.

[0335] The media pipeline 2130 may include a media engine 2137 and a video front-end 2134. Video front-end 2134 may receive pipeline commands from the command streamer 2103. The media pipeline 2130 may include a separate command streamer. Video front-end 2134 may process media commands before sending the command to the media engine 2137. Media engine 2137 may include thread spawning functionality to spawn threads for dispatch to thread execution logic 2150 via thread dispatcher 2131.

[0336] The graphics processor 2100 may include a display engine 2140. This display engine 2140 may be external to processor 2100 and may couple with the graphics processor via the ring interconnect 2102, or some other interconnect bus or fabric. Display engine 2140 may include a 2D engine 2141 and a display controller 2143. Display engine 2140 may contain special purpose logic capable of operating independently of the 3D pipeline. Display controller 2143 may couple with a display device (not shown), which may be a system integrated display device, as in a laptop computer, or an external display device attached via a display device connector.

[0337] The geometry pipeline 2120 and media pipeline 2130 may be configurable to perform operations based on multiple graphics and media programming interfaces and are

not specific to any one application programming interface (API). A driver software for the graphics processor may translate API calls that are specific to a particular graphics or media library into commands that can be processed by the graphics processor. Support may be provided for the Open Graphics Library (OpenGL), Open Computing Language (OpenCL), and/or Vulkan graphics and compute API, all from the Khronos Group. Support may also be provided for the Direct3D library from the Microsoft Corporation. A combination of these libraries may be supported. Support may also be provided for the Open Source Computer Vision Library (OpenCV). A future API with a compatible 3D pipeline would also be supported if a mapping can be made from the pipeline of the future API to the pipeline of the graphics processor.

Graphics Pipeline Programming

[0338] FIG. 22A is a block diagram illustrating a graphics processor command format **2200** used for programming graphics processing pipelines, such as, for example, the pipelines described herein in conjunction with FIG. 16A, 17, 21. FIG. 22B is a block diagram illustrating a graphics processor command sequence **2210** according to an embodiment. The solid lined boxes in FIG. 22A illustrate the components that are generally included in a graphics command while the dashed lines include components that are optional or that are only included in a sub-set of the graphics commands. The exemplary graphics processor command format **2200** of FIG. 22A includes fields to identify a client **2202**, a command operation code (opcode) **2204**, and a data field **2206** for the command. A sub-opcode **2205** and a command size **2208** are also included in some commands.

[0339] Client **2202** may specify the client unit of the graphics device that processes the command data. A graphics processor command parser may examine the client field of each command to condition the further processing of the command and route the command data to the appropriate client unit. The graphics processor client units may include a memory interface unit, a render unit, a 2D unit, a 3D unit, and a media unit. Each client unit may have a corresponding processing pipeline that processes the commands. Once the command is received by the client unit, the client unit reads the opcode **2204** and, if present, sub-opcode **2205** to determine the operation to perform. The client unit performs the command using information in data field **2206**. For some commands an explicit command size **2208** is expected to specify the size of the command. The command parser may automatically determine the size of at least some of the commands based on the command opcode. Commands may be aligned via multiples of a double word. Other command formats can also be used.

[0340] The flow diagram in FIG. 22B illustrates an exemplary graphics processor command sequence **2210**. Software or firmware of a data processing system that features an exemplary graphics processor may use a version of the command sequence shown to set up, execute, and terminate a set of graphics operations. A sample command sequence is shown and described for purposes of example only and is not limited to these specific commands or to this command sequence. Moreover, the commands may be issued as batch of commands in a command sequence, such that the graphics processor will process the sequence of commands in at least partially concurrence.

[0341] The graphics processor command sequence **2210** may begin with a pipeline flush command **2212** to cause any active graphics pipeline to complete the currently pending commands for the pipeline. Optionally, the 3D pipeline **2222** and the media pipeline **2224** may not operate concurrently. The pipeline flush is performed to cause the active graphics pipeline to complete any pending commands. In response to a pipeline flush, the command parser for the graphics processor will pause command processing until the active drawing engines complete pending operations and the relevant read caches are invalidated. Optionally, any data in the render cache that is marked 'dirty' can be flushed to memory. Pipeline flush command **2212** can be used for pipeline synchronization or before placing the graphics processor into a low power state.

[0342] A pipeline select command **2213** may be used when a command sequence requires the graphics processor to explicitly switch between pipelines. A pipeline select command **2213** may be required only once within an execution context before issuing pipeline commands unless the context is to issue commands for both pipelines. A pipeline flush command **2212** may be required immediately before a pipeline switch via the pipeline select command **2213**.

[0343] A pipeline control command **2214** may configure a graphics pipeline for operation and may be used to program the 3D pipeline **2222** and the media pipeline **2224**. The pipeline control command **2214** may configure the pipeline state for the active pipeline. The pipeline control command **2214** may be used for pipeline synchronization and to clear data from one or more cache memories within the active pipeline before processing a batch of commands.

[0344] Commands related to the return buffer state **2216** may be used to configure a set of return buffers for the respective pipelines to write data. Some pipeline operations require the allocation, selection, or configuration of one or more return buffers into which the operations write intermediate data during processing. The graphics processor may also use one or more return buffers to store output data and to perform cross thread communication. The return buffer state **2216** may include selecting the size and number of return buffers to use for a set of pipeline operations.

[0345] The remaining commands in the command sequence differ based on the active pipeline for operations. Based on a pipeline determination **2220**, the command sequence is tailored to the 3D pipeline **2222** beginning with the 3D pipeline state **2230** or the media pipeline **2224** beginning at the media pipeline state **2240**.

[0346] The commands to configure the 3D pipeline state **2230** include 3D state setting commands for vertex buffer state, vertex element state, constant color state, depth buffer state, and other state variables that are to be configured before 3D primitive commands are processed. The values of these commands are determined at least in part based on the particular 3D API in use. The 3D pipeline state **2230** commands may also be able to selectively disable or bypass certain pipeline elements if those elements will not be used.

[0347] A 3D primitive **2232** command may be used to submit 3D primitives to be processed by the 3D pipeline. Commands and associated parameters that are passed to the graphics processor via the 3D primitive **2232** command are forwarded to the vertex fetch function in the graphics pipeline. The vertex fetch function uses the 3D primitive **2232** command data to generate vertex data structures. The vertex data structures are stored in one or more return

buffers. The 3D primitive **2232** command may be used to perform vertex operations on 3D primitives via vertex shaders. To process vertex shaders, 3D pipeline **2222** dispatches shader execution threads to graphics processor execution resources.

[0348] The 3D pipeline **2222** may be triggered via an execute **2234** command or event. A register may write trigger command executions. An execution may be triggered via a ‘go’ or ‘kick’ command in the command sequence. Command execution may be triggered using a pipeline synchronization command to flush the command sequence through the graphics pipeline. The 3D pipeline will perform geometry processing for the 3D primitives. Once operations are complete, the resulting geometric objects are rasterized and the pixel engine colors the resulting pixels. Additional commands to control pixel shading and pixel back-end operations may also be included for those operations.

[0349] The graphics processor command sequence **2210** may follow the media pipeline **2224** path when performing media operations. In general, the specific use and manner of programming for the media pipeline **2224** depends on the media or compute operations to be performed. Specific media decode operations may be offloaded to the media pipeline during media decode. The media pipeline can also be bypassed and media decode can be performed in whole or in part using resources provided by one or more general-purpose processing cores. The media pipeline may also include elements for general-purpose graphics processor unit (GPGPU) operations, where the graphics processor is used to perform SIMD vector operations using computational shader programs that are not explicitly related to the rendering of graphics primitives.

[0350] Media pipeline **2224** may be configured in a similar manner as the 3D pipeline **2222**. A set of commands to configure the media pipeline state **2240** are dispatched or placed into a command queue before the media object commands **2242**. Commands for the media pipeline state **2240** may include data to configure the media pipeline elements that will be used to process the media objects. This includes data to configure the video decode and video encode logic within the media pipeline, such as encode or decode format. Commands for the media pipeline state **2240** may also support the use of one or more pointers to “indirect” state elements that contain a batch of state settings.

[0351] Media object commands **2242** may supply pointers to media objects for processing by the media pipeline. The media objects include memory buffers containing video data to be processed. Optionally, all media pipeline states must be valid before issuing a media object command **2242**. Once the pipeline state is configured and media object commands **2242** are queued, the media pipeline **2224** is triggered via an execute command **2244** or an equivalent execute event (e.g., register write). Output from media pipeline **2224** may then be post processed by operations provided by the 3D pipeline **2222** or the media pipeline **2224**. GPGPU operations may be configured and executed in a similar manner as media operations.

Graphics Software Architecture

[0352] FIG. 23 illustrates an exemplary graphics software architecture for a data processing system **2300**. Such a software architecture may include a 3D graphics application **2310**, an operating system **2320**, and at least one processor

2330. Processor **2330** may include a graphics processor **2332** and one or more general-purpose processor core(s) **2334**. The processor **2330** may be a variant of the processor **1402** or any other of the processors described herein. The processor **2330** may be used in place of the processor **1402** or any other of the processors described herein. Therefore, the disclosure of any features in combination with the processor **1402** or any other of the processors described herein also discloses a corresponding combination with the graphics processor **2332** but is not limited to such. Moreover, the elements of FIG. 23 having the same or similar names as the elements of any other figure herein describe the same elements as in the other figures, can operate or function in a manner similar to that, can comprise the same components, and can be linked to other entities, as those described elsewhere herein, but are not limited to such. The graphics application **2310** and operating system **2320** are each executed in the system memory **2350** of the data processing system.

[0353] 3D graphics application **2310** may contain one or more shader programs including shader instructions **2312**. The shader language instructions may be in a high-level shader language, such as the High-Level Shader Language (HLSL) of Direct3D, the OpenGL Shader Language (GLSL), and so forth. The application may also include executable instructions **2314** in a machine language suitable for execution by the general-purpose processor core **2334**. The application may also include graphics objects **2316** defined by vertex data.

[0354] The operating system **2320** may be a Microsoft® Windows® operating system from the Microsoft Corporation, a proprietary UNIX-like operating system, or an open source UNIX-like operating system using a variant of the Linux kernel. The operating system **2320** can support a graphics API **2322** such as the Direct3D API, the OpenGL API, or the Vulkan API. When the Direct3D API is in use, the operating system **2320** uses a front-end shader compiler **2324** to compile any shader instructions **2312** in HLSL into a lower-level shader language. The compilation may be a just-in-time (JIT) compilation or the application can perform shader pre-compilation. High-level shaders may be compiled into low-level shaders during the compilation of the 3D graphics application **2310**. The shader instructions **2312** may be provided in an intermediate form, such as a version of the Standard Portable Intermediate Representation (SPIR) used by the Vulkan API.

[0355] User mode graphics driver **2326** may contain a back-end shader compiler **2327** to convert the shader instructions **2312** into a hardware specific representation. When the OpenGL API is in use, shader instructions **2312** in the GLSL high-level language are passed to a user mode graphics driver **2326** for compilation. The user mode graphics driver **2326** may use operating system kernel mode functions **2328** to communicate with a kernel mode graphics driver **2329**. The kernel mode graphics driver **2329** may communicate with graphics processor **2332** to dispatch commands and instructions.

IP Core Implementations

[0356] One or more aspects may be implemented by representative code stored on a machine-readable medium which represents and/or defines logic within an integrated circuit such as a processor. For example, the machine-readable medium may include instructions which represent

various logic within the processor. When read by a machine, the instructions may cause the machine to fabricate the logic to perform the techniques described herein. Such representations, known as “IP cores,” are reusable units of logic for an integrated circuit that may be stored on a tangible, machine-readable medium as a hardware model that describes the structure of the integrated circuit. The hardware model may be supplied to various customers or manufacturing facilities, which load the hardware model on fabrication machines that manufacture the integrated circuit. The integrated circuit may be fabricated such that the circuit performs operations described in association with any of the embodiments described herein.

[0357] FIG. 24A is a block diagram illustrating an IP core development system 2400 that may be used to manufacture an integrated circuit to perform operations according to an embodiment. The IP core development system 2400 may be used to generate modular, re-usable designs that can be incorporated into a larger design or used to construct an entire integrated circuit (e.g., an SOC integrated circuit). A design facility 2430 can generate a software simulation 2410 of an IP core design in a high-level programming language (e.g., C/C++). The software simulation 2410 can be used to design, test, and verify the behavior of the IP core using a simulation model 2412. The simulation model 2412 may include functional, behavioral, and/or timing simulations. A register transfer level (RTL) design 2415 can then be created or synthesized from the simulation model 2412. The RTL design 2415 is an abstraction of the behavior of the integrated circuit that models the flow of digital signals between hardware registers, including the associated logic performed using the modeled digital signals. In addition to an RTL design 2415, lower-level designs at the logic level or transistor level may also be created, designed, or synthesized. Thus, the particular details of the initial design and simulation may vary.

[0358] The RTL design 2415 or equivalent may be further synthesized by the design facility into a hardware model 2420, which may be in a hardware description language (HDL), or some other representation of physical design data. The HDL may be further simulated or tested to verify the IP core design. The IP core design can be stored for delivery to a 3rd party fabrication facility 2465 using non-volatile memory 2440 (e.g., hard disk, flash memory, or any non-volatile storage medium). Alternatively, the IP core design may be transmitted (e.g., via the Internet) over a wired connection 2450 or wireless connection 2460. The fabrication facility 2465 may then fabricate an integrated circuit that is based at least in part on the IP core design. The fabricated integrated circuit can be configured to perform operations in accordance with at least one embodiment described herein.

[0359] FIG. 24B illustrates a cross-section side view of an integrated circuit package assembly 2470. The integrated circuit package assembly 2470 illustrates an implementation of one or more processor or accelerator devices as described herein. The package assembly 2470 includes multiple units of hardware logic 2472, 2474 connected to a substrate 2480. The logic 2472, 2474 may be implemented at least partly in configurable logic or fixed-functionality logic hardware and can include one or more portions of any of the processor core(s), graphics processor(s), or other accelerator devices described herein. Each unit of logic 2472, 2474 can be implemented within a semiconductor die and coupled with

the substrate 2480 via an interconnect structure 2473. The interconnect structure 2473 may be configured to route electrical signals between the logic 2472, 2474 and the substrate 2480, and can include interconnects such as, but not limited to bumps or pillars. The interconnect structure 2473 may be configured to route electrical signals such as, for example, input/output (I/O) signals and/or power or ground signals associated with the operation of the logic 2472, 2474. Optionally, the substrate 2480 may be an epoxy-based laminate substrate. The substrate 2480 may also include other suitable types of substrates. The package assembly 2470 can be connected to other electrical devices via a package interconnect 2483. The package interconnect 2483 may be coupled to a surface of the substrate 2480 to route electrical signals to other electrical devices, such as a motherboard, other chipset, or multi-chip module.

[0360] The units of logic 2472, 2474 may be electrically coupled with a bridge 2482 that is configured to route electrical signals between the logic 2472, 2474. The bridge 2482 may be a dense interconnect structure that provides a route for electrical signals. The bridge 2482 may include a bridge substrate composed of glass or a suitable semiconductor material. Electrical routing features can be formed on the bridge substrate to provide a chip-to-chip connection between the logic 2472, 2474.

[0361] Although two units of logic 2472, 2474 and a bridge 2482 are illustrated, embodiments described herein may include more or fewer logic units on one or more dies. The one or more dies may be connected by zero or more bridges, as the bridge 2482 may be excluded when the logic is included on a single die. Alternatively, multiple dies or units of logic can be connected by one or more bridges. Additionally, multiple logic units, dies, and bridges can be connected together in other possible configurations, including three-dimensional configurations.

[0362] FIG. 24C illustrates a package assembly 2490 that includes multiple units of hardware logic chiplets connected to a substrate 2480 (e.g., base die). A graphics processing unit, parallel processor, and/or compute accelerator as described herein can be composed from diverse silicon chiplets that are separately manufactured. In this context, a chiplet is an at least partially packaged integrated circuit that includes distinct units of logic that can be assembled with other chiplets into a larger package. A diverse set of chiplets with different IP core logic can be assembled into a single device. Additionally, the chiplets can be integrated into a base die or base chiplet using active interposer technology. The concepts described herein enable the interconnection and communication between the different forms of IP within the GPU. IP cores can be manufactured using different process technologies and composed during manufacturing, which avoids the complexity of converging multiple IPs, especially on a large SoC with several flavors IPs, to the same manufacturing process. Enabling the use of multiple process technologies improves the time to market and provides a cost-effective way to create multiple product SKUs. Additionally, the disaggregated IPs are more amenable to being power gated independently, components that are not in use on a given workload can be powered off, reducing overall power consumption.

[0363] In various embodiments a package assembly 2490 can include fewer or greater number of components and chiplets that are interconnected by a fabric 2485 or one or more bridges 2487. The chiplets within the package assem-

bly **2490** may have a 2.5D arrangement using Chip-on-Wafer-on-Substrate stacking in which multiple dies are stacked side-by-side on a silicon interposer that includes through-silicon vias (TSVs) to couple the chiplets with the substrate **2480**, which includes electrical connections to the package interconnect **2483**.

[0364] In one embodiment, silicon interposer is an active interposer **2489** that includes embedded logic in addition to TSVs. In such embodiment, the chiplets within the package assembly **2490** are arranged using 3D face to face die stacking on top of the active interposer **2489**. The active interposer **2489** can include hardware logic for I/O **2491**, cache memory **2492**, and other hardware logic **2493**, in addition to interconnect fabric **2485** and a silicon bridge **2487**. The fabric **2485** enables communication between the various logic chiplets **2472**, **2474** and the logic **2491**, **2493** within the active interposer **2489**. The fabric **2485** may be an NoC interconnect or another form of packet switched fabric that switches data packets between components of the package assembly. For complex assemblies, the fabric **2485** may be a dedicated chiplet enables communication between the various hardware logic of the package assembly **2490**.

[0365] Bridge structures **2487** within the active interposer **2489** may be used to facilitate a point-to-point interconnect between, for example, logic or I/O chiplets **2474** and memory chiplets **2475**. In some implementations, bridge structures **2487** may also be embedded within the substrate **2480**.

[0366] The hardware logic chiplets can include special purpose hardware logic chiplets **2472**, logic or I/O chiplets **2474**, and/or memory chiplets **2475**. The hardware logic chiplets **2472** and logic or I/O chiplets **2474** may be implemented at least partly in configurable logic or fixed-functionality logic hardware and can include one or more portions of any of the processor core(s), graphics processor(s), parallel processors, or other accelerator devices described herein. The memory chiplets **2475** can be DRAM (e.g., GDDR, HBM) memory or cache (SRAM) memory. Cache memory **2492** within the active interposer **2489** (or substrate **2480**) can act as a global cache for the package assembly **2490**, part of a distributed global cache, or as a dedicated cache for the fabric **2485**.

[0367] Each chiplet can be fabricated as separate semiconductor die and coupled with a base die that is embedded within or coupled with the substrate **2480**. The coupling with the substrate **2480** can be performed via an interconnect structure **2473**. The interconnect structure **2473** may be configured to route electrical signals between the various chiplets and logic within the substrate **2480**. The interconnect structure **2473** can include interconnects such as, but not limited to bumps or pillars. In some embodiments, the interconnect structure **2473** may be configured to route electrical signals such as, for example, input/output (I/O) signals and/or power or ground signals associated with the operation of the logic, I/O and memory chiplets. In one embodiment, an additional interconnect structure couples the active interposer **2489** with the substrate **2480**.

[0368] The substrate **2480** may be an epoxy-based laminate substrate, however, it is not limited to that and the substrate **2480** may also include other suitable types of substrates. The package assembly **2490** can be connected to other electrical devices via a package interconnect **2483**. The package interconnect **2483** may be coupled to a surface of

the substrate **2480** to route electrical signals to other electrical devices, such as a motherboard, other chipset, or multi-chip module.

[0369] A logic or I/O chiplet **2474** and a memory chiplet **2475** may be electrically coupled via a bridge **2487** that is configured to route electrical signals between the logic or I/O chiplet **2474** and a memory chiplet **2475**. The bridge **2487** may be a dense interconnect structure that provides a route for electrical signals. The bridge **2487** may include a bridge substrate composed of glass or a suitable semiconductor material. Electrical routing features can be formed on the bridge substrate to provide a chip-to-chip connection between the logic or I/O chiplet **2474** and a memory chiplet **2475**. The bridge **2487** may also be referred to as a silicon bridge or an interconnect bridge. For example, the bridge **2487** is an Embedded Multi-die Interconnect Bridge (EMIB). Alternatively, the bridge **2487** may simply be a direct connection from one chiplet to another chiplet.

[0370] FIG. 24D illustrates a package assembly **2494** including interchangeable chiplets **2495**, according to an embodiment. The interchangeable chiplets **2495** can be assembled into standardized slots on one or more base chiplets **2496**, **2498**. The base chiplets **2496**, **2498** can be coupled via a bridge interconnect **2497**, which can be similar to the other bridge interconnects described herein and may be, for example, an EMIB. Memory chiplets can also be connected to logic or I/O chiplets via a bridge interconnect. I/O and logic chiplets can communicate via an interconnect fabric. The base chiplets can each support one or more slots in a standardized format for one of logic or I/O or memory/cache.

[0371] SRAM and power delivery circuits may be fabricated into one or more of the base chiplets **2496**, **2498**, which can be fabricated using a different process technology relative to the interchangeable chiplets **2495** that are stacked on top of the base chiplets. For example, the base chiplets **2496**, **2498** can be fabricated using a larger process technology, while the interchangeable chiplets can be manufactured using a smaller process technology. One or more of the interchangeable chiplets **2495** may be memory (e.g., DRAM) chiplets. Different memory densities can be selected for the package assembly **2494** based on the power, and/or performance targeted for the product that uses the package assembly **2494**. Additionally, logic chiplets with a different number of type of functional units can be selected at time of assembly based on the power, and/or performance targeted for the product. Additionally, chiplets containing IP logic cores of differing types can be inserted into the interchangeable chiplet slots, enabling hybrid processor designs that can mix and match different technology IP blocks.

Exemplary System on a Chip Integrated Circuit

[0372] FIG. 25-26B illustrate exemplary integrated circuits and associated graphics processors that may be fabricated using one or more IP cores. In addition to what is illustrated, other logic and circuits may be included, including additional graphics processors/cores, peripheral interface controllers, or general-purpose processor cores. The elements of FIG. 25-26B having the same or similar names as the elements of any other figure herein describe the same elements as in the other figures, can operate or function in a manner similar to that, can comprise the same components,

and can be linked to other entities, as those described elsewhere herein, but are not limited to such.

[0373] FIG. 25 is a block diagram illustrating an exemplary system on a chip integrated circuit 2500 that may be fabricated using one or more IP cores. Exemplary integrated circuit 2500 includes one or more application processor(s) 2505 (e.g., CPUs), at least one graphics processor 2510, which may be a variant of the graphics processor 1408, 1508, 2510, or of any graphics processor described herein and may be used in place of any graphics processor described. Therefore, the disclosure of any features in combination with a graphics processor herein also discloses a corresponding combination with the graphics processor 2510 but is not limited to such. The integrated circuit 2500 may additionally include an image processor 2515 and/or a video processor 2520, any of which may be a modular IP core from the same or multiple different design facilities. Integrated circuit 2500 may include peripheral or bus logic including a USB controller 2525, UART controller 2530, an SPI/SDIO controller 2535, and an I²S/I²C controller 2540. Additionally, the integrated circuit can include a display device 2545 coupled to one or more of a high-definition multimedia interface (HDMI) controller 2550 and a mobile industry processor interface (MIPI) display interface 2555. Storage may be provided by a flash memory subsystem 2560 including flash memory and a flash memory controller. Memory interface may be provided via a memory controller 2565 for access to SDRAM or SRAM memory devices. Some integrated circuits additionally include an embedded security engine 2570.

[0374] FIG. 26A-26B are block diagrams illustrating exemplary graphics processors for use within an SoC, according to embodiments described herein. The graphics processors may be variants of the graphics processor 1408, 1508, 2510, or any other graphics processor described herein. The graphics processors may be used in place of the graphics processor 1408, 1508, 2510, or any other of the graphics processors described herein. Therefore, the disclosure of any features in combination with the graphics processor 1408, 1508, 2510, or any other of the graphics processors described herein also discloses a corresponding combination with the graphics processors of FIG. 26A-26B but is not limited to such. FIG. 26A illustrates an exemplary graphics processor 2610 of a system on a chip integrated circuit that may be fabricated using one or more IP cores, according to an embodiment. FIG. 26B illustrates an additional exemplary graphics processor 2640 of a system on a chip integrated circuit that may be fabricated using one or more IP cores, according to an embodiment. Graphics processor 2610 of FIG. 26A is an example of a low power graphics processor core. Graphics processor 2640 of FIG. 26B is an example of a higher performance graphics processor core. For example, each of graphics processor 2610 and graphics processor 2640 can be a variant of the graphics processor 2510 of FIG. 25, as mentioned at the outset of this paragraph.

[0375] As shown in FIG. 26A, graphics processor 2610 includes a vertex processor 2605 and one or more fragment processor(s) 2615A-2615N (e.g., 2615A, 2615B, 2615C, 2615D, through 2615N-1, and 2615N). Graphics processor 2610 can execute different shader programs via separate logic, such that the vertex processor 2605 is optimized to execute operations for vertex shader programs, while the one or more fragment processor(s) 2615A-2615N execute frag-

ment (e.g., pixel) shading operations for fragment or pixel shader programs. The vertex processor 2605 performs the vertex processing stage of the 3D graphics pipeline and generates primitives and vertex data. The fragment processor(s) 2615A-2615N use the primitive and vertex data generated by the vertex processor 2605 to produce a frame-buffer that is displayed on a display device. The fragment processor(s) 2615A-2615N may be optimized to execute fragment shader programs as provided for in the OpenGL API, which may be used to perform similar operations as a pixel shader program as provided for in the Direct 3D API.

[0376] Graphics processor 2610 additionally includes one or more memory management units (MMUs) 2620A-2620B, cache(s) 2625A-2625B, and circuit interconnect(s) 2630A-2630B. The one or more MMU(s) 2620A-2620B provide for virtual to physical address mapping for the graphics processor 2610, including for the vertex processor 2605 and/or fragment processor(s) 2615A-2615N, which may reference vertex or image/texture data stored in memory, in addition to vertex or image/texture data stored in the one or more cache(s) 2625A-2625B. The one or more MMU(s) 2620A-2620B may be synchronized with other MMUs within the system, including one or more MMUs associated with the one or more application processor(s) 2505, image processor 2515, and/or video processor 2520 of FIG. 25, such that each processor 2505-2520 can participate in a shared or unified virtual memory system. Components of graphics processor 2610 may correspond with components of other graphics processors described herein. The one or more MMU(s) 2620A-2620B may correspond with MMU 245 of FIG. 2C. Vertex processor 2605 and fragment processor 2615A-2615N may correspond with graphics multiprocessor 234. The one or more circuit interconnect(s) 2630A-2630B enable graphics processor 2610 to interface with other IP cores within the SoC, either via an internal bus of the SoC or via a direct connection, according to embodiments. The one or more circuit interconnect(s) 2630A-2630B may correspond with the data crossbar 240 of FIG. 2C. Further correspondence may be found between analogous components of the graphics processor 2610 and the various graphics processor architectures described herein.

[0377] As shown FIG. 26B, graphics processor 2640 includes the one or more MMU(s) 2620A-2620B, cache(s) 2625A-2625B, and circuit interconnect(s) 2630A-2630B of the graphics processor 2610 of FIG. 26A. Graphics processor 2640 includes one or more shader cores 2655A-2655N (e.g., 2655A, 2655B, 2655C, 2655D, 2655E, 2655F, through 2655N-1, and 2655N), which provides for a unified shader core architecture in which a single core or type or core can execute all types of programmable shader code, including shader program code to implement vertex shaders, fragment shaders, and/or compute shaders. The exact number of shader cores present can vary among embodiments and implementations. Additionally, graphics processor 2640 includes an inter-core task manager 2645, which acts as a thread dispatcher to dispatch execution threads to one or more shader cores 2655A-2655N and a tiling unit 2658 to accelerate tiling operations for tile-based rendering, in which rendering operations for a scene are subdivided in image space, for example to exploit local spatial coherence within a scene or to optimize use of internal caches. Shader cores 2655A-2655N may correspond with, for example, graphics multiprocessor 234 as in FIG. 2D, or graphics

multiprocessors 325, 350 of FIGS. 3A and 3B respectively, or multi-core group 365A of FIG. 3C.

Tensor Acceleration Logic for Graphics and Machine Learning Workloads

[0378] FIG. 27 is a block diagram of a data processing system 2700, according to an embodiment. The data processing system 2700 is a heterogeneous processing system having a processor 2702, unified memory 2710, and a GPGPU 2720 including machine learning acceleration logic. The processor 2702 and the GPGPU 2720 can be any of the processors and GPGPU/parallel processors as described herein. For example, with additional reference to FIG. 1, processor 2702 can be a variant of and/or share an architecture with a processor of the illustrated one or more processor(s) 102 and the GPGPU 2720 can be a variant of and/or share an architecture with a parallel processor of the illustrated one or more parallel processor(s) 112. With additional reference to FIG. 14, processor 2702 can be a variant of and/or share an architecture with one of the illustrated processor(s) 1402 and the GPGPU 2720 can be a variant of and/or share an architecture with one of the illustrated graphics processor(s) 1408.

[0379] The processor 2702 can execute instructions for a compiler 2715 stored in system memory 2712. The compiler 2715 executes on the processor 2702 to compile source code 2714A into compiled code 2714B. The compiled code 2714B can include instructions that may be executed by the processor 2702 and/or instructions that may be executed by the GPGPU 2720. Compilation of instructions to be executed by the GPGPU can be facilitated using shader or compute program compilers, such as a back-end shader compiler 2327 and/or front-end shader compiler 2324 as in FIG. 23. During compilation, the compiler 2715 can perform operations to insert metadata, including hints as to the level of data parallelism present in the compiled code 2714B and/or hints regarding the data locality associated with threads to be dispatched based on the compiled code 2714B. The compiler 2715 can include the information necessary to perform such operations or the operations can be performed with the assistance of a runtime library 2716. The runtime library 2716 can also assist the compiler 2715 in the compilation of the source code 2714A and can also include instructions that are linked at runtime with the compiled code 2714B to facilitate execution of the compiled instructions on the GPGPU 2720. The compiler 2715 can also facilitate register allocation for variables via a register allocator (RA) and generate load and store instructions to move data for variables between memory and the register assigned for the variable.

[0380] The unified memory 2710 represents a unified address space that may be accessed by the processor 2702 and the GPGPU 2720. The unified memory can include system memory 2712 as well as GPGPU memory 2718. The GPGPU memory 2718 is memory within an address space of the GPGPU 2720 and can include some or all of system memory 2712. In one embodiment, compiled code 2714B stored in system memory 2712 can be mapped into GPGPU memory 2718 for access by the GPGPU 2720. The GPGPU memory 2718 also includes GPGPU local memory 2728 of the GPGPU 2720. The GPGPU local memory 2728 can include, for example, HBM or GDDR memory.

[0381] The GPGPU 2720 includes multiple compute blocks 2724A-2724N, which can include one or more of a

variety of processing resources described herein. The processing resources can be or include a variety of different computational resources such as, for example, execution units, compute units, streaming multiprocessors, graphics multiprocessors, or multi-core groups. In one embodiment the GPGPU 2720 additionally includes a tensor accelerator 2723 (e.g., matrix accelerator), which can include one or more special function compute units that are designed to accelerate a subset of matrix operations (e.g., dot product, etc.). The tensor accelerator 2723 may also be referred to as a tensor accelerator or tensor core. In one embodiment, logic components within the tensor accelerator 2723 may be distributed across the processing resources of the multiple compute blocks 2724A-2724N.

[0382] The GPGPU 2720 can also include a set of resources that can be shared by the compute blocks 2724A-2724N and the tensor accelerator 2723, including but not limited to a set of registers 2725, a power and performance module 2726, and a cache 2727. In one embodiment the registers 2725 include directly and indirectly accessible registers, where the indirectly accessible registers are optimized for use by the tensor accelerator 2723. The power and performance module 2726 can be configured to adjust power delivery and clock frequencies for the compute blocks 2724A-2724N to power gate idle components within the compute blocks 2724A-2724N. In various embodiments the cache 2727 can include an instruction cache and/or a lower-level data cache.

[0383] The GPGPU 2720 can additionally include an L3 data cache 2730, which can be used to cache data accessed from the unified memory 2710 by the tensor accelerator 2723 and/or the compute elements within the compute blocks 2724A-2724N. In one embodiment the L3 data cache 2730 includes shared local memory 2732 that can be shared by the compute elements within the compute blocks 2724A-2724N and the tensor accelerator 2723.

[0384] In one embodiment the GPGPU 2720 includes instruction handling logic, such as a fetch and decode unit 2721 and a scheduler controller 2722. The fetch and decode unit 2721 includes a fetch unit and decode unit to fetch and decode instructions for execution by one or more of the compute blocks 2724A-2724N or the tensor accelerator 2723. The instructions can be scheduled to the appropriate functional unit within the compute block 2724A-2724N or the tensor accelerator via the scheduler controller 2722. In one embodiment the scheduler controller 2722 is an A SIC configurable to perform advanced scheduling operations. In one embodiment the scheduler controller 2722 is a micro-controller or a low energy-per-instruction processing core capable of executing scheduler instructions loaded from a firmware module.

[0385] In one embodiment some functions to be performed by the compute blocks 2724A-2724N can be directly scheduled to or offloaded to the tensor accelerator 2723. In various embodiments the tensor accelerator 2723 includes processing element logic configured to efficiently perform matrix compute operations, such as multiply and add operations and dot product operations used by 3D graphics or compute shader programs. In one embodiment the tensor accelerator 2723 can be configured to accelerate operations used by machine learning frameworks. In one embodiment the tensor accelerator 2723 is an application specific integrated circuit explicitly configured to perform a specific set of parallel matrix multiplication and/or addition operations.

In one embodiment the tensor accelerator **2723** is a field programmable gate array (FPGA) that provides fixed function logic that can be updated between workloads. In one embodiment, the set of compute operations that can be performed by the tensor accelerator **2723** may be limited relative to the operations that can be performed by the compute block **2724A-2724N**. However, the tensor accelerator **2723** can perform parallel tensor operations at a significantly higher throughput relative to the compute block **2724A-2724N**.

[0386] FIG. **28A-28B** illustrate a matrix operation **2805** performed by an instruction pipeline **2800**, according to embodiments. FIG. **28A** illustrates the instruction pipeline **2800** when configured with a systolic array **2808** within the tensor accelerator **2723**. FIG. **28B** illustrates the instruction pipeline when configured with graphics processor cores **2810A-2810N** that each include matrix engines **2812A-2812N**.

[0387] As shown in FIG. **28A**, the instruction pipeline **2800** can be configured to perform a matrix operation **2805**, such as, but not limited to a dot product operation. The dot product of two vectors is a scalar value that is equal to sum of products of corresponding components of the vectors. The dot product can be calculated as shown in equation (1) below.

$$\vec{a} \cdot \vec{b} = \sum_{i=1}^n a_i b_i = a_1 b_1 + \dots + a_n b_n \quad (1)$$

[0388] The dot product can be used in a convolution operation for a convolutional neural network (CNN). While 2D convolution is illustrated, N-dimensional convolution can be performed on an N-dimensional volume using N-dimensional filters. A receptive field tile **2802** highlights a portion of an input volume in an input volume buffer **2804**. The input volume buffer can be stored in memory **2830**. A dot product matrix operation **2805** can be performed between the data within the receptive field tile **2802** and a convolutional filter to generate a data point within output buffer **2806**, which can also be stored in memory **2830**. The memory **2830** can be any of the memory described herein, including system memory **2712**, GPGPU memory **2718**, cache **2727**, or L3 data cache **2730** as in FIG. **27**.

[0389] The combination of the data points within the output buffer **2806** represents an activation map generated by the convolution operation. Each point within the activation map is generated by sliding the receptive field tile across the input volume buffer **2804**. The activation map data can be input to an activation function to determine an output activation value. In one embodiment, convolution of the input volume buffer **2804** can be defined within a framework as high-level matrix operation **2805**. The high-level matrix operations can be performed via primitive operations, such as a basic linear algebra subprogram (BLAS) operation. The primitive operations can be accelerated via hardware instructions executed by the instruction pipeline **2800**.

[0390] The instruction pipeline **2800** used to accelerate hardware instructions can include the instruction fetch and decode unit **2721**, which can fetch and decode hardware instructions, and the scheduler controller **2722** which can schedule decoded instructions to one or more processing resources within the compute blocks **2724A-2724N** and/or

the tensor accelerator **2723**. In one embodiment, a hardware instruction can be scheduled to the compute blocks **2724A-2724N** and offloaded to the tensor accelerator **2723**. The one or more hardware instructions and associated data to perform the matrix operation **2805** can be stored in the memory **2830**. Output of the hardware instruction can also be stored in the memory **2830**.

[0391] In one embodiment, the tensor accelerator **2723** can execute one or more hardware instructions to perform the matrix operation **2805** using a systolic array **2808** of processing elements. The systolic array **2808** includes a combination of programmable and fixed function hardware that is configurable to perform matrix-matrix and matrix-vector dot product operations, as well as other operations, such as matrix-matrix and matrix-vector fused multiply-add operations.

[0392] In various embodiment, as an alternative or in addition to the tensor accelerator **2723**, matrix acceleration logic can also be included within the processing resources of the compute blocks **2724A-2724N**. For example, as shown in FIG. **28B**, in one embodiment each compute block (e.g., compute block **2724N**) includes an array of graphics cores **2810A-2810N**. Each graphics core in the array of graphics core **2810A-2810N** can include a matrix accelerator **2812A-2812N**. In one embodiment, the graphics cores **2810A-2810N** are graphics core **1815A-1815N** as in FIG. **18A** and the matrix accelerators **2812A-2812N** include a version of the matrix engine **1803** of FIG. **18C**. The scheduler controller **2722** can schedule matrix operations (dot products, fused multiply-adds, etc.) to available matrix accelerator **2812A-2812N** within the graphics cores **2810A-2810N** of the various compute blocks **2724A-2724N**.

[0393] While in one embodiment each of the compute blocks **2724A-2724N** include an array of graphics cores **2810A-2810N**, in another embodiment the compute blocks **2724A-2724N** share an architecture with the processing clusters **214A-214N** of the processing cluster array in FIG. **2A**. In such embodiment, the compute blocks **2724A-2724N** include multiple graphics multiprocessors **234** as in FIG. **2C**, which include internal components as illustrated in FIG. **2D**. Thus, the graphics multiprocessors within the compute blocks can include a load/store unit **266**, GPGPU cores **262**, and tensor/RT cores **263**. In one embodiment the compute blocks **2724A-2724N** can include multi-core group **365A-365N** of the GPU **380** of FIG. **3C** and include multiple sets of GFX cores **370**, tensor cores **371**, and ray tracing cores **372**. In such embodiment, the scheduler controller **2722** can schedule instructions to perform matrix operations to the tensor/RT cores **263** and/or tensor cores **371** within the compute blocks **2724A-2724N**. Accelerated matrix operations include dot product operations, matrix multiply operations, and/or fused multiply-add operations, which can be performed on integer or floating-point matrix elements and various levels of precision. Additionally, in one embodiment the compute blocks **2724A-2724N** can include a variant of the compute units **1560A-1560N** of FIG. **15C**, where such variants include matrix acceleration logic as described herein (e.g., systolic array, tensor core, systolic tensor core) that can execute integer or floating-point matrix acceleration instructions.

[0394] FIG. **29** illustrates a systolic array **2900** including multiplier and adder circuits organized in a pipelined fashion. In one embodiment, systolic array **2900** is representative of the physical pipeline stages included in the accelera-

tors **2812A-2812N** and includes capabilities described in relation to that matrix engine **1803** of FIG. **18C**, including support for sparse and block sparse operations, and may additionally be configured to support structured sparsity within a vector of elements or across a set of channels. Inputs **2912A-2912H** for the first input matrix are represented by the data elements contained in the inputs labeled Src1 and Src1+1 through Src1+7. Inputs **2910A-2910H** correspond to the second input matrix and are labeled as Src2. Inputs **2902A-2902B**, which may include initial accumulator values, can be provided as Src0. An array of processing elements makes up the physical pipeline stages **2911A-2911H** of the systolic array **2900**. Matrix-Matrix or Matrix-Vector operations, including fused multiply-add and/or dot product operations, can be performed at each pipeline stage **2911A-2911H** during each clock cycle. On each cycle, every pipeline stage can receive a new Src2 input can be used by the processing elements of the pipeline stage to compute a value using either the new Src1 input or an older Src1 input that was previously read, although during initial startup it may take several cycles before all of the pipeline stages **2911A-2911H** become active as the initial set of computed values propagate through the stages.

[0395] Input **2902A** can provide a Src0 value to processing element of pipeline stage **2911A**, for use as an initial accumulator value. Alternatively, input **2902B** can provide the Src0 value to be added to the values computed by pipeline stage **2911H** of the systolic array, which enables partial pass operation for systolic array **2900** using the lower stages of the array while the unused upper stages are power gated. During operation, the data elements of a selected channel of the Src2 input are broadcast across all channels of the processing elements of the pipeline stages **2911A-2911H**, where each channel represents a vector of multiple elements. The number of elements per channel can vary based on the size of the elements. The processing elements of a stage then perform operations using the selected Src2 channel and all channels of a given Src1 input. A Src2 input operates with eight Src1 inputs (e.g., one Src1 input per stage). The data elements of a channel of the Src2 input are broadcast across all channels of processing elements **2911A-2911H**. The processing elements then operate the Src2 channel with all channels of a Src1 input. In a first clock cycle, a Src1 input is operated with data elements of the first channel of Src2. In the next cycle, a second Src1 (labeled as Src1+1) operates with the data elements of the second channel of Src2. This sequence repeats on the eight stages of the pipeline. Each stage adds its operation to the output of the previous stage. A cross the pipeline stages, multiple Src2 inputs are operated in a pipelined fashion. As successive channels of a first Src2 input are pushed through the pipeline stages, a new Src2 input can be provided at the first stage.

[0396] Output **2922** from the final stage is labeled as Dst. Where d=the systolic depth and e=the number of data elements per channel, the output of a channel is described by equation (2) below:

$$Dst_i = Src0_i + \sum_{j=0}^d \sum_{k=0}^e (Src1 + j)_{\text{element } k \text{ of channel } i} * Src2_{\text{element } k \text{ of channel } j} \quad (2)$$

[0397] As shown in equation (2), each channel can include multiple data elements on which operations are performed in parallel. In one embodiment, each channel represents a four element data vector, although a different number of elements can be configured for each channel. In one embodiment, the number of data elements within a channel can vary based on the size of each data element. Dot products can be performed using, for example, four element vectors with 8-bit data types per element, two element vectors with 16-bit data types, eight element vectors with 4-bit data types (e.g., INT4), or 16 element vectors with 2-bit data types (e.g., INT2). The number of channels can be automatically adjusted depending on the datatype of Src1 and Src2. An instruction can also specify a required systolic depth to be used for the instruction.

[0398] In one embodiment the processing elements **2911A-2911H** may read inputs **2910A-2910H**, **2912A-2912H** directly from the general-purpose register file. In one embodiment systolic array **2900** includes logic to read inputs **2910A-2910H**, **2912A-2912H** from the general-purpose register file and store input data in registers, buffers, or memory that is internal to the systolic array. Internal logic can then feed the input data elements to the processing elements **2911A-2911H** for processing. Output **2922** can be written to internal registers or memory of the systolic array **2900** and/or written directly to the general-purpose register file.

[0399] FIG. **30A-30B** illustrates the use of a systolic array **3000** that can be configured to execute operations at an arbitrary systolic depth. In the illustrated example, the systolic array **3000** has a physical depth of four, which corresponds with four physical pipeline stages. The systolic array can be configured to operate using an arbitrary number of logical stages, including four, eight, twelve, or sixteen logical stages, or other numbers of logical stages that are not divisible by the number of physical stages using partial-pass operations as in FIG. **31** described below. FIG. **30A** shows the array receiving Src0 inputs from an external source and processing the first four stages with Src1 and Src2 inputs. The output of this array is fed back into the second step shown in FIG. **30B**. FIG. **30B** shows that the next four stages are calculated using the loopback data that includes the already processed values and the Src1 and Src2 inputs.

[0400] As shown in FIG. **30A**, systolic array **3000** can accept input **2902**, as Src0 input, which is read (**3002**) via data selector **3004**. Data selector **3004** selects between the input **2902** and loopback input **3006**. Processing elements **2911A-2911D** can process inputs **2910A-2910D** and **2912A-2912D** in a similar manner as systolic array **2900**. If four stages are sufficient to complete an operation, pipeline stage **2911D** can write (**3022**) output **2922** to a specified Dst register or memory via data selector **3024**. Where further stages are required, data selector **3024** can write loopback output **3026**, which is provided as loopback input **3006** to processing elements of pipeline stage **2911A**.

[0401] As shown in FIG. **30B**, in one embodiment, loopback input **3006** can be further processed by processing elements **2911A-2911D**. Loopback input **3006** includes the already processed values. In one embodiment, loopback input **3006** can also include input **2910E-2910H**, input **2912E-2912H**, which can be pre-fetched while processing the first four stages. Data selector **3004** select loopback input **3006** for input by pipeline stage **2911A**. Processing elements of the pipeline stages **2911A-2911D** can then process inputs

2910E-2910H and **2912E-2912H**. Data selector **3024** can then write (**3022**) the eighth stage result as output **2922** to the specified Dst register.

[0402] In one embodiment, the systolic array **3000** is modified to exclude the loopback output **3026** and loopback input **3006** and instead include intermediate storage **3025**, as shown in FIG. **30A-30B**. The intermediate storage **3025** may be a memory device or register that is internal to the systolic array **3000** or may be a register in a register file that is external to the systolic array **3000**. During the operations shown in FIG. **30A**, output from pipeline stage **2911D** can be stored in the intermediate storage **3025** instead of being output by loopback output **3026** and read by loopback input **3006** before the operations shown in FIG. **30B**. During the operations shown in FIG. **30B**, output from pipeline stage **2911D** can be added to the data stored in the intermediate storage **3025** and written to output **2922**. The systolic array **3000** can also be configured to perform multi-pass operations using at least one partial pass, as described below, to enable logical depths that are not divisible by the physical depth of the array.

Scalable Matrix Multiply Accelerator with Feedback Inputs

[0403] A second embodiment enables increased throughput using simultaneous instructions executed using parallel units. Several instances or paths of the multiply accelerator are run in parallel. These instances can share Src1, or they can have independent Src1 inputs. Each path will have their own Src2 and Src0 inputs. These instances will have their own src2 and src0 inputs. A version showing two paths with a depth of four stages is shown in FIG. **31**. Alternatively, a version using four paths of depth of two stages is shown in FIG. **32**.

[0404] FIG. **31** illustrates a two-path matrix multiply accelerator **3100** in which each path has a depth of four stages. The two-path matrix multiply accelerator **3100** includes input logic **3102A-3102B** for Src0 inputs, input buffers **3111A-3111B** to store data elements received from input logic **3110A-3110B**, and input buffers **3113A-3113B** to store data elements received from shared input logic **3112** for Src1. Each stage includes a pair of processing elements, which may operate in parallel. Stage one includes processing elements **3131A-3131B**, stage two includes processing elements **3132A-3132B**, stage three includes processing elements **3133A-3133B**, stage four includes processing elements **3134A-3134B**. Hardware logic of each of the processing elements **3131A-3131B**, **3132A-3132B**, **3131A-3133B**, **3134A-3134B** can be the same as or similar to the hardware logic of processing elements of systolic array **2900** or systolic array **3000** and may be manufactured with the same process technology or a more advanced process technology. The processing elements of the two-path matrix multiply accelerator **3100** may also operate at a higher frequency relative to implementations of systolic array **2900**. The processing elements and may be manufactured using more advanced process technology.

[0405] Feedback may be implemented using data selectors that are the same as or similar to data selectors **3004**, **3024**. Depending on the configuration of the read logic, input data can be pre-fetched into the input buffer in advance or read from registers or a cache within the two-path matrix multiply accelerator **3100** one or more cycles before input into the processing elements **3131A-3131B**. Processing elements **3134A-3134B** of stage four can feed back into the corresponding processing elements **3131A-3131B** stage one.

Dynamic logical depth may be enabled in multiples of four. After a configured number of logical stages, results may be written by output logic **3122A-3122B** to a specified destination.

[0406] FIG. **32** illustrates a four-path matrix multiply accelerator **3200** in which each path has a depth of two stages. Four-path matrix multiply accelerator **3200** includes the same number of processing elements as two-path matrix multiply accelerator **3100**, with the processing elements configured with twice as many paths, but each path is half as deep. Four-path matrix multiply accelerator **3200** includes input logic **3202A-3202D** for Src0, input buffers **3211A-3211D** to store input elements read by input logic **3210A-3210D** for Src2, and input buffers **3213A-3213D** to store input elements read by shared input logic **3212** for Src1. Processing elements **3231A-3231B** enable parallel processing for stage 1. Processing elements **3232A-3232B** enable parallel processing for stage 2. Stage 2 of each path can feed back into stage 1 or write results via output logic **3222A-3222D** to a specified destination. Processing elements **3231A-3231B**, **3232A-3232B** may include hardware logic similar to that of processing elements **3131A-3131B**, **3132A-3132B**, **3131A-3133B**, **3134A-3134B** and can implement loopback functionality using similar hardware logic.

[0407] The advantages of a two-path matrix multiply accelerator **3100** or a four-path matrix multiply accelerator **3200** include scalability, software compatibility, and throughput. The modular architecture of these accelerators enables more efficient scaling relative to an 8-deep systolic array. Different configurations of a matrix multiply accelerator can be tailored for different product needs or use cases without redesign. Additionally, the same software model that is used is independent of the hardware implementation. Algorithms designed for an instruction intended to be executed by a systolic pipeline of eight stages can be used in an implementation using a Matrix Multiply accelerator of four stages. Hardware will use feedback to simulate a pipeline of eight stages in a way that is transparent to the software. Multiple paths can be used in a design requiring high DPAS instruction throughput. Implementations with a greater number of paths can be coupled with higher bandwidth input logic and output logic. In one embodiment, the two-path matrix multiply accelerator **3100** and a four-path matrix multiply accelerator **3200** are configured to bypass inputs with block sparsity at a greater efficiency and/or finer granularity than possible with an 8-deep systolic array.

Sparse Multiplications on the Scalable Matrix Multiply Accelerator

[0408] A third embodiment facilitates increased instruction throughput when processing for data with irregular sparsity. Elements of Src1 and Src2 inputs can be individually selected via input multiplexer logic and processing can be performed using only non-zero values.

[0409] FIG. **33** illustrates a scalable sparse matrix multiply accelerator **3300** using systolic arrays with feedback inputs. Scalable sparse matrix multiply accelerator **3300** can include processing elements **3231A-3231D** as in four-path matrix multiply accelerator **3200**, or any other processing elements described herein. Processing elements **3231A-3221B** at the beginning of each path include input logic for Src0. Each stage of each path of scalable sparse matrix multiply accelerator **3300** can receive any element of an independent or

shared Src1 via input selectors 3312A-3312D. Each stage of each path can also receive any element of a Src2. Independent Src2 inputs are provided via separate input element selectors (e.g., Src2A via input selector 3310A and input selector 3311A, Src2B via input selector 3310B and input selector 3311B). The separate Src2 input enables the separate paths to compute different instructions. Separate output logic 3322A-3322B is present for each path to enable output for the different instructions.

[0410] FIG. 34 shows a scalable sparse matrix multiply accelerator 3400 using systolic arrays with feedback inputs and outputs on each stage. Scalable sparse matrix multiply accelerator 3400 includes similar hardware logic as scalable sparse matrix multiply accelerator 3300, along with additional input and output logic to enable Src0 elements to be provided to each stage of each path and to provide separate outputs for each stage of each path. In addition to input selectors 3310A and 3311A to select Src2A elements for the first path and input selectors 3310A and 3311B to select Src2B input for the second path, an input splitter 3403A-3403B is added for each path for Src0 input. Each input splitter 340A-3402B can include a demultiplexer or similar hardware logic to enable Src0 input elements that are read by input logic 3402A-3402B to be sent to each stage. Input selectors 3312A-3312D are also included to enable Src1 input to be elected by each stage of each path. In addition to output logic 3322A-3322B from the second stage of each path (processing element 3431C-3431D), additional output logic 3422A-3422B is provided to enable output from the first stage of each path (3431A-3431B). The processing elements 3431A-3431C may be otherwise similar to other processing elements described herein.

[0411] During operation, scalable sparse matrix multiply accelerator 3400 is configurable to accept groups of only one element. Given Src2 input {B0, 0, B2, B3, 0, 0, 0, 0}, two groups ([B0,B2], [B3,0]) are made for the non-zero elements on Src2 for the third embodiment (e.g., scalable sparse matrix multiply accelerator 3300), with the second group including a zero padding. The optimizations shown in FIG. 34 enable the groups to be formed as [B0,B2], [B3]. B0 and B2 will be assigned to the first and second stage of a path (e.g., either of a first set including of processing element 3431A and processing element 3431C or a second set including processing element 3431B and processing element 3431D). After the feedback, B3 will be assigned to the first stage of that path. As the first stage of a path can provide output (e.g., via either output logic 3422A or 3422B), there is no need to consume the second stage of the path (either of processing element 3431C or processing element 3431D). Moreover, the next Src2 input accepted for that path can start from the second stage, so a group of two elements will be assigned to the second and first stage respectively. Src0 for processing the new Src2 input can be assigned to the second stage of the path (e.g., via either output logic 3422A or 3422B).

[0412] In addition to the hardware logic of scalable sparse matrix multiply accelerator 3300 illustrated in FIG. 33 and scalable sparse matrix multiply accelerator 3400 illustrated FIG. 34, some embodiments additionally include input and output hardware memory buffers. Input memory buffers can be used to store and have ready groups of Src0 and Src2 inputs, which reduces the need for high bandwidth input logic. The output buffer allows Dst outputs generated in a

same cycle to be steadily written to memory at a slower rate, which reduces the need for high bandwidth output logic.

[0413] Additionally, some embodiments include a bypass for inputs in which all elements are zero. The bypass allows a direct write of Src0 as by output logic without passing through the systolic array. This bypass is used in concert with a data dependency strategy to prevent read-after-write (RAW) risks among instructions can damage the integrity of the data.

Matrix Accelerator Having a Dual Pipeline Parallel Systolic Array

[0414] FIG. 35 illustrates a dual pipeline parallel systolic array 3500 for a matrix accelerator, according to an embodiment. A matrix accelerator as described herein (e.g., tensor accelerator 2723, tensor/RT cores 263, tensor cores 371, matrix engine 1803) can include a dual pipeline parallel systolic array 3500 that includes two systolic array pipelines (systolic pipeline 3502, systolic pipeline 3504) that operate in parallel to execute instructions. The dual pipeline parallel systolic array 3500 enables the row data that is provided as Src2 input to be partitioned, with the partitions being processed in parallel using a common Src1 input. Such configuration enables increased throughput for matrix operations without incurring the power and area costs associated with two separate and fully independent systolic arrays.

[0415] Input for matrix operations can be read from a register file (e.g., register file(s) 258, 334A-334B, 369, vector registers 1561, GRF 1821, etc.) that is associated with the matrix accelerator. The dual pipeline parallel systolic array 3500 includes an input 3521 for a Src1 operand that is shared between the two systolic array pipelines. The Src1 input inputs column data that is used by the two systolic array pipelines to perform matrix multiply operations in which two sets of matrix row data (Src2 input 3522A-3522B) are multiplied by a single set of column data. A single Src2 register can store input for two stages of operation. For example, data from inputs 3522A-3522B can be read in 64-bit blocks, with the lower 32-bits being used for operations at a stage of the systolic array and the upper 32-bits being used for operations at the next successive stage of the systolic array. As one Src2 read can be used for two operations on an array, the second cycle of a pair of Src2 read cycles can be used to read a new Src2 for the second array. The common input 3521 for Src1 data and the use of Src2 register data for multiple operations reduces read demand on the GRF relative to two fully independent systolic arrays. The reduce register read demand relative to the use of independent systolic arrays can reduce the potential negative impact on performance caused for other processing elements that share the register file with the systolic array when those processing elements are operating concurrently with the systolic arrays.

[0416] Separate inputs 3520A-3520B are provided for Src0 (accumulator value) inputs. The data from inputs 3520A-3520B is stored in a Src0 data buffer 3530A-3530B and added to output from the systolic array pipelines, as opposed to being added at Stage 0 as in other systolic array designs. Output from each array can be stored in accumulator/adder circuits that include memory (e.g., an accumulator register) and an adder circuit. Accumulator/adder circuit 3532 can store output from systolic pipeline 3502 and add the output to data stored in Src0 data buffer 3530A.

Accumulator/adder circuit **3534** can store output from systolic pipeline **3504** and add the output to data stored in Src0 data buffer **3530B**.

[0417] In one embodiment, multi-pass operation is enabled, such that the eight physical stages of the array operate as sixteen logical states. The eight stages of each of systolic pipeline **3502** and systolic pipeline **3504** can operate as sixteen logical stages by respectively storing the output of a first pass to the first accumulator/adder circuit **3532** and second accumulator/adder circuit **3534**. The values stored in the circuits can be accumulated with output generated by a second pass through each of systolic pipeline **3502** and systolic pipeline **3504**. For a given stage *i*, the stage operates as stage *i* during a first pass and stage *i*+8 during a second pass. The appropriate input data is provided to the arrays depending on whether the array is performing first pass operations or second pass operation. In one embodiment, operations for instructions of any number of logical stages may be supported via single pass and/or multiple or partial pass operation. A selector circuit **3536** enables data within the first accumulator/adder circuit **3532** and second accumulator/adder circuit **3534** to be output to a destination register.

[0418] FIG. **36** illustrates a stage pair **3600** for a channel of a systolic array. In one embodiment the physical pipeline stages for each array of the dual pipeline parallel systolic array **3500** of FIG. **35** are grouped as a stage pair **3600**. A stage pair **3600** for Stage **0** (**3610**) and Stage **1** (**3611**) is illustrated, with other pairs of stages (e.g., [2,3], [4,5], [6,7]) being configured similarly. Each channel of each stage includes a pair of multipliers (e.g., multipliers **3612A-3612B** for Stage **0**, multipliers **3613A-3613B** for Stage **1**) and a common adder **3604**. The accumulator input **3620** (Src0) is passed through to Src0 data buffer **3530A-3530B** shown in FIG. **35** and is not operated on by the stage pair **3600**. The appropriate Src1 register data is provided as input to the appropriate stage. A single Src2 register read can store data for both stages in the stage pair **3600**.

[0419] FIG. **37** illustrates a systolic array **3700** including partial sum loopback and circuitry to accelerate sparse matrix multiply. In the systolic array **2808** described above, operands that include weight data may be stationary within the array and a partial sum is propagated throughout the array structure. While other details with respect to systolic array **2808** may be applicable, in systolic array **3700** a partial sum is recirculated instead of being propagated to a next systolic layer. In one embodiment a systolic array **3700** can be configured with *M* rows and *N* columns of processing elements (PE **3712AA-PE 3712MN**). The processing elements can access registers storing input data in the form of row and column data for input matrices. The registers may be stored in a register file that is local to the systolic array **3700** or in a register file of a processing resource that is coupled with or includes the systolic array **3700**. The registers may store row elements of matrix *A* **3702A-3702M**, which are to be multiplied by column elements of matrix *B* **3701A-3702N**.

[0420] In one embodiment a fused multiply-add (FMA) can be performed at each processing element PE **3712AA-PE 3712MN** each clock cycle. An element of matrix *A* is multiplied by a corresponding element of matrix *B* and then added to an accumulator value or, for the first cycle, an optional initial input value (e.g., SRC0). Partial sum loopback can be configured at each processing element. After

each cycle, the accumulator value may be looped back within the processing element and used as input for the next cycle. Once operations are performed for an entire row, the result may be stored to a register file. Data movement between the processing elements PE **3712AA-PE 3712MN** after a set of computational cycles can vary based on the instruction or macro-operation being performed.

Data Aware Sparsity with Compression

[0421] Embodiments described herein provide an encoding layout that enables sample blocks of sparse neural network data to be encoded in a reduced-bit format that reduces the amount of data that is required to be transmitted or stored when processing neural networks associated with the data. The number of non-zero values in a sample block is indicated in a header, followed by a significance map indicating a map of the non-zero values within the block. The non-zero values of the sample are encoded in order of appearance within the stream. In one embodiment, compression can be based on other values beyond zero values. For example, a specified value within a data set may be encoded and excluded from a compressed data stream, enabling compression based on ones, twos, or other specified values. In one embodiment compression is enabled based on near values. Values within a data set that are within a threshold of zero, or within a threshold of a specified value, may be compressed as though those values were zero or within a threshold of the specified value. Data aware sparsity with compression can be enabled via codec logic coupled with or within matrix accelerator logic.

[0422] FIG. **38A-38B** illustrate matrix acceleration circuitry including codecs to enable the reading of sparse data in a compressed format. FIG. **38A** illustrates a compute block **3800** including codec enabled disaggregated systolic logic. FIG. **38B** illustrates processing elements within a systolic array that are coupled with codecs to decompress input data.

[0423] As shown in FIG. **38A**, instead of including a systolic array **2808** in a separate tensor accelerator **2723**, as in FIG. **28A**, or including a matrix engine **1803A-1803N** in each graphics core **1815A-1815N**, a disaggregated set of systolic arrays **3812A-3812B** can be included in a compute block **3800** that is analogous to one of the compute blocks **2724A-2724N** of FIG. **27**. The compute block **3800** can include multiple interconnected processing resources (PR **3808A-3808O**), which may be similar to any processing resource architecture described herein, such as but not limited to processing resources described herein, including that may be similar to EU **1808A-1808N** or any other processing resource as described herein. In one embodiment the systolic arrays **3812A-3812B** include codecs **3824A-3824B** that enable the encoding and decoding of input and output data that is received for processing.

[0424] The systolic arrays **3812A-3812B** include a *W* wide and *D* deep network of data processing units that can be used to perform vector or other data-parallel operations in a systolic manner, similar to other systolic arrays described herein. In one embodiment the systolic arrays **3812A-3812B** can be configured to perform matrix operations, such as matrix dot product operations. In one embodiment the systolic arrays **3812A-3812B** support 16-bit floating point operations, as well as 8-bit and 4-bit integer operations. In one embodiment the systolic arrays **3812A-3812B** can be configured to accelerate machine learning operations. In such embodiments, the systolic arrays **3812A-3812B** can be

configured with support for the bfloat 16-bit floating point format. By including systolic arrays **3812A-3812B** within the compute block **3800** but outside of the PRs **3808A-3808O**, the size and number of systolic arrays **3812A-3812B** can be scaled independently from the number of PRs **3808A-3808O**. Additionally, communication bandwidth within an PR that would otherwise be consumed by systolic array activity may be preserved. Furthermore, the systolic arrays **3812A-3812B** may be clock/power gated when matrix workloads are not being performed.

[0425] Communication between the systolic arrays **3812A-3812B** and the PRs **3808A-3808O** may be performed via a cache or shared local memory (cache/SLM **3810**) and/or a shared register file **3814**. In one embodiment, instead of a distinct shared register file **3814**, the cache/SLM **3810** may be partitioned for use as a shared register file. The shared register file **3814** may be structured similarly to other GPGPU register files, such as register file **369** as in FIG. 3C. The shared register file may also include a set of special purpose registers that are used to configure the interaction between the systolic arrays **3812A-3812B** and the PRs **3808A-3808O**. The cache/SLM **3810** may be an L1 cache, an L2 cache, and/or a block of explicitly addressable on-die memory.

[0426] Matrix data for processing by the systolic arrays **3812A-3812B** may be stored in the cache/SLM **3810**. Processing commands or instructions can be provided to the systolic arrays **3812A-3812B** via the shared register file **3814**. Processing results may be read from the cache/SLM **3810** by the PRs **3808A-3808O** or from destination/output registers within the shared register file. During operation, instead of consuming bus/fabric bandwidth within the PRs **3808A-3808O**, communication traffic may be localized to the systolic arrays **3812A-3812B**, the cache/SLM **3810**, and/or shared register file **3814**. Any of the PRs **3808A-3808O** within the compute block **3800** may offload a matrix workload to one or both systolic arrays **3812A-3812B**. A message may be sent from a PR to a systolic array with a command that specifies an operation to be performed and operands for the operation. The systolic arrays **3812A-3812B** can perform the requested operations (multiply/add, fused multiply/add, multiply/accumulate, dot product, etc.) and output the results to the shared register file **3814**. Input, intermediate and/or output data for requested operations may be stored in the cache/SLM **3810** and multiple dependent operations may be chained. In one embodiment when processing operations for training or inference for a neural network are performed, the systolic arrays **3828A-3828B** may also perform activation functions including but not limited to sigmoid, ReLU, and hyperbolic tangent (TanH) activations. In such embodiment, operations for neural networks may be offloaded to the systolic arrays **3812A-3812B** at coarse granularity.

[0427] The PRs **3808A-3808O** can provide input data to the systolic arrays **3812A-3812B** in a compressed format and the codecs **3824A-3824B** can be used to decompress the data. When output data is ready to be provided to the PRs **3808A-3808O**, the data may remain decompressed if the PRs will perform operations and the data and do not support the direct read of compressed data. If the PRs **3808A-3808O** support the reading of compressed data or will not perform additional operations on the data, the output data may be re-encoded. Zero-based encoding may be used and compression may be enabled or disabled based on the degree of

data sparsity. Alternatively, other forms of encoding may be used based on the distribution of the data set to be processed or output. For example, the codecs **3824A-3824B** can be configured to decode sparse data that is encoded based on zero-based compression or using another form of compression described herein (e.g., one-based, two-based, near-zero, near-one, near-two, etc.).

[0428] As shown in FIG. 38B, system **3850** illustrates processing elements of systolic array **3700**, where the systolic array is configured to decode compressed sparse data. As described with respect to FIG. 37, each PE **3712AA-3713MN** includes hardware logic to perform computations for matrix operations. A (A_0, A_1 , through A_M) and B (B_0, B_1 , through B_N) are elements of input matrices with associated with dot product, matrix multiply, multiply/add, or multiply accumulate operations. In one embodiment each PE **3712AA-3713MN** is associated with codecs (**3851a**, **3851b**, . . . , **3851m**; **3852a**, **3852b**, . . . , **3852n**) to decode compressed input operands associated with operations to be performed. The codecs can be configured to decode sparse data that is encoded based on zero-based compression or using another form of compression described herein.

[0429] Sparse neural network data can be encoded (e.g., compressed) using a variety of encoding techniques, such as but not limited to unique absolute value (UAV) table encoding, significance map (SM) encoding, table encoding (TE), unique value coordinate (UVC) encoding, and mean encoding (ME). Metadata for the encoded data indicates the type of encoding format used for the data. In one embodiment, specific encoding formats can be selected for specific types of data, such as kernel data or feature data. In one embodiment, statistical analysis is performed on the data prior to encoding to enable an appropriate encoder to be selected for each block of data. The encoding may be zero-based encoding, near-zero encoding or based on other values (ones, twos, etc.).

[0430] In one embodiment data generated during SM encoding can be used to facilitate provision of compressed data to a systolic tensor array. In zero-based SM encoding mode, only non-zero values in a block are encoded. The number of non-zero values in a sample block is indicated in the header, followed by a significance map indicating a map of the non-zero values within the block. The non-zero values of the sample are then encoded in order of appearance within the stream.

Temporally Amortized Supersampling Using Mixed Precision Convolutional Neural Network

[0431] Described herein are embodiments that provide a machine learning-based temporally amortized supersampling technique that replaces temporal anti-aliasing (TAA). A mixed low precision convolutional neural network is used that applied different computational precisions at different stages to enable the high performance generation of high quality images based on source images rendered at a relatively lower resolution than the target output resolution. The network model enables anti-aliasing and upscaling with support for multiple scale factors, including fractional scale factors such as, but not limited to 1.3x, 1.5x, 1.7x, 2x, or 2.2x. Other scale factors are also possible. Temporally stable upsampled output can be generated that has an image quality that is better than or equal to native rendering at the target resolution. In various embodiments, different versions are provided that can be implemented on a variety of different

graphics processing architectures, including architectures with matrix acceleration hardware as described above in FIG. 28A through FIG. 34, as well as graphics processor architectures that lack dedicated matrix acceleration hardware.

[0432] FIG. 39 illustrates a conventional renderer **3900** with Temporal Anti-aliasing (TAA). The renderer within the rasterization and lighting stage **3910** can jitter (**3905**) the camera **3902** during rendering for every frame to sample different coordinates in screen space **3904**. Different pixels can be sampled from different frames over time. The TAA stage **3916** accumulates these samples temporally to produce a supersampled image. A warping operation **3924** is applied to the previously accumulated frame (History **3923**) using renderer generated velocity/motion vectors **3922** to align the previously accumulated frame with the current frame **3912** (frame N) before accumulation. Optional upscaling **3914** can be performed on the current frame before input to the TAA stage **3916**, such that the current frame can be rendered at a lower resolution than the target resolution. The output frame can then be added to the history **3923** for use in processing the next frame. Post processing operations **3918** can then be performed at the upscaled target resolution. While applying upscaling with TAA can improve rendering performance, the output images are of lower quality than images rendered natively at the target resolution. Some TAA implementations can use heuristics **3915** such as but not limited to neighborhood color clamping, object identifier comparisons, and depth value comparisons to detect mismatches between current and history frames and reject the history pixels. However, these heuristics often fail and produce a noticeable amount of ghosting, over-blurring and/or flickering.

[0433] FIG. 40 illustrates a renderer **4000** that replaces the TAA stage with a temporally amortized supersampling stage, according to embodiments provided herein. Renderer **4000** differs from renderer **3900** of FIG. 39 in that, in renderer **4000**, temporally amortized supersampling is performed using a convolutional neural network model **4050**. The convolutional neural network model **4050** is a mixed, low-precision model that replaces the TAA stage in the game renderer, achieving significantly better image quality than conventional TAA-based techniques, as well as providing a performance boost by enabling rendering to be performed at lower resolution. The renderer **4000** can render the current frame **3912** at a lower than target resolution. An upscaling filter **4014** is applied to the rendered image to upscale the image to the target resolution. In one embodiment, the upscaling filter **4014** is applied by the renderer **4000** before the current frame **3912** is provided to the supersampling stage. In one embodiment, the upscaling filter is performed by the neural network model **4050** during pre-processing operations. The upscaling filter **4014** can include optimizations to enhance the image quality of temporal stability of images that result from the processing performed by the neural network model **4050**. Warping operations **4024** on the history **3923** can be performed by an input block of the neural network model **4050**. In one embodiment the history **3923** is a multi-frame history that includes data from multiple previous frames.

[0434] The mixed, low-precision convolutional neural network is implemented via a neural network model **4050** that consists of multiple convolution layers, as well as other operations that are performed at low precisions, such as

INT8, mixed with operations performed at a higher precision, such as FP16. The mix of precisions enable the network to achieve a fast computational speed while generating high quality output images. The lower precision values are not limited to INT8 and different low-precision data formats (e.g., INT4, binary, bipolar binary, ternary, etc.) can be used for variations. The majority of the neural network model **4050** and the operations associated with the neural network model are performed at the lower precision to enable high inference performance. A computationally smaller part is performed at a relatively higher precision to preserve output quality. In addition to using FP16 for higher precision operations, other floating-point precisions may also be used, such as FP8, BF16 or TF32. Additionally, the majority of the neural network model **4050** is also in a reduced spatial dimension to provide fast inference performance by shuffling input pixels from the spatial (width, height) dimension to a depth or feature map channel dimension with no pixel information loss. The spatial dimension is shuffled back from the channel dimension during generating an output image.

[0435] Temporally amortized supersampling is performed by combining the current frame and the previous output frame warped with the current motion vectors. The neural network model **4050** determines the manner in which to combine the upscaled current frame **3912** and the history **3923**. In various embodiments, multiple different approaches are applied to preserve output quality. In one embodiment, a high precision combination of the upscaled current frame **3912** and the history is generated using 1x1 or 3x3 output convolution. In another embodiment, pixel prediction and high precision filtering of the upscaled image is performed to generate a high-quality upscaled image. The neural network model **4050** is used to generate input that is provided to the kernel prediction and filtering operations.

[0436] During training of the neural network model **4050**, both perceptual and temporal loss functions are optimized to enhance both the image quality and the temporal stability of the upsampling and anti-aliasing. In one embodiment, generalized training is sufficient to enable high quality output across a variety of games without requiring extensive per-game, per-upscale factor, or per-target resolution training.

[0437] FIG. 41 illustrates an implementation of a neural network model **4100**, according to an embodiment. The neural network model **4100** is an implementation of the neural network model **4050** of FIG. 40. In one embodiment, the neural network model **4100** is composed of three components: an input block **4108**, a feature extraction network **4110**, and an output block **4120**. Lower precision (e.g., Integer) operations are used for the majority of the neural network model to achieve fast inference performance. Output of the neural network model is generated using higher precision (e.g., floating-point) operations to enable the generation of high-quality output images. For example, the encoders (encoder block 1 through encoder block N), bottleneck block, and decoder blocks (decoder block 1 through decoder block N) in the feature extraction network **4110** are executed with relatively lower precision (e.g., INT8) compared to the output block **4120**, which is executed at a relatively higher precision (e.g., FP16). Utilizing lower precision in the feature extraction network **4110** significantly reduces the complexity of computation and improves memory bandwidth for fast inference performance. Utilizing higher precision in the output block **4120** enables the

generation of output images having an image quality that is as good as, or in some cases better than images that are natively rendered at the target resolution. As noted above, other precisions or data types in addition to INT8 and FP16 can be used, such as but not limited to INT4 for lower precision operations and BF 16 or TF32 for higher precision operations.

[0438] The input block **4108** receives, as input, history data **4102**, velocity data **4104**, the current frame **4106**, and a jitter offset **4107** for the camera. The history data **4102** includes previously generated output. The previously generated output includes at least the immediate previous frame (frame N-1), which is warped using the velocity data **4104** to align the frame with the current frame **4106** for temporal accumulation. In various embodiments, in addition to the previous frame, the history data **4102** can also include one or more additional frames of previous generated output (e.g., frame N-2, etc.), which can also be provided as input to the feature extraction network **4110**. The jitter offset **4107** is the camera offset that is applied to jitter the scene, with different jitter values being used for successive frames. The jitter offset **4107**, in one embodiment, is a sub-pixel offset. The input block generates both lower and higher precision tensors. Lower precision tensors are provided to the feature extraction network **4110**. Higher precision tensors are provided to the output block **4120**. Further details on the input block **4108** are shown in FIG. 42.

[0439] The feature extraction network **4110** is built upon a U-shaped network architecture, such as, for example, the U-net architecture. The feature extraction network **4110** differs from the conventional U-net architecture in that the feature extraction network **4110** includes an asymmetric structure in the encoder **4112** and decoder **4116**. The encoder **4112** of the feature extraction network **4110** includes a series of encoder blocks that downsample the spatial dimension of an input tensor while increasing the number of channels (depth or feature maps) until the production of a latent representation **4114** at a bottleneck block in the middle of the network. The latent representation **4114** is the abstract multi-dimensional space that encodes the meaningful features of the input data. The decoder blocks of the decoder **4116** reverse this process by upsampling spatial dimension and decreasing the number of channels. The encoder blocks have a skip connection to a corresponding decoder block, which enables high-frequency details to be relayed between the encoder **4112** and the decoder **4116**. Output from encoder block **1** is provided to decoder block **2** to be processed in conjunction with output from decoder block **3**. Output from encoder block **2** is provided to encoder block **3** to be processed in conjunction with output from the previous decoder block in the network. The input for encoder block N is provided to decoder block N. Decoder block **1**, the final decoder block, receives input from the input block **4108** and decoder block **2**. The decoder **4116**, from decoder block **1**, provides data to the output block **4120** in either a higher precision format or a lower precision format depending on the implementation approach used for the output block **4120**. Further details on the output block are shown in FIG. 43A and FIG. 43B.

[0440] FIG. 42 illustrates further details for the input block **4108** of the neural network model **4100**, according to embodiments. The input block **4108** receives input including history data **4102**, velocity data **4104**, the current frame **4106**, and the jitter offset **4107**. The input block **4108**

includes a warping unit **4202** to warp the previous output within the history data **4102** using motion vectors within the velocity data **4104**. The input block **4108** also include an upscaling unit **4203** to upscale the current frame **4106**. In one embodiment, the upscaling filter applied by the upscaling unit **4203** is an adaptive filter that adjusts the upscaling based on the jitter offset **4107**. A space to channel/depth shuffle unit **4204** shuffles pixels from the spatial dimension (width, height) to a channel (e.g., feature map) or depth dimension, which facilitates high performance inferencing via reduction of numerical precision and spatial dimension during feature extraction. For example, for an input image having (channel, height, width) pixels of data in the spatial dimension, the pixel data can be shuffled to

$$\left(\text{channel} \times r^2, \frac{\text{height}}{r}, \frac{\text{width}}{r} \right),$$

which reduces the spatial dimension in which the feature extraction is performed, which improves the performance of the feature extraction network **4110**. The input block **4108** generates both lower precision (e.g., INT8) and higher precision (e.g., FP16) tensors. The lower precision tensors are provided as input to the feature extraction network **4110**, while the higher precision tensors are passed to the output block **4120**, **4320A**-**4320B**. The input block **4108** also include an optional convolution/activation layer **4206** that can be applied before data is output to the feature extraction network.

[0441] FIG. 43A-43B illustrates output block variants for the neural network model, according to embodiments. FIG. 43A illustrates a decoder block **4320** and a variant of the output block **4320A** that is configured to perform direct generation of pixel data for the output image. FIG. 43B illustrates a decoder block **4320** and a variant of the output block **4320B** that is configured as a kernel prediction network that applies kernel pixel prediction and filtering to generate the output image. In FIG. 43A-43B, a decoder block **4320** (decoder block **1**) is shown as an example. While each encoder block of the encoder **4112** includes a downsample block and one or more convolution/activation layers that facilitate feature extraction, each decoder block of the decoder **4116** includes an upsample block **4322** to increase spatial dimension and one or more convolution/activation layer(s) **4324**, **4326** to restore features. Decoder block **1** receives data from decoder block **2** as well as skip connection data from the input block. For the output block **4320A**-**4320B**, two different approaches can be taken to preserve quality with higher precision. One embodiment provides an output block **4320A**, as shown in FIG. 43A, which configures the neural network model **4100** to operate as a direct reconstruction network. One embodiment provides an output block **4320B**, as shown in FIG. 43B, which configures the neural network model **4100** to operate as a kernel prediction network.

[0442] For output block **4320A** of FIG. 43A, data from the input block **4108** and the feature extraction network **4110** is combined using a 1x1 or 3x3 output convolution layer **4330** to directly generate data for the output image. The output convolution layer **4330** receives, as input, higher precision (e.g., FP16) output from the convolution/activation layer(s) **4326** of the final decoder block **4320**, as well as higher precision input from the input block **4108**. Data generated by

the output convolution layer **4330** is provided to the depth/channel to space shuffle unit **4332**, which shuffles the data back into the spatial dimension to generate an output image **4340**. The output image **4340** can be output via a display or further post-processed before output via the display.

[**0443**] For output block **4320B** of FIG. **43B**, kernel prediction and filtering are performed. Instead of directly generating an output image, per-pixel kernel values (e.g., weights) are predicted by a kernel prediction layer **4334**. Lower precision (INT8) tensors are output by the decoder block **4320** for use by the kernel prediction layer **4334**, which uses the lower precision tensors in combination with the higher precision tensors provided by the input block **4108**. The depth/channel to space shuffle unit **4332** shuffles frame data back into the spatial dimension to generate an intermediate output image. The intermediate output image is then filtered by the filter/blend layer **4346** using the per-pixel kernel values generated by the kernel prediction layer **4334** and blending with the previous output using blend weights generated by the kernel prediction layer **4334**. The filtered and blended image is then provided as the output image **4340**.

[**0444**] FIG. **44** illustrates a method **4400** to perform temporally amortized supersampling. The method **4400** includes to receive, at an input block of a neural network model described herein (e.g., neural network model **4050**), history data, velocity data, and current frame data (**4402**). The history data includes one or more previously generated frames. The velocity data includes renderer generated motion vectors that are used to align the one or more previously generated frames with the pixel data of the current frame. The current frame data includes a frame of a 3D graphics program, such as a 3D game application, that is output by a raster and lighting stage of the render pipeline of the graphics processor. In one embodiment, the current frame is an upscaled frame that has been upscaled by an upscaling filter from an initial rendering resolution to a target resolution. In one embodiment, the current frame is upscaled to the target resolution during pre-processing. The input block provides output at multiple precisions, with a first set of output being provided to the output block at high precision and a second set of output being provided to the feature extraction network at a relatively lower precision. In one embodiment, the first set of output is provided as floating-point data (e.g., FP16, BF16), while the second set of output is provided as integer data (e.g., INT4, INT8).

[**0445**] The neural network model can then pre-process the history data, velocity data, and current frame data at the input block and provide the pre-processed data to a feature extraction network (**4404**). The pre-processed data that is provided to the feature extraction network includes aligned history data and current frame data. The history data is warped using the velocity data to generate warped history data. The warped history data is then aligned with the current frame data to generate aligned history data. The aligned history data provides additional sample data that can be used to generate a supersampled anti-aliased output image via temporal accumulation. In one embodiment, the pre-processing includes upscaling the current frame data from the resolution output by the raster and lighting stage to the target resolution.

[**0446**] The neural network model processes the pre-processed data at the feature extraction network via one or more encoder stages and one or more decoder stages (**4406**). The

encoder stages reduce the spatial resolution of the input data and extracts the most salient features within the input data. The spatial resolution is then expanded via the decoder stages to generate tensor data that is used to process the current upscaled frame in view of the aligned history to generate a high quality upscaled frame that has an image quality that is, at the least, equal to an image that is natively rendered at the target resolution. The features extracted are used to determine an optimized combination of the current and previous frames during temporal accumulation.

[**0447**] The neural network model can then generate an output frame via an output block of the neural network model via temporal accumulation using direct reconstruction or kernel prediction (**4408**). The output frame is an anti-aliased image that has a higher resolution than the rendering resolution of the render pipeline, with additionally generated pixels to enhance the image quality beyond that of the originally upscaled image. In one embodiment, the neural network model is configured as a direct reconstruction network which, via one or more convolution layers, generates a high-quality output image for display. When configured as a direct reconstruction network, the feature extraction network provides higher precision tensors (e.g., FP16, BF16) as input to the output block. The output block uses the higher precision output from the feature extraction network in combination with the higher precision output from the input block to generate the output image. In one embodiment, the neural network model is configured as a kernel prediction network that generates per-pixel kernel values that applied to a high-precision filter. When configured as a kernel prediction network, the feature extraction network provides lower precision tensors (e.g., INT4, INT8, FP8) to the output block. The output block uses the lower precision output from the feature extraction network in combination with the higher precision output from the input block to predict the pre-pixel kernels/blend weights used to filter the upscaled input and blend the filtered input with the previous output.

[**0448**] FIG. **45** illustrates exemplary rendering performance comparisons for multiple rendering techniques described herein. Rendering time for a low-quality rendering **4505**, for example, at 1080p resolution, is significantly lower than the rendering time for a high-quality rendering **4501**, for example, at 4K resolution. Traditional upscaling **4504** (TAA Upsampling, Temporal Super Resolution, FidelityFX Super Resolution) renders frames at low resolution and the low-resolution image is upsampled to the target display resolution to achieve performance boost and potentially an image quality improvement over low-quality rendering **4505**.

[**0449**] One implementation of temporally amortized supersampling using a mixed precision convolutional neural network is X^e SS provided by Intel® Incorporated. X^e SS can be performed on hardware that includes a matrix accelerator (e.g., tensor accelerator **2723**) via the use of Intel X^e Matrix Extensions (XMX). Rendering via X^e SS+XMX **4502** can produce an image that is significantly higher quality than low-quality rendering **4505** or traditional upscaling **4504** and with significantly lower rendering times than high-quality rendering **4501** at native 4K resolutions. Rendering via X^e SS+DP4a **4503** replaces XMX with a dot product instruction (DP4a) that can be executed by a variety of graphics processor architectures from a variety of vendors and results in a high-quality image and a rendering time that

is still significantly lower than high-quality rendering **4501** at native 4K resolutions. In one embodiment, X^e SS+XMX **4502** is performed using direct reconstruction via output block **4320A** of FIG. **43A**, while X^e SS+DP4a **4503** is performed using kernel prediction and filtering via output block **4320B** of FIG. **43B**.

Motion Vector Refinement Via Learned Residual Optical Flow

[0450] The temporally amortized supersampling techniques described herein can be configured make use of motion vectors provided by the 3D rendering engine of a 3D application that is processed on a graphics processing architecture described herein. A rendered 3D scene has an associated motion field that reflects the 3D motion of the objects within the scene. An optical flow field is a 2D projection of the motion field, such that a 2D optical flow vector is representative of the 3D motion vector when translated into a 2D image. Pixel-by-pixel optical flow vectors may be substituted with block-based motion vectors, which provide an estimate or approximation of the optical flow between frames. The motion vectors are used during temporally amortized supersampling to warp and align a previous frame with a current frame. However, motion vectors may not be supplied for some portions of a frame. For example, animated textures, mirrored/reflected objects, and/or moving objects without motion information may reside within a frame. Motion vectors may not be provided for those regions within the frame.

[0451] Described herein is a technique to refine the motion vectors used to warp previous frames via learned residual optical flow. A residual network is trained to predict a set of residual motion vectors based on an existing set of motion vectors, an upsampled current frame, and a previous frame. The residual motion vectors provide additional motion data for portions of the frame for which motion vectors are not provided. The set of residual motion vectors are combined with the existing set of motion vectors to generate a set of refined motion vectors. The refined motion vectors can be used to enable a higher quality transformation (e.g., warping) of the previous frame.

[0452] FIG. **46** illustrates data flow for image reconstruction and upscaling, according to an embodiment. In one embodiment, the data flow for image reconstruction and upscaling is implemented using the neural network model **4100** of FIG. **41**, as executed using graphics and/or compute accelerator hardware described herein. A current frame **4602** that is rendered below a target resolution is received for processing. Depth data **4613**, jittered samples **4614**, and motion vectors **4612** for the current frame **4602** are provided as input to circuitry **4616** that is configured to upsample the motion vectors. The motion vectors **4612** can be generated by and received from a game engine. For scenarios in which a game engine is not configured to provide motion vectors, the graphics processor may be configured to generate at least a partial or sparse set of motion vectors using video encode hardware (e.g., within video codec engine **1606**). Color data for the current frame **4602** is provided to circuitry **4604** that is configured to upsample the color data to the target resolution. The upsampled motion vectors and a previous output frame **4622** are provided to circuitry **4618** that is configured to warp the previous output frame **4622** to generate a warped previous output frame **4620**. The warped previous output frame **4620** and the upsampled color data

are provided to an image reconstruction network **4606** described herein. The image reconstruction network **4606** is configured to either 1) directly reconstruct an anti-aliased image at the target resolution or 2) predict per-pixel kernels/blend weights to use to blend the upsampled color data and the warped previous output frame. During training operations, a reference frame **4608** that is rendered at the target resolution is used to generate an output loss **4610**. Stochastic gradient descent and backpropagation may be used to minimize the output loss **4610** during training.

[0453] FIG. **47** illustrates an extended flow that includes a residual motion refinement network **4724** and warp loss optimization, according to an embodiment. The residual motion refinement network **4724** is a neural network that is trained to predict residual motion vectors for portions of an image without motion vectors based on the motion vectors that are provided for other parts of the image. A warp loss **4722** is optimized during joint training of the residual motion refinement network **4724** and the image reconstruction network **4606**. In this extended flow, the previous output frame **4622**, the upsampled motion vectors generated by circuitry **4616**, and the upsampled color data generated by circuitry **4604**, are provided to the residual motion refinement network **4724** that is executed via the graphics and/or compute acceleration hardware described herein. The residual motion refinement network **4724** is trained to predict a set of residual motion vectors **4726** that provide additional motion data for portions of the frame for which motion vectors are not provided or available. For example, in some configurations, motion vectors are not generated by the game engine for animated textures and mirrored/reflected objects. Depending on the engine, other types of moving objects may not have associated motion information.

[0454] The graphics processor includes circuitry **4702** that is configured to combine the upsampled motion vectors and the residual motion vectors **4726** to generate refined motion vectors **4712**. Circuitry **4718** is configured to use the refined motion vectors **4712** to warp the previous output frame **4622** to generate a warped previous output frame **4720**. The warped previous output frame **4720** is provided to the image reconstruction network **4606**, which generates an output frame based on the warped previous output frame **4720** and the upsampled color data for the current frame **4602**. During training, circuitry **4718** is configured to output warp loss **4722** that is used to train the residual motion refinement network **4724**.

[0455] In some embodiments, the residual motion refinement network **4724** can be used to enable temporally amortized supersampling in scenarios where motion vectors are not provided by the game engine. In one embodiment, the trained residual motion refinement network **4724** may be configured to generate at least a partial set of motion vectors based on the upsampled color data generated by circuitry **4604** and the previous output frame **4622**. The residual motion refinement network **4724** may also be configured to refine a set of sparse motion vectors that are generated by processing the upsampled color data and previous output frame **4622** via video encoder hardware. The video encoder hardware (e.g., video codec engine **1606**) may be configured to rapidly generate a sparse set of motion vectors that are less detailed than the motion vectors generally used during video encoding. Those sparse motion vectors may be refined using the residual motion refinement network **4724** to gen-

erate refined motion vectors **4712** that may be used by circuitry **4718** to generate the warped previous output frame **4720**. The refined motion vectors **4712** can be used to generate more precisely warped frames, which improves the sharpness of the final image.

[**0456**] The residual motion refinement network **4724** may be implemented using a variety of neural network architectures. In one embodiment, the residual motion refinement network **4724** is implemented as a custom convolutional neural network. In one embodiment, the residual motion refinement network **4724** can be implemented using a version of an existing neural network architecture including but not limited to the Res-Net or U-Net architectures. In one embodiment, the residual motion refinement network **4724** is a version of the neural network model **4100** of FIG. **41**.

[**0457**] FIG. **48** illustrates a method **4800** of motion vector refinement via a residual motion refinement network, according to an embodiment. The method **4800** can be implemented in part by temporally amortized supersampling logic executed by and/or associated with a graphics processor having an architecture described herein, such as but not limited to the GPGPU **2720** of the data processing system **2700** of FIG. **27**. Operations of the method may be accelerated using matrix processing hardware described herein. The method **4800** enables operations to generate additional motion vectors to improve the quality of previous frame warping and alignment during temporally amortized supersampling, which enables the generation of a higher quality output image.

[**0458**] In one embodiment, the method **4800** includes to perform operations to train a neural network (e.g., residual motion refinement network **4724**) to generate residual motion vectors based on an existing set of motion vectors, a current frame, and a previous frame (**4802**). The neural network may be a second neural network that is trained jointly with a first neural network (e.g., image reconstruction network **4606**). The training can be performed to optimize or minimize a warp loss that is generated by image warping logic. The image warping logic can generate the warp loss based on warping and alignment operations that are performed to warp the previous frame and align the warped previous frame with the current frame. The training process enables the neural network to predict residual motion vectors for portions of the scene for which motion vectors are not provided. The training process can be performed prior to the deployment of neural network. The neural network can be deployed as a portion of a graphics driver, software development kit, or runtime library to enable or facilitate temporally amortized supersampling as described herein.

[**0459**] During rendering of a current frame of a rendered scene, temporally amortized supersampling logic can provide the current frame, the previous frame, and a first set of motion vectors to the neural network (**4804**). The temporally amortized supersampling logic can then generate, via the neural network, a second set of motion vectors (**4806**). The second set of motion vectors augment the first set of motion vectors by providing motion information for portions of the scene that are not covered by the first set of motion vectors, such as but not limited to animated textures or reflected objects. The temporally amortized supersampling logic can then combine the second set of motion vectors with the first set of motion vectors to generate a combined set of motion vectors (**4808**). The temporally amortized supersampling can then warp the previous frame via the combined set of

motion vectors (**4810**). In one embodiment, the previous frame can be warped to generate history data, which in one embodiment can include multiple warped previous frames. The warped history data can be aligned with the current frame and provided as input to the image reconstruction network.

Additional Exemplary Computing Device

[**0460**] FIG. **49** is a block diagram of a computing device **4900** including a graphics processor **4904**, according to an embodiment. Versions of the computing device **4900** may be or be included within a communication device such as a set-top box (e.g., Internet-based cable television set-top boxes, etc.), global positioning system (GPS)-based devices, etc. The computing device **4900** may also be or be included within mobile computing devices such as cellular phones, smartphones, personal digital assistants (PDAs), tablet computers, laptop computers, e-readers, smart televisions, television platforms, wearable devices (e.g., glasses, watches, bracelets, smartcards, jewelry, clothing items, etc.), media players, etc. For example, in one embodiment, the computing device **4900** includes a mobile computing device employing an integrated circuit (“IC”), such as system on a chip (“SoC” or “SOC”), integrating various hardware and/or software components of computing device **4900** on a single chip. The computing device **4900** can be a computing device including components illustrated in the data processing system **2700** as in FIG. **27**.

[**0461**] The computing device **4900** includes a graphics processor **4904**. The graphics processor **4904** represents any graphics processor described herein. In one embodiment, the graphics processor **4904** includes a cache **4914**, which can be a single cache or divided into multiple segments of cache memory, including but not limited to any number of L1, L2, L3, or L4 caches, render caches, depth caches, sampler caches, and/or shader unit caches. In one embodiment the cache **4914** may be a last level cache that is shared with the application processor **4906**.

[**0462**] In one embodiment the graphics processor **4904** includes a graphics microcontroller that implements control and scheduling logic for the graphics processor. The control and scheduling logic can be firmware executed by the graphics microcontroller **4915**. The firmware may be loaded at boot by the graphics driver logic **4922**. The firmware may also be programmed to an electronically erasable programmable read only memory or loaded from a flash memory device within the graphics microcontroller **4915**. The firmware may enable a GPU OS **4916** that includes device management logic **4917** and driver logic **4918**, and a scheduler **4919**. The GPU OS **4916** may also include a graphics memory manager **4920** that can supplement or replace the graphics memory manager **4921** within the graphics driver logic **4922**.

[**0463**] The graphics processor **4904** also includes a GPGPU engine **4944** that includes one or more graphics engine(s), graphics processor cores, and other graphics execution resources as described herein. Such graphics execution resources can be presented in the forms including but not limited to execution units, shader engines, fragment processors, vertex processors, streaming multiprocessors, graphics processor clusters, or any collection of computing resources suitable for the processing of graphics resources or image resources or performing general purpose computational operations in a heterogeneous processor. The process-

ing resources of the GPGPU engine 4944 can be included within multiple tiles of hardware logic connected to a substrate, as illustrated in FIG. 24B-24D. The GPGPU engine 4944 can include GPU tiles 4945 that include graphics processing and execution resources, caches, samplers, etc. The GPU tiles 4945 may also include local volatile memory or can be coupled with one or more memory tiles, such as memory tiles 1626A-1626D as in FIG. 16B-16C.

[0464] The GPGPU engine 4944 can also include and one or more special tiles 4946 that include, for example, a non-volatile memory tile 4956, a network processor tile 4957, and/or a general-purpose compute tile 4958. The GPGPU engine 4944 also includes a matrix multiply accelerator 4960. The general-purpose compute tile 4958 may also include logic to accelerate matrix multiplication operations. The non-volatile memory tile 4956 can include non-volatile memory cells and controller logic. The controller logic of the non-volatile memory tile 4956 may be managed by one of device management logic 4917 or driver logic 4918. The network processor tile 4957 can include network processing resources that are coupled to a physical interface within the input/output (I/O) sources 4910 of the computing device 4900. The network processor tile 4957 may be managed by one or more of device management logic 4917 or driver logic 4918.

[0465] In one embodiment, the matrix multiply accelerator 4960 is a modular scalable sparse matrix multiply accelerator. The matrix multiply accelerator 4960 can include multiple processing paths, with each processing path including multiple pipeline stages. Each processing path can execute a separate instruction. In various embodiments, the matrix multiply accelerator 4960 can have architectural features of any one of more of the matrix multiply accelerators described herein. For example, in one embodiment, the matrix multiply accelerator 4960 is a systolic array 3000 that is configurable to operate with a multiple of four number of logical stages (e.g., four, eight, twelve, sixteen, etc.). In one embodiment the matrix multiply accelerator 4960 includes one or more instances of a two-path matrix multiply accelerator 3100 with a four-stage pipeline or a four-path matrix multiply accelerator 3200 with a two-stage pipeline. In one embodiment the matrix multiply accelerator 4960 includes processing elements configured as a scalable sparse matrix multiply accelerator. The matrix multiply accelerator 4960 can be used to accelerate matrix operations performed via XMV extensions, or another compute library that facilitates the acceleration of matrix compute operations. For example, the matrix multiply accelerator 4960 can perform tensor computations for training or inference of the neural network models 4050, 4100 described herein.

[0466] As illustrated, in one embodiment, and in addition to the graphics processor 4904, the computing device 4900 may further include any number and type of hardware components and/or software components, including, but not limited to an application processor 4906, memory 4908, and input/output (I/O) sources 4910. The application processor 4906 can interact with a hardware graphics pipeline to share graphics pipeline functionality. Processed data is stored in a buffer in the hardware graphics pipeline and state information is stored in memory 4908. The resulting data can be transferred to a display controller for output via a display device. The display device may be of various types, such as Cathode Ray Tube (CRT), Thin Film Transistor (TFT), Liquid Crystal Display (LCD), Organic Light Emitting

Diode (OLED) array, etc., and may be configured to display information to a user via a graphical user interface.

[0467] The application processor 4906 can include one or processors, such as processor(s) 102 of FIG. 1 and may be the central processing unit (CPU) that is used at least in part to execute an operating system (OS) 4902 for the computing device 4900. The OS 4902 can serve as an interface between hardware and/or physical resources of the computing device 4900 and one or more users. The OS 4902 can include driver logic for various hardware devices in the computing device 4900. The driver logic can include graphics driver logic 4922, which can include the user mode graphics driver 2326 and/or kernel mode graphics driver 2329 of FIG. 23. The graphics driver logic can include a graphics memory manager 4921 to manage a virtual memory address space for the graphics processor 4904. The graphics memory manager 4921 can facilitate a unified virtual address space that may be accessed by the application processor 4906 and the graphics processor 4904.

[0468] It is contemplated that in some embodiments the graphics processor 4904 may exist as part of the application processor 4906 (such as part of a physical CPU package) in which case, at least a portion of the memory 4908 may be shared by the application processor 4906 and graphics processor 4904, although at least a portion of the memory 4908 may be exclusive to the graphics processor 4904, or the graphics processor 4904 may have a separate store of memory. The memory 4908 may comprise a pre-allocated region of a buffer (e.g., framebuffer); however, it should be understood by one of ordinary skill in the art that the embodiments are not so limited, and that any memory accessible to the lower graphics pipeline may be used. The memory 4908 may include various forms of random-access memory (RAM) (e.g., SDRAM, SRAM, etc.) comprising an application that makes use of the graphics processor 4904 to render a desktop or 3D graphics scene. A memory controller hub, such as memory controller 1416 of FIG. 14, may access data in the memory 4908 and forward it to graphics processor 4904 for graphics pipeline processing. The memory 4908 may be made available to other components within the computing device 4900. For example, any data (e.g., input graphics data) received from various I/O sources 4910 of the computing device 4900 can be temporarily queued into memory 4908 prior to their being operated upon by one or more processor(s) (e.g., application processor 4906) in the implementation of a software program or application. Similarly, data that a software program determines should be sent from the computing device 4900 to an outside entity through one of the computing system interfaces, or stored into an internal storage element, is often temporarily queued in memory 4908 prior to its being transmitted or stored.

[0469] The I/O sources can include devices such as touchscreens, touch panels, touch pads, virtual or regular keyboards, virtual or regular mice, ports, connectors, network devices, or the like, and can attach via a platform controller hub 1430 as referenced in FIG. 14. Additionally, the I/O sources 4910 may include one or more I/O devices that are implemented for transferring data to and/or from the computing device 4900 (e.g., a networking adapter); or, for a large-scale non-volatile storage within the computing device 4900 (e.g., SSD/HDD). User input devices, including alphanumeric and other keys, may be used to communicate information and command selections to graphics processor 4904. Another type of user input device is cursor control,

such as a mouse, a trackball, a touchscreen, a touchpad, or cursor direction keys to communicate direction information and command selections to GPU and to control cursor movement on the display device. Camera and microphone arrays of the computing device 4900 may be employed to observe gestures, record audio and video and to receive and transmit visual and audio commands.

[0470] The I/O sources 4910 can include one or more network interfaces. The network interfaces may include associated network processing logic and/or be coupled with the network processor tile 4957. The one or more network interface can provide access to a LAN, a wide area network (WAN), a metropolitan area network (MAN), a personal area network (PAN), Bluetooth, a cloud network, a cellular or mobile network (e.g., 3rd Generation (3G), 4th Generation (4G), 5th Generation (5G), etc.), an intranet, the Internet, etc. Network interface(s) may include, for example, a wireless network interface having one or more antenna (e). Network interface(s) may also include, for example, a wired network interface to communicate with remote devices via network cable, which may be, for example, an Ethernet cable, a coaxial cable, a fiber optic cable, a serial cable, or a parallel cable.

[0471] Network interface(s) may provide access to a LAN, for example, by conforming to IEEE 802.11 standards, and/or the wireless network interface may provide access to a personal area network, for example, by conforming to Bluetooth standards. Other wireless network interfaces and/or protocols, including previous and subsequent versions of the standards, may also be supported. In addition to, or instead of, communication via the wireless LAN standards, network interface(s) may provide wireless communication using, for example, Time Division, Multiple Access (TDM A) protocols, Global Systems for Mobile Communications (GSM) protocols, Code Division, Multiple Access (CDMA) protocols, and/or any other type of wireless communications protocols.

[0472] It is to be appreciated that a lesser or more equipped system than the example described above may be preferred for certain implementations. Therefore, the configuration of the computing devices described herein may vary from implementation to implementation depending upon numerous factors, such as price constraints, performance requirements, technological improvements, or other circumstances. Examples include (without limitation) a mobile device, a personal digital assistant, a mobile computing device, a smartphone, a cellular telephone, a handset, a one-way pager, a two-way pager, a messaging device, a computer, a personal computer (PC), a desktop computer, a laptop computer, a notebook computer, a handheld computer, a tablet computer, a server, a server array or server farm, a web server, a network server, an Internet server, a work station, a mini-computer, a main frame computer, a supercomputer, a network appliance, a web appliance, a distributed computing system, multiprocessor systems, processor-based systems, consumer electronics, programmable consumer electronics, television, digital television, set top box, wireless access point, base station, subscriber station, mobile subscriber center, radio network controller, router, hub, gateway, bridge, switch, machine, or combinations thereof.

[0473] The techniques described herein relate to a graphics processor including a set of processing resources configured to perform a supersampling anti-aliasing operation

via a mixed precision convolutional neural network. The set of processing resources include circuitry configured to receive, at an input block of a neural network model, a set of data including previous frame data, current frame data, and velocity data. The previous frame data includes one or more previously generated output frames, the current frame data includes output of a raster and lighting stage of a render pipeline, and the velocity data includes motion vectors generated by the render pipeline. The circuitry can pre-process the set of data to generate pre-processed data. To pre-process the set of data includes to generate residual motion vectors via a second neural network model. The residual motion vectors are generated for areas of the previous frame data that do not have associated motion vectors that indicate how to warp those regions. The circuitry can then provide first pre-processed data to a feature extraction network of the first neural network model and second-processed data to an output block of the first neural network model, the first pre-processed data at a first precision and the second pre-processed data at a second precision that is higher than the first precision. The circuitry can then process the pre-processed data at the feature extraction network via one or more encoder stages and one or more decoder stages, output tensor data from the feature extraction network to the output block, and generate an output frame via the output block based on the second pre-processed data from the input block and the tensor data output from the feature extraction network. The output frame is an anti-aliased frame. The circuitry can include a matrix accelerator that is configured to perform matrix operations for the neural network model. The matrix accelerator can include a systolic array.

[0474] In one embodiment, to generate the pre-processed data includes to generate the residual motion vectors via the second neural network based on the current frame data, previous frame data, and the motion vectors, combine the motion vectors and the residual motion vectors to create refined motion vectors, warp the previous frame data based on the refined motion vectors to generate warped history data, align the warped history data with the current frame data to generate aligned history data, and shuffle the aligned history data and current frame data from a spatial dimension to a channel dimension. The channel dimension can include a depth channel or multiple feature map channels. To generate an output frame via the output block includes to generate the output frame via one or more convolution layers of the output block. The generate an output frame via the output block includes to predict a set of per-pixel kernel values and blend weights, filter the current frame data via the per-pixel kernel values, and blend the aligned history data with filtered current frame data. The circuitry is configured to upscale the output of the raster and lighting stage from a first resolution to a second resolution that is higher than the first resolution before the current frame data is provided to the input block of the first neural network model. In one embodiment, to generate the pre-processed data includes to upscale the output of the raster and lighting stage from a first resolution to a second resolution that is higher than the first resolution. In one embodiment, to generate the pre-processed data additionally includes to apply a convolution and an activation function to shuffled aligned history data and current frame data.

[0475] An additional embodiment provides a method to perform the operations of the graphics processor described

above. A further embodiment provides a non-transitory machine-readable medium that stores instructions to perform the operations of the graphics processor described above. A further embodiment provides a data processing system including the graphics processor described above.

[0476] The foregoing description and drawings are to be regarded in an illustrative rather than a restrictive sense. Persons skilled in the art will understand that various modifications and changes may be made to the embodiments described herein without departing from the broader spirit and scope of the features set forth in the appended claims.

What is claimed is:

1. A graphics processing unit, comprising:
memory interface; and
a graphics processing cluster including a plurality of streaming multiprocessors coupled via an interconnect, a streaming multiprocessor of the plurality of streaming multiprocessors including a tensor core configured to perform mixed precision matrix operations, wherein the streaming multiprocessor is configured to:
receive a set of input data for a first frame used by the streaming multiprocessor to generate multiple additional frames, the set of input data including first color data and motion vector data;
execute first matrix operations to upsample the first color data from a first resolution to a second resolution to generate upsampled color data; and
execute second matrix operations to generate at least one of the multiple additional frames based at least in part on the upsampled color data and an optical flow.
2. The graphics processing unit of claim 1, wherein the interconnect is configured to enable exchange of data between the plurality of streaming multiprocessors.
3. The graphics processing unit of claim 1, wherein the set of input data for the first frame includes depth data and the streaming multiprocessor is configured to generate at least one of the multiple additional frames based at least in part on the depth data.
4. The graphics processing unit of claim 1, wherein the first color data includes sample data with a sub-pixel jitter.
5. The graphics processing unit of claim 1, wherein the multiple additional frames comprise at least a second frame.
6. The graphics processing unit of claim 1, wherein the optical flow corresponds to one or more residual motion vectors.
7. The graphics processing unit of claim 6, wherein the motion vector data is to be received from a renderer.
8. The graphics processing unit of claim 7, wherein the one or more residual motion vectors are to provide motion data for a portion of at least one frame for which motion vector data is not provided.
9. The graphics processing unit of claim 8, wherein the streaming multiprocessor is configured to execute third matrix operations to generate the one or more residual motion vectors based on the motion vector data and at least one frame.
10. The graphics processing unit of claim 9, wherein the first matrix operations are associated with a first neural network, the second matrix operations are associated with a second neural network, the third matrix operations are associated with a third neural network, and the third neural network is an image reconstruction network.

11. A method comprising:

receiving, at a streaming multiprocessor of a graphics processing unit, a set of input data for a first frame to generate multiple additional frames, the set of input data including first color data and motion vector data;
executing, using a tensor core of the streaming multiprocessor, first matrix operations to upsample the first color data from a first resolution to a second resolution to generate upsampled color data; and
executing, using the tensor core, second matrix operations to generate at least one of the multiple additional frames based at least in part on the upsampled color data and an optical flow.

12. The method of claim 11, wherein the set of input data for the first frame includes depth data and the streaming multiprocessor is configured to generate at least one of the multiple additional frames based at least in part on the depth data.

13. The method of claim 11, wherein the first color data includes sample data with a sub-pixel jitter.

14. The method of claim 11, wherein the multiple additional frames comprise at least a second frame.

15. The method of claim 11, wherein the optical flow corresponds to one or more residual motion vectors, the motion vector data is to be received from a renderer, and the one or more residual motion vectors are to provide motion data for a portion of at least one frame for which motion vector data is not provided.

16. A graphics processing system, comprising:

a memory device; and
a graphics processing unit coupled with the memory device, the graphics processing unit including memory interface coupled with the memory device and a graphics processing cluster including a plurality of streaming multiprocessors coupled via an interconnect, wherein the interconnect is configured to enable exchange of data between the plurality of streaming multiprocessors, a streaming multiprocessor of the plurality of streaming multiprocessors including a tensor core configured to perform mixed precision matrix operations, wherein the streaming multiprocessor is configured to:
receive a set of input data for a first frame used by the streaming multiprocessor to generate multiple additional frames, the set of input data including first color data and motion vector data;
execute first matrix operations to upsample the first color data from a first resolution to a second resolution to generate upsampled color data; and
execute second matrix operations to generate at least one of the multiple additional frames based at least in part on the upsampled color data and an optical flow.

17. The graphics processing system of claim 16, wherein the set of input data for the first frame includes depth data and the streaming multiprocessor is configured to generate at least one of the multiple additional frames based at least in part on the depth data.

18. The graphics processing system of claim 16, wherein the first color data includes sample data with a sub-pixel jitter.

19. The graphics processing system of claim 16, wherein the multiple additional frames comprise at least a second frame.

20. The graphics processing system of claim 16, wherein the optical flow corresponds to one or more residual motion vectors, the motion vector data is to be received from a renderer, and the one or more residual motion vectors are to provide motion data for a portion of at least one frame for which motion vector data is not provided.

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